Understanding Health Insurance Cost Predictions Using Deep Neural Networks, Monte Carlo Simulation and Shapley Values

by

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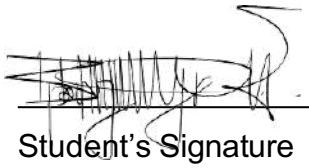
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Finally, I acknowledge the personal discipline, resilience and perseverance required to complete this dissertation. This work represents not only an academic contribution, but also an important milestone in my growth as a researcher in mathematical statistics and actuarial science.

Xitsonga xiri, xihlovo xa le mahlweni imadyisa mbitsi.

Abstract

This dissertation develops a three-stage statistical framework for predicting health insurance costs. Although each technique is individually well established, the combination of DNN prediction, parametric Monte Carlo simulation of policyholder inputs and threshold-conditional Shapley attribution has not previously been demonstrated in health insurance cost prediction; this dissertation addresses that gap by studying the fitted network as a statistical object in its own right. The analysis uses the Kaggle Insurance Data for Machine Learning benchmark dataset, which contains one million policyholder records generated from pre-specified distributions with a deterministic cost formula. Because the dataset has a clean, algorithmically constructed structure, the predictive accuracy reported here is substantially higher than would be expected on real-world insurance data, and the specific performance figures should not be generalised beyond this benchmark setting. The methodology, however, is designed to be transferable to other datasets and domains.

In the first stage, a DNN is trained to predict insurance charges from policyholder characteristics. Its accuracy is compared against three established statistical models: a traditional insurance pricing model, an RF regressor and a gradient-boosted tree ensemble. A five-phase selection process evaluates different network configurations, and the final model is assessed on held-out data to ensure the results reflect genuine predictive ability rather than memorisation.

In the second stage, Monte Carlo simulation generates a large population of policyholder profiles by drawing each attribute independently from a parametric distribution fitted to that attribute in the benchmark data. The simulated profiles therefore follow the marginal behaviour of each input variable but not the dependence between inputs, and the simulation is not intended to reproduce the joint structure of the observed data; it defines a modelling input space over which the fitted network is evaluated, including attribute combinations that do not occur in the data. The trained network then predicts costs for each simulated profile, producing a full distribution of predicted costs rather than predictions at individual data points alone. The consequences of the independence assumption are quantified through a suite of joint-distribution diagnostics, which confirm that the departure is small and that the attribution results are unaffected by it. Because the network achieves a near-perfect fit and each input is drawn from its own fitted marginal, the simulated distribution of predicted costs closely follows the original charge distribution; the primary purpose of this stage is therefore to define the high-cost regions used in the next stage.

In the third stage, each predicted cost is broken down into the individual contributions of policyholder characteristics such as age, smoking status, coverage level and medical

history. This decomposition is performed both across the full simulated population and specifically among the highest-cost profiles, using a method rooted in cooperative game theory that fairly allocates each prediction among the contributing variables.

On this benchmark dataset, the DNN captures over 99% of the variation in insurance costs, outperforming all three comparison models. The contribution analysis identifies smoking status, coverage level and medical history as the strongest overall cost drivers. Among the highest-cost profiles, however, the ranking changes: coverage level, medical history and family medical history become more important than smoking status. Part of this shift is mechanical, because smoking status is near-universal among the highest-cost profiles and therefore loses its ability to distinguish between them; part reflects the genuine explanatory role of the remaining variables within the upper tail of the cost distribution. This finding demonstrates that the factors driving extreme predicted costs are not necessarily the same as those most important on average.

All conclusions are internal to the fitted model and this benchmark dataset. No causal claims are made, and the modelling assumptions are stated explicitly throughout the dissertation.

Keywords: deep neural network, health insurance cost prediction, Monte Carlo simulation, Shapley values, explainable machine learning, generalised linear model, random forest, XGBoost, statistical learning.

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List of Acronyms / Abbreviations

Acronym	Meaning
Adam	Adaptive Moment Estimation
ANN	Artificial Neural Network
BMI	Body Mass Index
CELU	Continuously Differentiable Exponential Linear Unit
DNN	Deep Neural Network
EVT	Extreme Value Theory
GELU	Gaussian Error Linear Unit
GLM	Generalised Linear Model
GP	Gaussian Process
ICE	Individual Conditional Expectation
IRLS	Iteratively Reweighted Least Squares
LOOCV	Leave-One-Out Cross-Validation
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MC	Monte Carlo
MCSE	Monte Carlo Standard Error
MPS	Metal Performance Shaders
MSE	Mean Squared Error
NAdam	Nesterov-accelerated Adaptive Moment Estimation
PCA	Principal Component Analysis
PDP	Partial Dependence Plot
R^2	Coefficient of Determination

Acronym	Meaning
ReLU	Rectified Linear Unit
RF	Random Forest
RMSE	Root Mean Squared Error
SHAP	SHapley Additive exPlanations
XGBoost	eXtreme Gradient Boosting

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1 Introduction

1.1 Background and Problem Statement

Health insurance cost prediction lies at the intersection of statistical learning, actuarial modelling and explainable machine learning. The empirical task involves estimating a continuous monetary outcome from policyholder characteristics such as age, BMI, smoking status, coverage level and medical history. In the insurance industry, these estimates underpin individual pricing, reserve adequacy and portfolio risk management. A single fitted value or prediction-error metric, however, does not fully characterise the behaviour of the fitted model. Understanding how predicted costs vary across a simulated population and identifying which input variables contribute most to high predicted costs are questions that require methods beyond point prediction.

It is therefore important to establish the modelling landscape within which this dissertation operates: the classical actuarial framework based on GLMs, the tree-based ensemble methods that occupy an intermediate position between linear and fully flexible models, the introduction of DNNs into insurance pricing, and the growing demand for post-hoc model explanation. Each of these themes is introduced below, before the specific research problem and analytical framework are stated.

1.1.1 Classical actuarial pricing models

The statistical modelling of insurance costs has a long history. GLMs have served as the standard framework for actuarial pricing since the seminal work of Nelder and Wedderburn (1972) and the framework's subsequent adoption by the insurance profession. In health insurance, the Gamma distribution with a logarithmic link function is commonly used (McCullagh and Nelder, 1989; Wüthrich and Merz, 2023) to model the right-skewed, positive-valued distribution of claim amounts. The Gamma family is well suited to this setting because it is supported on the positive real line, matching the strictly positive nature of insurance costs, and its variance increases with the mean, accommodating the greater dispersion typically observed among larger claims. The logarithmic link function complements this distributional choice by ensuring that predicted mean costs remain strictly positive and by inducing a multiplicative tariff structure in which each covariate enters as a proportional adjustment to the base cost. Under this specification, the conditional mean of the response is given by

$$\log \mu_i = \mathbf{x}_i^\top \boldsymbol{\beta}, \tag{1.1.1}$$

where $\mu_i = E[Y_i \mid \mathbf{x}_i]$ is the conditional mean cost for policyholder i, $\mathbf{x}_i$ is the vector of covariate values and $\boldsymbol{\beta}$ is the coefficient vector. The multiplicative structure of the log-link allows each coefficient to be interpreted as a relative risk factor: exponentiating β_j yields the multiplicative effect of a unit increase in the j-th covariate on the expected cost, holding all other covariates constant. This interpretive simplicity has made GLMs the dominant modelling tool in actuarial practice for several decades.

Despite their advantages, GLMs impose a pre-specified linear predictor structure that restricts the functional form of the relationship between covariates and the response. Interaction effects must be specified manually, and nonlinear relationships can only be captured through explicit basis expansions or spline terms. When the true relationship between policyholder attributes and cost is complex, involving higher-order interactions among multiple categorical and continuous variables, these structural constraints limit the predictive performance of the model. The gap between what a linear predictor can represent and what the data contain motivates the exploration of more flexible model classes.

A further limitation concerns the distributional assumption itself. The Gamma distribution implies a specific variance-mean relationship ($\text{Var}(Y_i) \propto \mu_i^2$), which may not hold uniformly across all segments of the policyholder population. While robust standard errors and alternative distributional assumptions can mitigate this concern, the fundamental limitation of the linear predictor structure remains. As Goodfellow et al. (2016) note, the capacity of a model class to approximate a target function is determined by the richness of the hypothesis space, and the hypothesis space of a GLM is inherently constrained by its additive structure.

Between the constrained linear predictor of the GLM and the fully flexible hidden-layer representations of a DNN lie tree-based ensemble methods, which approximate nonlinear relationships through collections of decision trees. RFs (Breiman, 2001) aggregate predictions across independently grown trees, each trained on a bootstrap sample with a random feature subset at each split, reducing variance while capturing interactions that the GLM cannot represent. Gradient-boosted trees, exemplified by the XGBoost implementation (Chen and Guestrin, 2016), construct an additive ensemble of shallow trees sequentially, with each new tree fitted to the residuals of the preceding ensemble. Both methods offer greater flexibility than the GLM but remain fundamentally piecewise-constant approximators: unlike neural networks, they do not learn smooth, continuous mappings through differentiable transformations. Including an RF and an XGBoost benchmark alongside the GLM and DNN provides a graded comparison across model classes of increasing flexibility. The core comparison remains the actuarial-standard GLM against the DNN, since this dissertation is situated within an actuarial modelling context where the GLM is the established industry baseline and the DNN represents the class of model whose behaviour

the MC and Shapley analyses are designed to examine. The tree-based benchmarks complement this pairing by establishing where piecewise-constant nonlinear methods fall on the performance spectrum.

1.1.2 DNNs for insurance regression

DNNs provide a flexible alternative to the GLM framework. The universal approximation theorem (Hornik et al., 1989; Cybenko, 1989) establishes that a sufficiently wide DNN with a non-polynomial activation function can approximate any continuous function on a compact set to arbitrary precision; the formal statement and its conditions are presented in Section 2.4. The compactness condition is met naturally in this dissertation: all numerical inputs are rescaled to a bounded interval during preprocessing and all categorical inputs are coded as zero-one indicators, so the network always operates within a bounded input space. In practice, modern deep networks with multiple hidden layers and nonlinear activations learn hierarchical representations that capture complex interactions without requiring the modeller to enumerate them in advance (Goodfellow et al., 2016).

This capacity for automatic feature interaction has led to increasing adoption of neural networks in insurance applications. The survey by Richman (2021) and applications such as the deep loss-reserving model of Kuo (2019) demonstrate that neural networks can improve upon classical actuarial models across pricing and reserving tasks, particularly when the data contain nonlinear covariate-response relationships that the GLM cannot capture. The DNN architecture, in which each layer computes $\mathbf{h}^{(\ell)} = \sigma(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)})$ for an activation function σ, weight matrix $\mathbf{W}^{(\ell)}$ and bias vector $\mathbf{b}^{(\ell)}$, is a widely employed architecture for tabular regression problems (Goodfellow et al., 2016; Richman, 2021).

The flexibility of DNNs, however, comes at the expense of direct interpretability. The parameters of a trained network, which may number in the tens of thousands, do not admit the same coefficient-level reading as a linear model. A policyholder with specific attributes receives a predicted cost, but the relative contribution of each attribute to that prediction is not immediately transparent from the network weights. This lack of transparency is a practical concern in regulated industries such as insurance, where stakeholders require not only accurate predictions but also explanations of the factors driving those predictions. The tension between predictive flexibility and model transparency has motivated a substantial body of research on post-hoc explanation methods, of which Shapley-value attribution is one of the most principled.

1.1.3 Beyond point prediction: distributional analysis and feature attribution

Prediction accuracy alone provides an incomplete picture of model behaviour. A test-set metric such as R^2 or RMSE summarises average predictive performance but does not reveal how the predicted cost varies across the input space, what the distributional shape of predicted costs looks like under population-level input variation, or which variables are most important in specific regions of the output distribution (Molnar, 2022; Duncan et al., 2016). Answering these questions requires supplementary analyses that go beyond the standard train-test evaluation paradigm, such as the propagation of an explicit input law through the fitted model by Monte Carlo simulation (Glasserman, 2004; Saltelli, 2002).

The challenge is compounded by the heterogeneity of policyholder populations. An insurance portfolio typically contains a large number of low-cost policyholders and a smaller number of high-cost policyholders whose claims dominate the aggregate loss (Embrechts et al., 1997; Denuit et al., 2007; Bertsimas et al., 2008). From a risk-management perspective, understanding the drivers of predicted costs in this upper tail is at least as important as understanding the average prediction (McNeil et al., 2015; Coles, 2001). Global feature-importance summaries, which aggregate attributions across all predictions, may obscure the factors that are most relevant in the high-cost region (Lundberg et al., 2020; Ribeiro et al., 2016; Covert et al., 2021). A conditional analysis that focuses specifically on the upper tail of the predicted-cost distribution is therefore needed to complement the global view.

This tension between predictive flexibility and interpretability, and between point-prediction accuracy and distributional understanding, motivates the present study. A framework is needed that combines flexible prediction with structured distributional analysis and principled feature attribution, so that the fitted model can be examined not only for its accuracy but also for its behaviour under simulated input variation and for the variables to which it assigns large contributions in the upper tail of predicted costs.

1.2 Positioning within Mathematical Statistics

This dissertation is positioned within statistical learning, simulation-based inference, model-implied predictive distributions, and explainable statistical modelling. The statistical contribution is not only the fitting of a predictive model, but the construction of a model-implied distribution under an explicit input law and the conditional analysis of feature attributions in the upper tail of that distribution. Each of the three methodological pillars is discussed below to situate the work within the broader mathematical-statistics tradition.

1.2.1 Supervised regression and function approximation

The predictive component draws on the theory of supervised regression and neural-network function approximation. Supervised regression seeks a function $\hat{f} : \mathbb{R}^p \to \mathbb{R}$ that minimises a loss functional, typically the expected squared error $E[(Y - \hat{f}(\mathbf{X}))^2]$, over a distribution of input-output pairs $(\mathbf{X}, Y)$, whose minimiser over all measurable functions is the conditional mean $E[Y \mid \mathbf{X}]$ (Hastie et al., 2009; Vapnik, 1998). The DNN is one particular parameterisation of this function class, distinguished by its capacity to represent highly nonlinear mappings through compositions of affine transformations and elementwise activation functions. The universal approximation theorems establish that a feedforward network with a single hidden layer and a suitable nonpolynomial activation can approximate any continuous function on a compact domain to arbitrary accuracy (Cybenko, 1989; Hornik et al., 1989; Leshno et al., 1993), although these results are existence statements and are silent on the network width, the training algorithm and the sample size required to attain a given accuracy in practice (Goodfellow et al., 2016).

The bias-variance decomposition provides the theoretical framework for understanding the trade-off between model complexity and generalisation performance (Hastie et al., 2009; James et al., 2021). A model that is too simple (high bias) will underfit the data, while a model that is too complex (high variance) will overfit (Abu-Mostafa et al., 2012; Bishop, 2006). This classical account is an incomplete description of overparameterised neural networks, which can interpolate the training data and still generalise (Zhang et al., 2021), so the trade-off is treated here as a practical guide to model selection rather than as an exact characterisation of network behaviour. The five-phase hyperparameter comparison employed in this dissertation is designed to navigate this trade-off by selecting the configuration that minimises validation-set error, while the held-out test set provides an unbiased estimate of the generalisation performance (Hastie et al., 2009; Kadra et al., 2021).

1.2.2 MC simulation in statistical inference

The simulation component uses parametric MC methods, one of the foundational tools of computational statistics. MC simulation allows the analyst to explore the distribution of a quantity of interest by drawing random samples from a specified input distribution and propagating them through a deterministic or stochastic transformation. In the classical setting, MC methods are used to study the sampling distribution of estimators, the power of test statistics and the coverage properties of confidence intervals.

In this dissertation, the transformation is the trained neural network prediction function $\hat{f}$, and the input distribution is a product of parametric marginals fitted to the training

data. The resulting distribution of predicted costs is a model-implied distribution that characterises the range, central tendency, dispersion and shape of the network's output under the specified input law. Formally, if $\mathbf{X} \sim F_{\mathbf{X}}$ denotes the input distribution and $\hat{Y} = \hat{f}(\mathbf{X})$ denotes the predicted cost, then the distribution of $\hat{Y}$ under $F_{\mathbf{X}}$ is the object of inferential interest. This construction draws on the simulation-based inference tradition within mathematical statistics, where the behaviour of estimators, test statistics and predictive quantities is routinely studied through MC experiments. A caveat is warranted: because the marginals are fitted to the same data on which the network was trained, and because the network achieves near-perfect fit on the Kaggle benchmark dataset (Kaggle, 2023), the model-implied predicted-cost distribution necessarily recovers the empirical charge distribution almost by construction. The primary role of the MC stage in this application is therefore not to reveal unexpected distributional properties, but to define the high-cost conditioning regions on which the Shapley analysis operates.

1.2.3 Shapley values and cooperative game theory

The explanation component applies Shapley-value decomposition, a concept originating in cooperative game theory (Shapley, 1953), to attribute fitted predictions to individual input variables. In a cooperative game with p players, the Shapley value assigns each player a share of the total payoff by averaging the player's marginal contribution across all possible coalitions. It is the unique additive decomposition satisfying four fairness axioms; the formal definition, the axioms and the equation are presented in Section 2.10.

In the machine-learning interpretability literature, this decomposition has been operationalised through the SHAP framework (Lundberg and Lee, 2017), which interprets each prediction as a cooperative game in which the "players" are input features and the "payoff" is the predicted value. A gradient-based implementation is adopted in this dissertation; the computational considerations behind this choice are set out in Section 3.10.

A terminological note is warranted. Throughout this dissertation, "Shapley values" refers to the theoretical decomposition from cooperative game theory, while "SHAP" refers specifically to the computational framework and software library used to approximate those values. Because the theoretical properties of the Shapley value underpin the interpretation of all attribution results, the term "Shapley values" is used as the default throughout, and "SHAP" appears only when referring to the specific approximation algorithm or implementation.

1.2.4 The integration as the statistical contribution

The choice to combine these three components reflects a broader methodological perspective: that a fitted prediction model is a statistical object whose properties extend beyond its point-prediction accuracy. Just as the sampling distribution of an estimator is studied through MC experiments to understand its variability, the distribution of a prediction function's outputs under a specified input law can be studied to understand the range and shape of its predictions. And just as the influence of individual observations on an estimator can be decomposed through influence functions, the contribution of individual input variables to a prediction can be decomposed through Shapley values.

Each of these components has a well-established theoretical basis in mathematical statistics. The contribution of this dissertation lies in their integration within a single empirical workflow applied to health insurance cost prediction. The MC distribution defines the regions of interest for the Shapley analysis, and the conditional Shapley analysis decomposes the model behaviour within those regions. This integration produces insight that none of the three components could provide in isolation.

1.3 Conceptual Framework

The analytical workflow of this dissertation is organised around three sequentially connected stages: prediction, simulation and explanation. Each stage produces outputs that serve as inputs to the next, creating an integrated framework in which the fitted model, the simulated input population and the feature attributions are jointly analysed.

1.3.1 Research design

The study follows a quantitative, non-experimental research design applied to secondary data. No treatment is assigned, no intervention is observed and no counterfactual is constructed, so every quantity reported is a property of a fitted statistical model rather than of the data-generating process itself; the interpretive consequences of this choice are stated in Section 1.8. The object of study is the fitted prediction function, and the design is therefore computational: conclusions are drawn from out-of-sample performance on a held-out partition, from MC propagation of an explicit input law through the fitted function, and from an axiomatic attribution of the resulting predictions to input variables.

The empirical base is the Kaggle Insurance Data for Machine Learning benchmark dataset (Kaggle, 2023), comprising $n = 1\,000\,000$ policyholder records with eleven predictors and the continuous monetary response `charges`. Reference-category dummy encoding expands the categorical predictors to a model-ready input dimension of $p = 22$. The unit of

analysis is the individual policyholder record at the prediction stage and the synthetic policyholder profile at the simulation and explanation stages. The dataset is partitioned into training, validation and test subsets in an approximate 52/24/24 split, with the training partition fixed at $2^{19} = 524\,288$ records for batch alignment. Every preprocessing quantity, encoding map, standardisation constant, fitted marginal and Shapley background sample is estimated on the training partition alone and applied without refitting to the remaining partitions, so that the validation and test data remain uncontaminated throughout all three stages. The full specification is given in Chapter 3; the present section states only the methodological logic that connects the stages.

1.3.2 The three stages

In the first stage, a feedforward DNN (Goodfellow et al., 2016) is trained on the encoded training partition by minimising mean squared error, using backpropagation with an adaptive first-order optimiser. Because the fitted surface depends materially on the training configuration, and because tuning is what determines whether a network is competitive on tabular data (Kadra et al., 2021), the configuration is selected by a five-phase sequential comparison in which hidden activation, learning rate, batch size, optimiser and momentum parameters are compared one factor at a time, with the best choice carried forward and all other factors held fixed. Selection is made on validation-set error and the retained configuration is scored once on the held-out test set, so that the reported generalisation estimate is not contaminated by the selection process (Hastie et al., 2009). The trained network defines a prediction function $\hat{f} : \mathbb{R}^p \to \mathbb{R}$ that maps encoded policyholder attributes to predicted costs.

Three benchmark models are fitted on the same partitions to establish whether the added flexibility of the network is warranted: a Gamma GLM with a logarithmic link (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989), the standard actuarial model for positive, right-skewed claim severities (Denuit et al., 2007; Wüthrich and Merz, 2023); an RF regressor (Breiman, 2001); and an XGBoost gradient-boosted tree regressor (Chen and Guestrin, 2016). The tree ensembles are the appropriate comparators because they remain strong baselines on tabular problems (Grinsztajn et al., 2022; Shwartz-Ziv and Armon, 2022). Before the network is carried into the simulation stage it must pass a battery of predictive and structural diagnostics, comprising error metrics on all three partitions, actual-versus-predicted calibration, residual diagnostics, principal-component structure and correlation checks; this gate is what licenses the downstream use of $\hat{f}$ as the object of study. The output of this stage is a validated prediction function with documented out-of-sample performance across all four models.

In the second stage, a parametric marginal distribution is fitted to each raw input variable

on the training partition, with the family chosen from the exploratory diagnostics so that the fitted support respects the observed support of the variable: bounded numerical variables receive support-respecting continuous families, binary variables are generated as Bernoulli draws, and multi-level categorical variables are generated from the observed training proportions. An MC sample of $B = 100\,000$ synthetic policyholder profiles is then drawn from the product of these marginals (Rubinstein and Kroese, 2016; Robert and Casella, 2004), encoded under the same fixed preprocessing map as the training data, and passed through $\hat{f}$, yielding a sample of B predicted costs. Writing $\mathbf{X} \sim F_{\mathbf{X}}$ for the input law and $\hat{Y} = \hat{f}(\mathbf{X})$, the empirical distribution of the B draws is a consistent estimator of the law of $\hat{Y}$ under $F_{\mathbf{X}}$, and this model-implied distribution, not the data distribution, is the estimand of the stage.

Its mean, variance, skewness and tail behaviour are analysed through summary statistics, density estimation and convergence diagnostics. The simulation size is not assumed adequate but established: the simulation is repeated over a pilot ladder of increasing B, and the MC standard error for mean-type summaries is monitored against the $\mathcal{O}(B^{-1/2})$ rate (Caflisch, 1998; Glasserman, 2004; Owen, 2013), while quantile stability, which that formula does not govern, is assessed descriptively across the ladder. The product-of-marginals construction deliberately imposes independence, and sampling off the data manifold is a known source of distortion in model-based analysis (Aas et al., 2021; Hooker et al., 2021). This assumption is therefore treated as a claim to be tested rather than granted: it is examined against the training joint distribution through the diagnostics set out in Section 1.7, and a residual-augmented sensitivity analysis (Saltelli, 2002), in which scaled draws from the fitted residual distribution are added to the simulated predictions, tests whether the distributional conclusions survive the reintroduction of the variability the deterministic prediction function discards.

In the third stage, Shapley values (Shapley, 1953; Lundberg and Lee, 2017) are computed for each of the B simulated profiles, decomposing each predicted cost into additive contributions from the input variables. Exact evaluation requires the prediction function at all 2^p feature subsets, which is infeasible at $p = 22$, so a gradient-based explainer is used that exploits the differentiability of the fitted network against a background sample drawn from the training partition (Castro et al., 2009). The attributions are consequently approximate and conditional on that background sample and on the explainer algorithm, and their precise meaning depends on the chosen reference distribution (Sundararajan and Najmi, 2020; Covert et al., 2021); they are interpreted throughout as model-specific attributions rather than exact cooperative-game solutions. Contributions computed on the $p = 22$ encoded columns are aggregated back to the raw-variable level, so that the dummy indicators of a single categorical variable are reported as one attribution.

The attributions are summarised in two ways. The global summary is the mean absolute

contribution of each variable across all B profiles (Lundberg et al., 2020). The conditional summary restricts the same average to the high-cost set $\mathcal{H}_\tau$ of profiles whose predicted cost exceeds the τ-quantile of the simulated predicted-cost distribution, with $\tau = 0.95$ as the primary threshold and $\tau \in \{0.90,\ 0.99\}$ retained to test whether the pattern is an artefact of the cutoff (McNeil et al., 2015). Because the conditioning set at $\tau = 0.99$ contains only $0.01B$ profiles, a percentile bootstrap over the conditional attributions quantifies the resulting estimation uncertainty, so that any reported change in ranking is separated from sampling noise. Comparing the global and conditional summaries reveals how the feature-importance ranking changes as the focus shifts from average to extreme predicted costs.

This three-stage structure ensures that the dissertation answers not only "How well does the model predict?" but also "What does the model's output distribution look like under population-level input variation?" and "Which variables matter most, and does their relative importance change in the tail?"

The three stages are sequential rather than parallel, and the dependence is one-directional: the simulation stage is meaningful only if the prediction function has passed its diagnostic gate, and the attribution stage is meaningful only if the simulated input law has survived its validation diagnostics, since the high-cost conditioning regions are defined on the simulated predicted-cost distribution. A failure at an earlier stage therefore invalidates the stages that follow, which is why each stage is gated by explicit acceptance checks before its output is carried forward. Research Question 1 is answered at the prediction stage, Research Questions 2 to 4 at the simulation stage, and Research Questions 5 and 6 at the explanation stage, with the validation and robustness strategy of Section 1.7 cutting across all three. Chapter 3 gives the full specification of every step summarised here.

1.4 Significance of the Study

The study contributes to several intersecting areas within mathematical statistics and actuarial science.

From a methodological perspective, the three-stage workflow, whose integrated character is argued in Section 1.2.4, is demonstrated in a single reproducible framework that is applicable to any regression problem where distributional analysis and feature attribution in specific output regions are of interest. While each component is individually well established, and the three-stage combination has been applied in other domains (Santos et al., 2023), its application to health insurance cost prediction has received limited attention in the literature.

From a practical perspective, the conditional Shapley analysis at upper-tail thresholds ad-

dresses a question of direct relevance to actuarial practice: which variables drive the most expensive predicted claims? By computing conditional Shapley summaries at multiple thresholds, the study provides a graded view of how feature importance evolves into the tail. This information is relevant to risk selection, pricing segmentation and reinsurance analysis, where the focus is often on the upper tail of the cost distribution.

From a statistical-learning perspective, the construction of a model-implied predicted-cost distribution under an explicit input law connects the neural network prediction function to the broader tradition of simulation-based inference (Section 1.2.4). Rather than evaluating the model only at observed data points, the MC component generates a distribution of predicted costs that maps the range and shape of the model's output landscape. As noted in Section 1.2.2, this distribution is not claimed to uncover distributional information absent from the data: because the marginals are fitted to the training data and the network attains a near-perfect fit, the simulated predicted-cost distribution largely recovers the empirical charge distribution by construction. The contribution of the stage is that it defines, on an explicit and reproducible input law, the high-cost conditioning regions on which the attribution analysis operates.

Finally, from an explainability perspective, the study demonstrates that Shapley-value attribution, often presented as a single global summary, can be extended to conditional summaries that provide region-specific explanations. The ranking reversal observed in the upper tail of the predicted-cost distribution illustrates that a single global importance ranking may not adequately represent the model's behaviour in all regions of the output space. This finding contributes to the growing literature on context-dependent model explanation.

1.5 Research Questions

The background and problem statement outlined above motivate a set of research questions that span the three stages of the analytical framework. These questions are formulated to be answerable within the scope of the benchmark data and the fitted model, without requiring causal interpretation. Each question addresses a specific aspect of the prediction, simulation and explanation framework.

The dissertation addresses the following research questions:

1. How does a DNN regression model compare with a classical GLM, an RF regressor and an XGBoost gradient-boosted-tree regressor in predicting health insurance costs on a benchmark dataset?

2. How adequately does parametric MC simulation of policyholder profiles reproduce

the model-implied predicted-cost distribution, and how effectively does it define the high-cost conditioning regions required for tail-focused attribution?

3. Is the product-marginal MC input distribution a defensible approximation of the training-data joint distribution for the purpose of generating simulated profiles?

4. How sensitive is the predicted-cost distribution to residual perturbation from the fitted model?

5. Which input variables receive the largest Shapley-value attributions globally, and how does the feature-importance ranking change conditionally within high-cost regions of the predicted-cost distribution?

6. What explains any differences between the global and conditional feature-importance rankings in the upper tail?

Research Question 1 addresses the predictive stage. Research Questions 2 to 4 address the simulation stage, covering the adequacy of the simulated predicted-cost distribution, the input-distribution validity and the residual-perturbation robustness respectively. Research Questions 5 and 6 address the explanation stage, covering both the global and conditional Shapley analyses and the interpretation of any observed ranking changes.

1.6 Research Objectives

The research questions stated above are operationalised through the following objectives, each of which maps directly to one or more research questions:

1. To fit a DNN regression model to a health insurance benchmark dataset and to evaluate its predictive performance against a classical GLM, an RF regressor and an XGBoost gradient-boosted-tree regressor.

2. To design and implement a parametric MC simulation that generates synthetic policyholder profiles from fitted marginal distributions, propagates them through the trained network, and to assess how adequately the simulation reproduces the model-implied distribution of predicted costs and how effectively it defines the high-cost conditioning regions required for tail-focused attribution.

3. To validate the MC input distribution against the observed training data using joint-distribution diagnostics, bootstrap resampling, and a Shapley attribution invariance check.

4. To conduct a residual-augmented MC sensitivity analysis that examines the effect of residual perturbation on the predicted-cost distribution.

5. To compute Shapley-value attributions for simulated policyholder profiles and to summarise feature importance at the global level and conditionally within high-cost regions defined by upper-tail thresholds.

6. To examine how the conditional feature-importance ranking in high-cost regions differs from the global ranking, and to interpret these differences within the context of the fitted model and benchmark data.

1.7 Validation and Robustness Strategy

A central concern in any modelling study is whether the reported results are genuine reflections of the model's behaviour or artefacts of data contamination, distributional mismatch or estimation instability. This dissertation addresses this concern through a multi-layered validation strategy that spans all three stages of the analytical framework.

At the prediction stage, the validation strategy consists of three elements. First, a strict data-leakage prevention protocol ensures that all preprocessing parameters, including means, standard deviations, category mappings and normalisation constants, are computed exclusively on the training partition and applied without refitting to the validation and test partitions. This guarantees that the model has no access to information from the evaluation data during training. Second, the five-phase hyperparameter comparison uses validation-set performance as the selection criterion, while the final performance metrics are reported on the held-out test set that is never used for model selection. Third, the comparison of training-set and test-set performance metrics for both the DNN and the GLM provides evidence on the presence or absence of overfitting.

At the simulation stage, the validation strategy addresses the central question of whether the independent-marginal MC input distribution is a defensible approximation of the training-data joint distribution. This is assessed through four complementary diagnostics: pairwise correlation comparisons, categorical cross-tabulation comparisons, PCA projection overlays, and a formal energy distance permutation test. In addition, convergence diagnostics monitor the stability of the predicted-cost distribution summary statistics as the simulation size increases, confirming that $B = 100\,000$ is sufficient for the level of MC precision required. The residual-augmented sensitivity analysis adds a further robustness layer by examining whether the distributional conclusions change when scaled fitted residuals are incorporated into the predicted costs.

At the explanation stage, the validation strategy uses a Shapley attribution invariance check. This diagnostic compares the feature-importance rankings obtained when Shapley values are computed on training-data input profiles with those obtained on MC input profiles. Agreement between the two regimes confirms that the Shapley results are not

sensitive to the choice of input regime and that the MC input distribution produces attributions consistent with those obtained from observed data.

A final validation layer spans all three stages: the network is retrained under five independent random seeds and the complete simulation and attribution pipeline is replicated for each, so that the predictive metrics, the predicted-cost distribution and the Shapley rankings can be checked for sensitivity to the stochastic training process.

The details of each validation component are documented in the methods chapter (Chapter 3), and the corresponding results are reported in Chapter 4.

1.8 Scope and Delimitations

The following delimitations define the boundaries of the study:

- **Benchmark data.** The analysis uses the Kaggle Insurance Data for Machine Learning benchmark (Kaggle, 2023), a publicly available benchmark dataset containing one million policyholder records. The dataset was generated by sampling eleven predictor variables (age, sex, BMI, number of children, smoking status, region, exercise frequency, occupation, coverage level, medical history and family medical history) from pre-specified distributions and computing the response variable (insurance charges) through a deterministic formula that combines these predictors with fixed coefficients and interaction terms. Because the response is constructed algorithmically rather than observed from real claims, the covariate-response relationship is largely noise-free, which is why the fitted models achieve unusually high predictive accuracy. The results describe relationships learned by the fitted model within this benchmark setting and should not be interpreted as causal determinants of healthcare costs or as direct evidence of real-world pricing practice. The performance figures should not be extrapolated to real-world insurance portfolios, where data are noisier and covariate-response relationships are more complex.

- **Focused model comparison.** Three benchmark models (a Gamma GLM, an RF regressor and an XGBoost gradient-boosted-tree regressor) are included to contextualise the predictive performance of the DNN across model classes of increasing flexibility. However, the dissertation does not attempt an exhaustive model-selection study. Bayesian methods, support vector machines, other neural-network architectures and stacked-ensemble approaches are not evaluated. The comparison is designed to position the DNN within a spectrum from linear to piecewise-constant to smooth nonlinear models, rather than to identify the globally optimal model class.

- **Single primary network.** The main MC and Shapley analyses are applied to

a network trained with a fixed random seed, so the reported predicted-cost distribution and feature attributions are conditional on that fitted model. Sensitivity to the stochastic training process is assessed through a five-seed robustness analysis (Chapter 4), which retrains the network under independent initialisations and replicates the simulation and attribution pipeline; an exhaustive exploration of the initialisation space is not attempted.

- **Independent marginals.** The MC input distribution is constructed as a product of independent fitted marginals; copula-based joint distributions are not considered. The rationale and empirical justification for this choice are discussed in Sections 2.7 and 4.6.1.

- **Approximate Shapley values.** The reported attributions are gradient-based SHAP approximations, not exact game-theoretic Shapley values. They are conditional on the fitted network, the background sample, and the explainer settings. Exact Shapley computation is infeasible for the number of features and the sample size considered.

- **No causal inference.** The Shapley decomposition identifies variables to which the fitted model assigns large contributions. It does not establish causal relationships between input variables and healthcare expenditure. All attributions are internal to the fitted model and the benchmark data.

1.9 Ethical Considerations

The analysis is conducted on the Kaggle Insurance Data for Machine Learning benchmark dataset (Kaggle, 2023), which is publicly available and contains no personally identifiable information. The dataset does not represent actual policyholder records from any insurance company. No data collection involving human participants was conducted for this study, and no ethical clearance from a research ethics committee was required. The proposal was nonetheless reviewed by the Departmental Research Committee of the Department of Mathematical Statistics and Actuarial Science, which confirmed its scientific merit and recommended it for ethics approval or waiver (project number UFS-HSD2026/1253).

Notwithstanding the use of benchmark data, the dissertation acknowledges the broader ethical considerations associated with the use of machine learning in insurance pricing. Predictive models that use sensitive attributes such as gender, age or health status as input variables may raise fairness concerns if deployed in production pricing systems. The Shapley-value analysis presented in this dissertation quantifies the contribution of each variable to the fitted model's predictions, which could in principle inform fairness audits by identifying variables that exert disproportionate influence on predicted costs. However,

the study does not conduct a formal fairness analysis, and the Shapley attributions should not be interpreted as evidence of discriminatory pricing.

All modelling code, trained model artefacts and simulation outputs are documented to support reproducibility. The use of a fixed random seed throughout the framework ensures that all results can be replicated by independent researchers with access to the same dataset and software environment.

1.10 Dissertation Outline

The remainder of this dissertation is organised as follows.

Chapter 2 (Literature Review) reviews the theoretical and empirical foundations of the DNN, MC and Shapley framework. The review covers statistical learning theory and supervised regression, DNN architectures and their approximation properties, GLMs as actuarial baselines, health insurance cost prediction with machine learning, MC simulation methodology, Shapley-value attribution theory and the SHAP framework, and conditional feature-importance analysis. The chapter concludes with a formal identification of the research gap that motivates the present study.

Chapter 3 (Methods and Data) describes the benchmark dataset, the data preprocessing and encoding procedure, the exploratory data analysis, the network architecture and training protocol, the five-phase hyperparameter comparison, the predictive and structural diagnostics, the GLM and tree-based benchmark specifications, the MC simulation design and input-distribution validation, the residual-augmented sensitivity analysis, and the Shapley-value implementation. All methodological choices are documented with their statistical justification, and a consolidated table of modelling assumptions is provided.

Chapter 4 (Results) presents the empirical results of the three-stage workflow. The chapter reports the hyperparameter comparison outcomes, the benchmark comparison against the GLM, RF and XGBoost models in both default and tuned configurations, the multi-seed robustness analyses of the DNN and the benchmark models, the MC predicted-cost distribution and its convergence diagnostics, the joint-distribution validation results, the residual-augmented sensitivity analysis, the global and conditional Shapley-value attributions, the feature-importance ranking reversal in the upper tail, the Shapley attribution invariance check, and the SHAP interaction analysis.

Chapter 5 (Conclusion), the final chapter, summarises the key findings in relation to each research question, states the contributions and strengths of the study, discusses limitations, examines implications for actuarial practice and statistical methodology, and identifies directions for future research.

2 Literature Review

2.1 Introduction

Chapter 1 introduced the tension between predictive flexibility and interpretability in health insurance cost prediction and outlined a three-stage framework that combines DNN regression, MC simulation and Shapley-value attribution. This chapter develops the conceptual and theoretical basis for that framework by reviewing the relevant literature in a sequence that mirrors the framework itself: from prediction to simulation and then to explanation.

The review begins with the integrated DNN, MC and Shapley framework (Section 2.2), establishing why prediction accuracy alone provides only a partial account of model behaviour and why distributional analysis and feature-level attribution are needed as complementary tools. Subsequent sections develop each component in detail: supervised regression and neural-network approximation theory supply the predictive formulation; MC methods extend the fitted prediction function from individual point estimates to a model-implied cost distribution; and Shapley-value decomposition attributes individual predictions to input variables, with particular attention to profiles in the upper tail of the predicted-cost distribution.

The fitted prediction function maps a policyholder profile to a predicted cost. It is interpreted as a statistical learning function fitted to the preprocessed benchmark data, not as a causal model of healthcare expenditure and not as evidence of real-world insurer pricing rules. A DNN is useful in this setting because it can represent nonlinearities and interactions among policyholder attributes without requiring all interaction terms to be specified manually, as would be necessary in a GLM or polynomial regression (Goodfellow et al., 2016; Hastie et al., 2009). Such flexibility is relevant where predicted costs depend on combinations of demographic, behavioural and health-related variables rather than on isolated predictors alone (Orji and Ukwandu, 2024; Samiuddin et al., 2023). However, the use of a DNN does not imply that neural networks are automatically superior to GLMs, tree-based models or other actuarial models for tabular insurance data (Richman, 2021). Its role is to provide a nonlinear fitted prediction surface that can be evaluated, simulated and interpreted within the proposed framework.

The research gap that motivates the dissertation follows from the limited integration of these components in the existing literature. This gap is developed through the reviewed studies in Section 2.2 and stated formally in Section 2.11.

2.2 DNN-MC-Shapley Framework

Health insurance cost prediction provides a natural setting for the three-stage framework developed in this dissertation. The empirical task can be framed as a supervised statistical-learning problem in which a vector of policyholder attributes is used to approximate a continuous monetary outcome (Hastie et al., 2009; James et al., 2021). A DNN supplies a flexible nonlinear regression function for modelling the relationship between policyholder attributes and cost (Goodfellow et al., 2016; Hornik et al., 1989; Richman, 2021; Samiuddin et al., 2023). The use of a DNN in this role is motivated by the universal approximation property: DNNs with suitable activation functions can approximate broad classes of continuous functions (Hornik et al., 1989; Goodfellow et al., 2016), so that the nonlinear relationship between policyholder attributes and cost does not need to be specified in advance by the researcher. The formal statement of this property and its conditions is presented in Section 2.4.

Once the network has been fitted, its behaviour across a broad range of input configurations is examined through MC simulation. MC simulation is a long-established framework for analysing the behaviour of complex functions by generating samples from a specified input distribution and evaluating the function at each sampled point (Glasserman, 2004; Rubinstein and Kroese, 2016; Owen, 2013). Its central advantage is that direct analytical evaluation can be replaced with repeated numerical evaluation over a large synthetic sample, so that the function is studied indirectly through its simulated outputs. The resulting outputs form an empirical distribution that can be summarised using descriptive quantities such as the mean, variance, quantiles and upper-tail probabilities. This approach is particularly useful when the function of interest has no convenient closed-form representation, as is often the case for nonlinear fitted models such as DNNs (Goodfellow et al., 2016; Hornik et al., 1989). Although such models define an explicit computational mapping from inputs to outputs, their fitted response surfaces are typically too complex to be examined using closed-form analytical arguments. In the framework adopted here, MC simulation is used to generate a large synthetic sample of policyholder profiles from parametric distributions fitted to the observed training attributes. The fitted network is then evaluated at each simulated profile to obtain a model-implied distribution of predicted costs.

Shapley values provide the attribution layer for the simulated output space. Originally introduced in cooperative game theory by Shapley (1953) and subsequently adapted to the explanation of machine-learning predictions (Lundberg and Lee, 2017; Rozemberczki et al., 2022), Shapley values decompose an individual fitted prediction into additive contributions attributed to each input variable. This decomposition is particularly valuable when the fitted model is too complex for its parameters to admit direct interpretation.

In a linear regression, individual coefficients describe the contribution of each predictor to the fitted response. In a nonlinear model such as a DNN, the relationship between inputs and predictions is distributed across many layers and parameters (Goodfellow et al., 2016), so that no single parameter captures the role of a given input variable. Shapley values address this difficulty by providing a model-agnostic attribution that satisfies formal fairness axioms (stated in Section 2.10) regardless of the internal structure of the fitted function (Lundberg and Lee, 2017; Rozemberczki et al., 2022). In this dissertation, Shapley values are applied to the simulated predictions, with particular attention given to profiles in the high-cost region of the predicted-cost distribution, namely profiles whose fitted predictions exceed a chosen upper-tail threshold, as formalised in Section 2.2.4.

The hierarchical rationale for combining the three components was set out in the conceptual framework of Section 1.3; the precise distributional choices and simulation procedure are specified in Chapter 3.

Predictive accuracy provides only a partial description of model behaviour. Measures such as RMSE, MAE and R^2 are informative about point-prediction performance, but they do not show how fitted costs vary across a simulated population or which variables are associated with high predicted costs. A broader assessment requires both distributional analysis of fitted predictions and feature-level attribution of those predictions.

At the broader actuarial level, Richman (2021) surveys recent developments in artificial intelligence and deep learning in actuarial science and concludes that neural-network methods are increasingly being adopted across pricing, reserving and risk-classification tasks, typically supplementing rather than replacing the established generalised linear modelling tradition. The review supports the choice of a DNN as the prediction function in the framework adopted here, but it is positioned at the level of a methodological survey and does not prescribe how a fitted network should be examined once it has been trained. In particular, it does not consider the combination of simulation-based exploration of the fitted prediction surface with feature-level attribution as a single actuarial workflow, which is the methodological gap that the proposed framework addresses.

A closer comparison is provided by Orji and Ukwandu (2024), who apply machine-learning regressors together with SHAP and ICE plots to predict medical insurance premiums on the same family of policyholder attributes used here and identify smoker status, age and BMI as the strongest contributors to the predicted premium. Their use of SHAP for feature-level attribution is consistent with the attribution layer adopted here, and their finding that smoker status and age dominate the explanation of high premiums is in line with what would be expected for the same family of inputs in the present setting. The relevant difference lies in the input space over which attribution is performed. Their analysis is conducted on the observed training and test data alone, so the resulting attributions describe the fitted model's behaviour on profiles that actually appear in

the data. The framework adopted here retains SHAP for attribution but applies it to a larger synthetic policyholder population generated by MC simulation, which allows attribution to be examined at policyholder configurations that are not individually present in the training data, and in particular among the simulated profiles with the largest fitted predictions.

The closest deep-learning-based precedent in health insurance cost prediction is Samiuddin et al. (2023), who fit ANN and DNN models to the same family of attributes used here, namely age, gender, BMI, number of children, smoking behaviour and region, and report predictive performance comparable to benchmark regression models on held-out data. Their results support the use of a DNN as a viable prediction function for health insurance cost on policyholder-level attributes, which is consistent with the role assigned to the DNN in the predictive layer of the framework adopted here. The point of departure is the scope of the analysis. Their study is restricted to the point-prediction stage of the workflow: the fitted network is evaluated against held-out data using standard predictive metrics, but no simulation is conducted over a synthetic input space and no feature-level attribution of the fitted predictions is reported. The framework developed here builds on their use of a DNN for cost prediction and extends it with the simulation and attribution stages that their study does not consider.

Taken together, these studies establish that DNN-based prediction and Shapley-based attribution are individually well supported in the health insurance cost-prediction literature, and that the choice of a DNN for the prediction layer is justified by recent evidence. The integration of these components into a single simulation-based workflow has, however, received limited attention in the insurance pricing domain.

The closest methodological precedent appears in an applied project-management context. Santos et al. (2023) implement a three-stage structure combining MC simulation, machine-learning prediction and Shapley-based attribution in an earned-value project-control setting: MC simulation generates a large synthetic sample of input configurations representing plausible project states, a machine-learning model approximates the relationship between those inputs and a continuous project-performance outcome, and Shapley-based explanations attribute the resulting predictions to individual input variables. The framework adopted here applies the same three-stage logic to health insurance cost prediction: a fitted DNN provides the prediction surface, MC simulation generates synthetic policyholder profiles from parametric distributions fitted to the observed training attributes and propagates them through the network, and Shapley values decompose the resulting predictions into feature-level contributions, concentrating on the upper tail of the simulated output. The primary difference is the empirical domain and the choice of prediction model: the present study uses a DNN rather than a general machine-learning regressor and applies the workflow to policyholder-level health insurance cost data rather

than project-performance metrics. This adaptation remains subject to the benchmark-data limitations stated above, namely that the conclusions are internal to the modelling workflow and are not direct evidence of real-world pricing practice.

Figure 1 summarises the framework. The predictive layer fits the DNN to the preprocessed benchmark data. The distributional layer generates simulated policyholder profiles and propagates them through the fitted network. The attributional layer decomposes fitted predictions into Shapley contributions. These layers converge in the high-cost analysis, where simulated upper-tail predictions are interpreted through conditional Shapley summaries.

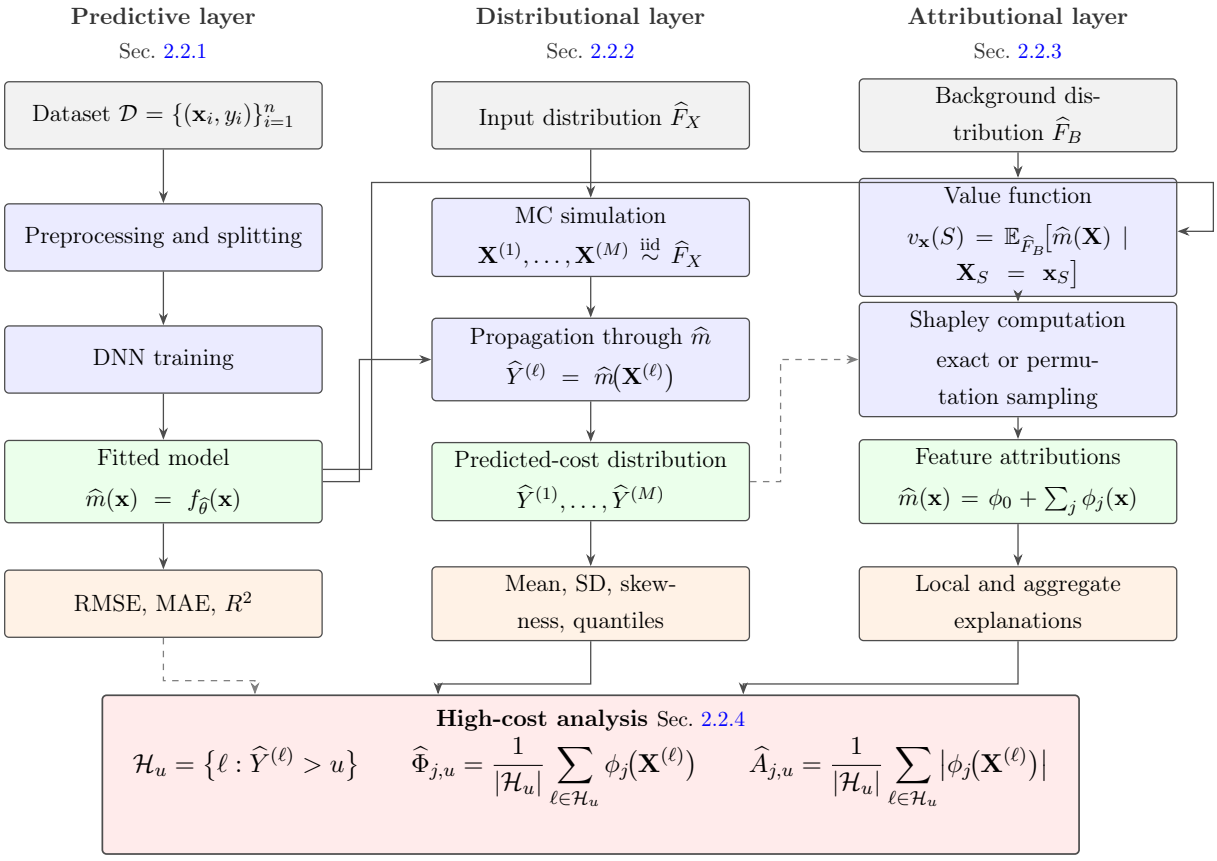


Figure 1: Prediction, Simulation and Explanation framework.

2.2.1 Predictive modelling

Let

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n \tag{2.2.1}$$

denote the preprocessed insurance benchmark dataset, where $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})^\top \in \mathbb{R}^p$ is the vector of policyholder attributes for observation i, and $y_i \geq 0$ is the corresponding

insurance cost, charge or premium. The regression formulation is

$$y_i = m(\mathbf{x}_i) + \varepsilon_i, \qquad \mathbb{E}(\varepsilon_i \mid \mathbf{x}_i) = 0, \tag{2.2.2}$$

where $m(\cdot)$ denotes the unknown regression function and ε_i represents variation not explained by the observed predictors (Hastie et al., 2009; James et al., 2021). A DNN with trainable parameter vector θ is fitted to approximate $m(\cdot)$, yielding

$$\widehat{m}(\mathbf{x}) = f_{\widehat{\theta}}(\mathbf{x}). \tag{2.2.3}$$

For a DNN with L hidden layers, the parameter vector θ collects all weight matrices $\mathbf{W}^{(1)}, \ldots, \mathbf{W}^{(L+1)}$ and, where included, bias vectors $\mathbf{b}^{(1)}, \ldots, \mathbf{b}^{(L+1)}$ across the layers (Goodfellow et al., 2016). The total number of trainable parameters depends on the layer widths and on whether bias terms are included. In this dissertation, bias vectors are omitted from every layer (see Table 6), so θ reduces to the weight matrices alone. This architectural meaning of bias is distinct from statistical bias in the bias-variance trade-off (Hastie et al., 2009) and from data bias arising from the dataset, sampling mechanism or preprocessing procedure.

2.2.2 MC simulation

The MC stage studies how the fitted prediction surface $\widehat{m}(\mathbf{x})$ defined in equation (2.2.3) responds to variation in the input attributes. Let $\mathcal{D}_{\text{train}}$ denote the training partition of the preprocessed benchmark data and define

$$\mathcal{X}_{\text{train}} = \left\{\mathbf{x}_i : (\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{train}}\right\} \tag{2.2.4}$$

as the corresponding set of training attribute vectors. To simulate over the input space, a probability distribution must be specified for generating synthetic policyholder profiles. This distribution, denoted $\widehat{F}_X$, is estimated from $\mathcal{X}_{\text{train}}$ and constructed as a product of independent fitted marginals,

$$\widehat{F}_X = \prod_{j=1}^{p} \widehat{F}_{X_j}, \tag{2.2.5}$$

where p denotes the number of input attributes, that is the dimension of the attribute vector $\mathbf{x}$, and each $\widehat{F}_{X_j}$ is a parametric distribution fitted to the observed values of the j-th attribute in $\mathcal{X}_{\text{train}}$. The independence assumption in equation (2.2.5) is a deliberate modelling choice: the purpose of the simulation is to explore how the fitted network responds across a broad range of covariate combinations, including combinations that may not appear in $\mathcal{X}_{\text{train}}$, rather than to reproduce the observed joint distribution of policyholders. $\widehat{F}_X$ is not assumed to coincide with the true population distribution of

policyholders; it is the modelling object that defines the simulated input space. The distributional family, fitted parameter values, support restrictions and encoding rules for each attribute are specified in Chapter 3.

Given $\widehat{F}_X$, the procedure draws M synthetic policyholder profiles,

$$\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(M)} \overset{\text{iid}}{\sim} \widehat{F}_X, \tag{2.2.6}$$

and evaluates the fitted network at each profile to obtain the corresponding predicted costs,

$$\widehat{Y}^{(\ell)} = \widehat{m}\big(\mathbf{X}^{(\ell)}\big), \qquad \ell = 1, \ldots, M. \tag{2.2.7}$$

The collection $\{\widehat{Y}^{(\ell)}\}_{\ell=1}^{M}$ is the model-implied predicted-cost distribution under $\widehat{F}_X$ (Glasserman, 2004; Rubinstein and Kroese, 2016; Owen, 2013).

Two points of interpretation follow directly from the construction above. First, the output of the simulation is conditional on both $\widehat{m}$ and $\widehat{F}_X$. The simulation does not generate realised healthcare costs; it generates fitted predictions obtained by evaluating a deterministic trained network at synthetic input profiles. Variation in $\widehat{Y}^{(1)}, \ldots, \widehat{Y}^{(M)}$ therefore reflects variation in the generated attributes and the nonlinear response of $\widehat{m}$ to those attributes. It does not represent residual variation around the fitted surface or parameter uncertainty in $\widehat{\theta}$. Second, because $\widehat{F}_X$ is constructed under the independence assumption in equation (2.2.5), the simulated profiles populate the input space more broadly than the training data, so $\mathcal{X}_{\text{sim}}$ will contain attribute combinations that $\widehat{m}$ did not encounter during training. Residual-augmented MC analyses that extend beyond the fitted surface are developed in Chapter 3, although these address only the residual component of predictive uncertainty. The parameter-uncertainty component would be quantified by the standard deep-learning approaches of MC dropout, which interprets dropout applied at prediction time as approximate Bayesian inference (Gal and Ghahramani, 2016), and deep ensembles of independently trained networks (Lakshminarayanan et al., 2017); both are natural extensions of the present design.

2.2.3 Shapley attribution

MC input simulation must be distinguished from MC Shapley estimation. Whereas input simulation studies the predicted-cost distribution under $\widehat{F}_X$, Shapley estimation is a computational device for approximating exact Shapley values when the required summation over all 2^p feature coalitions is infeasible. Because the exact calculation is equivalent to averaging over all $p!$ feature orderings, sampling-based estimators are often employed. Castro et al. (2009) proposed a polynomial-time scheme based on randomly sampled permutations, and Maleki et al. (2013) derived confidence bounds for such estimators.

Let

$$\mathcal{P} = \{1, \ldots, p\} \tag{2.2.8}$$

denote the set of input features. For an observed or simulated profile $\mathbf{x}$, the Shapley framework defines a value function $v_{\mathbf{x}}(S)$ for each coalition $S \subseteq \mathcal{P}$. In a model-prediction setting, $v_{\mathbf{x}}(S)$ represents the fitted prediction available when the feature values in S are fixed at their values in $\mathbf{x}$, while the remaining features are integrated over or replaced using a background distribution. Let $\widehat{F}_B$ denote the background distribution used for Shapley-value evaluation. A conditional form of the value function is

$$v_{\mathbf{x}}(S) = \mathbb{E}_{\widehat{F}_B}[\widehat{m}(\mathbf{X}) \mid \mathbf{X}_S = \mathbf{x}_S]\,. \tag{2.2.9}$$

Equation (2.2.9) is read as follows. The coalition S is the set of features whose values are treated as known for the profile being explained. The conditioning event $\mathbf{X}_S = \mathbf{x}_S$ fixes those features at the values they take in $\mathbf{x}$, and the expectation is taken over the features in the complement $\overline{S} = \mathcal{P} \setminus S$, whose values are unknown and are therefore averaged out under the background distribution $\widehat{F}_B$. The value function is consequently not a prediction for a single input vector but the mean prediction over all background profiles that agree with $\mathbf{x}$ on S. Two boundary cases fix the scale of the decomposition: when $S = \mathcal{P}$ nothing remains to be averaged and $v_{\mathbf{x}}(\mathcal{P}) = \widehat{m}(\mathbf{x})$, while when $S = \varnothing$ the conditioning event is vacuous and $v_{\mathbf{x}}(\varnothing) = \mathbb{E}_{\widehat{F}_B}[\widehat{m}(\mathbf{X})]$, which is the baseline ϕ_0 appearing in the decomposition below. The difference $v_{\mathbf{x}}(S \cup \{j\}) - v_{\mathbf{x}}(S)$ in equation (2.2.11) is then the change in that mean prediction caused by additionally revealing feature j, which is the marginal contribution the Shapley average weights over coalitions.

The word *conditional* in equation (2.2.9) is a substantive modelling choice rather than notational convenience, because an alternative convention replaces the conditional expectation with a marginal, or interventional, one:

$$v_{\mathbf{x}}^{\mathrm{marg}}(S) = \mathbb{E}_{\widehat{F}_B}[\widehat{m}(\mathbf{x}_S, \mathbf{X}_{\overline{S}})]\,. \tag{2.2.10}$$

In equation (2.2.10) the features outside S are drawn from their background *marginal* distribution irrespective of the values held fixed in S, whereas in equation (2.2.9) they are drawn from their distribution *conditional* on those values. The two coincide when $\mathbf{X}_{\overline{S}}$ is independent of $\mathbf{X}_S$ under $\widehat{F}_B$, and diverge as the dependence among the inputs strengthens (Aas et al., 2021; Sundararajan and Najmi, 2020). The choice carries an interpretive consequence. The conditional form respects the dependence structure of

the data and never evaluates the fitted model on combinations of feature values that the background distribution regards as implausible; against this, it can assign a non-zero contribution to a feature the fitted model never uses, purely because that feature is correlated with one the model does use, so the attribution describes the joint distribution as much as the model (Janzing et al., 2020; Covert et al., 2021). The marginal form keeps the attribution on the features the model actually reads, at the cost of evaluating $\widehat{m}$ at off-manifold combinations where the fitted surface is an extrapolation and is not constrained by training data (Hooker et al., 2021).

A further difference is computational. The marginal form requires only draws from the background marginals, which are available by resampling the background data, whereas the conditional form requires the conditional laws of $\mathbf{X}_{\overline{S}}$ given $\mathbf{X}_S = \mathbf{x}_S$ for every coalition S, and these are generally unknown and must themselves be estimated (Aas et al., 2021). For this reason most software implementations, including the explainer used in this dissertation, operate under the marginal convention or an approximation to it, so that equation (2.2.9) states the population object of interest while equation (2.2.10) is closer to what is computed in practice.

This notation also separates the input distribution used for MC simulation, $\widehat{F}_X$, from the background distribution used for Shapley evaluation, $\widehat{F}_B$. These distributions may coincide, but they need not be identical, and the attributions are defined only relative to whichever background is chosen: a different $\widehat{F}_B$ shifts the baseline ϕ_0 and rescales every contribution. The empirical implementation therefore reports the background data and the SHAP convention used.

The exact Shapley contribution of feature j for profile $\mathbf{x}$ is

$$\phi_j(\mathbf{x}) = \sum_{S \subseteq \mathcal{P} \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \left[v_{\mathbf{x}}(S \cup \{j\}) - v_{\mathbf{x}}(S) \right]. \tag{2.2.11}$$

The fitted prediction can then be expressed as

$$\widehat{m}(\mathbf{x}) = \phi_0 + \sum_{j=1}^{p} \phi_j(\mathbf{x}), \tag{2.2.12}$$

where ϕ_0 is the baseline prediction determined by the chosen background distribution, commonly corresponding to $v_{\mathbf{x}}(\varnothing)$, and $\phi_j(\mathbf{x})$ is the contribution attributed to feature j for that profile.

When exact Shapley values are not computed, a permutation-based MC approximation may be used. Let $\pi^{(1)}, \ldots, \pi^{(R)}$ be sampled permutations of $\mathcal{P}$, and let $S_j(\pi^{(r)})$ denote the set of features appearing before feature j in permutation $\pi^{(r)}$. The permutation-sampling

estimator is

$$\widehat{\phi}_{j,R}(\mathbf{x}) = \frac{1}{R}\sum_{r=1}^{R}\left[v_{\mathbf{x}}\big(S_j(\pi^{(r)})\cup\{j\}\big) - v_{\mathbf{x}}\big(S_j(\pi^{(r)})\big)\right]. \tag{2.2.13}$$

Increasing R generally improves the stability of the approximation but also increases computational cost. Whenever sampling-based explanations are used, the number of Shapley samples, the background dataset, the random seed or replication strategy, and any stability checks are reported to support reproducibility.

A gradient-based approximation provides an alternative to coalition sampling. The permutation estimator of equation (2.2.13) treats $\widehat{m}$ as a black box and requires two value-function evaluations per sampled permutation. When the fitted model is differentiable in its inputs, as a DNN with differentiable activations is, a second route becomes available that replaces coalition enumeration with the integration of gradients along a path in input space. This is the class of explainer used in this dissertation, and it is therefore set out here in the same notation.

Let $\mathbf{x}'$ denote a baseline profile representing the absence of information, drawn from the background distribution $\widehat{F}_B$, and consider the straight-line path from $\mathbf{x}'$ to the profile $\mathbf{x}$ being explained, parameterised by $\alpha \in [0,1]$ as $\mathbf{x}' + \alpha(\mathbf{x}-\mathbf{x}')$. The integrated-gradients attribution of feature j is the path integral of the partial derivative of the fitted model with respect to that feature, scaled by the distance travelled in the jth coordinate:

$$\phi_j^{\mathrm{IG}}(\mathbf{x};\,\mathbf{x}') = \big(x_j - x'_j\big)\int_0^1 \frac{\partial\,\widehat{m}(\mathbf{x}' + \alpha(\mathbf{x}-\mathbf{x}'))}{\partial x_j}\,\mathrm{d}\alpha. \tag{2.2.14}$$

The two factors in equation (2.2.14) carry distinct meanings. The partial derivative $\partial\widehat{m}/\partial x_j$ measures the local sensitivity of the fitted prediction to an infinitesimal change in feature j, which is exactly the quantity backpropagation already computes, and the displacement $(x_j - x'_j)$ measures how far feature j has moved from its baseline value. A feature receives a large attribution only when the model is sensitive to it *and* it differs appreciably from the baseline; a feature that is influential but sits at its baseline value, or one that differs from the baseline in a flat region of the prediction surface, receives little. Integrating over α rather than evaluating the gradient at $\mathbf{x}$ alone is what makes the attribution complete: by the gradient theorem the contributions sum to the full prediction difference,

$$\sum_{j=1}^{p}\phi_j^{\mathrm{IG}}(\mathbf{x};\,\mathbf{x}') = \widehat{m}(\mathbf{x}) - \widehat{m}(\mathbf{x}'), \tag{2.2.15}$$

which is the path-integral counterpart of the efficiency axiom satisfied by equation (2.2.12) (Sundararajan et al., 2017). A single gradient evaluated at $\mathbf{x}$ would satisfy no such identity, and would be zero wherever the activation function saturates, even for a feature

the model relies on heavily.

Equation (2.2.14) is defined relative to one baseline $\mathbf{x}'$, which reintroduces the background-dependence discussed above. Averaging the attribution over baselines drawn from $\widehat{F}_B$, and over the path position $\alpha \sim \text{Uniform}(0,1)$, yields the expected-gradients form

$$\phi_j^{\text{EG}}(\mathbf{x}) = \mathbb{E}_{\mathbf{x}' \sim \widehat{F}_B,\, \alpha \sim \text{Uniform}(0,1)} \left[(x_j - x'_j) \, \frac{\partial \, \widehat{m}(\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}'))}{\partial x_j} \right], \tag{2.2.16}$$

which restates the path integral as a single expectation over the background distribution and so admits a direct MC estimator (Erion et al., 2021). Drawing K independent pairs $\left(\mathbf{x}'^{(k)}, \alpha^{(k)}\right)$ gives

$$\widehat{\phi}_{j,K}^{\text{EG}}(\mathbf{x}) = \frac{1}{K} \sum_{k=1}^{K} \left(x_j - x_j'^{(k)} \right) \frac{\partial \, \widehat{m}\left(\mathbf{x}'^{(k)} + \alpha^{(k)}(\mathbf{x} - \mathbf{x}'^{(k)})\right)}{\partial x_j}. \tag{2.2.17}$$

Taking the expectation over baselines replaces $\widehat{m}(\mathbf{x}')$ in equation (2.2.15) with $\mathbb{E}_{\widehat{F}_B}[\widehat{m}(\mathbf{X})]$, so the attributions sum to $\widehat{m}(\mathbf{x}) - \mathbb{E}_{\widehat{F}_B}[\widehat{m}(\mathbf{X})]$, recovering the baseline $\phi_0 = v_{\mathbf{x}}(\varnothing)$ of equation (2.2.12) and making the gradient-based attributions directly comparable with the coalitional ones on the same scale.

The computational gain is the reason for the construction: the cost of equation (2.2.17) is K gradient evaluations, each obtained in one backward pass, in place of the 2^p value-function evaluations required by equation (2.2.11), and K does not grow with p. The gain is not free. Equation (2.2.16) is an Aumann-Shapley path value rather than the discrete Shapley value of equation (2.2.11); the two agree when $\widehat{m}$ is linear along the path, and differ in general, so the gradient-based quantities are approximations whose accuracy is not guaranteed by the axioms that motivate them (Sundararajan and Najmi, 2020). The linear interpolation also passes through intermediate points $\mathbf{x}' + \alpha(\mathbf{x} - \mathbf{x}')$ that need not be valid profiles, since a fractional value of a binary dummy indicator has no interpretation as a policyholder attribute, which places the method within the marginal convention of equation (2.2.10) and inherits the off-manifold caveat attached to it (Hooker et al., 2021). Finally, equation (2.2.17) is itself a MC average, so its sampling variability is governed by K and the dispersion of the background sample. The practical consequence is that the reported attributions are conditional on the fitted network, the background sample and the number of draws, and the completeness identity should be verified numerically on the fitted model rather than assumed, since it holds exactly for equation (2.2.14) but only up to MC error for equation (2.2.17).

2.2.4 High-cost profiles

Distributional and attributional results meet in the analysis of high predicted costs. Let u denote a high-cost threshold. The threshold may be specified as a fixed monetary amount or as an empirical upper-tail quantile of the simulated predicted-cost distribution, for example

$$u = q_\tau\left(\widehat{Y}^{(1)}, \ldots, \widehat{Y}^{(M)}\right), \qquad 0 < \tau < 1. \tag{2.2.18}$$

The simulated high-cost set is

$$\mathcal{H}_u = \left\{\ell \in \{1, \ldots, M\} : \widehat{Y}^{(\ell)} > u\right\}. \tag{2.2.19}$$

The relative size of this set,

$$\widehat{p}_u = \frac{|\mathcal{H}_u|}{M}, \tag{2.2.20}$$

summarises the simulated proportion of profiles whose fitted predictions exceed the chosen threshold. Conditional on $|\mathcal{H}_u| > 0$, the average signed Shapley contribution of feature j in the high-cost region is

$$\widehat{\Phi}_{j,u} = \frac{1}{|\mathcal{H}_u|} \sum_{\ell \in \mathcal{H}_u} \phi_j\left(\mathbf{X}^{(\ell)}\right), \tag{2.2.21}$$

and the corresponding average absolute contribution is

$$\widehat{A}_{j,u} = \frac{1}{|\mathcal{H}_u|} \sum_{\ell \in \mathcal{H}_u} \left|\phi_j\left(\mathbf{X}^{(\ell)}\right)\right|. \tag{2.2.22}$$

The signed quantity $\widehat{\Phi}_{j,u}$ indicates whether feature j tends to raise or lower predictions within the high-cost subset. The magnitude-based quantity $\widehat{A}_{j,u}$ measures the strength of the feature's contribution regardless of direction. Together, these summaries identify which variables the fitted model attributes to high predicted costs among simulated profiles.

The interpretation of high-cost Shapley summaries remains model-based. A large value of $\widehat{A}_{j,u}$ means that feature j has a large average model-attributed contribution among simulated profiles satisfying $\widehat{Y}^{(\ell)} > u$, and a positive $\widehat{\Phi}_{j,u}$ indicates that the fitted model tends to associate the feature with higher predictions in the simulated high-cost region. As noted in the delimitations (Section 1.8), these attributions do not establish causal relationships.

The scope of the conclusions is determined by the data source, fitted model, input distribution and Shapley background distribution. Data bias may therefore affect both the simulated predicted-cost distribution and the Shapley summaries. The framework provides a coherent route from health insurance cost prediction to distributional and attributional

understanding of fitted predictions, while remaining bounded by the benchmark nature of the data.

2.3 Statistical Learning and Supervised Regression

The predictive component of the DNN-MC-Shapley workflow is a supervised statistical-learning problem. Statistical learning concerns the estimation of relationships between observed variables for purposes such as prediction, interpretation and model assessment (James et al., 2021; Hastie et al., 2009; Vapnik, 1998). In supervised learning, each observation contains both predictors and an observed response. In unsupervised learning, by contrast, the aim is to identify structure in the predictor space without an observed response (James et al., 2021). The insurance data considered in the empirical analysis pair policyholder attributes $\mathbf{x}_i$ with an observed cost outcome y_i, placing the prediction task within the supervised regression setting.

The fitted value for observation i is $\widehat{y}_i = \widehat{m}(\mathbf{x}_i)$, following the regression formulation in equations (2.2.2) and (2.2.3). The response y_i is a continuous monetary outcome, and the problem is therefore a regression problem rather than a classification problem. Classification methods are designed for categorical responses and are commonly assessed using quantities such as accuracy, sensitivity, specificity or cross-entropy. Regression methods are evaluated on a continuous scale using prediction-error measures (James et al., 2021; Hastie et al., 2009). The specific measures employed in this dissertation are defined below.

For a fitted regression function $\widehat{m}$ evaluated on an evaluation sample $\mathcal{I}_{\text{eval}}$ of size n_{eval}, the mean squared error is

$$\text{MSE} = \frac{1}{n_{\text{eval}}} \sum_{i \in \mathcal{I}_{\text{eval}}} \left(y_i - \widehat{m}(\mathbf{x}_i)\right)^2 . \tag{2.3.1}$$

The RMSE is the square root of the MSE,

$$\text{RMSE} = \sqrt{\frac{1}{n_{\text{eval}}} \sum_{i \in \mathcal{I}_{\text{eval}}} \left(y_i - \widehat{m}(\mathbf{x}_i)\right)^2} = \sqrt{\text{MSE}}. \tag{2.3.2}$$

Thus, RMSE measures the typical size of the prediction error after squaring errors, averaging them and returning the result to the original scale of the response variable. The MAE is

$$\text{MAE} = \frac{1}{n_{\text{eval}}} \sum_{i \in \mathcal{I}_{\text{eval}}} \left|y_i - \widehat{m}(\mathbf{x}_i)\right| . \tag{2.3.3}$$

The R^2 is

$$R^2 = 1 - \frac{\sum_{i \in \mathcal{I}_{\text{eval}}} \left(y_i - \widehat{m}(\mathbf{x}_i)\right)^2}{\sum_{i \in \mathcal{I}_{\text{eval}}} \left(y_i - \bar{y}_{\text{eval}}\right)^2}, \qquad \bar{y}_{\text{eval}} = \frac{1}{n_{\text{eval}}} \sum_{i \in \mathcal{I}_{\text{eval}}} y_i. \tag{2.3.4}$$

The R^2 statistic measures the relative reduction in squared prediction error compared with the mean-only baseline $\bar{y}_{\text{eval}}$ on the same evaluation sample. On held-out data, R^2 may be negative if the fitted model performs worse than this baseline. For this reason, RMSE, MAE and R^2 are reported together rather than treated as interchangeable summaries of predictive performance (James et al., 2021; Hastie et al., 2009).

Out-of-sample evaluation is central to statistical learning. A flexible model may achieve low training error while performing poorly on unseen observations, which corresponds to overfitting. Conversely, a model that is too restrictive may have high training and test error, which corresponds to underfitting. The bias-variance trade-off formalises this tension between systematic approximation error and sensitivity to the training sample (James et al., 2021; Hastie et al., 2009).

Classical regression models provide important reference points for actuarial and insurance applications. Ordinary least squares regression represents the conditional mean through a linear predictor,

$$\widehat{m}_{\text{lin}}(\mathbf{x}_i) = \widehat{\beta}_0 + \sum_{j=1}^{p} \widehat{\beta}_j \, x_{ij}. \tag{2.3.5}$$

GLMs extend this formulation by specifying a distributional family for the response and linking the conditional mean

$$\mu_i = \mathbb{E}(Y_i \mid \mathbf{X}_i = \mathbf{x}_i) \tag{2.3.6}$$

to a linear predictor through a link function:

$$g(\mu_i) = \beta_0 + \sum_{j=1}^{p} \beta_j \, x_{ij}. \tag{2.3.7}$$

These models remain attractive in actuarial work because their assumptions, coefficients and diagnostics are relatively transparent, and because GLM modelling has a long history in insurance pricing and risk modelling (Dobson and Barnett, 2008; Denuit et al., 2007). A purely linear predictor, however, cannot represent nonlinear predictor effects without explicitly specifying transformations or basis expansions in advance.

Because GLMs remain the industry-standard framework for health insurance pricing, a gamma GLM with a logarithmic link function serves as a natural benchmark against which the predictive performance of the more flexible DNN can be compared. The benchmark is intended to quantify the gain in accuracy achieved by replacing a purely linear predictor

with a non-linear neural network, and to provide a familiar reference point for readers accustomed to classical actuarial modelling. The specification of the benchmark model is given in Chapter 3.

Machine-learning methods offer a more flexible way to approximate complex predictor-response relationships. In actuarial contexts, Richman (2021) notes that neural-network methods can capture nonlinearities and interactions in risk relationships, learn internal feature representations from mixed-type inputs, and reduce the need for manual specification of predictor transformations. These properties are relevant when predictor effects are complex and involve both continuous and categorical variables, as is the case in the health insurance cost data considered here. Related evidence from population health-cost prediction indicates that flexible models can be useful when healthcare expenditure depends on nonlinear covariate relationships (Drewe-Boss et al., 2022). The DNN serves as one such flexible estimator of $m(\cdot)$.

The methodological role of the DNN rests on two properties. First, DNNs possess the universal approximation property discussed in Section 2.2. Secondly, neural networks have been used in actuarial science as efficient surrogate models for computationally demanding valuation and prediction problems (Hejazi and Jackson, 2016). These properties make a DNN suitable for estimating a fitted prediction function $\widehat{m}$ that can be evaluated repeatedly in the MC and Shapley-value stages. Under smooth activation choices, the fitted function is differentiable; under common piecewise-linear activations, it is piecewise differentiable. In either case, it provides a flexible parametric prediction surface compatible with the simulation and interpretation workflow.

The use of a DNN should not be read as a dismissal of competing tabular-data methods; tree-based ensembles such as RF and gradient-boosted trees are often strong competitors (Grinsztajn et al., 2022; Orji and Ukwandu, 2024) and are fitted here as benchmarks. The DNN is used because it provides a differentiable prediction surface that can be repeatedly evaluated and then connected to MC input simulation and Shapley-value explanation.

The supervised regression formulation also clarifies the roles of the Monte Carlo and Shapley stages. The learning task is to estimate $m(\cdot)$ from labelled observations in $\mathcal{D}$, producing the fitted prediction function $\widehat{m}(\cdot)$. MC simulation subsequently studies the distribution of fitted predictions under input variation. Shapley values then attribute fitted predictions to input variables. Predictive accuracy, distributional behaviour and attributional interpretation therefore address related but distinct aspects of model behaviour.

The resulting taxonomy is summarised in Figure 2. The empirical problem belongs to machine learning, falls under supervised learning because responses are observed, belongs to regression because the response is continuous, and is implemented through DNN regression for the fitted prediction component.

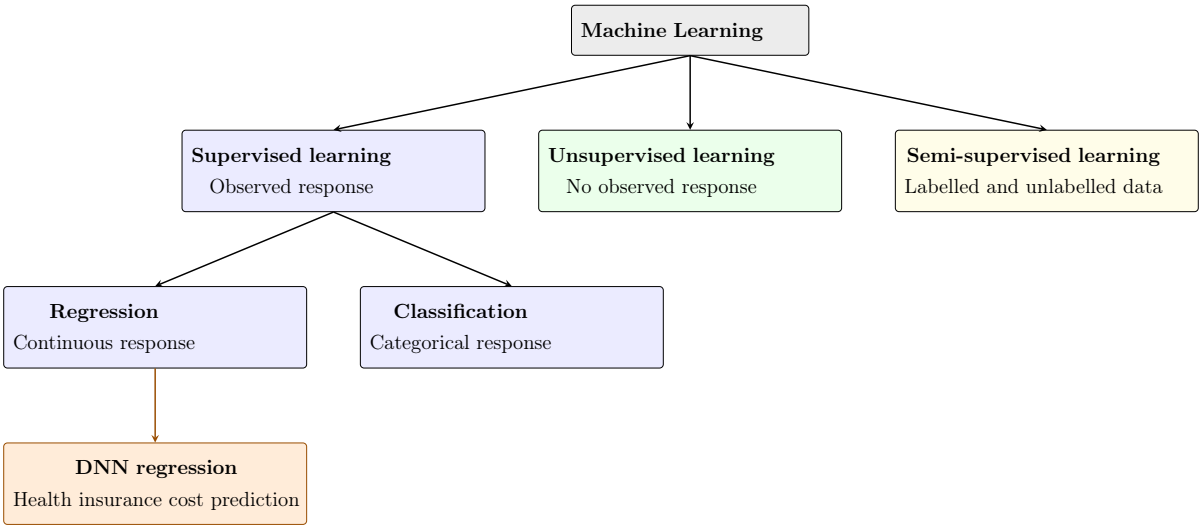


Figure 2: Health insurance cost prediction within the machine-learning taxonomy.

The central empirical task is not unsupervised because the objective is not to discover clusters or latent structure without a response variable. A semi-supervised formulation is also unnecessary because the training observations contain both policyholder attributes and cost outcomes. Supervised regression is therefore the appropriate learning formulation, with the fitted DNN serving as the prediction function to be evaluated, simulated and interpreted.

2.4 DNNs for Regression

A DNN is a composition of layers that maps an input vector to an output prediction. In regression problems, the aim is to approximate an unknown function that maps explanatory variables to a continuous response (Goodfellow et al., 2016; Bishop, 2006). The theoretical basis for this approximation role is the universal approximation theorem. Cybenko (1989) established the result for sigmoidal activation functions and Hornik et al. (1989) for general squashing functions, while the sharpest characterisation is due to Leshno et al. (1993), who showed that a single-hidden-layer network can approximate continuous functions on compact sets to arbitrary accuracy if and only if the activation function is not a polynomial. In this form the theorem states that for any continuous function $f : [0,1]^p \to \mathbb{R}$ and any $\varepsilon > 0$, there exists a single-hidden-layer network F with a non-polynomial activation function such that $\sup_{\mathbf{x}} |F(\mathbf{x}) - f(\mathbf{x})| < \varepsilon$. The domain $[0,1]^p$ requires every input variable to be confined to a finite, closed range; the specific choice of $[0,1]$ is a notational convention, since any bounded range $[a,b]$ can be mapped to $[0,1]$ by a linear rescaling, so the theorem applies to any compact input domain. Two qualifications are important. First, compactness is essential: uniform approximation is guaranteed only on bounded input sets, not over an unbounded domain. Secondly, the theorem is existential rather than a statement about a network of fixed size; the hidden-layer width required to achieve a given tolerance ε generally grows as ε shrinks, so the guarantee concerns the family of single-hidden-layer networks rather than any particular architecture. Recent studies have examined the conditions under which DNNs are competitive with tree-based models on tabular data, finding that well-tuned simple networks can match complex architectures (Kadra et al., 2021) and that the relative advantage of deep learning over gradient boosting depends on dataset characteristics such as sample size and feature cardinality (Shwartz-Ziv and Armon, 2022). Understanding deep learning generalisation remains an active research area; Zhang et al. (2021) demonstrate that conventional statistical learning theory, which predicts poor generalisation for overparameterised models, requires rethinking in the deep learning context where networks with more parameters than training examples can still generalise well. In the empirical application, the input vector consists of encoded policyholder characteristics and the response is a health insurance charge or premium.

Let $\mathbf{x}_i \in \mathbb{R}^p$ denote the preprocessed input vector for observation i, and let $y_i \in \mathbb{R}_+$ denote the corresponding observed insurance cost. A DNN constructs a sequence of layer activations from this input. Let $d_0 = p$ denote the input dimension, let d_l denote the number of nodes in layer l, and let $d_L = 1$ for a single continuous regression output. For observation i, the input layer is defined as

$$\mathbf{a}_i^{(0)} = \mathbf{x}_i \in \mathbb{R}^{d_0}. \tag{2.4.1}$$

For layers $l = 1, \ldots, L$, the pre-activation vector and activation vector are defined by

$$\mathbf{z}_i^{(l)} = \left(W^{(l)}\right)^\top \mathbf{a}_i^{(l-1)} + \mathbf{b}^{(l)}, \tag{2.4.2}$$

and

$$\mathbf{a}_i^{(l)} = \sigma_l\left(\mathbf{z}_i^{(l)}\right), \tag{2.4.3}$$

where

$$W^{(l)} \in \mathbb{R}^{d_{l-1} \times d_l}, \qquad \mathbf{b}^{(l)} \in \mathbb{R}^{d_l},$$

and therefore

$$\mathbf{z}_i^{(l)}, \mathbf{a}_i^{(l)} \in \mathbb{R}^{d_l}.$$

The activation function $\sigma_l(\cdot)$ is applied componentwise. For the hidden layers, $\sigma_l(\cdot)$ is nonlinear. For the regression output layer, a linear activation is used, so that

$$\sigma_L(z) = z.$$

The fitted prediction for observation i is therefore

$$\widehat{y}_i = f_\theta(\mathbf{x}_i) = a_{i1}^{(L)}, \tag{2.4.4}$$

where $a_{i1}^{(L)}$ is the single output activation. A linear output is appropriate because the response is a continuous monetary variable rather than a probability or class label. Classification tasks would instead use logistic or softmax output functions.

The network parameters are collected in

$$\theta = \left\{W^{(1)}, \mathbf{b}^{(1)}, \ldots, W^{(L)}, \mathbf{b}^{(L)}\right\}. \tag{2.4.5}$$

When the network is evaluated for all training observations simultaneously, let

$$A^{(0)} = \left[\mathbf{a}_1^{(0)}, \ldots, \mathbf{a}_n^{(0)}\right] = X^\top \in \mathbb{R}^{d_0 \times n},$$

where $X \in \mathbb{R}^{n \times p}$ is the model-ready design matrix. The matrix form of the layer recursion is

$$Z^{(l)} = \left(W^{(l)}\right)^\top A^{(l-1)} + \mathbf{b}^{(l)} \mathbf{1}_n^\top, \qquad A^{(l)} = \sigma_l\left(Z^{(l)}\right), \tag{2.4.6}$$

where

$$Z^{(l)}, A^{(l)} \in \mathbb{R}^{d_l \times n}, \qquad \mathbf{1}_n \in \mathbb{R}^n.$$

This notation makes clear that the network is evaluated recursively from the input activations $\mathbf{a}_i^{(0)}$ to the output activation $\mathbf{a}_i^{(L)}$, which supplies the fitted insurance-cost prediction.

2.4.1 Activation functions

The choice of activation function affects both the expressive capacity and the optimisation dynamics of the network. Activation functions introduce nonlinearity into the layer recursion; without them, a composition of affine layers would reduce to a single affine transformation. Several activation functions may be considered for the hidden layers, including the hyperbolic tangent function, the ReLU, the CELU and the GELU (Goodfellow et al., 2016; Lee and Wu, 2023).

For a scalar input $z \in \mathbb{R}$, the hyperbolic tangent activation is defined by

$$\tanh(z) = \frac{\exp(z) - \exp(-z)}{\exp(z) + \exp(-z)}. \tag{2.4.7}$$

The tanh function is smooth and bounded between -1 and 1. Its centred range may be useful in some network architectures, although its saturation for large positive or negative inputs can lead to small gradients in deeper networks.

The ReLU is defined by

$$\mathrm{ReLU}(z) = \max\{0, z\}. \tag{2.4.8}$$

The ReLU is widely used in DNNs because it is simple, computationally efficient and less prone to the saturation behaviour associated with some bounded activation functions (Nair and Hinton, 2010; Goodfellow et al., 2016). Its limitation is that negative inputs are mapped to zero, so some units may become inactive during training.

The CELU with parameter $\alpha > 0$ is defined by

$$\mathrm{CELU}_\alpha(z) = \begin{cases} z, & z \geq 0, \\ \alpha\left\{\exp(z/\alpha) - 1\right\}, & z < 0. \end{cases} \tag{2.4.9}$$

CELU is a smooth rectifier-type activation. It retains the linear positive branch of ReLU while assigning a smooth nonzero transformation to negative inputs (Barron, 2017).

The GELU is defined by

$$\mathrm{GELU}(z) = z\Phi(z), \tag{2.4.10}$$

where $\Phi(\cdot)$ is the cumulative distribution function of a standard normal random variable. A commonly used approximation is

$$\mathrm{GELU}(z) \approx \frac{1}{2} z \left[1 + \tanh\left\{\sqrt{\frac{2}{\pi}}\left(z + 0.044715 z^3\right)\right\}\right].$$

GELU is smooth and differentiable, and may be viewed as a soft probabilistic gating alternative to a hard rectifier (Hendrycks and Gimpel, 2016; Lee and Wu, 2023). These

activation functions are introduced as candidate nonlinear transformations. Their empirical suitability for health insurance cost prediction is assessed through the modelling procedure rather than assumed from their theoretical definitions alone.

2.4.2 Loss and optimisation

The DNN is trained by minimising a loss function. For continuous cost prediction, MSE is a natural objective because it penalises large deviations more heavily than small ones, directing the optimiser towards predictions that are accurate across the full range of observed costs. Its squared-error form also yields smooth, differentiable gradients that are well suited to gradient-based optimisation. The MSE loss is defined by

$$\mathcal{L}(\theta) = \frac{1}{n_{\text{train}}} \sum_{i \in \mathcal{I}_{\text{train}}} \left(y_i - f_\theta(\mathbf{x}_i)\right)^2, \tag{2.4.11}$$

where $\mathcal{I}_{\text{train}}$ is the training index set and $n_{\text{train}} = |\mathcal{I}_{\text{train}}|$. The fitted parameters are obtained by solving

$$\widehat{\theta} = \arg\min_{\theta} \ \mathcal{L}(\theta). \tag{2.4.12}$$

For multilayer neural networks, this optimisation problem is generally nonconvex. Closed-form estimation is therefore not available in the usual case, and iterative gradient-based methods are used to search for a set of parameters that gives good predictive performance.

Gradient descent provides the conceptual basis for this training procedure. At iteration t, the full-batch gradient descent update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta \mathcal{L}\left(\theta^{(t)}\right),$$

where $\eta > 0$ is the learning rate. The gradient points in the direction of steepest increase of the loss, so subtracting the gradient moves the parameter vector in a direction intended to reduce the loss. In neural-network training, this gradient is computed by backpropagation, which applies the chain rule through the sequence of layer transformations.

In practice, mini-batch stochastic gradient methods are often preferred over full-batch gradient descent because computing the full gradient at every iteration can be computationally expensive. If $\mathcal{B}_t$ denotes the mini-batch used at iteration t, the mini-batch gradient is

$$\mathbf{g}_t = \nabla_\theta \mathcal{L}_{\mathcal{B}_t}\left(\theta^{(t)}\right).$$

Stochastic gradient descent updates the parameters directly in the negative mini-batch gradient direction:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \mathbf{g}_t.$$

The use of mini-batches reduces computational cost and introduces stochasticity into the optimisation path. This is useful for training flexible models, but it also means that convergence behaviour must be monitored carefully.

Momentum methods modify stochastic gradient descent by using an exponentially weighted moving average of previous gradients:

$$\mathbf{m}_t = \beta \mathbf{m}_{t-1} + (1-\beta)\mathbf{g}_t, \qquad 0 < \beta < 1,$$

followed by the update

$$\theta^{(t+1)} = \theta^{(t)} - \eta \mathbf{m}_t.$$

The moving-average term can reduce the effect of isolated noisy gradients and can accelerate movement in directions that remain persistent across iterations.

Adaptive gradient methods rescale parameter updates using information from previous gradients. RMSProp uses an exponentially weighted average of squared gradients,

$$\mathbf{s}_t = \rho \mathbf{s}_{t-1} + (1-\rho)\mathbf{g}_t^2, \qquad 0 < \rho < 1,$$

and updates the parameters according to

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\mathbf{g}_t}{\sqrt{\mathbf{s}_t} + \epsilon},$$

where the square and square root are applied componentwise and $\epsilon > 0$ is included for numerical stability. RMSProp is useful when different parameters require different effective learning-rate scales (Ruder, 2016).

Adam combines a momentum-like first-moment estimate with an adaptive second-moment estimate (Kingma and Ba, 2015). It updates

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t, \qquad \mathbf{s}_t = \beta_2 \mathbf{s}_{t-1} + (1-\beta_2)\mathbf{g}_t^2,$$

with bias-corrected estimates

$$\widehat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1-\beta_1^t}, \qquad \widehat{\mathbf{s}}_t = \frac{\mathbf{s}_t}{1-\beta_2^t}.$$

The Adam update is

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\widehat{\mathbf{m}}_t}{\sqrt{\widehat{\mathbf{s}}_t} + \epsilon}.$$

Adam is widely used in applied deep learning because it combines adaptive parameter-wise scaling with a first-moment estimate of the gradient.

NAdam modifies Adam by incorporating a Nesterov-type momentum correction into the

first-moment component (Dozat, 2016). A common form replaces the Adam numerator by

$$\widetilde{\mathbf{m}}_t = \beta_1 \widehat{\mathbf{m}}_t + \frac{1-\beta_1}{1-\beta_1^t}\mathbf{g}_t,$$

so that

$$\theta^{(t+1)} = \theta^{(t)} - \eta \frac{\widetilde{\mathbf{m}}_t}{\sqrt{\widehat{\mathbf{s}}_t} + \epsilon}.$$

NAdam therefore retains Adam's adaptive second-moment scaling while modifying the momentum component through a Nesterov-style lookahead correction. These optimisers are introduced as candidate training mechanisms. The final optimiser choice is an empirical modelling decision and is justified in the methodology chapter using validation performance, convergence behaviour and held-out test results.

The optimiser is not a distributional assumption about insurance costs. It is the numerical mechanism used to estimate $\widehat{\theta}$. This distinction is important because the subsequent MC and Shapley analyses are applied to the fitted function $f_{\widehat{\theta}}$. The stability and predictive quality of the fitted DNN therefore affect not only point prediction, but also the model-implied predicted-cost distribution and the feature-attribution summaries derived later in the framework.

The empirical application considers a DNN architecture for health insurance cost prediction. After categorical variables are encoded and numerical variables are standardised, the model-ready input dimension is $d_0 = p$. Under the preprocessing design used here, this dimension is $p = 22$. A candidate funnel architecture is shown in Figure 3.

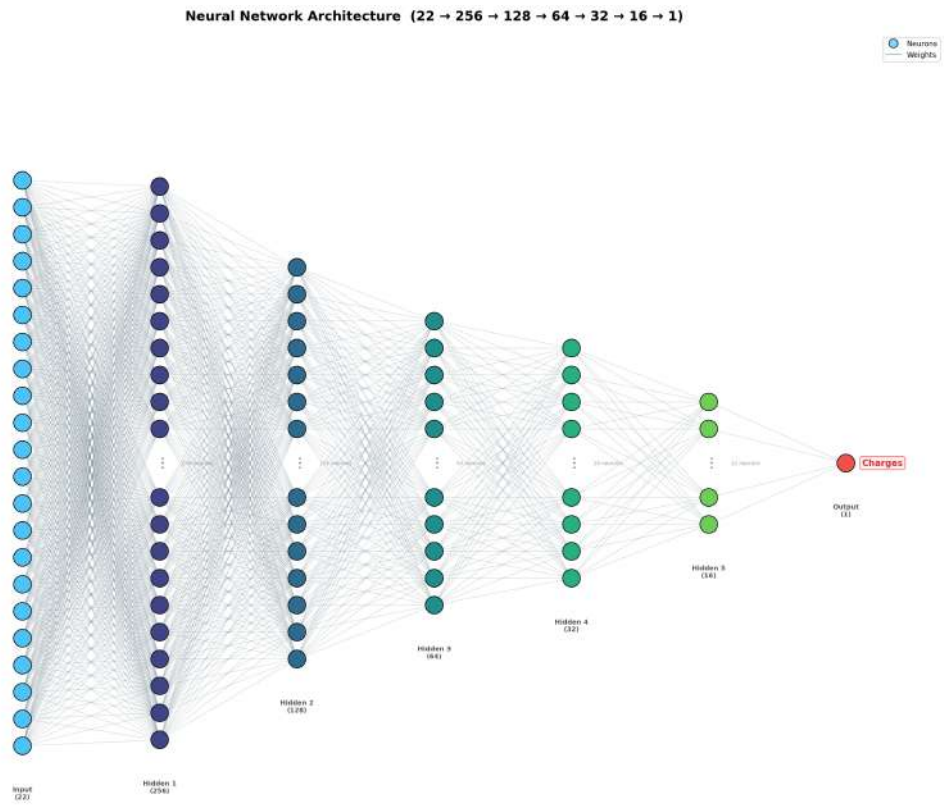


Figure 3: Candidate DNN architecture.

The final output neuron represents the predicted health insurance charge or premium. This architecture starts with a wide hidden layer and progressively compresses the representation through successive layers, allowing the network to learn nonlinear representations before reducing them to a scalar prediction.

The architecture used in the empirical chapter may be implemented without explicit bias vectors, depending on the final modelling choice confirmed for the study. In that case, the layer recursion in equations (2.4.2)-(2.4.3) simplifies to

$$\mathbf{z}_i^{(l)} = \left(W^{(l)}\right)^\top \mathbf{a}_i^{(l-1)}, \qquad \mathbf{a}_i^{(l)} = \sigma_l\left(\mathbf{z}_i^{(l)}\right), \qquad l = 1, \ldots, L, \tag{2.4.13}$$

with $\mathbf{a}_i^{(0)} = \mathbf{x}_i$ as in equation (2.4.1). The output prediction is then

$$f_\theta(\mathbf{x}_i) = a_{i1}^{(L)}.$$

This bias-free specification removes additive layer-bias parameters from the network architecture; the distinction between architectural, statistical and data-related meanings of bias is noted above. Its adequacy is assessed empirically through training, validation and test performance, and where appropriate through comparison with a bias-inclusive network. One theoretical qualification should be recorded. The approximation theorems

reviewed in Section 2.4 assume affine layer maps comprising both weights and biases; a bias-free network whose activation satisfies $\sigma(0) = 0$ is positively homogeneous and constrained to pass through the origin of the standardised input space, so the universal approximation guarantee does not directly apply to the implemented architecture.

The final architecture should not be selected purely by intuition; it should be determined using validation performance and evaluated on a held-out test set. This is necessary because neural networks can be highly flexible. A network that is too small may underfit, while a network that is too large may fit the training data closely without generalising to unseen observations. The DNN is therefore treated as a statistical learning model whose usefulness is judged by out-of-sample prediction performance and by the quality of the subsequent simulation and interpretability analysis (James et al., 2021; Hastie et al., 2009).

The dataset is partitioned as

$$\mathcal{D} = \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, \tag{2.4.14}$$

where the three components are disjoint and exhaust $\mathcal{D}$. The proportions allocated to each component are a design choice rather than a fixed convention; the split used in this dissertation is stated in Section 3.3. An epoch is one complete pass through the training data. For a fitted parameter vector $\widehat{\theta}^{(e)}$ after epoch e, the training and validation losses are

$$\mathcal{L}_{\text{train}}^{(e)} = \frac{1}{n_{\text{train}}} \sum_{i \in \mathcal{I}_{\text{train}}} \left(y_i - f_{\widehat{\theta}^{(e)}}(\mathbf{x}_i)\right)^2 \tag{2.4.15}$$

and

$$\mathcal{L}_{\text{val}}^{(e)} = \frac{1}{n_{\text{val}}} \sum_{i \in \mathcal{I}_{\text{val}}} \left(y_i - f_{\widehat{\theta}^{(e)}}(\mathbf{x}_i)\right)^2. \tag{2.4.16}$$

The candidate number of epochs is selected by comparing validation loss over the epoch grid. If $\mathcal{E}$ denotes the set of candidate epoch values, the selected training duration is

$$e^* = \arg\min_{e \in \mathcal{E}} \mathcal{L}_{\text{val}}^{(e)}. \tag{2.4.17}$$

This validation step is necessary because training loss typically decreases as the optimiser continues updating the model, but validation loss may eventually increase if the network begins to adapt too closely to the training sample. A decreasing training loss with a rising validation loss is a common indication of overfitting, while weak training and validation performance may indicate underfitting or unsuitable training settings.

LOOCV is not adopted as the primary validation strategy. It would require refitting the neural network once for each observation.

Although K-fold cross-validation is often useful when a separate validation set is unavailable or when dependence on a single random split must be reduced, it is not required for the main analysis because the data are already supplied with the fixed partition in equation (2.4.14). Applying K-fold cross-validation to the full dataset would mix observations assigned to validation and test components into the training process, weakening the interpretation of the final test performance. Applying K-fold cross-validation only to the training set would avoid this leakage, but would require repeated DNN training across folds and candidate epoch values, while the supplied validation set already provides the intended mechanism for selecting the training duration. The validation analysis therefore uses the pre-specified validation set to select e^*, and the test set is used only once for final model assessment.

2.4.3 Generalisation, validation analysis and bias-variance trade-off

The tension between overfitting and underfitting is particularly acute for flexible models such as DNNs, which may contain many trainable parameters and can represent complex nonlinear relationships. The formal bias-variance decomposition below makes this trade-off precise.

Expressing the regression formulation in equation (2.2.2) in population-level notation,

$$Y = f(\mathbf{X}) + \varepsilon, \tag{2.4.18}$$

where $f(\mathbf{X})$ denotes the true regression function $m(\cdot)$ from equation (2.2.2) and ε represents irreducible noise with $\mathbb{E}(\varepsilon \mid \mathbf{X}) = 0$. In the homoscedastic case, the irreducible variation is denoted σ_ε^2.

If $\widehat{f}_{\mathcal{D}}$ is a model trained on a dataset $\mathcal{D}$, then prediction error on a new observation depends not only on how well the model fits the observed data, but also on how stable the fitted function is across different possible training samples. This motivates the bias-variance trade-off in statistical learning (Hastie et al., 2009; James et al., 2021; Bishop, 2006).

For squared-error loss, the expected prediction error at a point $\mathbf{x}$ can be decomposed into three components:

$$\mathbb{E}_{\mathcal{D},Y}\left[\left(Y - \widehat{f}_{\mathcal{D}}(\mathbf{x})\right)^2\right] = \underbrace{\left(f(\mathbf{x}) - \mathbb{E}_{\mathcal{D}}\left[\widehat{f}_{\mathcal{D}}(\mathbf{x})\right]\right)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}\left[\left(\widehat{f}_{\mathcal{D}}(\mathbf{x}) - \mathbb{E}_{\mathcal{D}}\left[\widehat{f}_{\mathcal{D}}(\mathbf{x})\right]\right)^2\right]}_{\text{Variance}} + \underbrace{\sigma_\varepsilon^2}_{\text{Noise}}. \tag{2.4.19}$$

The bias term measures the systematic error introduced by the choice of model class, while the variance term measures the instability of the fitted model across different training

samples. The noise term σ_ε^2 represents irreducible variation in the response that cannot be removed by any model based on the available predictors.

The bias-variance trade-off can therefore be summarised as

$$\text{Expected test error} = \text{Variance} + \text{Bias}^2 + \text{Irreducible noise}.$$

Equivalently, in terms of model complexity,

$$\text{Model complexity} \uparrow \implies \text{Bias}^2 \downarrow, \qquad \text{Variance} \uparrow,$$

whereas

$$\text{Model complexity} \downarrow \implies \text{Bias}^2 \uparrow, \qquad \text{Variance} \downarrow .$$

This relationship explains why a highly flexible model may fit the training data closely but perform poorly on unseen observations, while an overly simple model may fail to capture the underlying pattern in both training and unseen data. The aim is not to minimise training error alone, but to find a level of model complexity that gives a favourable balance between bias and variance.

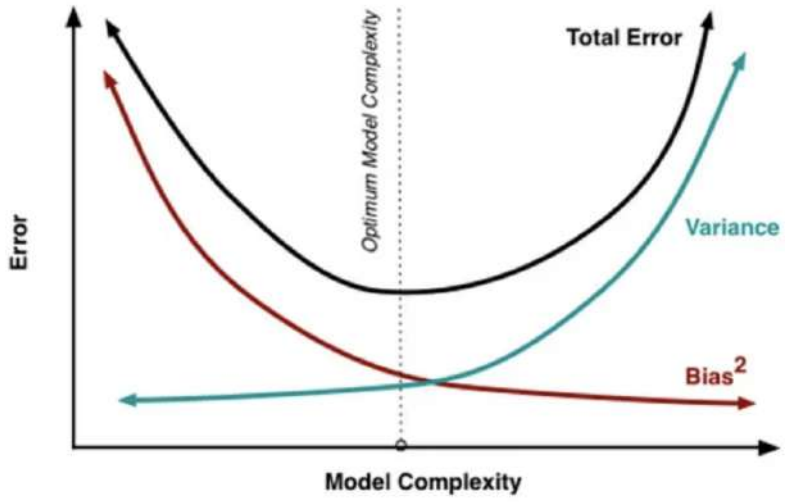


Figure 4: Bias-variance trade-off illustration (James et al., 2021).

Figure 4 provides a schematic illustration of this relationship. The diagram emphasises that neither a very simple model nor an excessively flexible model is generally desirable for prediction. Instead, the preferred model complexity is the one that achieves the best out-of-sample performance by balancing bias and variance. This principle motivates the epoch-selection criterion defined in equation (2.4.17).

The pointwise squared-error decomposition in equation (2.4.19) separates the expected prediction error at a fixed input value $\mathbf{x}$ into bias, variance and irreducible noise. The

same principle can also be expressed at the population level by considering expected out-of-sample error. Let $g^{(\mathcal{D})}$ denote a fitted model obtained from a training sample $\mathcal{D}$, and define the average fitted model by

$$\bar{g}(\mathbf{x}) = \mathbb{E}_{\mathcal{D}}\left[g^{(\mathcal{D})}(\mathbf{x})\right].$$

Also let

$$m(\mathbf{X}) = f(\mathbf{X}) = \mathbb{E}(Y \mid \mathbf{X})$$

denote the systematic component of the response, written with the regression-function symbol $m(\cdot)$ of equation (2.2.2) to avoid confusion with the bar notation used elsewhere for sample means. The expected out-of-sample error under squared loss can then be written as

$$\begin{aligned}
\mathbb{E}_{\mathcal{D}}\left[E_{\text{out}}\left(g^{(\mathcal{D})}\right)\right] &= \mathbb{E}_{\mathcal{D}}\left[\mathbb{E}_{\mathbf{X},Y}\left[\left(g^{(\mathcal{D})}(\mathbf{X}) - Y\right)^2\right]\right] \\
&= \underbrace{\mathbb{E}_{\mathbf{X}}\left[\mathbb{E}_{\mathcal{D}}\left[\left(g^{(\mathcal{D})}(\mathbf{X}) - \bar{g}(\mathbf{X})\right)^2\right]\right]}_{\text{Model variance}} \\
&\quad + \underbrace{\mathbb{E}_{\mathbf{X}}\left[(\bar{g}(\mathbf{X}) - f(\mathbf{X}))^2\right]}_{\text{Model bias}} + \underbrace{\mathbb{E}_{\mathbf{X}}\left[\mathbb{E}_{Y|\mathbf{X}}\left[(m(\mathbf{X}) - Y)^2\right]\right]}_{\text{Noise}}.
\end{aligned} \tag{2.4.20}$$

This form is useful because it links model complexity directly to expected out-of-sample performance, which is the quantity approximated by the validation and test-set analysis. Complex models may reduce model bias by representing richer patterns, but they may also increase model variance by becoming sensitive to the particular training sample. Simpler models may have lower variance, but they may miss important nonlinear structure. The practical aim is therefore to choose a model that captures stable structure without merely recalling noise in the training data.

The training partition $\mathcal{D}_{\text{train}}$ (equation (2.4.14)) is used to estimate model parameters through gradient-based optimisation. The validation partition $\mathcal{D}_{\text{val}}$ guides the choice of training duration via the epoch-selection criterion in equation (2.4.17). The test partition $\mathcal{D}_{\text{test}}$ is reserved for final out-of-sample evaluation after the model has been selected and its architecture fixed. Using the test set for any model-selection decision invalidates its role as an honest estimate of generalisation performance (Hastie et al., 2009; Abu-Mostafa et al., 2012).

Generalisation is assessed by comparing training, validation and test performance using RMSE, MAE and R^2. If training performance is substantially better than validation or test performance, this may indicate overfitting. If all three are weak, this may indicate underfitting. If training, validation and test results are similar and sufficiently strong, this supports the conclusion that the fitted model generalises reasonably well within the

available dataset.

The use of a benchmark health insurance dataset introduces an additional qualification: internal predictive validity within the dataset's distributional domain must be distinguished from external validity for real insurance populations with different risk profiles, benefit structures or demographics. The scope delimitations of Section 1.8 formalise this boundary.

2.5 Health Insurance Pricing and Cost Prediction

The supervised-regression formulation gives the statistical form of the empirical problem: the observed response y_i is a continuous monetary outcome, and the fitted function $\widehat{m}(\mathbf{x})$ maps policyholder attributes to a predicted cost. The health insurance setting gives this regression problem its actuarial meaning. A predicted monetary outcome is not merely an abstract response variable; it is connected to premium rating, claim-cost estimation and risk segmentation. Classical actuarial pricing often relies on transparent regression-based models, especially GLMs, while the DNN considered here plays the related role of a flexible nonlinear estimator of the cost surface rather than a replacement for actuarial reasoning.

The broader actuarial machine-learning literature provides extensive evidence that neural networks and related models can be embedded within traditional insurance frameworks. Wüthrich and Merz (2023) provide a comprehensive treatment of statistical foundations for actuarial learning, Noll et al. (2020) demonstrate neural-network pricing on French motor third-party liability data, and Schelldorfer and Wüthrich (2019) show how classical actuarial models can be nested within neural-network architectures. Blier-Wong et al. (2021) survey machine-learning applications across property and casualty insurance pricing and reserving. In the life insurance domain, Blomerus (2025) demonstrates the viability of deep learning for long-term life insurance valuations using commercial European portfolio data, achieving valuations sufficiently accurate to support capital calculations under Solvency II; the 99.5% figure in that framework is the confidence level of the one-year Value-at-Risk defining the Solvency Capital Requirement, rather than a prescribed valuation accuracy standard.

Within health insurance specifically, recent work has demonstrated that neural networks can be incorporated into actuarial pricing structures without abandoning the underlying frequency-severity logic. Laporta et al. (2025) propose neural-network models for pricing correlated health risks by embedding negative multinomial neural networks for claim frequency and gamma neural networks for claim severity within a classical frequency-severity structure. Their empirical results on real-world health insurance data show that neural-network models achieve lower out-of-sample deviance and better risk diversification than established regression methods such as GLMs, as measured by the ABC lift metric. The relevance of this work is that neural networks need not be treated as generic black-box alternatives; they can be incorporated within actuarial pricing logic to improve both accuracy and risk segmentation.

A complementary approach to neural-network pricing is to impose explicit shape constraints during training. Richman and Wüthrich (2024) develop the ICEnet, which adds smoothness and monotonicity penalty terms to the loss function so that the network's ICE

outputs vary in a manner consistent with actuarial domain knowledge. Applied to claims frequency prediction on the French motor third-party liability dataset, the ICEnet achieves a test-set Poisson deviance within 0.0001 of the unconstrained network, demonstrating that actuarial reasonableness can be enforced with negligible accuracy cost. Because the ICEnet targets count data under a Poisson deviance criterion, its results are not directly comparable with the continuous monetary cost predictions in the present dissertation; a methodological comparison is presented in Section 5.12.

Alongside neural-network approaches, tree-based ensemble methods have become widely used for insurance cost prediction on structured tabular data. Orji and Ukwandu (2024) apply RFs and gradient-boosted trees to health insurance cost data and report predictive performance comparable to, and in some configurations exceeding, that of neural networks. More broadly, Grinsztajn et al. (2022) provide systematic evidence that tree-based models, particularly gradient-boosted implementations such as XGBoost (Chen and Guestrin, 2016), building on the gradient boosting framework of Friedman (2001), frequently match or outperform DNNs on medium-sized tabular datasets of the kind encountered in actuarial practice. Because insurance cost prediction is fundamentally an actuarial problem defined on structured policyholder attributes rather than on unstructured data such as images or text, tree-based methods represent a credible alternative to the DNN and may in some settings be the preferred model class. The present dissertation nonetheless selects the DNN as the primary model for the MC and Shapley workflow because it provides a differentiable prediction surface compatible with gradient-based SHAP explainers and with continuous forward-pass evaluation over large simulated input samples; the specific gradient-based explainer employed is described in Chapter 3. An RF and an XGBoost regressor are fitted as benchmarks to contextualise the DNN's predictive performance within this spectrum of model classes.

Earlier work on algorithmic health cost prediction by Bertsimas et al. (2008) demonstrated the potential of machine-learning methods to outperform traditional actuarial approaches, while Duncan et al. (2016) tested alternative regression frameworks for predictive modelling of healthcare costs in the North American context. These studies confirm that health insurance cost prediction is a well-studied regression problem where model flexibility can yield meaningful improvements over simpler specifications.

The available Kaggle insurance data (Kaggle, 2023) do not provide separate claim-frequency and claim-severity processes. A separate actuarial claims model is therefore not fitted. Instead, the observed monetary response is treated as an aggregate health insurance charge or premium proxy, and cost prediction is formulated as a supervised regression problem. This gives a narrower empirical objective: to study how a fitted nonlinear prediction surface behaves on a benchmark health insurance dataset, without claiming to reproduce the complete pricing tariff or underwriting rules of a specific insurer.

The objective is not merely to obtain a point estimate of cost for each policyholder, but to understand the distribution and drivers of predicted costs under varying input profiles. This is important because actuarial modelling is concerned not only with average prediction, but also with uncertainty, high-cost scenarios and the interpretation of risk drivers (Denuit et al., 2007). In this sense, the fitted regression function $\widehat{m}$ is the starting point for a broader analysis of predicted-cost behaviour rather than the end of the modelling exercise.

Health expense projection literature provides a further actuarial motivation for studying predicted costs distributionally. Christiansen et al. (2018) develop practical models for projecting health insurance expenditures over short time horizons, which supports the view that health cost analysis should consider the behaviour of cost outcomes beyond isolated point estimates. The present dissertation differs from traditional health expense projection because it does not construct a demographic, utilisation or claims-process model. Instead, it trains a DNN on policyholder-level attributes and uses Monte Carlo input simulation to examine the distribution of costs implied by the fitted prediction function. This distinction keeps the empirical claim modest: the analysis concerns model-implied predicted costs under the selected benchmark-data and simulation design, rather than realised expenditure dynamics in an insurer's full portfolio.

Ordinary cost prediction is therefore extended in two ways. First, after fitting the DNN, MC input simulation is used to generate many possible policyholder profiles and examine the distribution of model-implied predicted costs. Second, Shapley-value interpretation is used to identify the variables that drive individual predictions and high-cost simulated profiles. This connects health insurance cost prediction with simulation-based risk analysis and explainable machine learning. The resulting analysis asks not only how accurately the fitted model predicts costs, but also how predicted costs vary across simulated profiles and which variables the model attributes to the upper tail of the predicted-cost distribution.

The interpretation remains model-based: the MC distribution describes variation in fitted predictions under the selected input distribution, and Shapley values explain the fitted DNN's predictions relative to the chosen background distribution. The causal and scope delimitations stated in Section 1.8 apply throughout. The health insurance application is therefore used as an actuarially motivated setting in which prediction, simulation and explanation can be combined while keeping the empirical claims within the limits of the data.

2.6 Input Distribution Diagnostics for Simulation

The validation analysis of Section 2.4.3 addresses whether the fitted DNN generalises within the supplied training, validation and test structure, but MC input simulation requires an additional diagnostic step. Even when a fitted prediction function $f_{\widehat{\theta}}$ has acceptable validation and test performance, the simulation results may be misleading if the input profiles are generated from an unrealistic or poorly specified input distribution. The diagnostic concern therefore shifts from the fitted model itself to the probability law used to generate simulated policyholder profiles.

A trained model $f_{\widehat{\theta}}$ can be evaluated at any admissible input vector, but simulation-based analysis requires a probability law for the random input vector $\mathbf{X}$. The distribution of simulated predictions depends not only on the fitted DNN, but also on the assumed or estimated distribution of policyholder attributes. Input distribution diagnostics therefore provide the empirical basis for specifying the input distribution $\widehat{F}_X$ used later in the MC component of the framework.

Let

$$\mathbf{X} = (X_1, \ldots, X_p)^\top$$

denote the vector of input variables before final model encoding, and let $\widetilde{\mathbf{X}}$ denote the corresponding model-ready vector after encoding and standardisation. Where no ambiguity arises, $\mathbf{X}$ is used as the generic random input vector. Before specifying an input distribution for $\mathbf{X}$, the empirical distribution of each variable must be examined. For numerical variables, this includes measures of location, scale, skewness, tail behaviour and possible outliers. For categorical variables, this includes category frequencies, rare levels and possible class imbalance.

For a numerical variable X_j, the sample mean and variance are

$$\bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}, \qquad s_j^2 = \frac{1}{n-1}\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2 .$$

Skewness can be estimated by

$$\widehat{\gamma}_{1j} = \frac{\frac{1}{n}\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^3}{\left[\frac{1}{n}\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2\right]^{3/2}},$$

and kurtosis by

$$\widehat{\gamma}_{2j} = \frac{\frac{1}{n}\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^4}{\left[\frac{1}{n}\sum_{i=1}^{n} (x_{ij} - \bar{x}_j)^2\right]^{2}}.$$

Following common convention, the sample variance s_j^2 uses the unbiased divisor $n-1$, whereas the skewness and kurtosis estimators are ratios of divisor-n central moments; this

mixed convention is adopted throughout. Values of $\widehat{\gamma}_{1j} > 0$ indicate right-skewness, while values of $\widehat{\gamma}_{2j}$ substantially above 3 indicate heavier-than-normal tails. These quantities help identify variables that are asymmetric, heavy-tailed or potentially unsuitable for simple symmetric distributional assumptions.

The response variable `charges` is also subject to diagnostic analysis, even though MC input simulation focuses primarily on the predictors. Insurance cost variables often display right-skewness, heterogeneity and upper-tail behaviour (Denuit et al., 2007). If the response is strongly skewed, a log transformation of the form

$$Y_i^* = \log(1 + Y_i)$$

may be considered during model training. If such a transformation is used, final predictions and performance metrics must be back-transformed and reported on the original monetary scale.

For categorical variables, the relevant diagnostic is the empirical frequency distribution. If X_j takes values in the finite set $\{c_1, \ldots, c_{K_j}\}$, the empirical proportion of category c_k is

$$\widehat{p}_{jk} = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}\{X_{ij} = c_k\}.$$

Rare categories may be poorly represented in the training set and may lead to unstable predictions or unreliable Shapley summaries. In the health insurance setting, categorical variables such as region, smoking status, medical history, exercise frequency, occupation and coverage level may all influence the structure learned by the fitted model.

Input diagnostics also inform preprocessing. Numerical variables are standardised before training a DNN to support numerical stability in gradient-based optimisation. The standardisation parameters must be computed from the training set only and then applied to the validation and test sets using those same training-set values, so that information from held-out observations does not leak into the training procedure. The specific standardisation formula and its implementation are detailed in Chapter 3.

Validation analysis assesses whether $f_{\widehat{\theta}}$ performs adequately on held-out observations from the available data distribution. Input distribution diagnostics assess whether the profiles generated for MC simulation remain compatible with that same empirical domain. Without both checks, the subsequent predicted-cost distribution and high-cost Shapley summaries could reflect unrealistic simulated profiles rather than meaningful behaviour of the fitted model.

2.7 Specification of the Input Distribution

The input distribution $\widehat{F}_X$ was defined as a product of independent fitted marginals (equation (2.2.5)), with the sampling and propagation procedure given by equations (2.2.6) and (2.2.7). Three main approaches are available for constructing $\widehat{F}_X$.

The simplest approach is empirical row resampling, in which complete observed covariate vectors are sampled from the dataset with replacement (Efron and Tibshirani, 1993):

$$\mathbf{X}^{(\ell)} \in \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}, \qquad \ell = 1, \ldots, M. \tag{2.7.1}$$

Because the full covariate vector is sampled as a unit, empirical row resampling retains whatever joint structure is present in the observed data. Its limitation is that it cannot generate policyholder profiles outside the observed data range.

A second approach is marginal parametric simulation, in which each input variable is assigned a fitted marginal distribution (Rubinstein and Kroese, 2016; Glasserman, 2004). For a numerical variable X_j, a parametric family $F_j(\,\cdot\,;\psi_j)$ may be fitted by maximum likelihood:

$$\widehat{\psi}_j = \arg\max_{\psi_j} \sum_{i=1}^{n} \log f_j(x_{ij};\psi_j), \tag{2.7.2}$$

where $f_j(\cdot;\psi_j)$ is the corresponding density or probability mass function. Simulated values are then drawn as

$$X_j^{(\ell)} \sim F_j(\,\cdot\,;\widehat{\psi}_j), \qquad \ell = 1, \ldots, M. \tag{2.7.3}$$

This approach is convenient and can generate values beyond the observed sample. It is the approach adopted in this dissertation, where independent fitted marginals are combined as specified in equation (2.2.5). The independence assumption is a deliberate modelling choice, as discussed in connection with equation (2.2.5). Two strands of literature qualify this choice. Evaluating or explaining a fitted model at covariate combinations that lie off the data manifold can produce extrapolation artefacts, because the model is queried in regions where it was never constrained by training data (Hooker et al., 2021). Relatedly, when features are dependent, the marginal (interventional) and conditional formulations of Shapley attribution diverge, and estimating the conditional expectations required by the latter is non-trivial (Aas et al., 2021; Janzing et al., 2020). Under the product-marginal design adopted here the two formulations coincide by construction, which is precisely the condition under which the marginal SHAP estimator used later is internally consistent. The generated profiles may nonetheless include jointly implausible attribute combinations, so attribution results in sparsely supported regions of the input space must be interpreted with care.

A third approach is nonparametric or semi-parametric simulation, using bootstrap-type resampling or smoothed empirical distributions (Efron and Tibshirani, 1993). These methods avoid imposing a strong parametric form on the marginal distributions, although they remain limited by the information contained in the observed sample. Their usefulness depends on whether the empirical data provide sufficient coverage of the regions of the input space that are relevant for the simulation exercise.

The choice of specification method determines the input profiles passed through the fitted prediction function. A poorly specified input distribution may create high-cost profiles that are artefacts of the simulation design rather than meaningful evaluations of the fitted model.

Figure 5 illustrates the marginal parametric construction adopted here. Each column of $\mathcal{X}_{\text{train}}$ is fitted independently to a parametric marginal distribution, and the product of these marginals defines $\widehat{F}_X$, from which M synthetic input vectors are drawn to form the simulation design matrix.

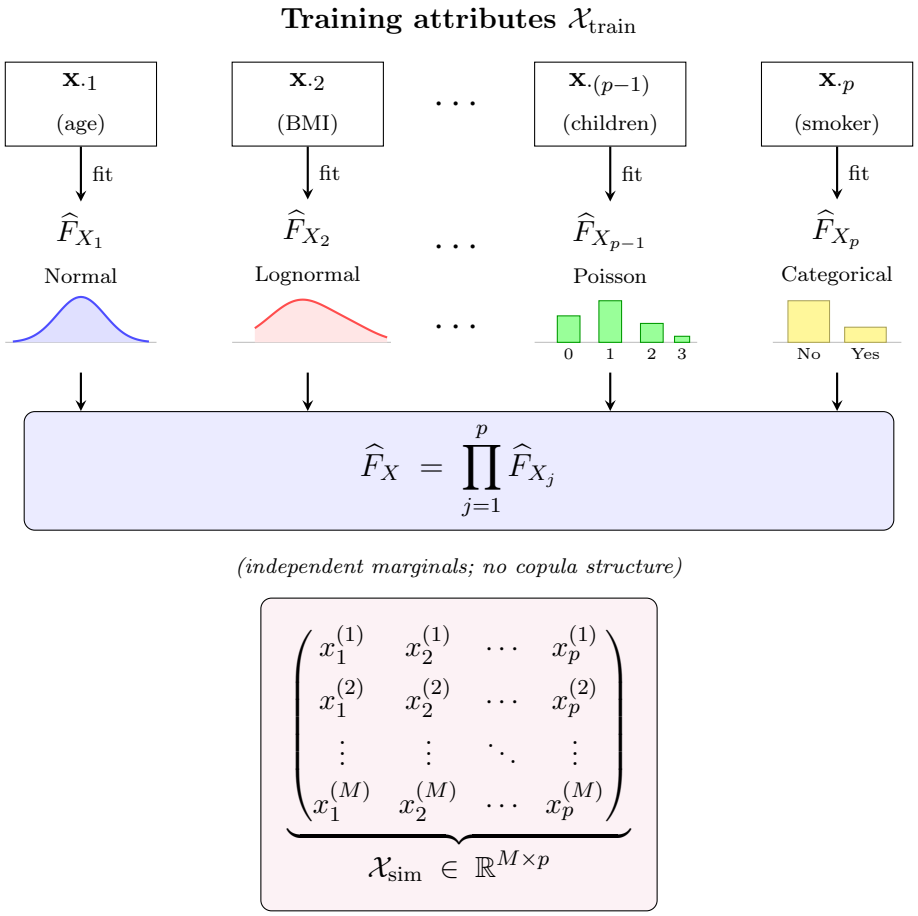


Figure 5: Construction of the input distribution $\widehat{F}_X$.

2.8 Foundations of MC

MC simulation rests on a small number of foundational results concerning the crude MC estimator, its convergence properties and the precision measures that underpin the analysis (Glasserman, 2004; Rubinstein and Kroese, 2016; Robert and Casella, 2004; Kroese et al., 2011).

Let X be a random variable with distribution F, and suppose that the quantity of interest is

$$\theta = \mathbb{E}[g(X)]. \tag{2.8.1}$$

If

$$X^{(1)}, \ldots, X^{(M)} \overset{\text{iid}}{\sim} F, \tag{2.8.2}$$

then the crude MC estimator of θ is

$$\widehat{\theta}_M = \frac{1}{M} \sum_{\ell=1}^{M} g\big(X^{(\ell)}\big)\,. \tag{2.8.3}$$

Under standard regularity conditions, the law of large numbers implies

$$\widehat{\theta}_M \longrightarrow \theta \qquad \text{as} \qquad M \longrightarrow \infty. \tag{2.8.4}$$

If

$$\sigma_g^2 = \mathrm{Var}\{g(X)\} < \infty, \tag{2.8.5}$$

then

$$\mathrm{Var}\Big(\widehat{\theta}_M\Big) = \frac{\sigma_g^2}{M}, \tag{2.8.6}$$

which yields the familiar $M^{-1/2}$ convergence rate of ordinary MC simulation (Caflisch, 1998; Glasserman, 2004).

The MCSE quantifies the simulation error associated with using a finite number of random draws. It is estimated by

$$\mathrm{MCSE}\Big(\widehat{\theta}_M\Big) = \frac{s_M}{\sqrt{M}}, \tag{2.8.7}$$

where

$$s_M^2 = \frac{1}{M-1} \sum_{\ell=1}^{M} \Big[g\big(X^{(\ell)}\big) - \widehat{\theta}_M\Big]^2. \tag{2.8.8}$$

By the central limit theorem,

$$\frac{\widehat{\theta}_M - \theta}{s_M/\sqrt{M}} \xrightarrow{\;d\;} N(0,1), \tag{2.8.9}$$

giving the approximate $100(1-\alpha)\%$ MC confidence interval

$$\widehat{\theta}_M \pm z_{1-\alpha/2}\frac{s_M}{\sqrt{M}}. \tag{2.8.10}$$

This interval describes uncertainty due to the finite number of MC draws. It does not, by itself, represent all sources of statistical uncertainty in the fitted DNN, the input distribution or the realised insurance cost.

In the health insurance framework, the function g is the fitted prediction function $\widehat{m}$, the distribution F is the fitted input distribution $\widehat{F}_X$, and the target θ corresponds to summaries of the model-implied predicted-cost distribution such as the mean, variance and upper-tail probabilities. The convergence, variance and confidence-interval results above therefore apply directly to those summaries.

2.9 Variance Reduction and Tail Simulation

For many quantities, ordinary MC simulation is sufficient when the simulation budget is large and the MCSE is reported. However, when the quantity of interest is a tail probability, a high quantile or a rare high-cost event, crude MC can become inefficient because few simulated draws fall in the region of interest. This issue is directly relevant to insurance because high-cost events may be financially important even when they are relatively rare (Glasserman, 2004). The statistical modelling of extreme events is well developed in the quantitative risk management literature (McNeil et al., 2015; Embrechts et al., 1997; Coles, 2001), where tail behaviour is characterised through extreme value distributions and related threshold exceedance models.

Two clarifications situate the present tail analysis within this literature. First, the quantity simulated here is the fitted conditional mean $\widehat{m}(\mathbf{X})$, so the upper tail examined is a tail of conditional means rather than of realised costs; since $\mathrm{Var}\{\mathbb{E}(Y \mid \mathbf{X})\} \leq \mathrm{Var}(Y)$, this tail is necessarily lighter than the tail of realised claim costs, and high-cost conditioning statements therefore concern expected costs rather than realised claim severities. Second, a fuller treatment of extremes would adopt the peaks-over-threshold framework, in which suitably normalised excesses over a high threshold converge to the generalised Pareto distribution (Pickands, 1975; Balkema and de Haan, 1974; Coles, 2001); fitting such threshold-exceedance models to the cost distribution is deferred to future work.

Let u denote a high-cost threshold, and suppose that the model-implied exceedance probability is

$$p_u = \mathbb{P}\left(\widehat{Y} > u\right),$$

where

$$\widehat{Y} = \widehat{m}(\mathbf{X}), \qquad \mathbf{X} \sim \widehat{F}_X.$$

Given simulated predictions $\widehat{Y}^{(1)}, \ldots, \widehat{Y}^{(M)}$, the crude MC estimator of p_u is

$$\widehat{p}_u = \frac{1}{M} \sum_{\ell=1}^{M} \mathbb{I}\left\{ \widehat{Y}^{(\ell)} > u \right\}.$$

Its estimated standard error is

$$\widehat{\mathrm{SE}}(\widehat{p}_u) = \sqrt{\frac{\widehat{p}_u \, (1 - \widehat{p}_u)}{M}}.$$

The relative standard error is approximately

$$\frac{\widehat{\mathrm{SE}}(\widehat{p}_u)}{\widehat{p}_u} \approx \sqrt{\frac{1 - \widehat{p}_u}{M \widehat{p}_u}}.$$

When p_u is small, the expected number of exceedances, $M p_u$, may remain small unless M is large. This explains why rare-event estimation can be difficult under crude MC sampling.

Importance sampling is one classical variance-reduction method. Suppose that samples are drawn from an alternative density q rather than the original density f. For a target expectation

$$\theta = \int g(x) f(x) \, dx,$$

the same quantity may be written as

$$\theta = \int g(x) \frac{f(x)}{q(x)} q(x) \, dx.$$

The corresponding importance-sampling estimator is

$$\widehat{\theta}_{\mathrm{IS}} = \frac{1}{M} \sum_{\ell=1}^{M} g\big(X^{(\ell)}\big) \, w\big(X^{(\ell)}\big), \qquad X^{(\ell)} \sim q,$$

where

$$w(x) = \frac{f(x)}{q(x)}$$

is the likelihood ratio, also called the importance weight. For the identity above to hold, the proposal density must satisfy $q(x) > 0$ wherever $g(x) f(x) \neq 0$, so that the integrand remains absolutely continuous with respect to the proposal. A well-chosen proposal distribution q concentrates simulation effort in the region that contributes most to the quantity being estimated, and can substantially reduce estimator variance (Glasserman, 2004; Ru-

binstein and Kroese, 2016). An ill-chosen proposal, however, yields unbounded likelihood ratios and can produce an estimator with severely inflated, or even infinite, variance, so the proposal distribution must be selected with care.

Stratified sampling is another variance-reduction approach. Suppose the input space is partitioned into disjoint strata

$$A_1, \dots, A_H,$$

with

$$p_h = \mathbb{P}(\mathbf{X} \in A_h)\,.$$

If M_h samples are drawn from stratum h, the stratified estimator of $\theta = \mathbb{E}\{g(\mathbf{X})\}$ is

$$\widehat{\theta}_{\text{strat}} = \sum_{h=1}^{H} p_h \overline{g}_h,$$

where $\overline{g}_h$ is the sample mean within stratum h. Its variance is

$$\operatorname{Var}\left(\widehat{\theta}_{\text{strat}}\right) = \sum_{h=1}^{H} \frac{p_h^2 \sigma_h^2}{M_h},$$

where σ_h^2 is the conditional variance of $g(\mathbf{X})$ within stratum h. In a health insurance application, natural strata may include smoking status, age band, BMI category, medical history group or coverage level. Stratification is useful when these groups are expected to contribute differently to the predicted-cost distribution (Rubinstein and Kroese, 2016).

Quasi-MC methods replace independent random points with low-discrepancy sequences designed to cover the unit cube more evenly. If

$$\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(M)} \in [0,1]^p$$

are low-discrepancy points and T maps uniform vectors to input profiles, the quasi-MC estimator is

$$\widehat{\theta}_{\text{QMC}} = \frac{1}{M} \sum_{\ell=1}^{M} g\big(T\big(\mathbf{u}^{(\ell)}\big)\big)\,.$$

For integrands of bounded variation in the sense of Hardy and Krause, the Koksma-Hlawka inequality implies that quasi-MC methods can improve convergence relative to ordinary Monte Carlo, with rates closer to

$$O\big(M^{-1}(\log M)^p\big)$$

than the usual

$$O\big(M^{-1/2}\big)$$

MC rate (Caflisch, 1998; Lemieux, 2009; Owen, 2013). At the input dimension used here, $p = 22$, the $(\log M)^p$ factor renders this worst-case bound practically vacuous for feasible simulation sizes, so any benefit of quasi-MC sampling in this setting would rest on low effective dimension of the integrand rather than on the formal rate.

In the present framework, variance-reduction methods are primarily relevant for interpreting high-cost simulated profiles. The main simulation exercise may use crude MC or empirical resampling, but the variance-reduction literature clarifies why tail quantities require care when reporting precision. If high-cost probabilities, upper quantiles or conditional Shapley summaries for extreme predicted costs are estimated, the simulation size and precision of those estimates should be reported and justified.

A practical criterion for selecting a simulation size is a pilot-based calculation. If s_0^2 is a pilot estimate of the variance of the simulated quantity and an absolute half-width ϵ is desired for an approximate confidence interval, then a rough simulation-size requirement is

$$M \gtrsim \left(\frac{z_{1-\alpha/2} s_0}{\epsilon}\right)^2.$$

This expression connects the simulation budget to the desired numerical precision of the reported MC summaries.

Tail simulation also connects to Shapley-value attribution. The simulated high-cost set $\mathcal{H}_u$ (equation (2.2.19)) contains the profiles for which feature-attribution summaries are most relevant to the upper tail of the fitted model. Shapley values can then be used to identify which variables receive large model-attributed contributions among simulated profiles whose fitted predictions exceed u. In this way, tail simulation provides the subset of simulated profiles on which the high-cost interpretability analysis is based.

2.10 Explainability, Shapley Values and Conditional Interpretation

DNNs are flexible nonlinear prediction models, but their flexibility often comes at the cost of interpretability. In insurance applications, this is a serious practical and regulatory concern. A model may produce accurate predictions, but insurers, regulators and policyholders may still require an explanation of why a particular profile receives a high predicted cost. This motivates the use of explainable artificial intelligence methods (Lundberg and Lee, 2017; Molnar, 2022).

Shapley values originate in cooperative game theory and provide a principled axiomatic method for allocating a total payoff among players (Shapley, 1953). In machine learning, the players are the input variables and the payoff is the model prediction. The Shapley value of feature j measures the average marginal contribution of that feature across all

possible subsets of features, as formalised in equation (2.2.11). The Shapley value is the unique solution satisfying four axioms: efficiency (contributions sum to the total prediction shift), symmetry (identically contributing features receive identical values), dummy (a feature that never changes the value receives zero), and additivity (Shapley, 1953).

The SHAP framework connects Shapley values to machine learning model interpretation (Lundberg and Lee, 2017). The additive decomposition in equation (2.2.12) provides a local explanation of an individual prediction. Within the class of additive feature attribution methods, the Shapley value is the unique attribution satisfying local accuracy (the machine-learning counterpart of the efficiency axiom above), missingness and consistency (Lundberg and Lee, 2017, Theorem 1), a result building on the uniqueness theorem of Young (1985), in which a monotonicity condition takes the place of the dummy and additivity axioms of the classical characterisation. This uniqueness result is why the Shapley value constitutes the theoretical foundation for the SHAP computational framework. Covert et al. (2021) provide a unifying removal-based framework that relates SHAP to other explanation methods, Sundararajan and Najmi (2020) examine the many distinct Shapley value formulations that arise under different baseline and feature-dependence assumptions, and Lundberg et al. (2020) extend Shapley-based explanations from local to global understanding for tree models.

Shapley-based explanations can be interpreted locally or globally. A local explanation focuses on a single policyholder profile and explains why the model assigned a particular predicted cost. A global explanation summarises feature contributions across many observations. A common global importance measure is the mean absolute Shapley value:

$$I_j = \frac{1}{n} \sum_{i=1}^{n} |\phi_j(\mathbf{x}_i)|.$$

Large values of I_j indicate that feature j tends to contribute strongly to predictions across the dataset.

In health insurance cost prediction, Shapley values can help identify whether variables such as smoking status, age, BMI, medical history, occupation, region or coverage level are driving predicted charges. The model-based interpretation caveat stated in Section 2.2.4 applies equally here: Shapley contributions reflect the fitted model's attributions, not causal effects. This distinction is particularly important when working with a benchmark dataset such as the Kaggle Insurance Data for Machine Learning (Kaggle, 2023), where Shapley values explain the relationships learned from the dataset's generating structure rather than from real-world pricing processes.

The dissertation extends ordinary Shapley analysis by connecting it to MC simulation.

Instead of computing Shapley values only on the observed test set, attributions are also examined for simulated policyholder profiles whose fitted predictions exceed a high-cost threshold, using the conditional summaries $\widehat{\Phi}_{j,u}$ and $\widehat{A}_{j,u}$ already formalised in Section 2.2.4 (equations (2.2.21) and (2.2.22)). This conditional interpretation moves beyond ordinary global feature ranking by focusing on the part of the prediction distribution that is most relevant for insurance risk analysis: it asks which variables contribute most strongly among precisely those profiles that the model regards as expensive.

2.11 Research Gap

Supervised regression provides the predictive formulation, the DNN supplies a fitted non-linear prediction surface, MC simulation propagates variation in the input profiles, and Shapley values provide feature-level attribution for fitted predictions. Each of these components is individually well established in the statistical and machine-learning literatures (Vapnik, 1998; Friedman, 2001; Robert and Casella, 2004; Lundberg and Lee, 2017; Molnar, 2022). The contribution of this dissertation is methodological: the conditional Shapley analysis computed within Monte-Carlo-simulated high-cost regions produces a form of threshold-conditional feature attribution that none of these individually mature tools can deliver in isolation. While each component is well known, their integration within a single, reproducible workflow provides a capability not previously demonstrated in the health insurance cost-prediction literature.

MC simulation has a long history in actuarial science and quantitative finance, where it is used primarily for loss reserving, aggregate loss distributions, solvency capital estimation and derivative pricing (Glasserman, 2004). In these applications the simulation target is typically a financial or insurance outcome variable, and the objective is to estimate a distributional quantity such as a quantile or expected shortfall. The use of MC simulation to generate a synthetic input space and propagate it through a fitted prediction function, so that the model-implied predicted-cost distribution can be examined, is less standard in the health insurance cost-prediction literature.

Shapley-based explanations have become widespread in applied machine learning (Rozemberczki et al., 2022), including insurance applications where SHAP is used to rank feature importance on observed test data (Kaushik et al., 2022). Conditional Shapley analysis, in which attributions are computed for profiles within a specific region of the predicted-cost distribution such as the simulated high-cost tail, is not a routine component of these analyses. The survey by Rozemberczki et al. (2022) catalogues a wide range of Shapley-value applications in machine learning but does not discuss conditioning Shapley attribution on simulated prediction regions.

At the prediction stage, several studies fit DNNs or comparable machine-learning models

to health insurance cost data and evaluate predictive performance using standard regression metrics (Samiuddin et al., 2023; Drewe-Boss et al., 2022; Kaushik et al., 2022). These studies demonstrate that neural networks and related estimators can approximate the cost surface on policyholder-level attributes, but the analysis typically concludes at the point-prediction stage. Neither simulation over a synthetic input space nor conditional feature attribution beyond global importance is reported.

The research gap is therefore the limited integration of these three individually established components for health insurance cost prediction. The literature supports each component in isolation (Wüthrich and Merz, 2023; Blier-Wong et al., 2021; Kroese et al., 2011; Covert et al., 2021), but gives less attention to the combined question of how a fitted prediction model behaves across many simulated policyholder profiles, how the resulting model-implied predicted-cost distribution behaves, and which variables the fitted model attributes to high-cost simulated profiles. Building on the deep-learning approach established by Blomerus (2025) in the life insurance domain, the present study extends this line of inquiry to health insurance cost prediction with an added simulation and attribution layer.

The framework addresses this gap by connecting the three components within a single empirical workflow (Figure 1). Chapter 3 translates the conceptual framework and theoretical foundations developed in this chapter into a concrete methodological workflow, specifying the benchmark dataset, preprocessing protocol, network architecture, MC simulation design and Shapley-value implementation.

3 Methods and Data

3.1 Introduction

This chapter translates the conceptual DNN, MC and Shapley framework of Chapter 2 into a concrete empirical workflow applied to the Kaggle Insurance Data for Machine Learning benchmark dataset (Kaggle, 2023). Every methodological choice made here, from the preprocessing protocol to the Shapley implementation settings, is documented with its statistical justification, so that the results of Chapter 4 can be traced back to an explicit and reproducible design.

The chapter is organised as follows. Section 3.2 describes the data source and variable structure, and Section 3.3 sets out the data audit, partitioning, preprocessing and leakage-prevention protocol. Section 3.4 presents the exploratory diagnostics that motivate the main modelling and simulation choices. Section 3.5 specifies the architecture and training protocol of the DNN that serves as the prediction function, together with the GLM and tree-based benchmark models, and Section 3.6 presents the predictive and structural checks that must be satisfied before the network is used in the simulation and attribution analyses. Section 3.7 sets out the parametric MC simulation design, Section 3.8 validates the resulting input distribution by diagnosing its joint-distribution difference from the training data, and Section 3.9 develops the residual-augmented sensitivity analysis. Section 3.10 details the Shapley-value implementation for simulated profiles, and Section 3.11 closes the chapter with a formal statement of the modelling assumptions and interpretive boundaries.

3.2 Data Source and Variable Structure

The raw dataset, taken from Kaggle under the title Insurance Data for Machine Learning (Kaggle, 2023), contains $n = 1\,000\,000$ observations and 12 original columns. Eleven of these are predictors; the remaining column, `charges`, is the continuous monetary outcome. The terms *cost* and *charge* are used interchangeably throughout the dissertation to denote this same modelled quantity, and a *predicted cost* is always the fitted value of `charges` returned by a model. *Premium*, the price actually quoted to a policyholder, is a distinct quantity that incorporates expense and profit loadings and is not modelled here; where the word appears it refers to the literature being reviewed rather than to the response of this study. Reference-category dummy encoding expands the categorical predictors, yielding a final model-ready input dimension of $p = 22$.

Let $Y =$ `charges` denote the response, and let $\mathbf{X}_{\text{raw}}$ be the vector of raw predictors before

encoding. The predictor set consists of three numerical variables (`age`, `bmi`, `children`), two binary variables (`gender`, `smoker`), and six multi-level categorical variables (`region`, `medical_history`, `family_medical_history`, `exercise_frequency`, `occupation`, `coverage_level`). Table 1 summarises the role, measurement scale, and model-ready representation of each variable. All categorical levels are well represented; none require merging.

Variable	Role and type	Description	Model-ready representation
`charges`	Response, continuous	Insurance charge; positive monetary outcome	Standardised during training; de-normalised for evaluation
`age`	Predictor, numerical	Age (years), 18 to 65	One numerical input, standardised
`bmi`	Predictor, numerical	BMI, 18.00 to 50.00	One numerical input, standardised
`children`	Predictor, count	Number of dependants, 0 to 5	One numerical input, standardised
`gender`	Predictor, binary	Policyholder gender	One binary indicator (reference: female)
`smoker`	Predictor, binary	Smoking status (yes/no)	One binary indicator (reference: no)
`region`	Predictor, categorical	Geographical region (4 levels)	Three dummy indicators (reference: northeast)
`medical_history`	Predictor, categorical	Personal medical history	Three dummy indicators (reference: none / no condition)
`family_medical_history`	Predictor, categorical	Family medical history	Three dummy indicators (reference: none / no condition)
`exercise_frequency`	Predictor, categorical	Exercise frequency (4 levels)	Three dummy indicators (reference: rarely)
`occupation`	Predictor, categorical	Occupation (4 levels)	Three dummy indicators (reference: unemployed)
`coverage_level`	Predictor, categorical	Coverage tier (3 levels)	Two dummy indicators (reference: basic)

3.3 Data Audit, Partitioning and Preprocessing

3.3.1 Data audit

A comprehensive audit was performed before modelling. The variables `medical_history` and `family_medical_history` contain 250 762 and 250 404 missing values, respectively; all other variables are complete. Missing entries are treated as the reference category ("no reported condition"), and no observations are discarded. This convention conflates "not reported" with the genuine absence of a condition; an explicit missing-indicator category, or a sensitivity analysis under alternative missing-data treatments, would be required to isolate the effect of this choice. The treatment is therefore recorded as a limitation and revisited in Chapter 5 (Section 5.9). No exact duplicate records exist, all numerical values fall within plausible empirical ranges, and every categorical level is adequately represented.

3.3.2 Training, validation and test partition

The model-ready dataset is split into three mutually exclusive subsets according to the partition defined in equation (2.4.14). The split was generated by the SamplingForML platform (Blomerus, 2026). The training set contains 524 288 observations, fixed at 2^{19} observations for batch-alignment convenience, and the validation and test sets contain 237 856 observations each, corresponding to an approximately 52/24/24 partition. The role of each partition is as described in Section 2.4.3.

3.3.3 Encoding and standardisation

Categorical variables are converted to numerical form using reference-category dummy encoding, as introduced in equation (2.4.2). For a categorical variable C with levels $c_1, \ldots, c_K$, the dummy for level c_k is $Z_{ik} = \mathbb{I}\{C_i = c_k\}$, $k = 1, \ldots, K-1$, with c_K as the reference. Table 2 records the full encoding scheme.

Variable	Encoded columns	Reference category
`gender`	One indicator (male = 1)	Female
`smoker`	One indicator (yes = 1)	No
`region`	`southwest`, `northwest`, `southeast`	Northeast
`medical_history`	`Heart_disease`, `High_blood_pressure`, `Diabetes`	None / no reported condition
`family_medical_history`	`Heart_disease`, `High_blood_pressure`, `Diabetes`	None / no reported condition
`exercise_frequency`	`Occasionally`, `Frequently`, `Never`	Rarely
`occupation`	`Student`, `Blue_collar`, `White_collar`	Unemployed
`coverage_level`	`Standard`, `Premium`	Basic

Table 2: Variable encoding conventions and reference categories.

All numerical predictors are standardised using the training-set means $\widehat{\mu}_{j,\text{train}}$ and standard deviations $\widehat{\sigma}_{j,\text{train}}$:

$$\widetilde{x}_{ij} = \frac{x_{ij} - \widehat{\mu}_{j,\text{train}}}{\widehat{\sigma}_{j,\text{train}}}. \tag{3.3.1}$$

All encoded columns, including the non-degenerate binary dummy indicators, pass through this same training-estimated standardisation map, so each dummy column is likewise centred by its training-set mean and scaled by its training-set standard deviation. For binary dummy variables that have zero variance on the training set, the standard deviation is set to one so that the column retains its original 0/1 coding. The response is standardised analogously during training:

$$\widetilde{y}_i = \frac{y_i - \widehat{\mu}_{Y,\text{train}}}{\widehat{\sigma}_{Y,\text{train}}}, \tag{3.3.2}$$

where $\widehat{\mu}_{Y,\text{train}} \approx 16\,744$ and $\widehat{\sigma}_{Y,\text{train}} \approx 4\,419$ on the training partition (the full-data descriptive values in Table 3 differ marginally because they are computed over all one million records). Predictions are mapped back to the original monetary scale by

$$\widehat{y}_i = \widehat{\mu}_{Y,\text{train}} + \widehat{\sigma}_{Y,\text{train}}\widehat{\widetilde{y}}_i. \tag{3.3.3}$$

Every preprocessing parameter is estimated from the training partition only and applied unchanged to the validation set, the test set, and the MC simulated profiles. We denote this complete preprocessing procedure by $T(\cdot)$. The treatment of categorical variables during MC simulation, including the requirement to simulate at the raw level before

encoding, is described in Section 3.7.

3.3.4 Data-leakage prevention

Data leakage occurs when information from outside the training partition influences the model fitting or evaluation process, leading to overly optimistic performance estimates that do not generalise to new data. Three safeguards are enforced throughout the analysis to prevent leakage:

1. **Split before preprocessing.** The training, validation and test partitions are fixed before any preprocessing is applied. Standardisation parameters ($\widehat{\mu}_{j,\text{train}}$, $\widehat{\sigma}_{j,\text{train}}$) and encoding schemes are estimated from the training set only.
2. **Training-only standardisation.** The validation and test sets are transformed using the training-set means and standard deviations (equations (3.3.1) and (3.3.2)). MC simulated profiles are likewise preprocessed through the same training-derived transformation $T(\cdot)$.
3. **No test-set feedback.** The test partition is never consulted during hyperparameter selection or model training. It is evaluated only once, after the final network configuration is selected.

Additionally, no variable derived from `charges` is used as an input feature, preventing target leakage. Duplicate observations were checked during the data audit (Section 3.3), and no exact duplicate records were found. The unusually high predictive performance reported in Chapter 4 is attributable to the strong simulated structure of the Kaggle benchmark dataset rather than to any leakage pathway. These protocols ensure that every reported performance metric reflects genuine out-of-sample predictive ability. The preprocessing procedure is audited in Annexure G.

3.4 Exploratory Diagnostics

Exploratory diagnostics are used to guide the preprocessing, modelling and simulation choices. The analysis focuses on four aspects: the marginal behaviour of the numerical variables, the balance of categorical variables, the empirical distribution of the response, and the dependence structure among the numerical predictors.

3.4.1 Numerical variables and the response

Table 3 reports descriptive statistics for the three numerical predictors (`age`, `bmi`, `children`) and the response variable (`charges`).

Variable	Mean	SD	Min	Q1	Median	Q3	Max	$\widehat{\gamma}_1$	$\widehat{\gamma}_2$
`age`	41.50	13.86	18	29	41	53	65	0.001	−1.201
`bmi`	34.00	9.23	18.00	26.02	34.00	41.99	50.00	−0.000	−1.198
`children`	2.50	1.71	0	1	2	4	5	0.000	−1.268
`charges`	16 735	4 416	3 445	13 600	16 622	19 781	32 562	0.130	−0.349

Table 3: Descriptive statistics and distributional shape for numerical variables.

The three numerical predictors are essentially symmetric, with skewness values close to zero and negative excess kurtosis. This platykurtic behaviour is consistent with bounded, uniform-type distributions: the observations are spread relatively evenly across their supports rather than concentrated around their means. This directly motivates the use of uniform marginal distributions in the MC simulation design. The empirical bounds `age` $\in [18, 65]$, `bmi` $\in [18, 50]$, and `children` $\in \{0, \ldots, 5\}$ also provide the support restrictions that will be enforced when synthetic profiles are generated.

The response variable `charges` has mean 16 735 and median 16 622; their close agreement indicates that the centre of the distribution is not strongly affected by extreme observations. The skewness of 0.130 indicates mild positive skewness, while the excess kurtosis of −0.349 confirms that the response is not heavy-tailed. No logarithmic transformation is therefore applied; the response is standardised during training for numerical stability and de-normalised for evaluation.

Figure 6 displays the empirical histograms. The variables `age` and `bmi` are approximately flat across their bounded ranges, with mean and median nearly coincident. The count variable `children` is discrete and approximately balanced across its six integer values. The response `charges` is unimodal with the mean lying slightly to the right of the median, reflecting the mild positive skewness.

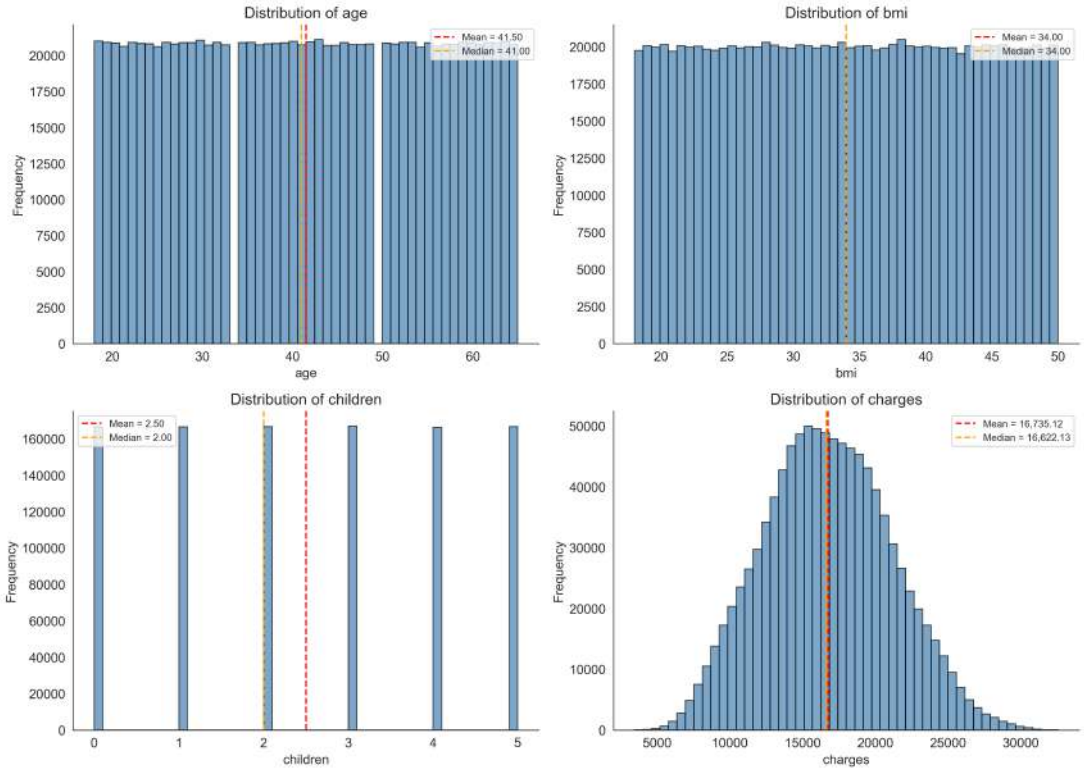


Figure 6: Empirical distributions of the numerical variables.

Selected percentiles of `charges` are reported in Table 4. The interquartile range is approximately 6 181, and the 90th, 95th, and 99th percentiles are 22 591, 24 224, and 27 016, respectively. These values serve as empirical scale references; in the later MC and Shapley stages, high-cost profiles are defined using upper-tail quantiles of the simulated predicted-cost distribution, not directly from these observed percentiles.

Percentile	P1	P5	P10	P25	P50	P75	P90	P95	P99
Value	7 520	9 558	10 947	13 600	16 622	19 781	22 591	24 224	27 016

Table 4: Selected percentiles of the response variable `charges`.

3.4.2 Dependence structure

The Pearson correlation matrix for the four numerical variables is given in Table 5.

	age	bmi	children	charges
age	1.000	0.001	−0.001	0.063
bmi	0.001	1.000	−0.002	0.104
children	−0.001	−0.002	1.000	0.077
charges	0.063	0.104	0.077	1.000

Table 5: Pearson correlation matrix for the numerical variables.

Pairwise correlations among the predictors are negligible ($|r| \leq 0.002$), and the marginal correlations with `charges` are weak. This does not imply that the predictors are unimportant (the deep network can learn complex nonlinear interactions invisible to a linear correlation matrix), but it does provide empirical support for the product-marginal independence assumption used in the MC simulation. The correlation heatmap in Figure 7 confirms the uniformly pale off-diagonal shading.

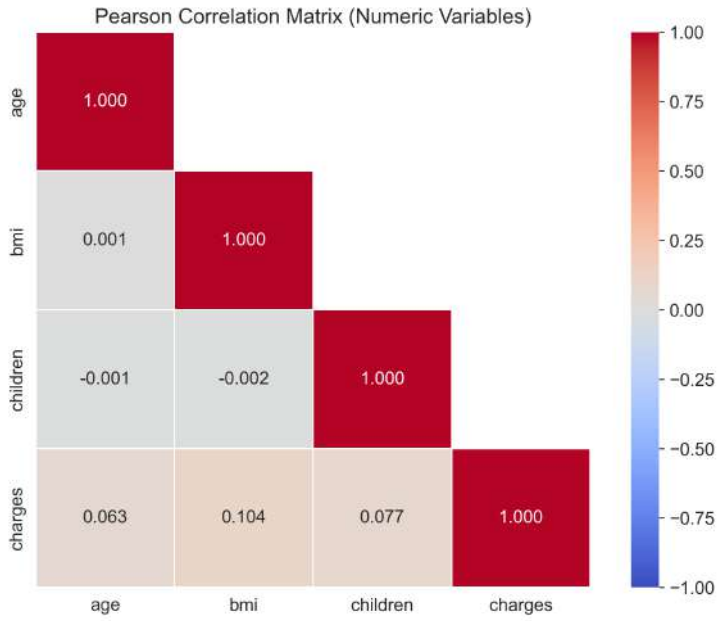
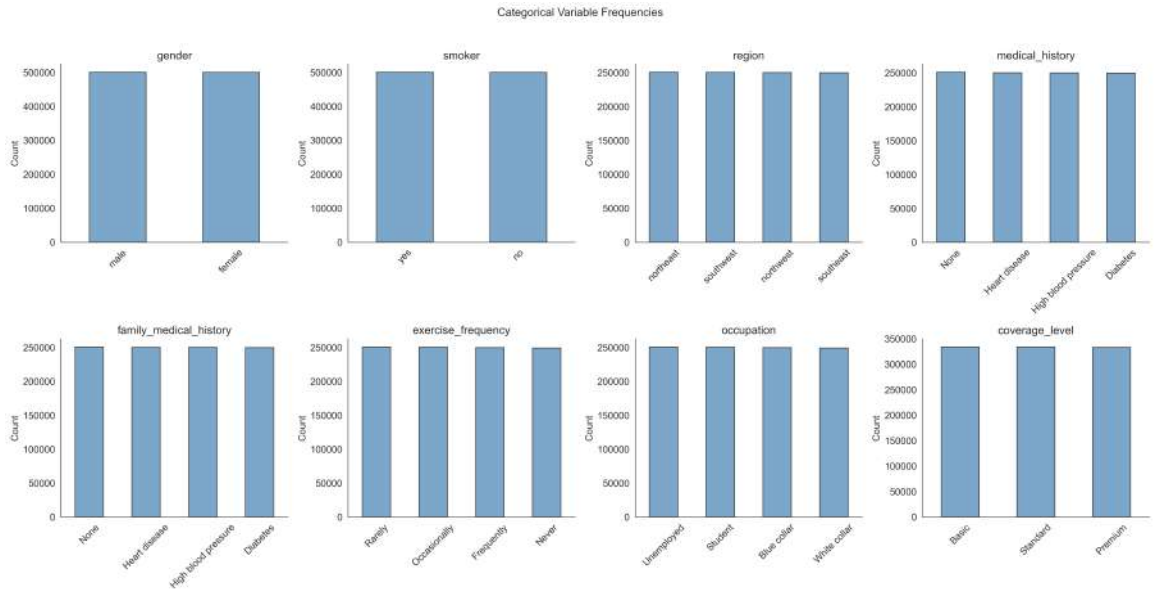
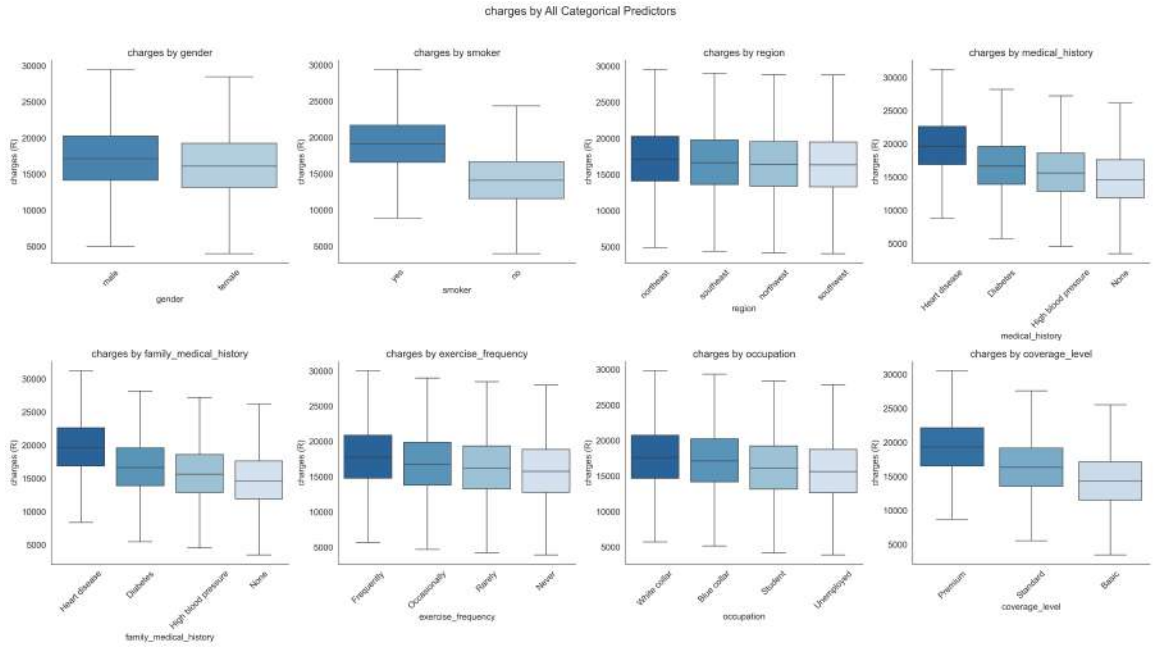


Figure 7: Pearson correlation heatmap for the numerical variables.

3.4.3 Categorical predictors and encoded features



(a) Frequency distributions of categorical and binary predictors.



(b) Boxplots of `charges` by categorical predictor.

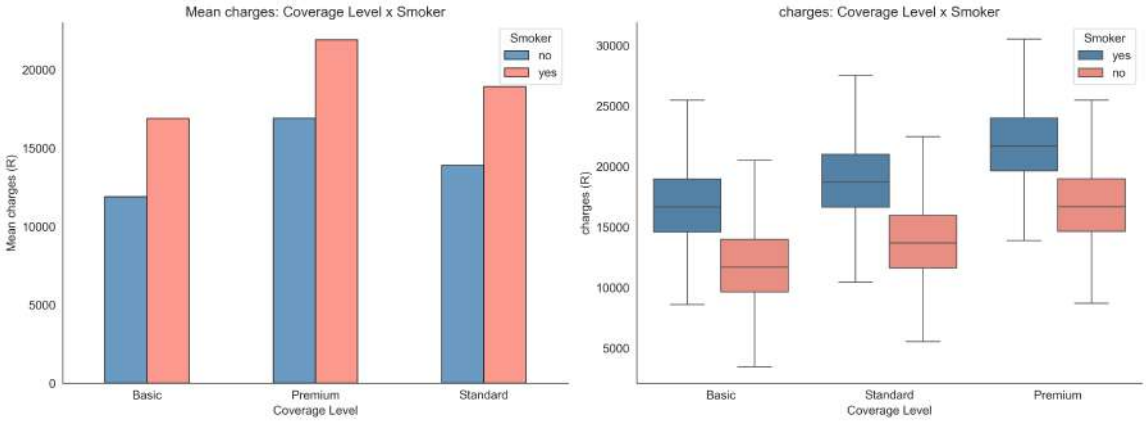
Figure 8: Categorical predictor diagnostics: frequency distributions and conditional boxplots.

Figure 8a confirms that every categorical and binary predictor is approximately uniformly distributed across its levels. The two binary variables (`gender` and `smoker`) each have close to 500 000 observations per level, the four-level variables (`region`, `medical_history`, `family_medical_history`, `exercise_frequency` and `occupation`) each have approximately 250 000 observations per level, and the three-level `coverage_level` vari-

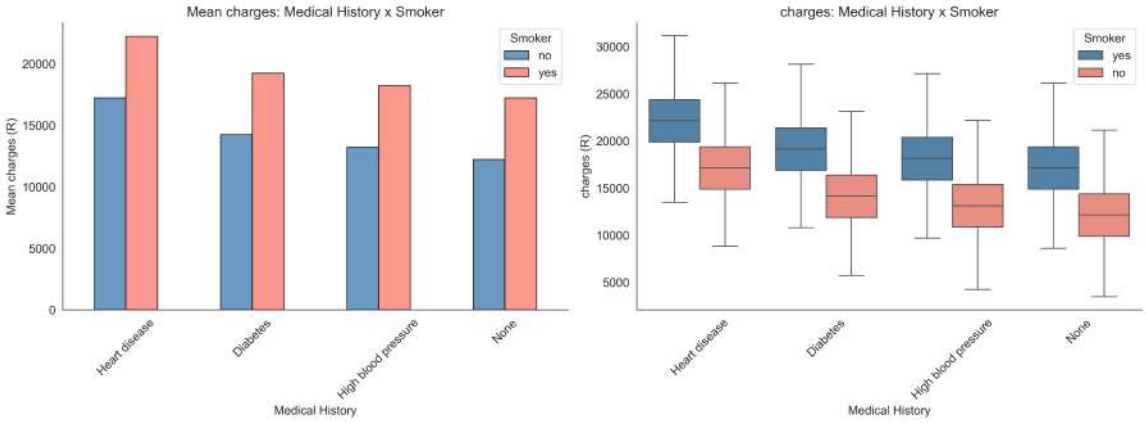
able has approximately 333 000 observations per tier. These balanced frequencies ensure that no category is underrepresented in the training data and that the fitted category proportions used in the MC simulation are estimated with high precision.

Figure 8b displays the conditional distribution of charges for each categorical predictor. The `smoker` panel shows the clearest separation: smokers have a median near R19 000 compared to approximately R14 000 for non-smokers. The `coverage_level` panel reveals a monotonic tier structure, with Premium policyholders having the highest median charges (approximately R20 000), followed by Standard (approximately R17 000) and Basic (approximately R15 000). Among `medical_history` categories, Heart disease is associated with the highest median charges (approximately R20 000), followed by Diabetes, High blood pressure and None. The `family_medical_history` panel shows the same ordering, with Heart disease again yielding the highest charges. The `exercise_frequency` panel shows a modest gradient, with Frequently exercising policyholders having marginally higher median charges than those who Never exercise. The `gender`, `region` and `occupation` panels display smaller unconditional differences, though males tend to have slightly higher charges than females across all panels.

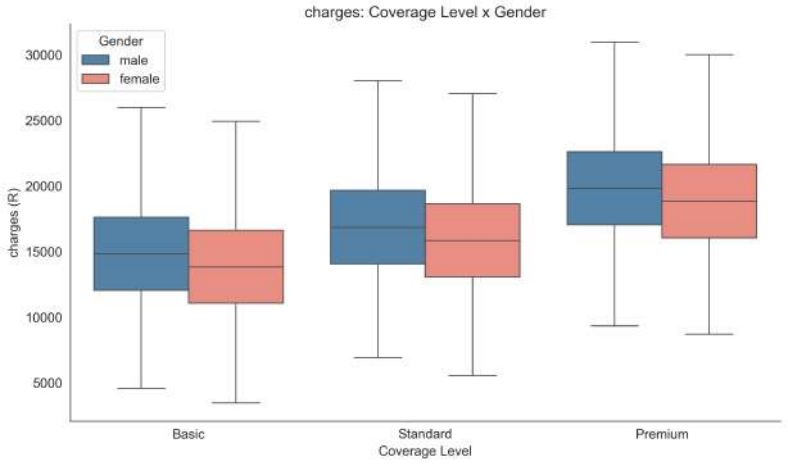
Two-way interactions were examined for `smoker` × `coverage_level`, `smoker` × `medical_history` and `coverage_level` × `gender` to assess whether the effect of one variable on charges depends on the level of another.



(a) `smoker` × `coverage_level` interaction effects on charges.



(b) `smoker` × `medical_history` interaction effects on charges.



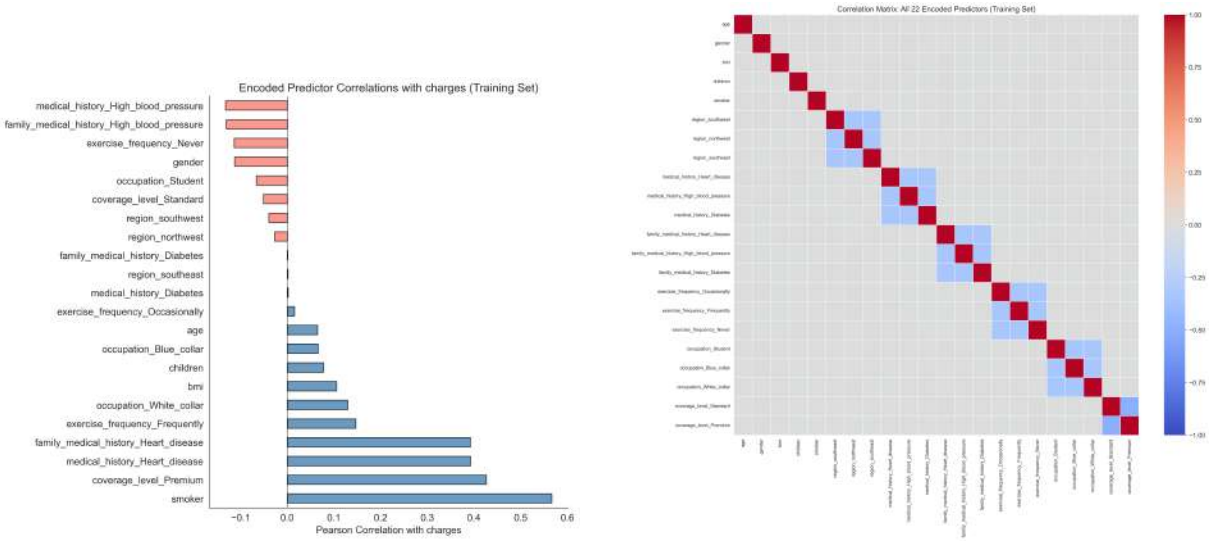
(c) `coverage_level` × `gender` interaction effects on charges.

Figure 9: Two-way interaction effects on `charges`.

Figure 9a shows that the smoker effect is present at every coverage tier but is amplified at higher coverage levels. At the Basic tier, the difference in mean charges between smokers and non-smokers is approximately R5 000 (R17 000 versus R12 000), whereas at the Premium tier the difference widens to approximately R5 000 to R6 000 on a higher

base (R22 000 versus R17 000). The boxplots confirm that smokers consistently occupy a higher distributional range within each tier. Figure 9b reveals a consistent smoker premium across all medical history categories. The largest combined effect occurs for smokers with Heart disease (mean charges approximately R22 500), while the lowest occurs for non-smokers with no reported condition (approximately R12 500). The smoker effect (approximately R5 000) is additive with the medical history effect, with the gap remaining relatively stable across all four levels. Figure 9c shows that males have slightly higher median charges than females at every coverage tier, with the gender gap widening modestly from Basic (approximately R1 000) to Premium (approximately R1 500). The interaction effects visible in Figure 9 support the use of a flexible DNN architecture capable of capturing nonlinear and interaction-based relationships.

After encoding, the Pearson correlations between the 22 encoded predictors and `charges` were computed on the training partition. The strongest linear associations are `smoker` ($r = 0.567$), `coverage_level_Premium` ($r = 0.427$), `medical_history_Heart_disease` ($r = 0.393$) and `family_medical_history_Heart_disease` ($r = 0.393$).



(a) Pearson correlations of encoded predictors with `charges`.

(b) Full 22×22 Pearson correlation matrix of encoded predictors.

Figure 10: Encoded predictor correlation analysis on the training partition.

Figure 10a ranks all 22 encoded predictors by their Pearson correlation with `charges`. The four strongest positive associations are `smoker` ($r = 0.567$), `coverage_level_Premium` ($r \approx 0.43$), `medical_history_Heart_disease` ($r \approx 0.39$) and `family_medical_history_Heart_disease` ($r \approx 0.39$). A second tier of moderate positive associations includes `exercise_frequency_Frequently` ($r \approx 0.15$), `occupation_White_collar` ($r \approx 0.12$), `bmi` ($r \approx 0.10$), `children` ($r \approx 0.08$) and `age` ($r \approx 0.06$). On the negative side, the largest associations are `medical_history_High_blood_pressure` ($r \approx -0.11$), `family_medical_`

`history_High_blood_pressure` ($r \approx -0.10$), `exercise_frequency_Never` ($r \approx -0.09$) and `gender` ($r \approx -0.08$). These negative values reflect the reference-category encoding: for example, a negative correlation for `medical_history_High_blood_pressure` indicates lower charges relative to the omitted reference category rather than a protective effect. Figure 10b shows the full inter-predictor correlation structure. The dominant feature is the block of negative correlations within each dummy group: dummy indicators derived from the same raw categorical variable are mechanically negatively correlated because only one indicator can be active at a time. Outside these within-group blocks, the off-diagonal correlations are uniformly near zero (grey shading), confirming the absence of problematic multicollinearity among predictors from different raw variables.

3.4.4 Partition consistency

Kernel density estimates of `charges` and of each numerical predictor were overlaid for the training, validation, and test partitions; the distributions are virtually indistinguishable. Proportions of each dummy variable were likewise compared, and no systematic shifts are present. The partitioning procedure therefore preserves the distributional characteristics of the data.

3.4.5 Decisions from the exploratory diagnostics

Four implementation decisions follow directly from the EDA:

1. Bounded numerical predictors are simulated under support restrictions that maintain their observed ranges.
2. Categorical variables are simulated at the raw-category level and only afterwards encoded.
3. The weak linear correlations among the numerical predictors, together with their uniform-like marginal shapes, motivate the product-marginal independence assumption adopted in the MC design (Section 3.7).
4. The response is standardised during training and de-normalised for evaluation; no further transformation is needed.

3.5 Network Specification and Training

The predictive component is a DNN for supervised regression, following the formulation of equations (2.2.2) and (2.2.3). The network maps the model-ready input vector $\mathbf{z}_i \in \mathbb{R}^{22}$

to a scalar prediction of the standardised response,

$$\widehat{\widetilde{y}}_i = f_{\widehat{\theta}}(\mathbf{z}_i), \tag{3.5.1}$$

which is subsequently de-normalised to the original monetary scale via equation (3.3.3).

3.5.1 Architecture

The network adopts the contracting funnel topology

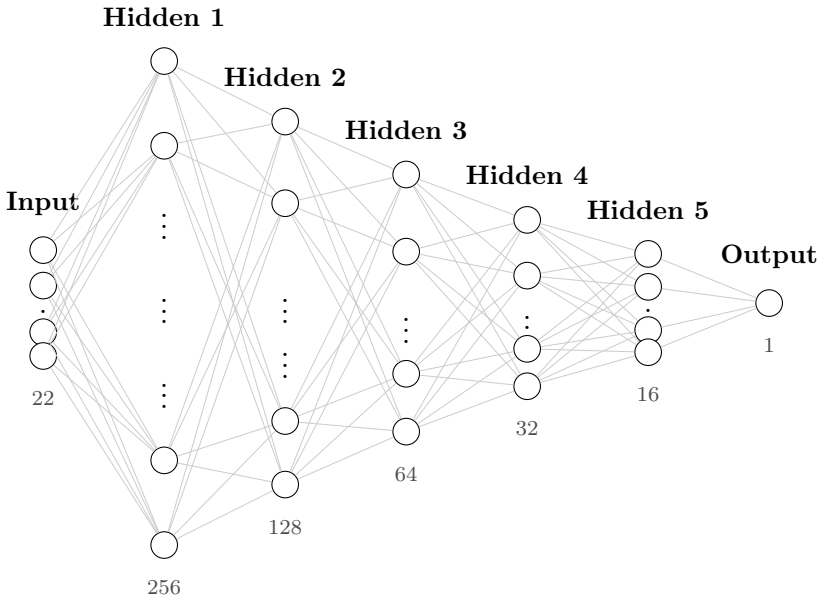


Figure 11: Funnel architecture of the DNN.

with the layer-wise recursion given by equation (2.4.2). All hidden layers use the CELU activation function (equation (2.4.9)). CELU was selected from among the candidate activations reviewed in Section 2.4.1 because it combines the computational efficiency of a rectifier with continuous differentiability across its entire domain, avoiding the dead-unit problem associated with ReLU while maintaining a linear positive branch that preserves gradient flow in deep architectures. The output layer uses an identity mapping, as required for continuous regression. The full five-hidden-layer depth, including the final 16-unit layer, is retained; its adequacy is assessed through the validation diagnostics of Section 3.6.

The network is constructed without bias vectors. This is a genuine restriction rather than a costless simplification: because the CELU activation satisfies $\sigma(0) = 0$, a bias-free network necessarily maps the origin to zero, so the fitted function is pinned to zero at the standardised data centroid and cannot represent arbitrary additive shifts (the corresponding universal-approximation caveat is discussed in Chapter 2). After standardisation, however, the data centroid maps near the origin, where the response surface lies

well within the support of the data, and the adequacy of the bias-free restriction is assessed empirically through the predictive diagnostics of Section 3.6 rather than guaranteed theoretically. The total number of trainable parameters is

$$\begin{aligned} N_{\text{par}} &= (22)(256) + (256)(128) + (128)(64) + (64)(32) + (32)(16) + (16)(1) \\ &= 49\,168. \end{aligned} \tag{3.5.2}$$

3.5.2 Forward and backward propagation

The operation of the contracting funnel specified in figure 11 can be described compactly by the forward- and backward-pass equations that underpin all gradient-based training. For a single input vector $\mathbf{x}$ (with $\mathbf{a}^{(0)} = \mathbf{x}$), the forward computation of layer l is

$$\mathbf{z}^{(l)} = \big(W^{(l)}\big)^{\top}\mathbf{a}^{(l-1)} + \mathbf{b}^{(l)}, \qquad W^{(l)} \in \mathbb{R}^{d_{l-1}\times d_l},\ \mathbf{b}^{(l)} \in \mathbb{R}^{d_l}, \tag{3.5.3}$$

$$\mathbf{a}^{(l)} = \sigma_l\big(\mathbf{z}^{(l)}\big), \tag{3.5.4}$$

where d_{l-1} and d_l are the widths of the adjacent layers. Because the input and response are standardised and the architecture omits bias vectors (Section 3.5), the bias term is not used; the forward pass simplifies to equation (2.4.13).

The network is trained by minimising the MSE loss $\mathcal{L} = \frac{1}{2}(y - \hat{y})^2$. Defining the gradient of the loss with respect to the output pre-activation as

$$\delta^{(L)} \equiv \frac{\partial \mathcal{L}}{\partial \mathbf{z}^{(L)}} = \hat{y} - y, \tag{3.5.5}$$

this error signal is propagated backward through the hidden layers. For layer $l = L - 1, L - 2, \ldots, 1$, the recursion is

$$\delta^{(l)} = \frac{\partial \mathcal{L}}{\partial \mathbf{z}^{(l)}} = \big(W^{(l+1)}\big)^{\top}\delta^{(l+1)} \odot \sigma_l'\big(\mathbf{z}^{(l)}\big), \tag{3.5.6}$$

where $\odot$ denotes element-wise multiplication and σ_l' is the derivative of the activation function. For the CELU activation employed in every hidden layer (equation (2.4.9)), the derivative is smooth and non-zero for all inputs, ensuring stable gradient flow.

The gradient of the loss with respect to the weight matrix of layer l is

$$\frac{\partial \mathcal{L}}{\partial W^{(l)}} = \mathbf{a}^{(l-1)}\big(\delta^{(l)}\big)^{\top}. \tag{3.5.7}$$

Because bias vectors are omitted, no bias-update term appears. The weight matrices

are then updated by the NAdam optimiser (Section 3.5.3), which combines Nesterov-accelerated momentum with adaptive learning rates.

These equations are standard in the deep learning literature (Goodfellow et al., 2016) and are reproduced here for completeness; they make the training procedure, including the bias-free simplification, explicit. In practice, all gradients are computed automatically by the deep learning framework (PyTorch) through reverse-mode automatic differentiation, so the closed-form expressions are never coded directly. Their inclusion ensures that the reader can relate the chosen architecture, loss function, and activation to the mechanics of gradient-based optimisation.

3.5.3 Training configuration

The network is trained by minimising the MSE on the standardised response scale (equation (2.4.11)). Weights are updated with the NAdam optimiser (Dozat, 2016), described in Section 2.4.2. Validation performance is evaluated every $v = 5$ epochs. For a fixed validation set, minimising the validation MSE is equivalent to maximising the validation R^2; the dot-product R^2 is therefore used as the operational model-selection criterion. The selected epoch is

$$e^* = \arg\max_{e \in \mathcal{E}_{\text{val}}} R^2_{\text{val}}(e), \qquad \mathcal{E}_{\text{val}} = \{v, 2v, 3v, \dots\} \cap \{1, \dots, E_{\max}\}. \tag{3.5.8}$$

Training stops early if the validation R^2 does not improve for $P = 10$ consecutive validation checks (Prechelt, 2012); the retained model is the checkpoint with the highest validation R^2 observed. The test set is reserved exclusively for final out-of-sample evaluation.

On the original monetary scale, the standard $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ (equation (2.3.4)) is reported alongside the MAE and RMSE.

Table 6 summarises the complete architecture and training configuration.

Component	Specification
Input dimension	$p = 22$ encoded predictor variables
Hidden layer widths	$256, 128, 64, 32, 16$
Output dimension	One scalar regression output
Architecture type	Fully connected DNN
Network shape	Contracting representation from 256 to 16 hidden units
Bias vectors	Omitted
Hidden activation	CELU (equation (2.4.9))
Output activation	Identity mapping
Training response	Standardised `charges`
Prediction scale	De-normalised to the original monetary scale
Training objective	MSE on the standardised response scale
Secondary evaluation metrics	MAE, RMSE and R^2 on the original response scale
Optimiser	NAdam (Dozat, 2016)
Momentum parameters	(β_1, β_2) selected by validation comparison from $\{(0.90, 0.999), (0.95, 0.999), (0.99, 0.999)\}$
Learning rate	Selected by validation comparison from $\{0.01, 0.001, 0.0005, 0.0001\}$
Batch size	Selected by validation comparison from $\{64, 128, 256, 512\}$
Maximum epochs	$E_{\max} = 200$
Validation frequency	Every $v = 5$ epochs
Model-selection criterion	Dot-product validation R^2
Early stopping	Patience $P = 10$ validation checks without R^2 improvement
Explicit regularisation	Not used
Dropout	Not used
Random seed	$s = 42$, fixed for reproducibility
Trainable parameters	49 168 weights

Table 6: DNN architecture and training configuration.

3.5.4 Hyperparameter comparison

The main training hyperparameters that materially affect the fitted prediction surface are compared sequentially:

1. **Hidden activation:** CELU, GELU and Tanh are evaluated with learning rate 0.001, batch size 256, and Adam held fixed.
2. **Learning rate:** Values $\{0.01, 0.001, 0.0005, 0.0001\}$ are tested with the best activation carried forward.
3. **Batch size:** Sizes $\{64, 128, 256, 512\}$ are tested with the best activation and learning rate.
4. **Optimiser:** Adam and NAdam are compared with all previous best choices carried forward.
5. **Momentum parameters:** (β_1, β_2). Since NAdam is selected in the previous step, its Nesterov-accelerated momentum parameters are compared over $\{(0.90, 0.999), (0.95, 0.999), (0.99, 0.999)\}$. The first coefficient β_1 controls the decay of the gradient mean estimate; the second, β_2, controls the decay of the uncentred variance estimate (Dozat, 2016).

At each stage the best choice is carried forward and all other factors are held constant. Explicit $\mathcal{L}_1$, $\mathcal{L}_2$, and dropout (Srivastava et al., 2014) regularisation are not included; the adequacy of this design is assessed through validation curves, out-of-sample error metrics, and structural diagnostics[1].

The training procedure consists of: (1) random weight initialisation using Kaiming uniform initialisation (He et al., 2015), which scales the initial weights according to the fan-in of each layer so that activations neither vanish nor explode during the early forward passes, a property that is particularly important in deeper architectures such as the five-hidden-layer network used here; (2) mini-batch gradient descent with the NAdam optimiser; (3) validation evaluation every 5 epochs using the dot-product R^2; (4) early stopping when the validation R^2 has not improved for 10 consecutive checks; (5) selection of the checkpoint with the highest validation R^2; and (6) final test-set evaluation on the original monetary scale using R^2, RMSE and MAE. The complete pseudocode is presented in Algorithm 33 (Annexure C.4).

[1] In this context, $\mathcal{L}_1$ and $\mathcal{L}_2$ regularisation refer to adding penalty terms to the training objective. If $\mathcal{L}_0(\theta)$ denotes the original training loss and $\theta = (\theta_1, \ldots, \theta_d)$ denotes the vector of trainable parameters, then $\mathcal{L}_1$ regularisation uses

$$\mathcal{L}_1(\theta) = \mathcal{L}_0(\theta) + \lambda \sum_{j=1}^{d} |\theta_j|,$$

whereas $\mathcal{L}_2$ regularisation uses

$$\mathcal{L}_2(\theta) = \mathcal{L}_0(\theta) + \lambda \sum_{j=1}^{d} \theta_j^2.$$

Here, $\lambda \geq 0$ controls the strength of the penalty. The $\mathcal{L}_1$ penalty encourages sparse parameter estimates, while the $\mathcal{L}_2$ penalty shrinks parameters towards zero without necessarily setting them exactly equal to zero.

3.5.5 Benchmark GLM

A classical actuarial benchmark is provided by a GLM with a gamma distribution and logarithmic link function (Dobson and Barnett, 2008; McCullagh and Nelder, 1989). For a policyholder with model-ready input vector $\mathbf{z}_i \in \mathbb{R}^{22}$ (the same encoded predictors used by the DNN), the GLM assumes

$$Y_i \sim \text{Gamma}(\mu_i, \phi), \qquad \log(\mu_i) = \beta_0 + \sum_{j=1}^{22} \beta_j z_{ij}, \tag{3.5.9}$$

where $\mu_i = \mathbb{E}[Y_i \mid \mathbf{z}_i]$ and ϕ is the dispersion parameter. The distributional and link-function rationale for this specification was set out in Section 1.1.1; under the log link the predicted mean $\mu_i = \exp(\beta_0 + \sum_j \beta_j z_{ij})$ is always positive.

The model includes no interaction terms or non-linear transformations. It is fitted to the training partition by maximum likelihood estimation. The log-likelihood for the gamma GLM is

$$\ell(\boldsymbol{\beta}, \phi) = \sum_{i=1}^{n} \left[\frac{1}{\phi} \left(-\frac{y_i}{\mu_i} - \log \mu_i \right) + \left(\frac{1}{\phi} - 1 \right) \log y_i - \log \Gamma\left(\frac{1}{\phi} \right) - \frac{1}{\phi} \log \phi \right], \tag{3.5.10}$$

where $\mu_i = \exp(\mathbf{z}_i^\top \boldsymbol{\beta})$ and $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_{22})^\top$. The parameter vector $\boldsymbol{\beta}$ is estimated by IRLS, which solves the score equations

$$\sum_{i=1}^{n} \frac{(y_i - \mu_i)}{\text{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} z_{ij} = 0, \qquad j = 0, 1, \ldots, 22, \tag{3.5.11}$$

where $\eta_i = \log(\mu_i)$ is the linear predictor and $\text{Var}(Y_i) = \phi \, \mu_i^2$ under the gamma family. Each IRLS iteration computes a weighted least squares regression of an adjusted dependent variable on the design matrix, with weights inversely proportional to the variance function evaluated at the current fitted values (Dobson and Barnett, 2008).

Algorithm 34 formalises the fitting procedure.

The GLM is fitted via IRLS, which computes working weights and adjusted dependent variables at each iteration until convergence. The dispersion parameter $\widehat{\phi}$ is estimated from the Pearson residuals, and predictions are obtained on the original monetary scale by applying the inverse link $\widehat{y}_i = \exp(\mathbf{z}_i^\top \widehat{\boldsymbol{\beta}})$. The complete IRLS algorithm is presented in Algorithm 34 (Annexure C.5).

No hyperparameter tuning is applied to the GLM; its role as a benchmark was motivated in Section 2.3.

3.5.6 Tree-based benchmarks

Two tree-based ensemble methods are included alongside the GLM to provide a more comprehensive model-class comparison. Both models use the same 22 encoded features and the same approximately 52/24/24 partition (Section 3.3) as the DNN and GLM.

3.5.7 RF benchmark

An RF regressor (Breiman, 2001) is fitted with 500 trees, maximum depth unrestricted, minimum samples per leaf of 5, and `random_state=42`. RF aggregates predictions across decorrelated decision trees, reducing variance through bootstrap aggregation (bagging) while maintaining low bias.

The RF constructs an ensemble of B regression trees, each grown on a bootstrap sample of the training data. At every candidate split the algorithm considers only a random subset of m out of p predictors, thereby decorrelating the individual trees and reducing the variance of the averaged prediction. For a regression tree T_b grown on bootstrap sample $\mathcal{D}_b^*$, the prediction for input $\mathbf{z}$ is $\hat{y}_b(\mathbf{z}) = T_b(\mathbf{z})$. The RF prediction is the average across all B trees:

$$\hat{f}_{\mathrm{RF}}(\mathbf{z}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{z}). \tag{3.5.12}$$

Each tree T_b is grown by recursively partitioning the feature space. At each internal node, the algorithm selects a random subset of $m = \lfloor p/3 \rfloor$ predictors (the default for regression) and chooses the split variable j^* and split point s^* that minimise the weighted sum of squared residuals in the resulting child nodes:

$$(j^*, s^*) = \arg\min_{j \in \mathcal{S},\, s} \left[\sum_{i:\, z_{ij} \le s} (y_i - \bar{y}_L)^2 + \sum_{i:\, z_{ij} > s} (y_i - \bar{y}_R)^2 \right], \tag{3.5.13}$$

where $\mathcal{S} \subset \{1, \ldots, p\}$ with $|\mathcal{S}| = m$ is the randomly selected predictor subset, and $\bar{y}_L$ and $\bar{y}_R$ denote the sample means in the left and right child nodes respectively. Trees are grown until each terminal node contains at least 5 observations.

The variance reduction mechanism of the RF can be understood through the decomposition of the variance of the ensemble average. If individual tree predictions have variance σ^2 and pairwise correlation ρ, then

$$\mathrm{Var}\big(\hat{f}_{\mathrm{RF}}\big) = \rho\, \sigma^2 + \frac{1 - \rho}{B}\, \sigma^2. \tag{3.5.14}$$

As B increases, the second term vanishes, and the ensemble variance is governed by

$\rho\,\sigma^2$. By restricting splits to a random predictor subset, the RF reduces ρ below the value obtained by standard bagging (which uses all p predictors at every split), yielding a lower ensemble variance (Hastie et al., 2009).

The complete training procedure is presented in Algorithm 35 (Annexure C.6), and Table 7 summarises the RF configuration.

Component	**Specification**
Number of trees (B)	500
Candidate predictors per split (m)	$\lfloor p/3 \rfloor = 7$
Maximum depth	Unrestricted
Minimum samples per leaf	5
Bootstrap sampling	With replacement, sample size = n_{train}
Split criterion	MSE (equation (3.5.13))
Hyperparameter tuning	Not applied (default values)
Random seed	42
Implementation	`scikit-learn RandomForestRegressor`

Table 7: RF configuration.

3.5.8 XGBoost benchmark

A gradient-boosted tree regressor is fitted using the XGBoost implementation (Chen and Guestrin, 2016) with 1 000 boosting rounds, a learning rate of 0.05, maximum depth of 8, and early stopping on the validation set with a patience of 50 rounds. XGBoost constructs an additive ensemble of shallow trees via gradient boosting, using approximate histogram-based splitting for computational efficiency.

Gradient boosting builds an additive model in a stage-wise fashion. The ensemble prediction after M boosting rounds is

$$\hat{f}_M(\mathbf{z}) = h_0(\mathbf{z}) + \sum_{m=1}^{M} \eta\, h_m(\mathbf{z}), \tag{3.5.15}$$

where $h_0(\mathbf{z}) = \bar{y}$ is the initial constant prediction (the training-set mean), h_m is the regression tree fitted at round m, and $\eta \in (0, 1]$ is the learning rate (shrinkage parameter) that scales each tree's contribution. The shrinkage applies only to the trees fitted at rounds $m \geq 1$; the initial constant enters unshrunk, so that the ensemble begins at the training-set mean and each subsequent tree adds a damped correction to it.

At each boosting round m, XGBoost fits a new tree h_m to minimise a regularised objective. For the squared-error loss used in regression, the objective at round m is

$$\mathcal{O}^{(m)} = \sum_{i=1}^{n} \ell\big(y_i,\ \hat{f}_{m-1}(\mathbf{z}_i) + h_m(\mathbf{z}_i)\big) + \Omega(h_m), \tag{3.5.16}$$

where $\ell(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$ is the squared-error loss and $\Omega(h_m)$ is a regularisation penalty on the complexity of tree h_m. The regularisation term is defined as

$$\Omega(h) = \gamma\, J + \tfrac{1}{2}\,\lambda \sum_{j=1}^{J} w_j^2, \tag{3.5.17}$$

where J is the number of terminal nodes (leaves), w_j is the predicted value in leaf j, $\gamma \geq 0$ penalises additional leaves, and $\lambda \geq 0$ applies $\mathcal{L}_2$ shrinkage to the leaf weights.

The key computational innovation of XGBoost is the use of a second-order Taylor expansion of the loss around the current prediction $\hat{f}_{m-1}(\mathbf{z}_i)$. Define the first- and second-order gradient statistics for observation i:

$$g_i = \frac{\partial\, \ell(y_i,\ \hat{y})}{\partial\, \hat{y}}\bigg|_{\hat{y}=\hat{f}_{m-1}(\mathbf{z}_i)}, \qquad h_i = \frac{\partial^2\, \ell(y_i,\ \hat{y})}{\partial\, \hat{y}^2}\bigg|_{\hat{y}=\hat{f}_{m-1}(\mathbf{z}_i)}. \tag{3.5.18}$$

For the squared-error loss, $g_i = \hat{f}_{m-1}(\mathbf{z}_i) - y_i$ (the negative residual) and $h_i = 1$. With these gradient statistics, the approximate objective simplifies to

$$\tilde{\mathcal{O}}^{(m)} = \sum_{j=1}^{J} \left[G_j\, w_j + \frac{1}{2}(H_j + \lambda)\, w_j^2 \right] + \gamma\, J, \tag{3.5.19}$$

where $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} h_i$ are the sums of gradient and Hessian statistics over the observations I_j assigned to leaf j. The optimal leaf weight and the corresponding optimal objective value are

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \qquad \tilde{\mathcal{O}}^* = -\frac{1}{2} \sum_{j=1}^{J} \frac{G_j^2}{H_j + \lambda} + \gamma\, J. \tag{3.5.20}$$

To evaluate a candidate split that partitions observations in a node into left (I_L) and right (I_R) child nodes, XGBoost computes the gain:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma, \tag{3.5.21}$$

where $G_L = \sum_{i \in I_L} g_i$, $G_R = \sum_{i \in I_R} g_i$, $H_L = \sum_{i \in I_L} h_i$, and $H_R = \sum_{i \in I_R} h_i$. A split is

made only if Gain > 0; the regularisation parameter γ acts as a minimum-gain threshold that prunes unprofitable splits (Chen and Guestrin, 2016).

The complete training procedure is presented in Algorithm 36 (Annexure C.7), and Table 8 summarises the XGBoost configuration.

Component	**Specification**
Maximum boosting rounds (M)	1 000
Learning rate (η)	0.05
Maximum tree depth ($d_{\max}$)	8
Booster	`gbtree` (tree-based)
Objective	`reg:squarederror`
Split evaluation	Approximate histogram-based
$\mathcal{L}_2$ regularisation (λ)	Default (1.0)
Minimum split gain (γ)	Default (0.0)
Early stopping patience	50 rounds on validation loss
Hyperparameter tuning	Not applied (default values)
Random seed	42
Implementation	`xgboost XGBRegressor`

Table 8: XGBoost configuration.

In the primary comparison, hyperparameters for both tree-based models are set to commonly recommended defaults rather than tuned via cross-validation, paralleling the no-tuning treatment of the GLM; this isolates model-class differences from tuning effort. As a complementary analysis, both tree-based models are subsequently re-fitted with tuned hyperparameters selected by randomised search with five-fold cross-validation on the training partition, quantifying how much of any remaining performance gap is attributable to tuning rather than to the model class itself (results in Section 4.4). All models are evaluated using R^2, RMSE and MAE on the original monetary scale across all three partitions.

3.6 Predictive and Structural Diagnostics

The predictive and structural diagnostics described below serve a single purpose: to verify that the fitted DNN is sufficiently stable and well-behaved to act as the prediction surface in the subsequent MC and Shapley-value analyses.

All predictive performance metrics are computed on the original monetary scale after de-

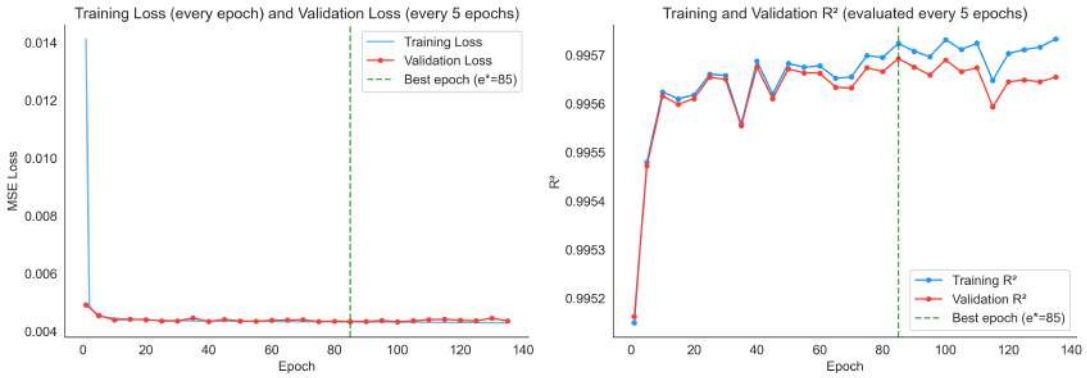
normalising the fitted predictions. The metrics are the MAE, RMSE and R^2 defined in equations (2.3.3), (2.3.2) and (2.3.4). For observation i the fitted prediction is denoted $\widehat{y}_i$ and the residual is

$$r_i = y_i - \widehat{y}_i. \tag{3.6.1}$$

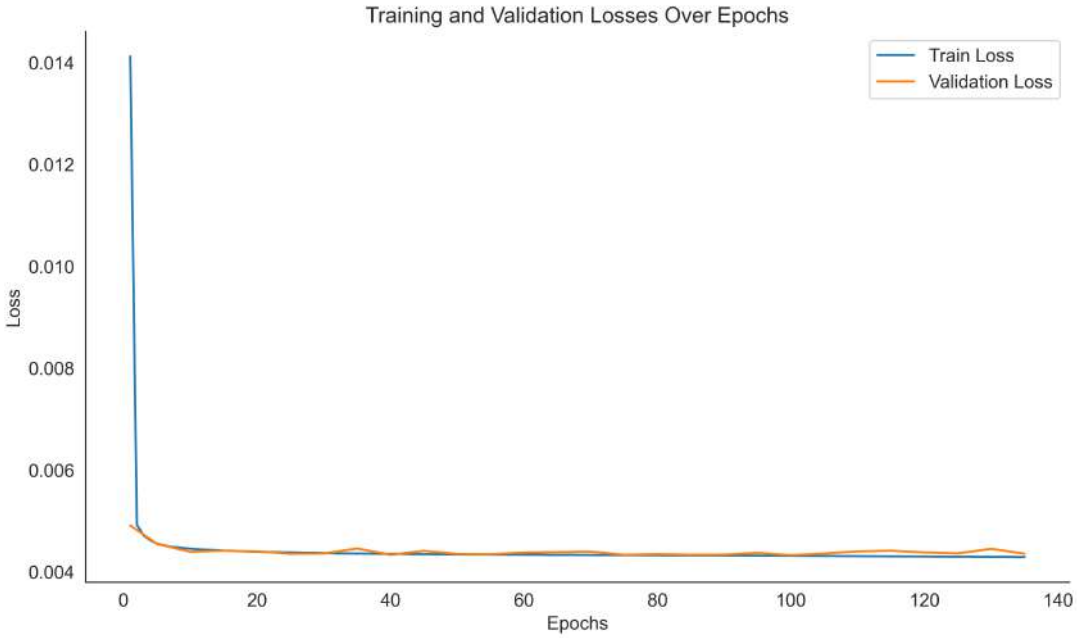
Residuals are computed separately for the training, validation and test partitions.

3.6.1 Predictive error assessment

Predictive error is assessed by comparing observed charges with fitted predictions on the original response scale. The training partition is used to evaluate in-sample fit, the validation partition to guide model-selection decisions, and the test partition to provide a final out-of-sample evaluation. The three scalar metrics are interpreted jointly: a suitable model should exhibit consistent performance across the validation and test partitions, without a large deterioration relative to the training partition.



(a) Training and validation loss with R^2 over epochs.



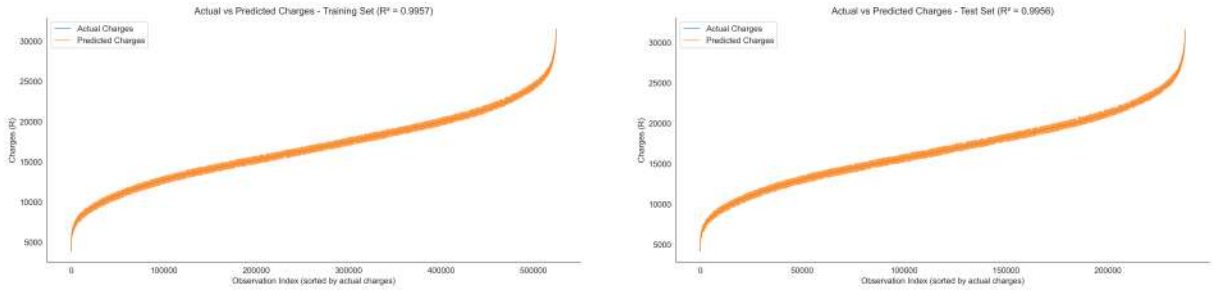
(b) Training and validation MSE losses over epochs for the final model.

Figure 12: Learning curves for the fitted DNN.

Figure 12a presents the learning curves for the fitted DNN. In the left panel, the training MSE loss drops sharply from approximately 0.014 at the first epoch to roughly 0.005 by epoch 5, after which it decreases gradually and stabilises near 0.0043. The validation MSE loss tracks the training loss closely throughout, with no visible divergence between the two curves. The selected epoch $e^* = 85$ (green dashed line) falls in the region where both losses have plateaued. In the right panel, both training and validation R^2 values rise rapidly from approximately 0.9951 to a plateau near 0.9957, with the gap between them remaining negligible. Figure 12b provides a complementary view: the training loss (blue) exhibits a single spike at epoch 1, consistent with random weight initialisation, before converging rapidly, and the validation loss (orange) overlaps with it from epoch 20 onwards. The absence of any sustained divergence confirms stable convergence without overfitting.

3.6.2 Actual versus predicted diagnostics

Two graphical summaries are used to compare observed charges with fitted predictions. First, the observed values y_i are plotted directly against the fitted values $\widehat{y}_i$ with a 45° reference line. Second, observations are sorted by their observed charge, and the actual and predicted values are displayed as overlaid curves; a well-fitted model produces a predicted curve that closely tracks the actual curve across the full range of the response. Both diagnostics are produced separately for the training, validation and test partitions, and together they indicate whether the fitted model behaves differently in low-, central- and high-cost regions.



(a) Training set ($R^2 = 0.9957$). (b) Test set ($R^2 = 0.9956$).

Figure 13: Sorted actual versus predicted charges on training and test sets.

Figure 13 shows the sorted actual versus predicted charges on both partitions. In the training set (Figure 13a), the predicted line (orange) tracks the actual line (blue) so closely that the two are nearly indistinguishable across the full range of the response, from approximately R4 000 at the lower end to R32 000 at the upper end. A modest increase in prediction scatter is visible in the upper tail above R25 000, but the deviations remain small relative to the response scale. The test set (Figure 13b) displays a virtually identical pattern. The negligible drop from training $R^2 = 0.9957$ to test $R^2 = 0.9956$ confirms that the model generalises to unseen observations without meaningful performance degradation.

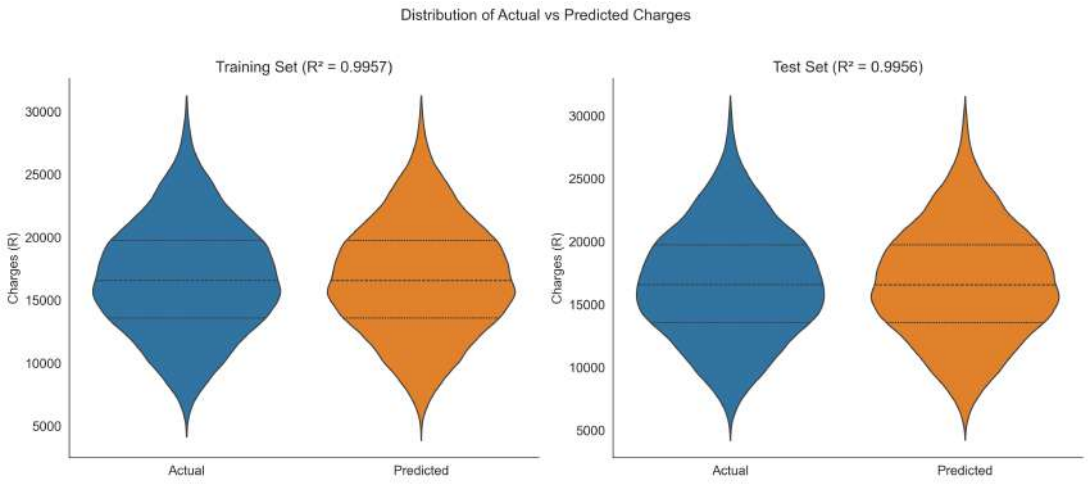


Figure 14: Violin plots of actual versus predicted charges on training and test sets.

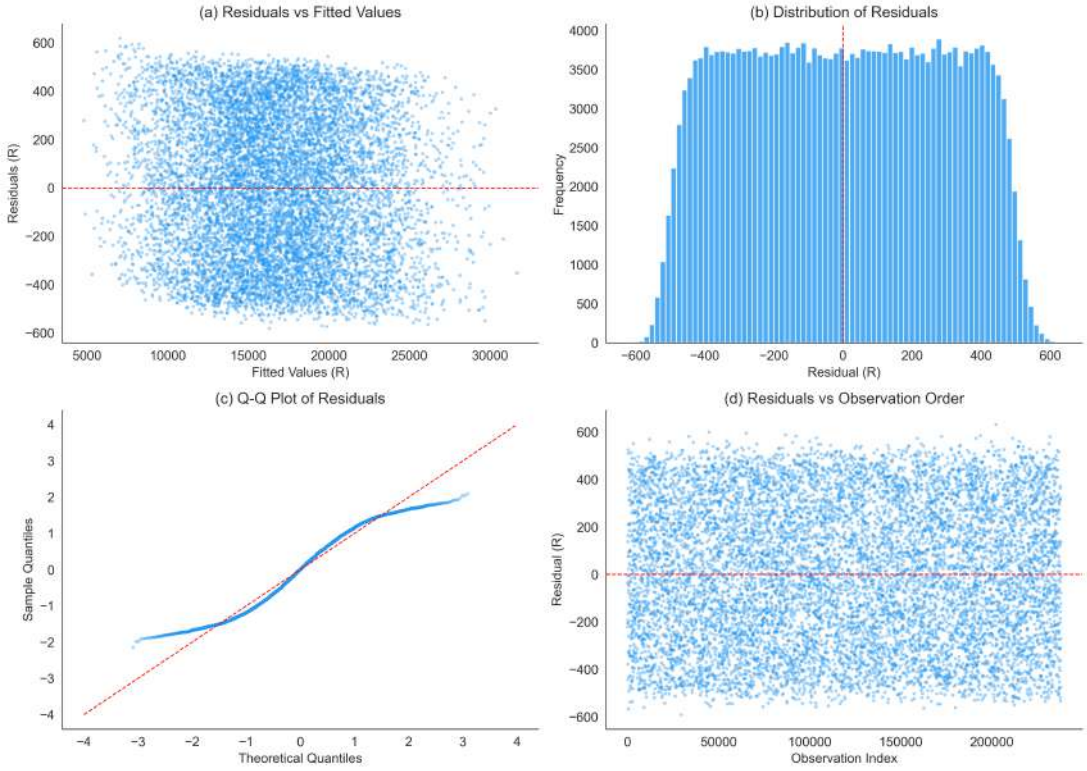
Figure 14 compares the distributional profiles of actual and predicted charges on both partitions. In the training set (left panel, $R^2 = 0.9957$), the actual (blue) and predicted (orange) violins exhibit closely matching shapes: both are unimodal with the bulk of density concentrated between approximately R10 000 and R25 000, and the dashed median lines fall at approximately the same level near R16 700. The predicted violin is marginally narrower at the extreme tails, indicating that the DNN compresses the most extreme values slightly, which is consistent with the smoothing behaviour of a regression model. The test set (right panel, $R^2 = 0.9956$) displays the same pattern, with the actual and predicted distributions virtually mirroring each other. The consistency between training and test panels confirms that the model's distributional calibration is preserved on unseen data.

3.6.3 Residual diagnostics

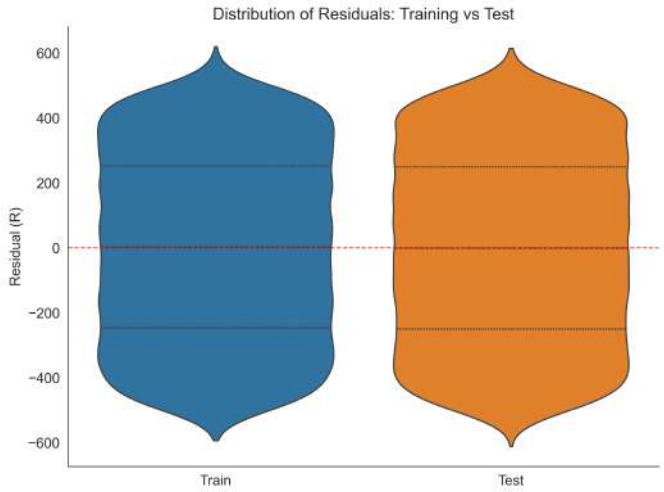
Residual diagnostics examine whether prediction errors contain systematic structure. Using the residuals defined in equation (3.6.1), the following plots are produced:

1. residuals versus fitted values, to detect systematic bias or changes in spread;
2. a residual histogram, to assess the empirical shape and centring of the prediction errors;
3. a Q-Q plot of standardised residuals, to inspect tail behaviour;
4. residuals versus observation index, to identify possible ordering or sequential patterns;
5. residual distribution comparisons across the training, validation and test partitions.

The DNN does not require normally distributed residuals; these checks are purely descriptive. The fitted residuals are also retained for the residual-augmented MC sensitivity analysis described later.



(a) Residual diagnostic panels for the fitted DNN.



(b) Violin plots of residuals on the training and test sets.

Figure 15: Residual diagnostics for the fitted DNN.

Figure 15a presents the four residual diagnostic panels for the test set. Panel (a) shows residuals versus fitted values: the residuals are centred at zero across the full range of fitted charges (approximately R5 000 to R30 000), with no visible funnel shape or system-

atic trend. The residual magnitudes range from approximately −600 to +600, which is small relative to the response scale. Panel (b) displays the residual histogram, which is symmetric and centred at zero but distinctly non-Gaussian: the distribution is platykurtic and plateau-shaped, with probability spread nearly evenly across a bounded interval and hard cutoffs near ±600 rather than smoothly decaying tails. Panel (c) shows the Q-Q plot of standardised residuals. The central portion follows the theoretical reference line, while both tails bend markedly away from it, confirming that the residual distribution is bounded, with far lighter tails than a Gaussian distribution. This bounded, near-uniform residual shape is the fingerprint of the benchmark's data-generating process: the response is produced by a deterministic cost formula with a bounded random perturbation, which explains both the hard residual cutoffs and the near-ceiling R^2 values observed on every partition. Consequently, the accuracy figures reported in this study characterise how well the network recovers an algorithmic generating formula, and no claim is made that these specific figures transfer to real insurance portfolios (see the delimitations in Section 1.8). The non-Gaussian residual shape is not itself a concern for the DNN, which does not assume normality of residuals. Panel (d) plots residuals against observation index: the points are scattered randomly around zero with no discernible sequential pattern. Figure 15b compares the residual distributions across partitions. Both the training and test violins are centred at zero and are nearly identical in width, spread and symmetry, confirming that the residual error does not systematically differ between the training and test data and that the fitted DNN does not overfit the training partition.

3.6.4 Principal component structural diagnostics

PCA is employed as a structural diagnostic of the standardised, model-ready input space. It is not used for dimension reduction, and no predictors are removed on the basis of the principal components. The purpose is twofold: to assess whether the data partition preserves the structure of the input space, and to check whether the fitted predictions reproduce the dominant target pattern when displayed on the same low-dimensional projection.

Let $\widetilde{\mathbf{X}} \in \mathbb{R}^{n_{\text{train}} \times p}$ denote the column-standardised training input matrix ($p = 22$). The sample covariance matrix is

$$\mathbf{S} = \frac{1}{n_{\text{train}} - 1} \widetilde{\mathbf{X}}^{\top} \widetilde{\mathbf{X}}. \tag{3.6.2}$$

Its spectral decomposition is

$$\mathbf{S} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^{\top}, \qquad \mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \ldots, \lambda_p), \qquad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0, \tag{3.6.3}$$

where $\mathbf{W} = [\mathbf{w}_1 \cdots \mathbf{w}_p]$ contains the principal loading vectors. The score of observation

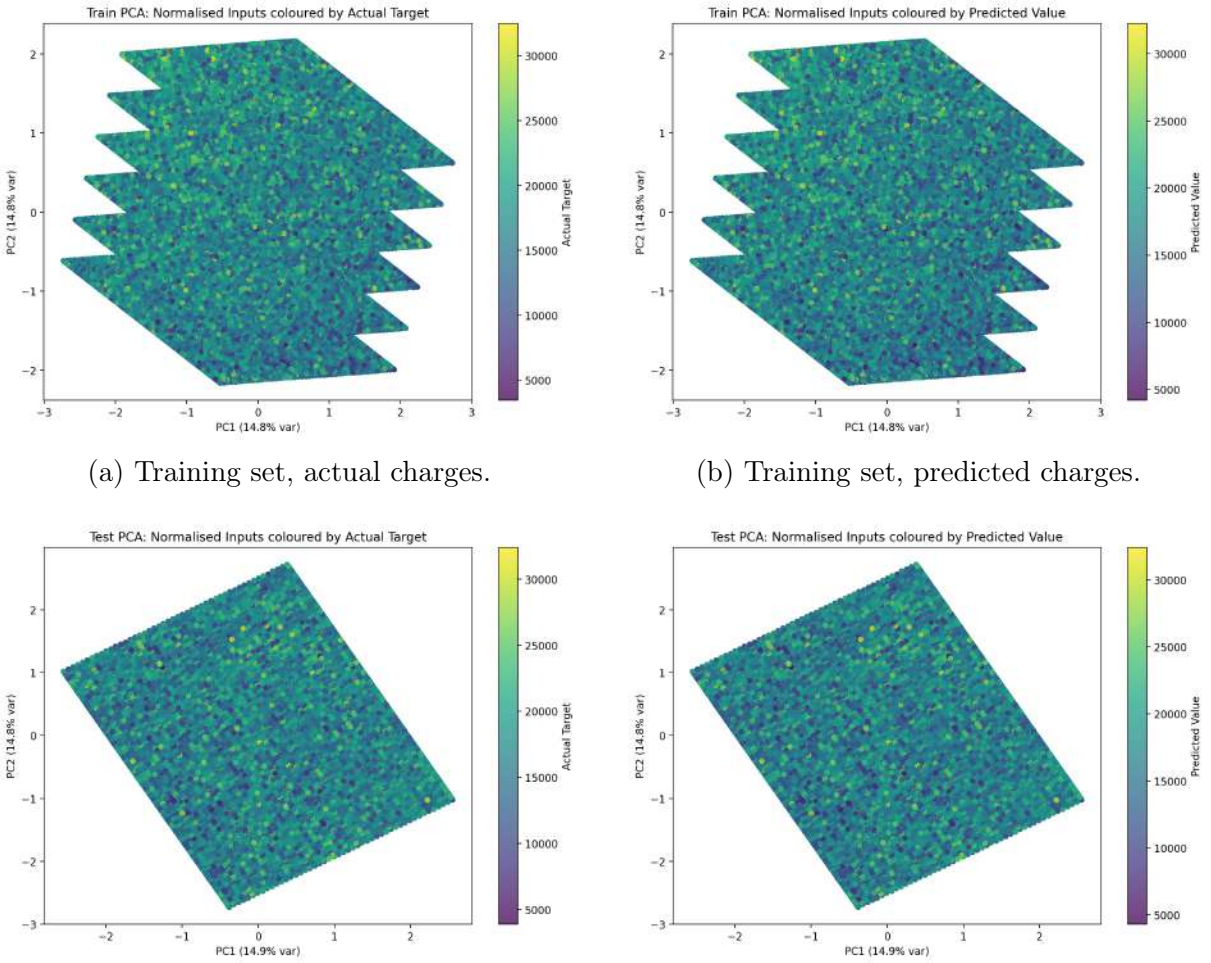
i on component k is

$$z_{ik} = \widetilde{\mathbf{x}}_i^{\top} \mathbf{w}_k, \qquad k = 1, \ldots, p. \tag{3.6.4}$$

The PCA decomposition is computed from the training partition only; the resulting eigenvectors are then used to project the validation and test observations, so that all sets are displayed in a common coordinate system. The first two principal components provide the two-dimensional diagnostic projection.

Two PCA comparisons are performed. First, the training, validation and test point clouds are overlaid; similar structure across partitions supports the view that the data split has not introduced a major structural shift. Second, the same PCA coordinates are coloured by observed charges and by fitted predictions; similar colour gradients indicate that the fitted DNN has captured the dominant target structure visible in the input space.

The loading vectors are inspected to verify that the first two principal components are not driven by a single variable and that they account for a moderate share of the total variance, so that the diagnostic is not dominated by one narrow direction (Hastie et al., 2009). This inspection is descriptive and is not used to modify the fitted model.



(a) Training set, actual charges.

(b) Training set, predicted charges.

(c) Test set, actual charges.

(d) Test set, predicted charges.

Figure 16: PCA projections coloured by actual and predicted charges on training and test sets.

Figure 16 displays the PCA diagnostic scatter plots. In the training set (Figures 16a and 16b), the first two principal components each explain approximately 14.8% of the total variance. The point cloud exhibits a layered structure with several horizontal shelves, reflecting the discrete levels of the encoded categorical predictors projected onto the PC axes. The colour gradient, representing charges from approximately R5 000 (purple) to R30 000 (yellow), is broadly mixed across the point cloud, with higher charges (yellow-green) tending to appear at specific regions within the layered structure. The predicted-target colouring (Figure 16b) is virtually indistinguishable from the actual-target version, confirming that the fitted DNN has captured the dominant target structure visible in the first two principal component directions. The test set (Figures 16c and 16d) shows the same point-cloud shape and colour distribution as the training set (PC1: 14.9%, PC2: 14.8%), and the actual and predicted colourings again match closely. This consistency across partitions and between actual and predicted targets jointly satisfies both the partition consistency and model capture diagnostic criteria.

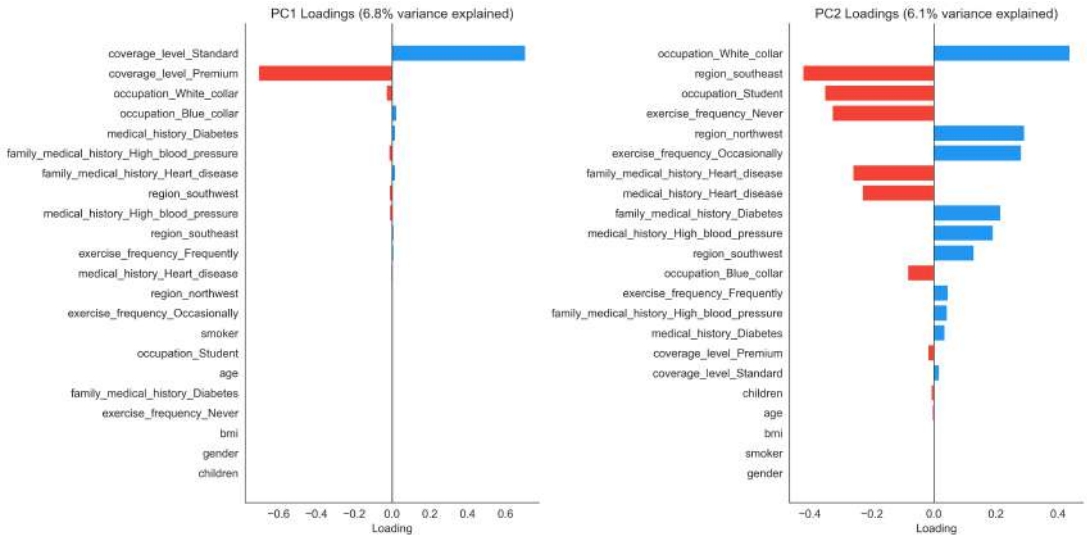


Figure 17: PCA loading vectors for all 22 encoded predictors.

Figure 17 displays the loading vectors for the first two principal components. In the left panel, PC1 (6.8% of variance) is dominated by `coverage_level_Standard` (positive loading ≈ 0.71) and `coverage_level_Premium` (negative loading ≈ -0.71), with all other loadings substantially smaller in magnitude. This confirms that the layered structure observed in the scatter plots is driven by the `coverage_level` variable. In the right panel, PC2 (6.1% of variance) has loadings spread across multiple categorical predictors: `occupation_White_collar` (≈ 0.44), `region_northwest` (≈ 0.33) and `exercise_frequency_Never` (≈ 0.30) contribute positively, while `region_southeast` (≈ -0.42) and `medical_history_Heart_disease` (≈ -0.20) contribute negatively. The more distributed loading structure of PC2 indicates that no single variable dominates the second component. Although PC1 is driven by a single raw variable, the first two components together account for only 13% of total variance, so the diagnostic is not determined by a single narrow direction in the 22-dimensional input space.

3.6.5 Correlation heatmap diagnostics

Correlation heatmaps provide an additional structural check. The full Pearson correlation matrix is computed for the standardised encoded predictors together with either the observed response or the fitted predictions, yielding two views:

1. correlations between predictors and observed charges;
2. correlations between predictors and fitted predictions.

Comparing these views indicates whether the fitted DNN has preserved the main linear feature-target relationships present in the data. The heatmaps are also produced for

the training, validation and test partitions to assess whether the data split preserves the linear dependence structure of the model-ready input space. These heatmaps are purely diagnostic; they do not capture the nonlinear relationships learned by the network, nor do they constitute a proof of model correctness.

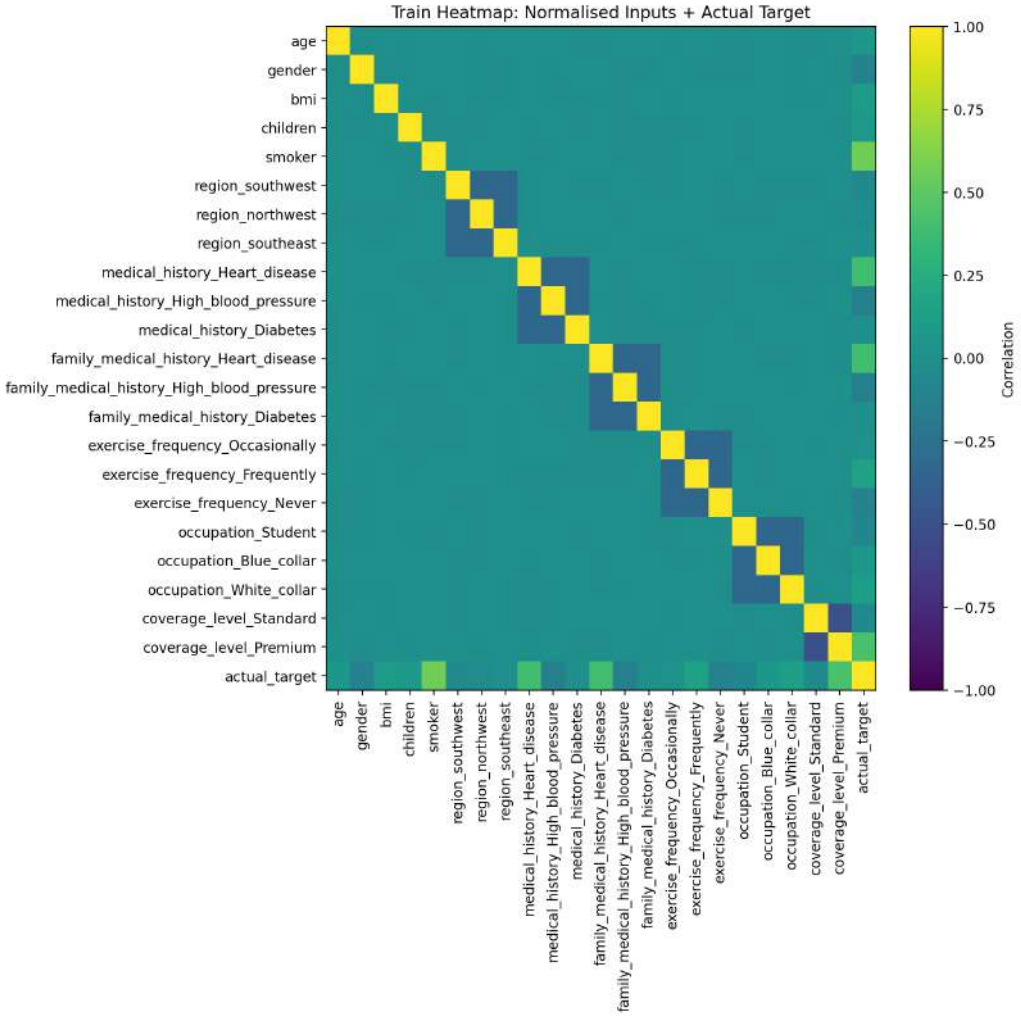


Figure 18: Input-target correlation heatmap: training set, actual charges.

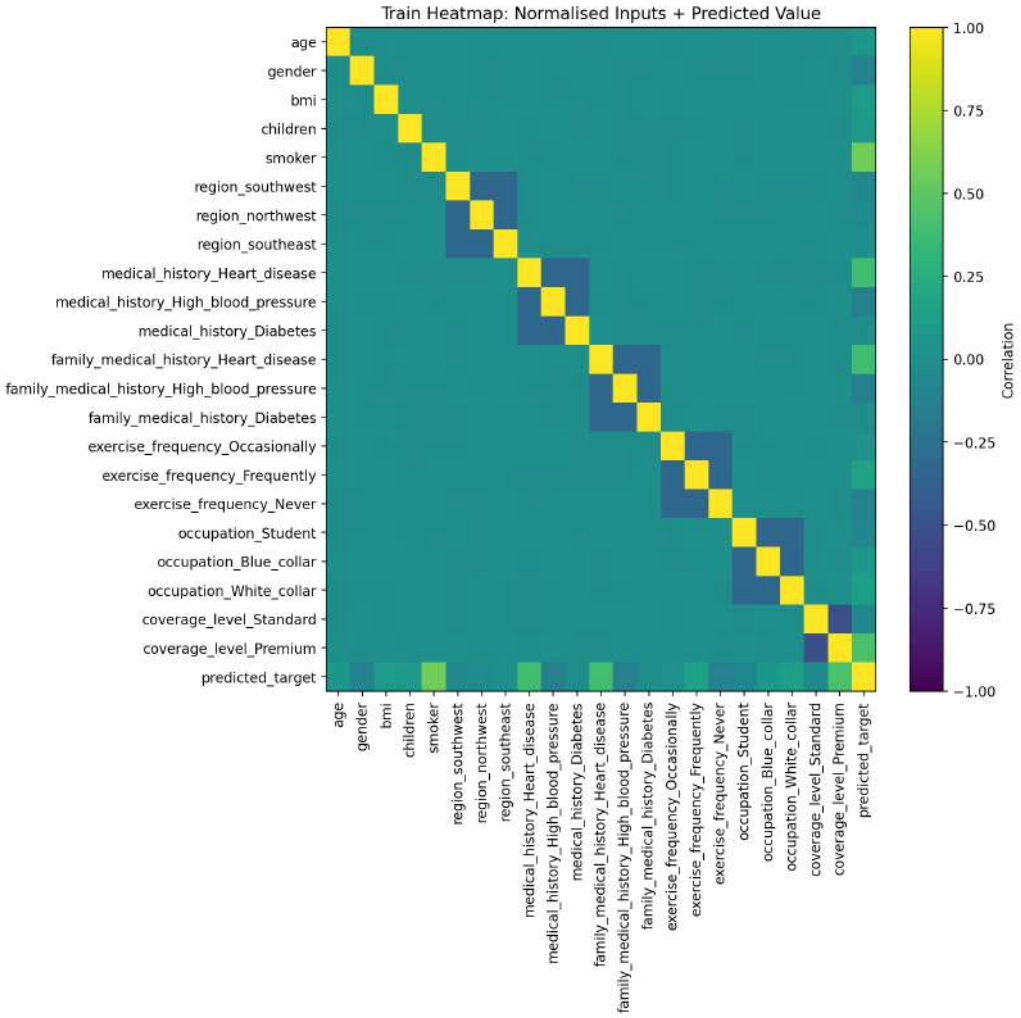


Figure 19: Input-target correlation heatmap: training set, predicted charges.

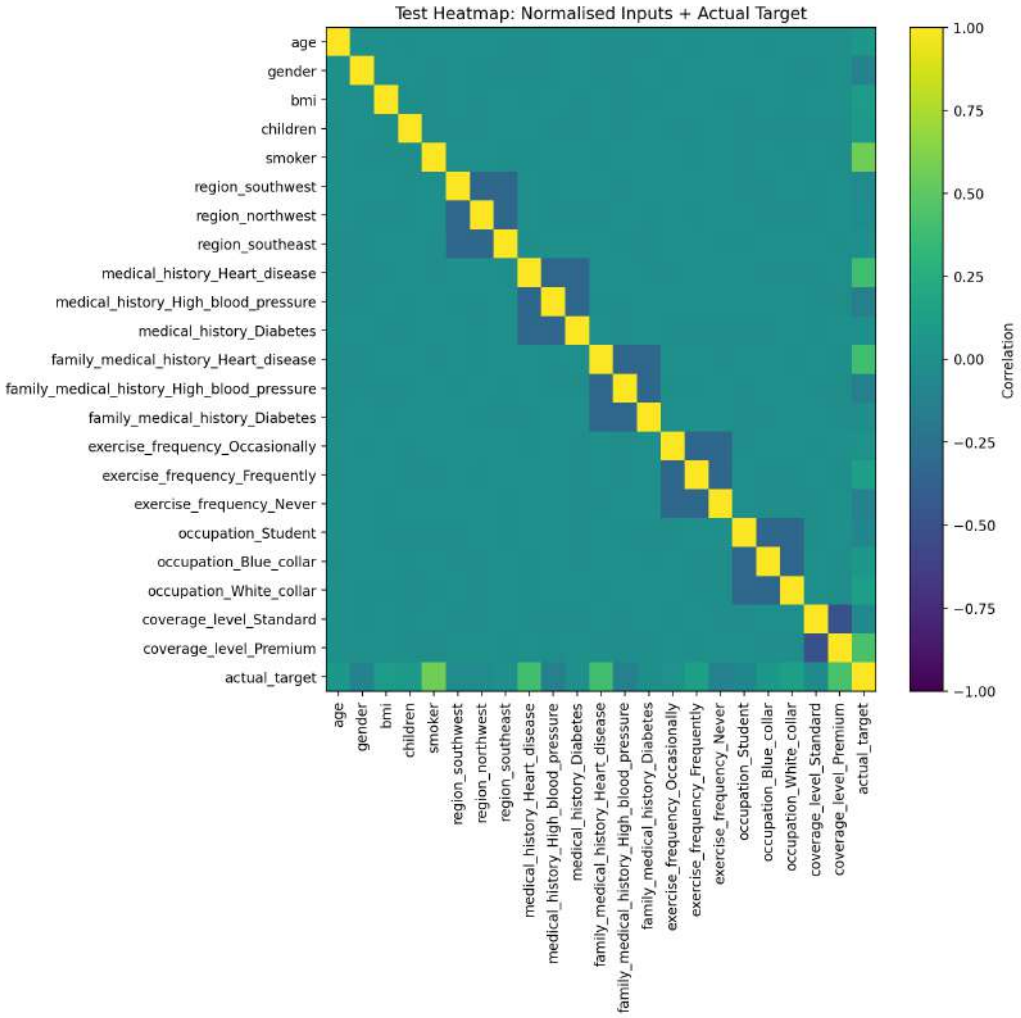


Figure 20: Input-target correlation heatmap: test set, actual charges.

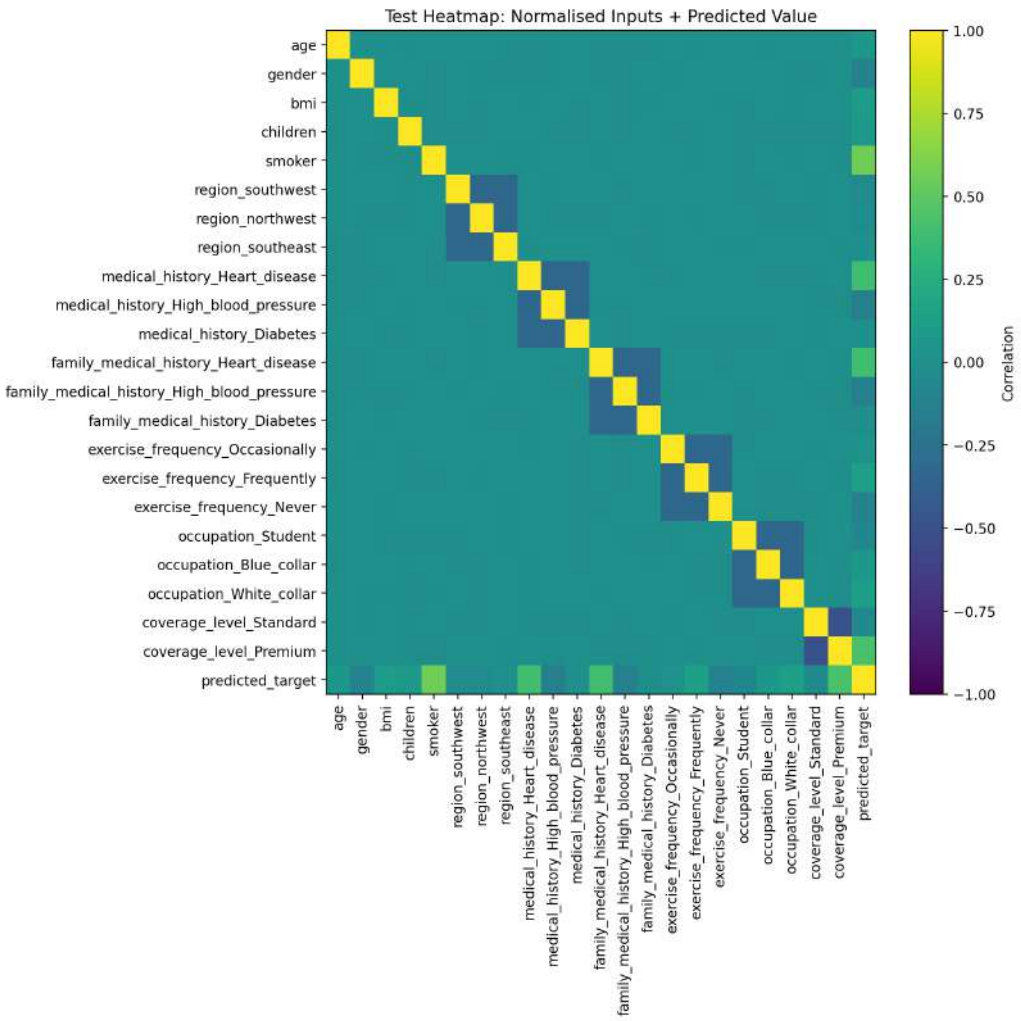


Figure 21: Input-target correlation heatmap: test set, predicted charges.

Figures 18 to 21 present the four correlation heatmaps. The inter-feature correlations are predominantly near zero (teal shading), with the expected negative correlations visible within dummy groups belonging to the same raw categorical variable (e.g. the `medical_history`, `region`, `occupation` and `coverage_level` dummy blocks). The bottom row of each panel shows the feature-target correlations. In the training set (Figure 18), `smoker` shows the largest positive correlation with actual charges (bright green-yellow), followed by `coverage_level_Premium`, `medical_history_Heart_disease` and `bmi`. The predicted version (Figure 19) displays a virtually identical target row, confirming that the fitted DNN has preserved the linear association structure. The test set (Figures 20 and 21) reproduces the same patterns, with the target-correlation rows near-identical across both the actual and predicted versions. These consistency findings complete the diagnostic verification; the fitted DNN is therefore ready to serve as the prediction surface for the MC and Shapley-value analyses that follow.

3.7 Parametric MC Simulation Design

The MC stage generates a synthetic policyholder input space and the corresponding model-implied distribution of predicted costs. Each raw predictor variable is assigned a fitted simulation distribution, synthetic policyholder profiles are drawn from these distributions, and the fitted DNN computes predicted costs for the generated profiles. The simulation is performed at the raw-variable level: categorical variables are simulated as valid categories first and are only then transformed into the same encoding used during model training. Dummy variables are never simulated independently, since that could produce impossible category combinations.

3.7.1 Joint input distribution

The joint simulation distribution is constructed as a product of fitted marginal distributions, following the principle of equation (2.2.5). Let $\mathbf{X}_{\text{raw}} = (X_1, X_2, \ldots, X_q)$ denote the vector of raw predictor variables, where $q = 11$. Applying the product-marginal construction of equation (2.2.5) to the raw variables before encoding yields

$$\widehat{F}_{X,\text{raw}} = \prod_{j=1}^{q} \widehat{F}_{X_j}. \tag{3.7.1}$$

As discussed in Section 2.2.2, the product-marginal construction is a modelling choice adopted for simulation purposes. Two empirical findings from the exploratory analysis support it in this dataset: the weak pairwise correlations among the numerical predictors reported in Table 5, and the observation that `age`, `bmi`, and `children` are closely consistent with independent uniform draws over their observed ranges (Section 3.4). The product-marginal construction is therefore not merely a computational convenience but a faithful reflection of the benchmark data's apparent generating design.

3.7.2 Fitted marginal distributions

The fitted marginal distributions are selected using the empirical summaries from the exploratory diagnostics. All distributional parameters, including the support bounds for the uniform distributions, are estimated from $\mathcal{D}_{\text{train}}$ only, ensuring that no validation- or test-set information enters the simulation design. Bounded numerical variables are assigned distributions that respect their observed support. Binary variables are generated from Bernoulli distributions. Multi-level categorical variables are generated using the observed training-set proportions. Table 9 specifies the distributional assumptions for each raw variable.

Variable	Type	Distribution used	Parameter estimation	Support
`age`	Bounded integer	Discrete uniform	$\widehat{a} = 18,\ \widehat{b} = 65$	$\{18, \ldots, 65\}$
`bmi`	Bounded continuous	Continuous uniform	$\widehat{a} = 18,\ \widehat{b} = 50$	$[18, 50]$
`children`	Bounded count	Discrete uniform	$\widehat{a} = 0,\ \widehat{b} = 5$	$\{0, \ldots, 5\}$
`gender`	Binary	Bernoulli	$\widehat{p}_{\text{male}} = n_{\text{male}}/n_{\text{train}}$	$\{0, 1\}$
`smoker`	Binary	Bernoulli	$\widehat{p}_{\text{yes}} = n_{\text{yes}}/n_{\text{train}}$	$\{0, 1\}$
`region`	Categorical	Categorical	$\widehat{p}_k = n_k/n_{\text{train}}$	Valid regions
`medical_history`	Categorical	Categorical	$\widehat{p}_k = n_k/n_{\text{train}}$	Valid categories
`family_medical_history`	Categorical	Categorical	$\widehat{p}_k = n_k/n_{\text{train}}$	Valid categories
`exercise_frequency`	Categorical	Categorical	$\widehat{p}_k = n_k/n_{\text{train}}$	Valid categories
`occupation`	Categorical	Categorical	$\widehat{p}_k = n_k/n_{\text{train}}$	Valid categories
`coverage_level`	Categorical	Categorical	$\widehat{p}_k = n_k/n_{\text{train}}$	Valid tiers

Table 9: MC input distributions for raw policyholder variables.

The fitted probabilities for categorical variables are estimated from the training partition only, following the marginal parametric approach in equation (2.7.2). If C is a categorical variable with levels $c_1, \ldots, c_K$, then

$$\widehat{p}_k = \frac{1}{n_{\text{train}}} \sum_{i \in \mathcal{D}_{\text{train}}} \mathbb{I}\{C_i = c_k\}, \qquad k = 1, \ldots, K. \tag{3.7.2}$$

A simulated categorical value is drawn as $C^{(b)} \sim \text{Categorical}(\widehat{p}_1, \ldots, \widehat{p}_K)$ and subsequently encoded using the reference-category scheme.

3.7.3 Goodness-of-fit and support checks

For continuous variables, empirical histograms and density overlays are inspected, and Kolmogorov-Smirnov (K-S) and Anderson-Darling (A-D) statistics are recorded. These two statistics summarise complementary aspects of marginal fit: the K-S statistic measures the maximum absolute deviation between the empirical and fitted cumulative dis-

tribution functions, while the A-D statistic places greater weight on the tails, making it particularly sensitive to distributional departures in the regions most relevant to the high-cost analysis. Because the parameters of the fitted marginals are estimated from the training data, the standard critical values of these tests are invalid (the Lilliefors problem); the K-S and A-D statistics are therefore reported as descriptive distance measures of marginal fit rather than as formal hypothesis tests with nominal critical values or p-values. Parametric-bootstrap calibration would be required to obtain valid p-values, and is not pursued because the statistics serve a purely descriptive role here. For the discrete and categorical variables (including `age` and `children`, to which continuous-distribution K-S theory does not apply), the comparison is instead made between the empirical and fitted probability mass functions, using frequency tables and chi-squared discrepancy summaries interpreted in the same descriptive spirit. Because the sample size is large, even negligibly small distributional differences would be flagged as significant by formal tests without being practically important for simulation, which further motivates the descriptive treatment. Support restrictions are treated as compulsory: simulated numerical predictors are confined to their observed ranges and categorical variables take only valid observed levels.

3.7.4 Sample size and convergence monitoring

The base simulation uses $B = 100\,000$ generated policyholder profiles. A pilot convergence check is conducted by repeating the simulation for increasing values of $B \in \{1\,000,\ 2\,500,\ 5\,000,\ 10\,000,\ 25\,000,\ 50\,000,\ 100\,000\}$. For each B, the running mean, standard deviation, and selected quantiles of the simulated predicted costs are recorded. The Monte Carlo standard error is estimated using equation (2.8.7), with s_B denoting the simulated standard deviation. This mean-based MCSE formula does not apply to quantiles: for a quantile level $u \in (0, 1)$, the Monte Carlo uncertainty is instead governed by the asymptotic variance $u(1-u)/\big(B\, f\big(F^{-1}(u)\big)^2\big)$, where f denotes the density of the predicted-cost distribution at the quantile of interest; percentile stability is therefore assessed descriptively across the increasing pilot sizes. The selected simulation size is retained when running summaries are stable relative to the scale of the response variable.

Quantity	Purpose
Running mean	Checks stability of the average simulated predicted cost
Running standard deviation	Checks stability of simulated output spread
Median	Checks central distributional stability
90th percentile	Checks upper-tail stability
95th percentile	Candidate high-cost threshold
99th percentile	Extreme upper-tail stability check
MCSE	Quantifies simulation precision for selected summaries

Table 10: MC convergence quantities recorded during the pilot run.

3.7.5 Base DNN MC scoring

For each simulated raw profile, the preprocessing map $T(\cdot)$ produces the model-ready input vector $\mathbf{z}^{(b)}$. The fitted DNN then calculates the base MC prediction, following the formulation in equation (2.2.7):

$$\widehat{Y}^{(b)} = \widehat{m}\left(\mathbf{z}^{(b)}\right), \qquad b = 1, \ldots, B. \tag{3.7.3}$$

The resulting collection $\{\widehat{Y}^{(1)}, \ldots, \widehat{Y}^{(B)}\}$ forms the base model-implied predicted-cost distribution introduced in equation (2.2.7). The generated input profiles are also summarised before interpreting the predicted-cost outputs, to confirm what type of simulated input space produced the DNN predictions.

Algorithm 3.1: Base parametric MC simulation

Input: fitted raw-variable distributions $\{\widehat{F}_{X_j}\}_{j=1}^{q}$, fitted model $\widehat{m}(\cdot)$, preprocessing map $T(\cdot)$, number of replications B, random seed s.

Output: simulated raw profiles $\{\mathbf{X}_{\text{raw}}^{(b)}\}_{b=1}^{B}$, model-ready profiles $\{\mathbf{z}^{(b)}\}_{b=1}^{B}$, and base predicted costs $\{\widehat{Y}^{(b)}\}_{b=1}^{B}$.

1. Set the random seed to s.
2. For $b = 1, \ldots, B$, simulate each raw variable from its fitted marginal distribution.
3. For categorical variables, simulate a valid category before applying reference-category encoding.
4. Apply the training-set preprocessing map $T(\cdot)$ to obtain $\mathbf{z}^{(b)}$.
5. Calculate $\widehat{Y}^{(b)} = \widehat{m}(\mathbf{z}^{(b)})$.
6. Transform predictions back to the original monetary scale if the network output is on the standardised scale.
7. Store the simulated inputs, encoded inputs and predicted costs.

Table 11: Base MC simulation algorithm.

3.8 Validation of the MC Input Distribution

The product-marginal simulation law in equation (3.7.1) constructs the joint input distribution as a product of independent fitted marginals. Even if every marginal distribution matches the training data closely, the joint distribution of the simulated profiles is structurally different from the joint distribution of the training data because all cross-variable dependencies are set to zero by construction. This section diagnoses the magnitude of that joint-distribution difference and then assesses whether it affects the Shapley-value attributions derived from the simulated profiles.

3.8.1 Joint distribution diagnostics

The first diagnostic examines the difference between correlation matrices. Because Pearson correlation is undefined for unordered nominal variables such as `region`, `occupation` and `medical_history`, this comparison is carried out on the model-ready encoded representation, in which the numerical variables enter directly and each categorical variable enters through its binary dummy indicators. The Pearson correlation matrix of the encoded columns is computed separately on the training partition and on the parametric MC sample. The element-wise difference

$$\Delta_{jk} = r_{jk}^{(\text{train})} - r_{jk}^{(\text{MC})}, \qquad j, k = 1, \ldots, p, \tag{3.8.1}$$

quantifies how the pairwise linear association structure of the simulated profiles deviates from that of the training data. Because the parametric MC sample draws each raw variable independently, the correlations between encoded columns derived from different raw variables are approximately zero by construction (the mechanical negative correlations within a dummy group arise in both datasets), so $\Delta_{jk} \approx r_{jk}^{(\text{train})}$ for column pairs from different raw variables. The resulting difference heatmap therefore reveals which variable pairs carry the strongest dependencies in the training data that are absent from the simulation. Linear correlation on binary indicators is a limited measure of association for nominal attributes, so this diagnostic is complemented by the Cramér's V comparison below, which measures categorical association directly.

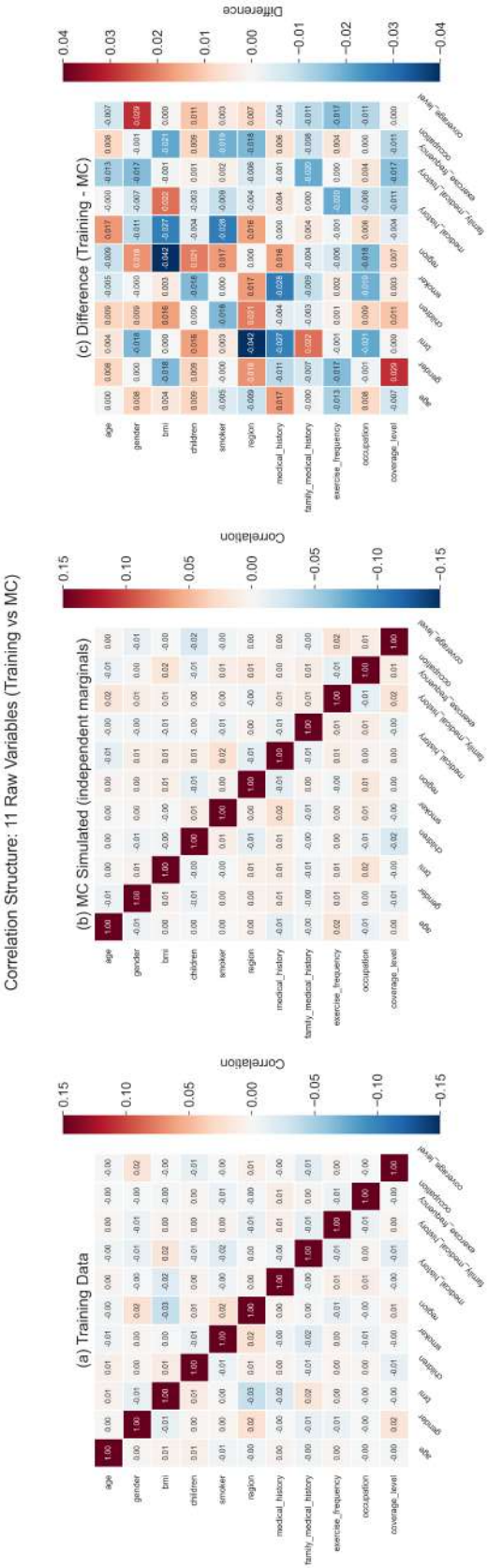


Figure 22: Correlation structure comparison: training data versus parametric MC sample.

The second diagnostic addresses the categorical predictors. For multi-level categorical predictors the strength of pairwise association is measured using Cramér's V statistic (Cramér, 1946). For each pair of categorical variables, V is computed on the training partition and on the parametric MC sample. All pairwise values of V are compared to confirm that cross-variable associations are below a weak-association threshold in both datasets and that the parametric MC sample, which has exactly zero association by construction, does not materially differ from the training data.

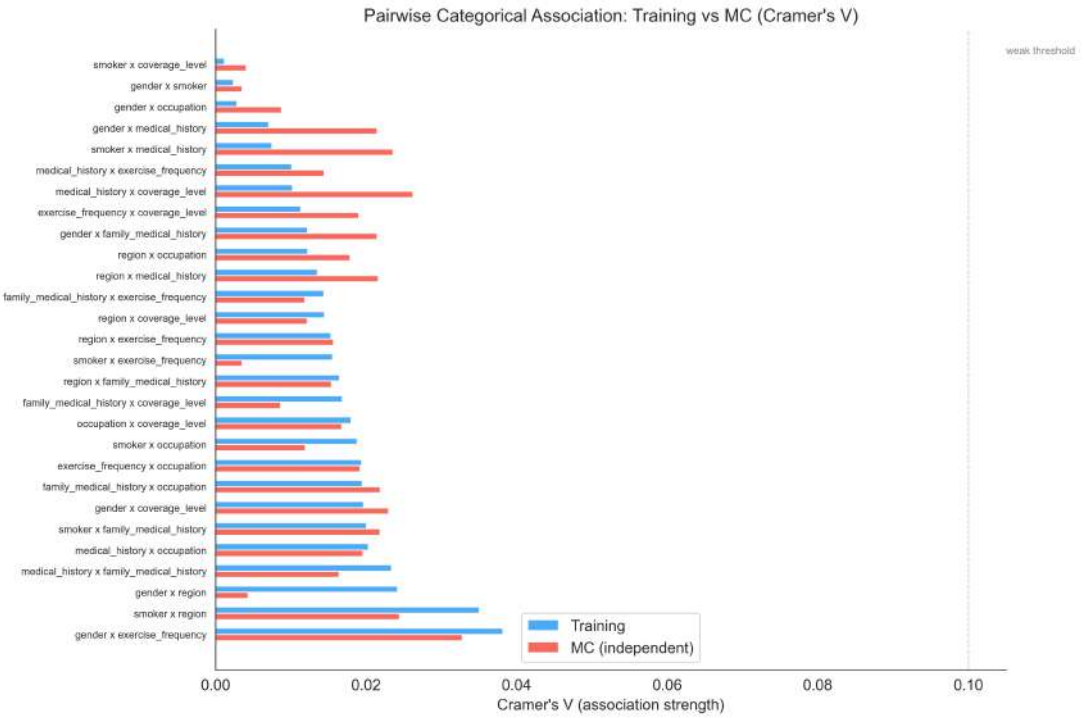


Figure 23: Cramér's V comparison: training versus parametric MC sample.

The third diagnostic is a PCA overlay of the joint input space. A principal component projection provides a visual comparison of the joint input structure. The first two principal components are computed from the standardised training inputs (using the eigenvectors from Section 3.6.4), and both the training set and the parametric MC sample are projected onto these components. The two point clouds are overlaid with distinct colours. If the independence assumption substantially alters the joint structure, the MC cloud will appear more spread out or exhibit less internal clustering than the training cloud. As reported in Section 3.6.4 (Figure 17), the first two principal components are dominated by the `coverage_level` variable yet together account for only about 13% of total variance, so no single direction dominates the 22-dimensional input space.

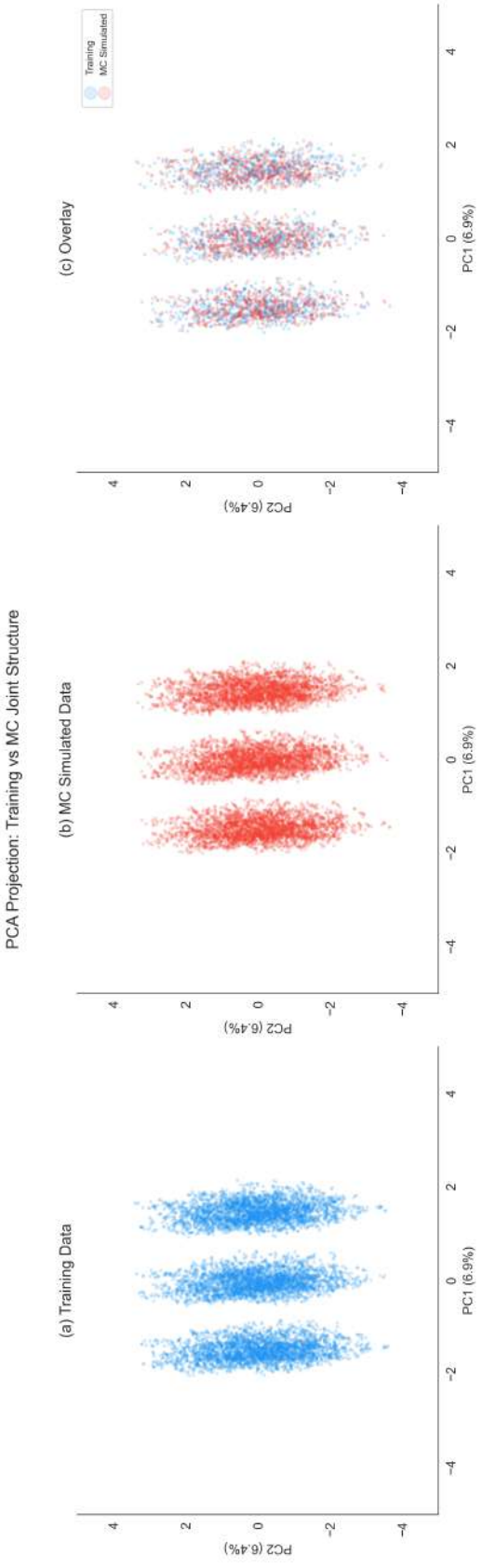


Figure 24: PCA projection of joint input structure: training, MC sample and overlay.

The MC sample populates the same region of principal component space as the training data. The vertical banding visible in both panels reflects the discrete structure introduced by the categorical predictors. The overlay in panel (c) confirms that the two point clouds are broadly co-located, with the MC cloud exhibiting a marginally wider spread consistent with the removal of the weak dependencies identified in the correlation analysis.

3.8.2 Bootstrap MC resampling

To separate the effect of the independence assumption from the effect of marginal distribution choice, a second MC sample of size B is drawn via bootstrap resampling with replacement from the training set:

$$\mathbf{X}_{\text{raw}}^{(b,\text{boot})} \sim \widehat{F}_{X,\text{emp}}, \qquad b = 1, \ldots, B, \tag{3.8.2}$$

where $\widehat{F}_{X,\text{emp}}$ is the empirical distribution of the training data. This sample preserves the empirical joint distribution exactly, including all cross-variable dependencies, while the parametric MC sample uses independent marginals. The bootstrap profiles are preprocessed and scored through the fitted DNN in the same manner as the parametric profiles. Shapley values are then computed for both sets of profiles using identical explainer settings, allowing a direct comparison of the attribution results under the two input regimes.

Property	**Parametric MC**	**Bootstrap MC**
Marginals	Fitted parametric (uniform, Bernoulli, categorical)	Empirical (training-set resampled)
Joint structure	Independent (product of marginals)	Empirical (preserves all dependencies)
Sample size	$B = 100{,}000$	$B = 100{,}000$
Purpose	Primary simulation for cost distribution and Shapley analysis	Validation check for Shapley invariance to input dependence

Table 12: Comparison of the two MC input regimes.

3.8.3 Shapley invariance assessment

The mean absolute Shapley values I_g are aggregated to the raw-variable level ($q = 11$ variables) and compared between the parametric MC and bootstrap MC profiles. If the Shapley attributions are driven by the fitted DNN's internal importance structure rather than by the particular dependence pattern in the input profiles, the two sets of attributions should be nearly identical.

The comparison is summarised by the Spearman rank correlation between the two feature-importance vectors:

$$\rho_{\mathrm{S}} = \mathrm{Spearman}\big(\{I_g^{(\mathrm{param})}\}_{g=1}^{q},\ \{I_g^{(\mathrm{boot})}\}_{g=1}^{q}\big)\,. \tag{3.8.3}$$

A rank correlation close to one indicates that the feature-importance ordering is preserved regardless of whether the input profiles carry training-data dependencies or assume full independence.

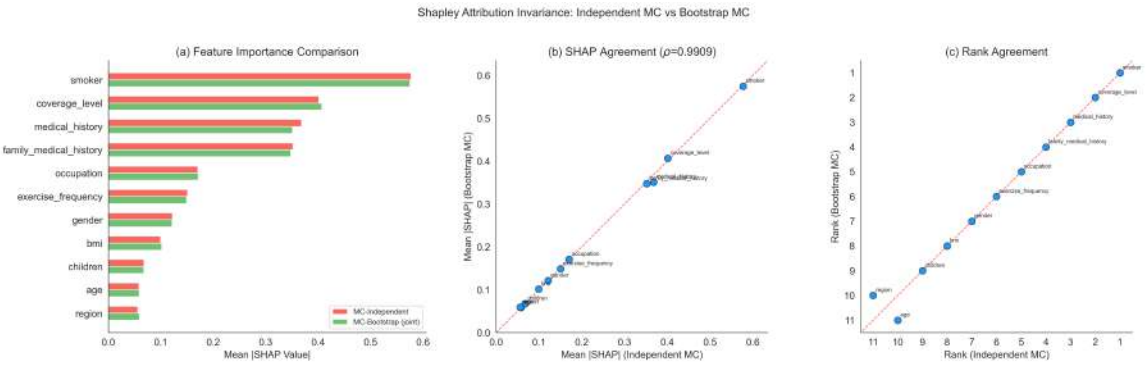


Figure 25: Shapley attribution invariance: parametric MC versus bootstrap MC input regimes.

3.8.4 Interpretation and conclusion

The joint-distribution diagnostics confirm that the parametric MC sample differs from the training data in its dependence structure: the parametric sample has zero cross-variable association by construction, whereas the training data carries weak but non-zero dependencies. However, the magnitude of these differences is small. The correlation matrix differences do not exceed 0.042 (Figure 22), all Cramér's V values fall below 0.04 (Figure 23), and the PCA overlays show that both datasets populate the same region of principal component space (Figure 24). The Shapley invariance comparison yields a Spearman rank correlation of $\rho_{\mathrm{S}} = 0.991$ between the parametric and bootstrap attribution vectors (Figure 25), with the feature-importance ordering preserved for all but the two weakest contributors.

Because the joint-distribution differences are small and the Shapley attributions are invariant to the input dependence structure, the product-marginal independence assumption in Section 3.7 is a defensible modelling choice. It does not introduce meaningful bias into the subsequent high-cost conditional Shapley analysis. The fitted DNN's internal importance structure, rather than the dependence pattern of the input profiles, determines the attribution results.

3.9 Residual-Augmented MC Sensitivity Analysis

A second MC design assesses sensitivity to fitted residual variation. The base MC design produces deterministic fitted predictions. The residual-augmented design adds a residual component to the predicted costs to examine how the simulated output distribution changes when fitted model error is introduced. The perturbation operates in the output space (predicted costs) rather than the input space (policyholder covariates). This is appropriate because the primary concern is whether the distributional conclusions drawn from the base MC analysis are sensitive to the magnitude of the fitted model's residual error, not to misspecification of the input distributions.

3.9.1 Fitted residual distribution

Let $e_i = y_i - \widehat{y}_i$ denote the fitted residual on the original response scale for observations in the residual estimation set $\mathcal{S}_\varepsilon$, taken as the training partition. Using the training set for residual estimation is appropriate in this context because the residuals serve only to characterise the magnitude and distributional shape of fitted prediction error for the sensitivity analysis, not to assess out-of-sample generalisation performance. The centred residuals are

$$e_i^c = e_i - \bar{e}, \qquad \bar{e} = \frac{1}{|\mathcal{S}_\varepsilon|} \sum_{i \in \mathcal{S}_\varepsilon} e_i. \tag{3.9.1}$$

The empirical residual distribution is $\widehat{F}_\varepsilon = |\mathcal{S}_\varepsilon|^{-1} \sum_{i \in \mathcal{S}_\varepsilon} \delta_{e_i^c}$, where $\delta_{e_i^c}$ denotes a point mass at e_i^c. The residual standard deviation is

$$\widehat{\sigma}_\varepsilon = \sqrt{\frac{1}{|\mathcal{S}_\varepsilon| - 1} \sum_{i \in \mathcal{S}_\varepsilon} (e_i - \bar{e})^2}. \tag{3.9.2}$$

3.9.2 Residual-augmented simulation

For each base MC prediction $\widehat{Y}^{(b)}$, a residual draw is sampled:

$$\varepsilon^{(b)} \sim \widehat{F}_\varepsilon, \qquad b = 1, \ldots, B. \tag{3.9.3}$$

The residual-augmented predicted cost is

$$Y_{\text{res}}^{(b)} = \widehat{Y}^{(b)} + \varepsilon^{(b)}, \qquad b = 1, \ldots, B. \tag{3.9.4}$$

Negative values are truncated at zero: $Y_{\text{res},+}^{(b)} = \max\{0, Y_{\text{res}}^{(b)}\}$.

3.9.3 Fixed residual perturbation check

A fixed perturbation check assesses sensitivity to the magnitude of residual augmentation:

$$Y_{+}^{(b)} = \widehat{Y}^{(b)} + 0.5\,\widehat{\sigma}_{\varepsilon}, \qquad Y_{-}^{(b)} = \max\left\{0,\, \widehat{Y}^{(b)} - 0.5\,\widehat{\sigma}_{\varepsilon}\right\}. \tag{3.9.5}$$

This is a deterministic sensitivity band around the fitted DNN predictions, not a probabilistic uncertainty interval. The residual-augmented simulation is interpreted as a sensitivity analysis based on fitted residual variability. It does not account for parameter uncertainty, model uncertainty or data-generating uncertainty.

Component	**Specification**
Base prediction	$\widehat{Y}^{(b)} = \widehat{m}(\mathbf{z}^{(b)})$
Residual source	Fitted residuals from the training partition
Residual centring	Residuals centred before resampling
Main residual design	$\varepsilon^{(b)} \sim \widehat{F}_{\varepsilon}$
Residual-augmented output	$Y_{\text{res}}^{(b)} = \widehat{Y}^{(b)} + \varepsilon^{(b)}$
Fixed perturbation check	$\widehat{Y}^{(b)} \pm 0.5\,\widehat{\sigma}_{\varepsilon}$
Non-negativity rule	Negative simulated costs truncated at zero

Table 13: Residual-augmented MC sensitivity design.

Algorithm 3.2: Residual-augmented MC sensitivity simulation

Input: base MC predictions $\{\widehat{Y}^{(b)}\}_{b=1}^{B}$, fitted residual distribution $\widehat{F}_{\varepsilon}$, residual standard deviation $\widehat{\sigma}_{\varepsilon}$, random seed s.

Output: residual-augmented outputs $\{Y_{\text{res}}^{(b)}\}_{b=1}^{B}$, upper perturbation outputs $\{Y_{+}^{(b)}\}_{b=1}^{B}$, and lower perturbation outputs $\{Y_{-}^{(b)}\}_{b=1}^{B}$.

1. Set the random seed to s.
2. Estimate residuals $e_i = y_i - \widehat{y}_i$ on the residual estimation set.
3. Centre the residuals and construct $\widehat{F}_{\varepsilon}$.
4. For $b = 1, \ldots, B$, draw $\varepsilon^{(b)} \sim \widehat{F}_{\varepsilon}$.
5. Calculate $Y_{\text{res}}^{(b)} = \widehat{Y}^{(b)} + \varepsilon^{(b)}$.
6. Apply the non-negativity rule if required.
7. Calculate the fixed perturbation outputs $Y_{+}^{(b)}$ and $Y_{-}^{(b)}$.
8. Store all outputs for comparison with the base MC distribution.

Table 14: Residual-augmented MC algorithm.

3.10 Shapley-Value Attribution for Simulated Profiles

Shapley-value methods are used to explain the fitted DNN predictions calculated for simulated policyholder profiles. The attribution analysis is connected to the MC predicted-cost distribution, with particular attention to the upper tail.

3.10.1 Prediction function and Shapley decomposition

The prediction function supplied to SHAP is the fitted DNN together with the response denormalisation step. Shapley values are therefore reported on the original monetary scale of `charges`. For a simulated model-ready profile $\mathbf{z}^{(b)}$, the fitted prediction is decomposed following equation (2.2.12), with $p = 22$ encoded features. The baseline prediction ϕ_0 and the feature-level contributions $\phi_j(\mathbf{z}^{(b)})$ are defined exactly as in Section 2.10.

3.10.2 Implementation settings

The background data are sampled from the model-ready training partition. The primary implementation uses a gradient-based SHAP explainer for the differentiable DNN. A model-agnostic Kernel SHAP calculation may be applied to a small validation subset as a computational check if required.

The gradient-based SHAP explainer computes approximate Shapley values, not exact game-theoretic Shapley values. Exact computation requires evaluating the prediction function at all 2^p feature subsets, which is computationally infeasible for $p = 22$ encoded features ($2^{22} = 4\,194\,304$ coalition evaluations per profile). The gradient-based approximation leverages the differentiability of the fitted DNN to estimate feature contributions efficiently, but the resulting attributions are conditional on the background sample, the explainer algorithm, and the fitted network parameters. All reported Shapley values should therefore be interpreted as model-specific, approximate attributions rather than exact cooperative-game solutions.

Item	Specification
Prediction function	Fitted DNN with response de-normalisation
Output scale	Original monetary scale of `charges`
Explainer	Gradient-based SHAP explainer
Background data	Sample from the model-ready training partition
Background size	$m_0 = 200$ observations, selected as the largest size that maintains computational feasibility for the gradient-based explainer
Explained profiles	MC simulated profiles $\mathbf{z}^{(b)}$
Primary summaries	Mean signed and mean absolute Shapley values
Conditional summaries	Shapley summaries for upper-tail simulated profiles
Categorical treatment	Dummy-level attributions aggregated to raw-variable level

Table 15: SHAP implementation settings.

3.10.3 Aggregation from encoded inputs to raw variables

The fitted DNN receives $p = 22$ encoded features, but some columns are dummy variables derived from a single raw categorical predictor. If G_g denotes the set of encoded columns associated with raw variable g, the raw-variable Shapley contribution is

$$\Phi_g\left(\mathbf{z}^{(b)}\right) = \sum_{j \in G_g} \phi_j\left(\mathbf{z}^{(b)}\right). \tag{3.10.1}$$

This aggregation prevents the interpretation from being fragmented across several dummy columns belonging to the same original variable.

The four Shapley axioms that uniquely characterise the Shapley value (Shapley, 1953) were discussed in Section 2.10. In the present context, the efficiency, symmetry, dummy, and additivity properties ensure that the aggregation from encoded features to raw variables is a linear repartitioning that preserves the complete decomposition. The raw-variable contributions Φ_g therefore retain the same axiomatic guarantees as the underlying feature-level values ϕ_j, and the total prediction remains fully decomposed into $q = 11$ raw-variable contributions without remainder.

3.10.4 Global Shapley summaries

Shapley values are computed for all B simulated profiles generated in the base MC simulation, so $B_{\text{shap}} = B$. Global feature importance I_j follows the mean absolute Shapley value defined in Section 2.10, computed over the simulated profiles. A mean signed summary is also reported. The mean signed Shapley contribution for feature j is

$$\overline{\phi}_j = \frac{1}{B_{\text{shap}}} \sum_{b=1}^{B_{\text{shap}}} \phi_j \left(\mathbf{z}^{(b)}\right), \tag{3.10.2}$$

and the mean absolute Shapley contribution is

$$I_j = \frac{1}{B_{\text{shap}}} \sum_{b=1}^{B_{\text{shap}}} \left|\phi_j \left(\mathbf{z}^{(b)}\right)\right|. \tag{3.10.3}$$

The signed summary describes the average direction of the contribution; the absolute summary measures average magnitude.

3.10.5 High-cost region and conditional Shapley summaries

The high-cost region is defined using the quantile-based threshold u_τ from equation (2.2.18) and the high-cost set $\mathcal{H}_\tau$ from equation (2.2.19), applied to the B simulated profiles with $\tau \in \{0.90, 0.95, 0.99\}$. The 95th percentile is used as the main high-cost threshold because it balances two competing requirements: it retains a sufficient number of upper-tail profiles ($|H_{0.95}| = 0.05B$) for stable conditional Shapley averages, while still isolating profiles that represent genuinely high predicted costs. The 90th and 99th percentiles are retained as sensitivity thresholds to assess whether the attribution patterns are robust to the choice of cutoff. The conditional mean signed contribution $\overline{\phi}_{j,\tau}$ and the conditional mean absolute contribution $I_{j,\tau}$ are computed as in equations (2.2.21) and (2.2.22), with the high-cost set $\mathcal{H}_\tau$ replacing the generic conditioning set and the summation taken over the B simulated profiles. These conditional summaries are also aggregated to the raw-variable level using equation (3.10.1). Comparing I_j with $I_{j,\tau}$ shows whether a feature is generally important across all simulated profiles or specifically important in the high-cost region. For the residual-augmented simulation, a separate high-cost set can be defined using the residual-augmented outputs to examine whether the profiles classified as high cost change when fitted residual variability is introduced. Since the residual component is added after the fitted DNN prediction, Shapley values explain the fitted DNN component $\widehat{m}(\mathbf{z}^{(b)})$ rather than the residual addition.

3.10.6 Bootstrap confidence intervals for conditional Shapley values

At τ_{99} the conditioning set contains only $|\mathcal{H}_{0.99}| = 0.01B$ profiles, so the conditional mean absolute Shapley values are estimated from a small effective sample. To quantify estimation uncertainty, a nonparametric bootstrap is applied. Let $\mathbf{\Phi}_\tau \in \mathbb{R}^{|\mathcal{H}_\tau| \times q}$ denote the matrix of raw-variable Shapley values for profiles in $\mathcal{H}_\tau$. The bootstrap procedure is:

1. Draw $R = 10\,000$ bootstrap resamples $\mathbf{\Phi}_\tau^{*(r)}$, each of size $|\mathcal{H}_\tau|$, with replacement from $\mathbf{\Phi}_\tau$.

2. For each resample r, compute the mean absolute Shapley value for each variable:

$$I_{j,\tau}^{*(r)} = \frac{1}{|\mathcal{H}_\tau|} \sum_{i=1}^{|\mathcal{H}_\tau|} \left|\phi_{ij}^{*(r)}\right|, \qquad j = 1, \ldots, q.$$

3. The 95% bootstrap confidence interval for $I_{j,\tau}$ is the interval $\left[I_{j,\tau}^{*(0.025)},\ I_{j,\tau}^{*(0.975)}\right]$, where $I_{j,\tau}^{*(\alpha)}$ is the α-quantile of the bootstrap distribution $\{I_{j,\tau}^{*(1)}, \ldots, I_{j,\tau}^{*(R)}\}$.

This percentile bootstrap is reported alongside the conditional importance values at τ_{99} in Chapter 4.

Summary	**Purpose**
Mean signed Shapley value	Average directional effect on fitted predictions
Mean absolute Shapley value	Average contribution magnitude across simulated profiles
Raw-variable aggregated Shapley value	Combines dummy-level attributions into original variables
High-cost conditional Shapley value	Identifies drivers of upper-tail predicted costs
Threshold comparison	Compares attribution at 90th, 95th and 99th percentiles
Base versus residual high-cost sets	Checks whether residual augmentation changes high-cost membership

Table 16: Shapley summaries calculated for simulated profiles.

3.11 Modelling Assumptions

The empirical framework depends on the modelling assumptions stated in Table 17. These assumptions define the boundary of the empirical claims: the analysis can support statements about how the fitted DNN behaves on the benchmark dataset and under the stated

Monte Carlo input assumptions, but cannot, without external validation, support causal claims about healthcare expenditure.

No.	Assumption	Interpretation
A1	Benchmark data	The dataset is a public benchmark and not a representative insurer portfolio
A2	Training-set preprocessing	Encoding, numerical scaling and response scaling are estimated from the training set and applied unchanged elsewhere
A3	Bias-free DNN	The constrained architecture is assessed empirically through validation and test diagnostics
A4	No explicit regularisation	Generalisation is assessed using validation behaviour, held-out performance and structural diagnostics
A5	Raw-variable MC simulation	Synthetic profiles are generated at the raw variable level and only then encoded
A6	Product-marginal input distribution	The joint MC input law is a practical simulation assumption and not a proof of covariate independence; validated by joint-distribution diagnostics and Shapley invariance comparison (Section 3.8)
A7	Joint-distribution validation	The Shapley attributions are shown to be invariant to whether input profiles preserve training-data dependencies (bootstrap MC) or assume full independence (parametric MC)
A8	Base MC interpretation	The base simulated output distribution represents DNN fitted predictions under the stated input assumptions, not realised healthcare costs
A9	Residual augmentation	Residual-augmented simulation is a sensitivity analysis and not a complete predictive uncertainty model
A10	Shapley interpretation	Shapley values explain the fitted DNN predictions and are not causal feature effects
A11	High-cost definition	High-cost profiles are defined by empirical upper-tail quantiles of the simulated predicted-cost distribution

Table 17: Summary of modelling assumptions and interpretive boundaries.

With the data, architecture, simulation design, attribution method and modelling assumptions now fully specified, the empirical results of the three-stage framework are presented in Chapter 4.

4 Results

4.1 Introduction

This chapter presents the empirical results of the three-stage methodological framework specified in Chapter 3. The presentation follows the same order as the workflow itself: the predictive stage is evaluated first and its robustness established, after which the simulation and attribution stages are built on the validated prediction function. The results are organised as follows: Section 4.2 reports the sequential hyperparameter comparison that produced the final DNN training configuration, together with the predictive performance of the retained model and a baseline comparison with the actuarial-standard GLM; Section 4.3 presents the multi-seed DNN training robustness analysis across five random initialisations; Section 4.4 reports the tuned benchmark comparison, including hyperparameter-optimised RF and XGBoost models; Section 4.5 evaluates cross-seed stability across all four model classes; Section 4.6 presents the MC predicted-cost distribution, its joint-distribution input validation, and convergence properties; Section 4.7 evaluates the residual-augmented sensitivity analysis; Section 4.8 reports the Shapley-value attribution results at the global level; Section 4.9 presents the high-cost conditional Shapley analysis; Section 4.10 presents the cross-seed Shapley attribution stability analysis; Section 4.11 reports the Shapley attribution correlation analysis; and Section 4.12 examines the correspondence between the MC distributional findings and the Shapley attribution patterns.

4.2 Hyperparameter Comparison Results

The five-phase sequential hyperparameter comparison (Section 3.5) was carried out on the architecture specified in Figure 11. The results of each phase are presented below.

4.2.1 Phase 1: Activation function

Three candidate activation functions were compared: CELU (equation (2.4.9)), GELU and Tanh, with all other hyperparameters fixed at learning rate 0.001, batch size 256 and Adam.

The validation loss and R^2 trajectories for this phase are presented in Figure 54 (Annexure H). CELU achieves the lowest validation loss (approximately 0.00435) and the highest validation R^2 (approximately 0.9957) from epoch 60 onwards. GELU tracks CELU closely but settles at a marginally higher loss plateau. Tanh converges more slowly and exhibits greater epoch-to-epoch volatility, reaching a best validation R^2 of approximately 0.9954.

The CELU activation was therefore retained.

4.2.2 Phase 2: Learning rate

Four learning rates were evaluated: $\eta \in \{0.01, 0.001, 0.0005, 0.0001\}$, with CELU carried forward.

The validation trajectories are shown in Figure 55 (Annexure H). $\eta = 0.01$ produces catastrophic divergence: the validation loss exceeds 1.0 and the validation R^2 drops to near zero, indicating that the gradient updates overshoot the loss surface at this step size. The three remaining learning rates (0.001, 0.0005 and 0.0001) all converge to validation $R^2 \approx 1.00$ on the standardised scale, with their loss curves overlapping on the visible scale. Among these, $\eta = 0.0005$ achieves the lowest best validation loss and was retained.

4.2.3 Phase 3: Batch size

Four batch sizes were compared: $|\mathcal{B}| \in \{64, 128, 256, 512\}$, with CELU and $\eta = 0.0005$ fixed.

The validation trajectories are shown in Figure 56 (Annexure H). All four batch sizes converge to similar validation performance, with best validation R^2 values between 0.9956 and 0.9957. The batch size $|\mathcal{B}| = 256$ was retained, as it achieves among the lowest validation loss values while maintaining stable convergence behaviour.

4.2.4 Phase 4: Optimiser

Adam and NAdam were compared with all previous best choices carried forward.

The validation trajectories are shown in Figure 57 (Annexure H). NAdam converges faster than Adam in the initial epochs and exhibits smoother validation loss trajectories throughout training. From approximately epoch 40 onwards, NAdam consistently achieves lower validation loss and higher validation R^2 than Adam, with fewer loss spikes. NAdam was therefore retained.

4.2.5 Phase 5: Momentum parameters

Three momentum configurations were compared for the NAdam optimiser: $(\beta_1, \beta_2) \in \{(0.90, 0.999),\ (0.95, 0.999),\ (0.99, 0.999)\}$.

The validation trajectories are shown in Figure 58 (Annexure H). All three momentum configurations converge to similar final validation performance. The $\beta_1 = 0.95$ configuration achieves marginally lower and more stable validation loss from epoch 40 onwards, with

the best validation R^2 reaching approximately 0.9957. The conservative setting $\beta_1 = 0.99$ converges slightly faster in early epochs but exhibits comparable final performance. The configuration $(\beta_1, \beta_2) = (0.95, 0.999)$ was retained.

4.2.6 Hyperparameter comparison overview

Figure 59 (Annexure H) presents the best validation loss achieved by each configuration across all five phases. The selected configuration (CELU, $\eta = 0.0005$, $|\mathcal{B}| = 256$, NAdam, $\beta_1 = 0.95$) achieves the lowest validation loss among all evaluated configurations. It should be noted, however, that several phase winners were separated only by validation-R^2 differences at the fourth decimal place; differences of this magnitude lie within the run-to-run variability documented in Section 4.3, so the selected configuration should be read as one member of a set of statistically equivalent configurations. Table 18 records the final training configuration.

Phase	**Candidates**	**Selected**
1. Activation function	CELU, GELU, Tanh	CELU
2. Learning rate	0.01, 0.001, 0.0005, 0.0001	0.0005
3. Batch size	64, 128, 256, 512	256
4. Optimiser	Adam, NAdam	NAdam
5. Momentum (β_1, β_2)	(0.90, 0.999), (0.95, 0.999), (0.99, 0.999)	(0.95, 0.999)

Table 18: Selected hyperparameters from the sequential comparison.

4.2.7 Final model predictive performance

The final DNN was retrained with the selected configuration and early stopping at epoch $e^* = 85$ (the checkpoint with the highest validation R^2). Table 19 reports the predictive performance metrics computed on the original monetary scale after de-normalising the fitted predictions, using the MAE (equation (2.3.3)), RMSE (equation (2.3.2)), and R^2 (equation (2.3.4)).

Partition	**MAE (R)**	**RMSE (R)**	$\boldsymbol{R^2}$
Training ($n = 524\,288$)	251	291	0.9957
Test ($n = 237\,856$)	251	290	0.9957

Table 19: Predictive performance of the final DNN on the original monetary scale.

The near-identical performance across partitions (MAE = R251 on both; RMSE within R290 to R291; $R^2 = 0.9957$ on both) confirms that the fitted model generalises to unseen data without overfitting. The training-set response statistics used for standardisation are $\overline{y}_{\text{train}}$ = R16 744 and $s_{y,\text{train}}$ = R4 419. On the standardised scale, the validation MSE at the selected epoch $e^* = 85$ is 0.00432 (Figure 59), and the original-scale residuals range from approximately −600 to +600 as characterised in the residual diagnostics of Section 3.6.

The model's predictive performance supports its use in the subsequent simulation and attribution analyses.

4.2.8 Baseline comparison: GLM, RF, XGBoost and DNN

The GLM benchmark specified in Section 3.5.5 and the two tree-based benchmarks specified in Section 3.5.6 were evaluated alongside the DNN. All four models were fitted on the same training partition ($n = 524\,288$) using the same encoded feature set and the same train/validation/test split, ensuring that the comparison reflects genuine model-class differences rather than data-handling artefacts. Table 20 reports the performance of all four models across training, validation and test partitions.

Model	Partition	MAE (R)	RMSE (R)	R^2
GLM (Gamma, log-link)	Training	623	911	0.9574
	Validation	622	913	0.9574
	Test	624	913	0.9571
RF	Training	261	324	0.9946
	Validation	359	443	0.9900
	Test	358	443	0.9899
XGBoost	Training	241	281	0.9959
	Validation	255	297	0.9955
	Test	255	297	0.9955
DNN (Funnel)	Training	251	291	0.9957
	Validation	251	291	0.9957
	Test	251	290	0.9957

Table 20: Predictive performance comparison across four model classes on the original monetary scale.

The DNN outperforms the GLM on every metric. On the test set, the DNN achieves

an R^2 of 0.9957 compared with 0.9571 for the GLM, an improvement of 3.86 percentage points. The test RMSE decreases from R913 to R290 (a 68.2% reduction), and the test MAE decreases from R624 to R251 (a 59.8% reduction).

The two tree-based benchmarks narrow the gap considerably. The RF achieves a test R^2 of 0.9899, reducing the GLM's test RMSE from R913 to R443 (51.5%) and its MAE from R624 to R358 (42.6%). However, its training R^2 of 0.9946 versus its test R^2 of 0.9899 indicates a degree of overfitting: with unrestricted depth (Table 7), the fully grown trees fit the training partition more closely than they generalise. Because the response is near-deterministic, this gap reflects the approximation and estimation variance of fully grown trees rather than the memorisation of noise, and the minimum-leaf-size constraint of five observations does not remove it. XGBoost performs substantially better, achieving a test R^2 of 0.9955 with a test RMSE of R297 and a test MAE of R255. Its training-to-test gap is negligible ($R^2_{\text{train}} = 0.9959$ versus $R^2_{\text{test}} = 0.9955$), indicating effective regularisation from the learning rate ($\eta = 0.05$) and subsampling parameters. The DNN marginally outperforms XGBoost on all three test metrics (R^2: 0.9957 versus 0.9955; RMSE: R290 versus R297; MAE: R251 versus R255), while showing training-to-test agreement to four rounded decimals. The DNN's advantage over the tree-based models, although modest in absolute terms, is consistent with its ability to learn smooth, continuous mappings through backpropagation, whereas tree-based models approximate these mappings through piecewise-constant splits.

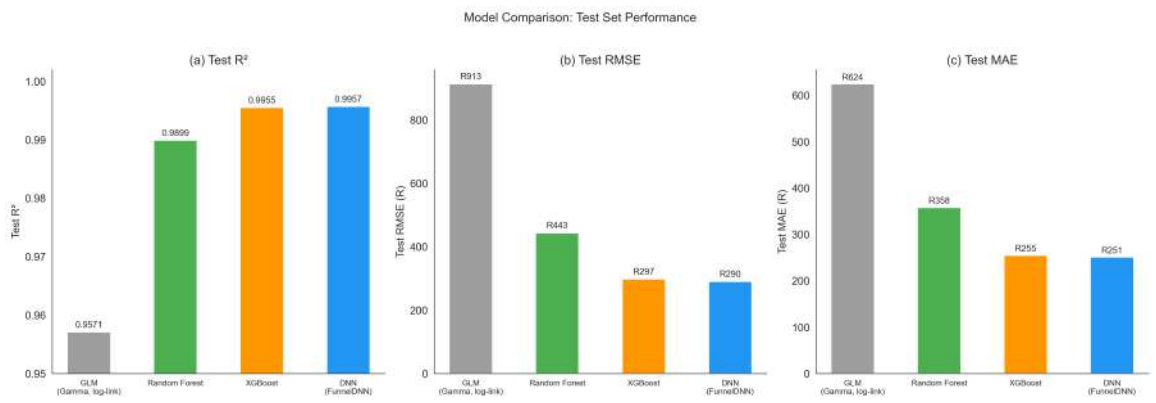


Figure 26: Test-set performance comparison across four model classes.

Figure 26 visualises the test-set comparison across all four model classes. The GLM's inferior fit is attributable to the log-link's multiplicative mean structure, which does not match the additive data-generating process underlying this dataset. The charges variable is generated by a deterministic, approximately additive cost formula with a bounded random perturbation; the GLM's exponential inverse-link therefore introduces systematic model misspecification. The RF and XGBoost close the gap substantially, with XGBoost nearly matching the DNN. The DNN learns the additive structure directly through its

flexible hidden-layer representations and achieves the highest test R^2 and lowest test RMSE and MAE.

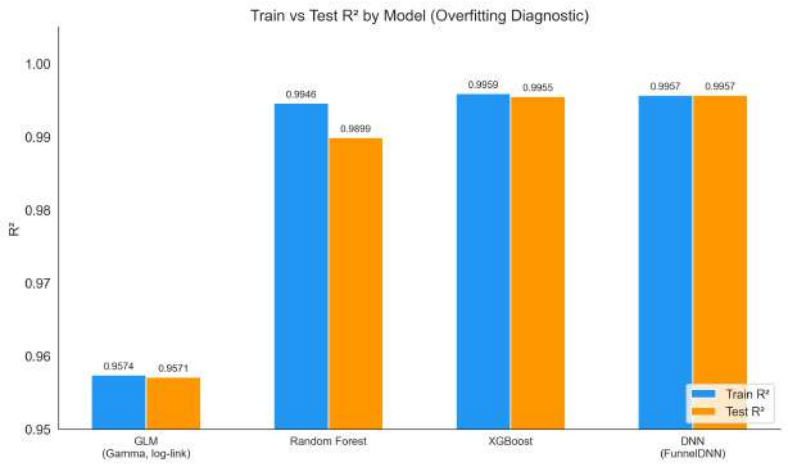


Figure 27: Overfitting diagnostic: training versus test R^2 for all four model classes.

Figure 27 compares the training and test R^2 for all four models. The GLM shows negligible generalisation loss ($R^2_{\text{train}} = 0.9574$ versus $R^2_{\text{test}} = 0.9571$), as expected for a parametric model with far fewer parameters than observations. The RF exhibits the largest train-test gap ($R^2_{\text{train}} = 0.9946$ versus $R^2_{\text{test}} = 0.9899$), indicating a degree of overfitting despite the depth constraint. XGBoost shows a negligible gap ($R^2_{\text{train}} = 0.9959$ versus $R^2_{\text{test}} = 0.9955$), confirming that the learning rate and subsampling regularisation are effective. The DNN likewise shows no evidence of overfitting ($R^2_{\text{train}} = 0.9957$ versus $R^2_{\text{test}} = 0.9957$), confirming that the early stopping criterion (Section 3.5) effectively regularises the 49 168 trainable parameters.

The benchmark comparison serves two purposes. First, it establishes that the DNN's high R^2 reflects genuine predictive superiority over the actuarial-standard GLM and competitive performance relative to the tree-based alternatives, not merely an artefact of the Kaggle benchmark dataset's (Kaggle, 2023) algorithmically generated structure. Second, because the Gamma/log-link GLM is the model an actuary would deploy as a first approach and XGBoost represents the current state of the art for tabular data, the comparison provides practical benchmark context that situates the DNN within the actuarial modelling landscape. The scope of both claims is bounded by the data: the deterministic cost formula with bounded perturbation underlying the benchmark (Kaggle, 2023) produces a strong, near-deterministic covariate-response structure, so the relative model ranking holds within this setting rather than universally.

The finding that a DNN outperforms a classical GLM on this benchmark is consistent with the broader health-cost prediction literature. Drewe-Boss et al. (2022) report that deep-learning models achieve lower prediction error than standard actuarial models on claims data, while Samiuddin et al. (2023) and Kaushik et al. (2022) similarly observe that

neural-network architectures improve upon traditional regression baselines for insurance cost estimation. As noted in Section 2.2, however, this superiority is not universal across all tabular insurance datasets (Richman, 2021).

A related line of actuarial research applies explicit shape constraints to neural networks during training, exemplified by the ICEnet of Richman and Wüthrich (2024), reviewed in Chapter 2. Because that study targets claims frequency under a Poisson deviance criterion rather than continuous monetary costs, a direct numerical comparison with the present results is not possible; the methodological differences are discussed in Section 5.12.

4.3 Multi-Seed DNN Training Robustness

The results reported above are conditional on a single trained network with a fixed random seed (seed 42). To assess the sensitivity of the predictive performance and subsequent analyses to the stochastic training process, the FunnelDNN was retrained from scratch under five independent random seeds (42, 123, 456, 789 and 1024), each producing a different weight initialisation and mini-batch ordering. All other training settings (architecture, learning rate, batch size, optimiser, momentum and early stopping criterion) were held fixed at the values in Table 18.

4.3.1 Training convergence across seeds

The training loss, validation loss and validation R^2 trajectories across all five seeds are presented in Annexure J.1 (Figures 60 to 62). All five seeds converge to the same loss plateau, with the validation R^2 trajectories virtually indistinguishable from epoch 60 onwards. No seed is trapped in a materially different local minimum, confirming that the optimisation landscape for this architecture and dataset is well-behaved.

4.3.2 Test-set performance across seeds

Table 21 reports the test-set predictive performance for each of the five seeds.

Seed	MAE (R)	RMSE (R)	R^2
42	251	291	0.99566
123	251	291	0.99565
456	251	291	0.99567
789	252	291	0.99564
1024	251	291	0.99566
Mean ± SD	251.2 ± 0.4	291.0 ± 0.0	0.99566 ± 0.00001

Table 21: Test-set predictive performance of the FunnelDNN across five random seeds (original monetary scale).

The cross-seed variability is negligible: the test R^2 ranges from 0.99564 to 0.99567 (standard deviation = 0.00001), the test RMSE rounds to R291 in all five seeds (R290 to R291 across identical-configuration runs, given the R290 test value reported for seed 42 in Table 19), and the test MAE varies by less than R1 across seeds. These results confirm that the predictive performance reported in Table 19 is not an artefact of a fortunate random initialisation.

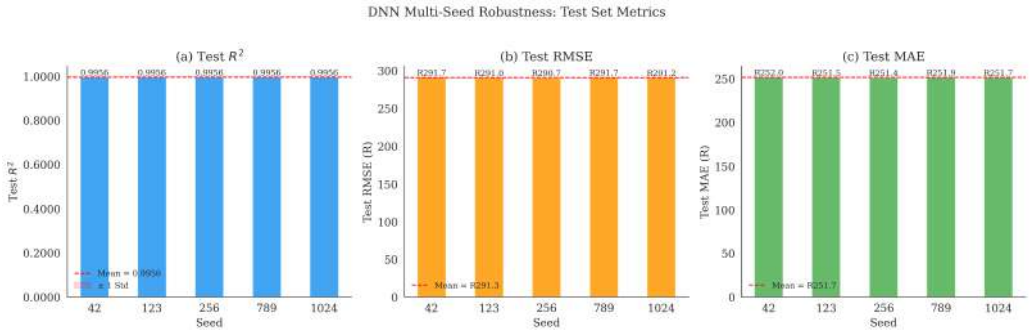


Figure 28: Test-set performance metrics across five random seeds.

Figure 28 visualises the test-set metrics across all five seeds. The near-identical bar heights confirm that the FunnelDNN's predictive accuracy is robust to the stochastic training process, with the cross-seed standard deviation of R^2 at 1.28×10^{-5}.

4.4 Tuned Benchmark Comparison

The baseline comparison in Section 4.2.8 used the RF and XGBoost benchmarks with default hyperparameter settings. To ensure a fair comparison, both tree-based models were subsequently tuned using the randomised-search protocol specified in Section 3.5.6.

4.4.1 Tuned RF

Table 22 compares the default and tuned RF performance.

Configuration	MAE (R)	RMSE (R)	R^2
Default	358	443	0.9899
Tuned	320	385	0.9924

Table 22: RF performance: default versus tuned hyperparameters (test set, original monetary scale).

Tuning improves the RF test R^2 from 0.9899 to 0.9924, reducing the test RMSE from R443 to R385 (13.1% reduction) and the test MAE from R358 to R320 (10.6% reduction). The tuned RF substantially narrows the gap to the DNN but remains below both XGBoost configurations.

4.4.2 Tuned XGBoost

Table 23 compares the default and tuned XGBoost performance.

Configuration	MAE (R)	RMSE (R)	R^2
Default	255	297	0.9955
Tuned	252	292	0.9956

Table 23: XGBoost performance: default versus tuned hyperparameters (test set, original monetary scale).

Tuning improves the XGBoost test R^2 from 0.9955 to 0.9956, matching the DNN's test R^2 to four decimal places. The test RMSE decreases from R297 to R292 and the test MAE from R255 to R252, both within R2 of the DNN's corresponding metrics.

4.4.3 Updated six-model comparison

Table 24 presents the updated comparison across all six model configurations.

Model	MAE (R)	RMSE (R)	R^2
GLM (Gamma, log-link)	624	913	0.9571
RF (default)	358	443	0.9899
RF (tuned)	320	385	0.9924
XGBoost (default)	255	297	0.9955
XGBoost (tuned)	252	292	0.9956
DNN (FunnelDNN)	251	290	0.9957

Table 24: Test-set performance comparison across six model configurations (original monetary scale).

The tuned XGBoost matches the DNN at $R^2 = 0.9956$ (versus 0.9957), with a test RMSE difference of only R2. This result confirms that the predictive advantage of the DNN over XGBoost on this benchmark dataset is marginal once hyperparameter tuning is applied. The DNN is nevertheless retained for the subsequent MC and Shapley analyses on the grounds of its differentiability, which enables gradient-based SHAP computation (Section 3.10).

4.5 Multi-Seed Benchmark Stability

To assess whether the cross-seed stability observed for the DNN extends to the benchmark models, all four model classes were retrained under the same five random seeds. Table 25 reports the cross-seed variability of the test R^2 for each model class.

Model class	Mean test R^2	SD of test R^2
GLM (Gamma, log-link)	0.9571	< 0.0001
RF (tuned)	0.9924	0.0001
XGBoost (tuned)	0.9956	< 0.0001
DNN (FunnelDNN)	0.9957	< 0.0001

Table 25: Cross-seed variability of test R^2 across four model classes (five random seeds each).

All four model classes exhibit negligible cross-seed variability. The GLM, being a deterministic convex optimisation, produces identical results regardless of initialisation. The RF's cross-seed standard deviation of 0.0001 reflects the stochastic subsampling inherent in bagging. Both XGBoost and the DNN show standard deviations below 0.0001, confirming that the model comparison in Table 24 is robust to the random seed.

4.6 MC Predicted-Cost Distribution

The base parametric MC simulation (Algorithm 3.1) was executed with $B = 100\,000$ synthetic policyholder profiles drawn from the product-marginal input distribution specified in equation (3.7.1). Each simulated profile was preprocessed, scored through the fitted DNN, and de-normalised to the original monetary scale.

4.6.1 Simulated input validation

The simulated inputs were validated following the joint-distribution diagnostic protocol in Section 3.8. The bivariate and multivariate dependence structure of the training data was compared with that of the MC sample using five complementary diagnostics (detailed figures in Annexure G.3). Pairwise scatter plots for the three continuous variables confirmed that all training-data Pearson correlations are negligible ($r \leq 0.012$), and the Monte Carlo sample shows even weaker correlations ($r \leq 0.002$), as expected under independence (Figure 44). Conditional cross-tabulations of six key categorical pairs show approximately uniform conditional proportions in both samples, confirming effective independence among the training-data categoricals (Figure 45). These bivariate diagnostics complement the correlation-matrix and Cramér's V comparisons reported in Section 3.8.

Three-way joint proportions computed for the four categorical variables most relevant to the high-cost analysis (`smoker`, `coverage_level`, `medical_history`, `family_medical_history`) yielded a maximum absolute difference of 0.012 and mean absolute differences between 0.003 and 0.004 (Figure 46), confirming that higher-order interactions are negligible. A permutation-based energy distance test (1 000 permutations) returned $p = 0.730$ (Figure 47), providing no evidence of a difference between the joint distributions under this non-parametric criterion. A t-SNE projection showed thoroughly intermixed point clouds across all identified clusters (Figure 48), providing nonlinear confirmation that the MC profiles do not occupy novel regions of the feature space.

The marginal distributions of all 11 raw predictor variables closely match the training-data marginals (Figure 49). The parametric simulation therefore faithfully implements the distributional assumptions stated in Table 9, at both the joint and marginal levels.

4.6.2 Model-implied cost distribution

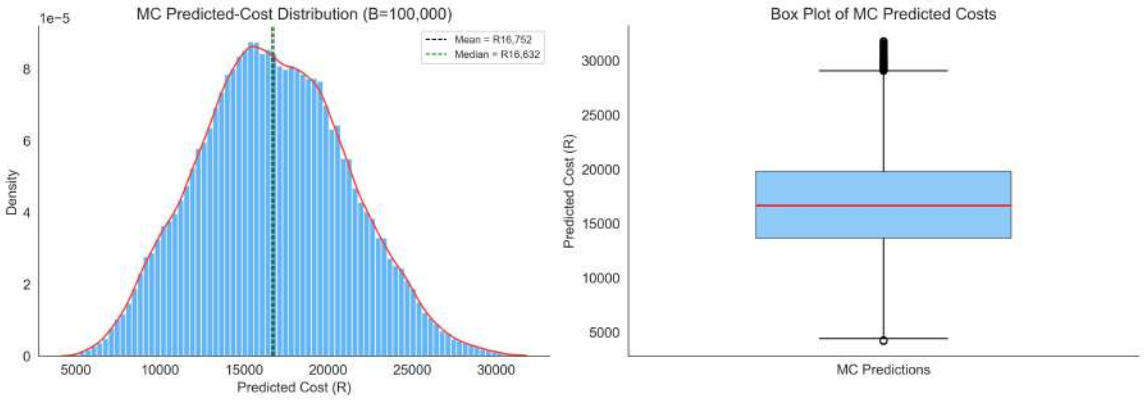


Figure 29: MC predicted-cost distribution ($B = 100\,000$).

Figure 29 displays the model-implied distribution of predicted insurance costs. Table 26 reports the distributional summary statistics.

Statistic	**Value**
Mean	R16 752
Standard deviation	R4 417
Skewness	0.132
Excess kurtosis	−0.337
Minimum	R4 172
25th percentile	R13 621
Median	R16 632
75th percentile	R19 792
90th percentile	R22 585
95th percentile	R24 249
99th percentile	R27 049
Maximum	R31 781

Table 26: Distributional summary of the MC predicted costs ($B = 100\,000$).

The distribution is approximately symmetric (skewness = 0.132) and slightly platykurtic (excess kurtosis = −0.337), with a mean of R16 752 and a standard deviation of R4 417. The interquartile range spans R13 621 to R19 792, and the upper tail extends to a maximum predicted cost of R31 781. The near-zero skewness and the close agreement between the mean and median (R16 752 versus R16 632) indicate that the predicted-cost distribution is centred, without heavy-tailed distortion.

4.6.3 Convergence diagnostics

The MCSE and running distributional summaries were recorded at increasing simulation sizes $B \in \{1\,000,\ 2\,500,\ 5\,000,\ 10\,000,\ 25\,000,\ 50\,000,\ 100\,000\}$ following the convergence monitoring procedure in Section 3.7.

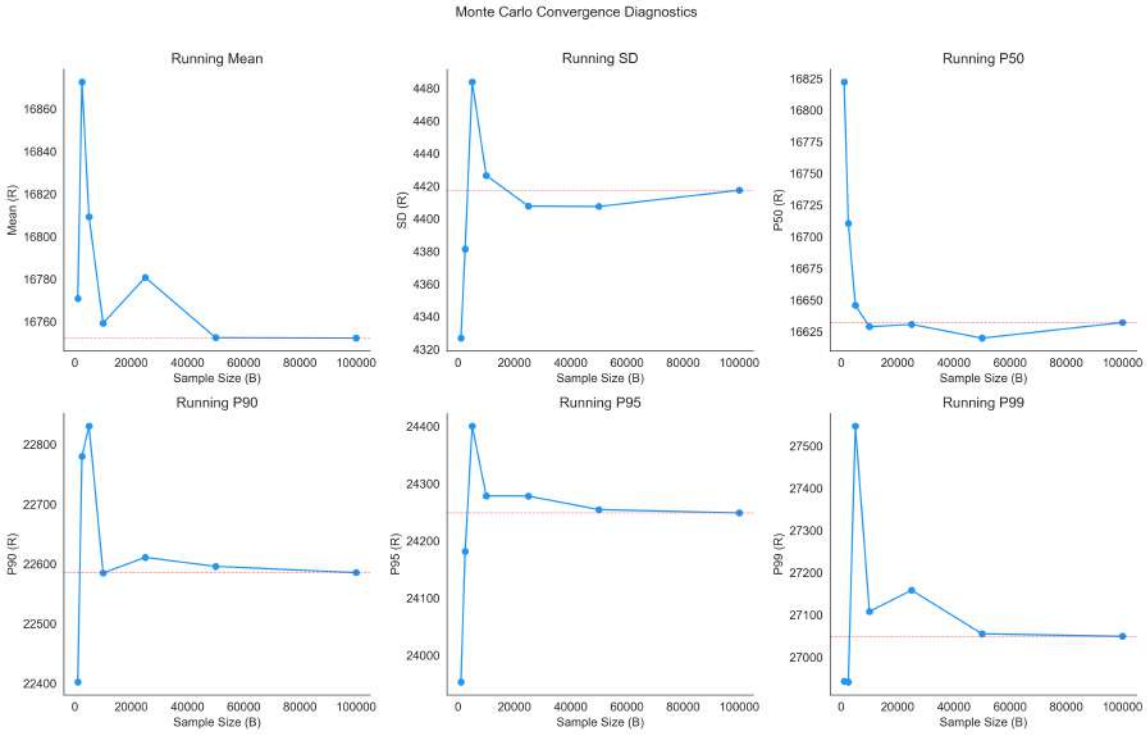


Figure 30: MC convergence diagnostics by simulation size B.

Figure 30 shows that the running mean stabilises within a narrow band (R16 752 to R16 873) across all simulation sizes. The running standard deviation converges from R4 327 at $B = 1\,000$ to R4 417 at $B = 100\,000$. The median and upper-tail quantiles exhibit similar stability, with the 95th percentile ranging from R23 953 to R24 400 across all sizes. Table 27 reports the MCSE at each simulation size.

B	Mean	SD	Median	P95	P99	MCSE
1 000	16 771	4 327	16 822	23 953	26 942	136.8
2 500	16 873	4 381	16 711	24 181	26 941	87.6
5 000	16 809	4 484	16 646	24 400	27 548	63.4
10 000	16 759	4 427	16 629	24 278	27 108	44.3
25 000	16 781	4 408	16 631	24 278	27 158	27.9
50 000	16 752	4 408	16 620	24 255	27 055	19.7
100 000	16 752	4 417	16 632	24 249	27 049	14.0

Table 27: MC convergence: running summaries and MCSE by simulation size.

The MCSE decreases from R136.8 at $B = 1\,000$ to R14.0 at $B = 100\,000$, representing 0.08% of the mean predicted cost. At the selected simulation size of $B = 100\,000$, the running summaries have stabilised and the simulation precision is adequate for the subsequent distributional and attribution analyses.

The theoretical MCSE for the sample mean of B independent draws from a distribution with standard deviation σ is $\sigma/\sqrt{B}$, which decreases at rate $\mathcal{O}(B^{-1/2})$. Because the reported MCSE is computed as the running standard deviation divided by $\sqrt{B}$, the ratios in Table 27 follow this scaling largely by construction: the ratio MCSE(B=1 000) / MCSE(B=10 000) = 136.8/44.3 = 3.09 agrees with the theoretical ratio $\sqrt{10\,000/1\,000} = \sqrt{10} \approx 3.16$, and MCSE($B$=1 000) / MCSE($B$=100 000) = 136.8/14.0 = 9.77 agrees with $\sqrt{100} = 10.0$, with the small departures reflecting movement in the running standard deviation. The table therefore verifies the expected $\mathcal{O}(B^{-1/2})$ scaling arithmetic; the substantive evidence for convergence is the practical stability of the estimated distributional summaries across pilot simulation sizes.

The running upper-tail percentiles in Table 27 provide an empirical assessment of quantile estimation precision. The 95th percentile varies within a band of R447 across simulation sizes (R23 953 to R24 400), and the 99th percentile within R606 (R26 941 to R27 548). The progressively wider variation at higher quantiles reflects the decreasing effective sample size in the tail: at τ_{99} with $B = 100\,000$, approximately 1 000 observations determine the percentile estimate, compared with 5 000 at τ_{95}. The running 99th percentile has stabilised to within $\pm 0.4\%$ of its final value from $B = 25\,000$ onwards, indicating that the tail percentile estimates are sufficiently precise for the conditional Shapley analysis that follows.

4.6.4 Cross-seed MC stability

To assess whether the MC predicted-cost distribution is sensitive to the DNN training seed, the simulation was repeated at a reduced size of $B = 10\,000$ using each of the five trained networks from Section 4.3. The cross-seed MC diagnostics are presented in Annexure J.2 (Figures 63 to 65).

The cross-seed mean predicted cost is R16 649 ± R15 across the five seeds, representing a coefficient of variation below 0.1%. The gap of approximately R100 between this cross-seed mean and the base-run mean of R16 752 (Table 26) is consistent with MC variability at the reduced simulation size together with seed-to-seed network differences. The predicted-cost density overlays are virtually identical across seeds (Figure 63), and the bootstrap 95% confidence intervals overlap within a narrow band of approximately R15 (Figure 64). The MCSE decay follows the canonical $\mathcal{O}(B^{-1/2})$ rate in all five seeds, dropping from approximately R200 at $B = 500$ to approximately R43 at $B = 10\,000$ (Figure 65). These results confirm that the model-implied predicted-cost distribution is invariant to the DNN training seed.

4.6.5 Comparison with observed training charges

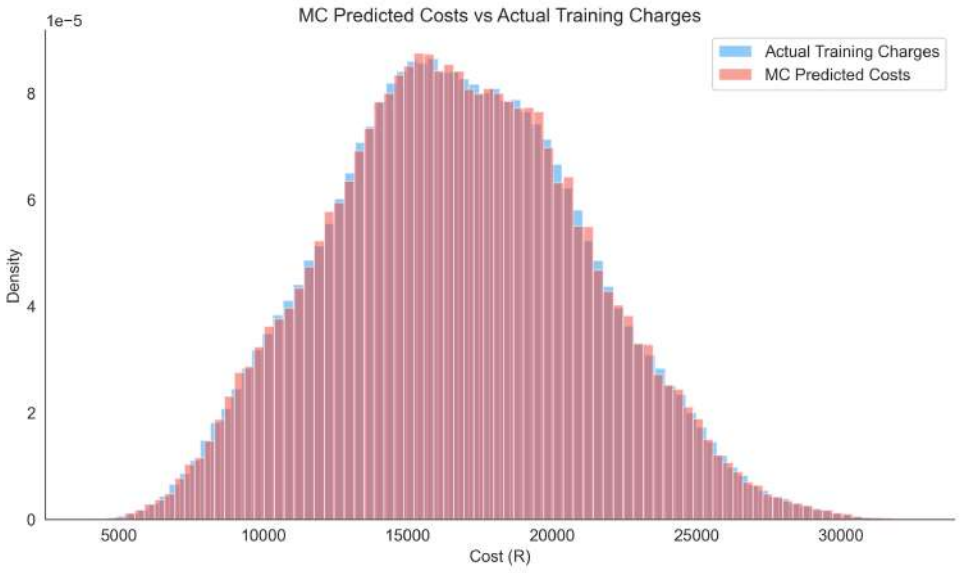


Figure 31: MC predicted-cost distribution overlaid with observed training-set charges.

Figure 31 overlays the MC predicted-cost histogram with the observed training-set charge distribution. The two distributions share similar location, spread and shape: both are unimodal, roughly symmetric, and span the range from approximately R5 000 to R32 000. The close agreement confirms that the fitted DNN, when evaluated on profiles drawn from the parametric input distribution, produces a predicted-cost landscape that is consistent

with the distributional characteristics of the observed training data. Minor differences in the tails reflect the smoothing behaviour of the regression model and the slight mismatch between the product-marginal simulation law and the empirical joint distribution.

4.7 Residual-Augmented MC Sensitivity

The residual-augmented sensitivity analysis (Section 3.9) was applied to the $B = 100\,000$ base MC predictions. Centred fitted residuals were drawn from the empirical residual distribution $\widehat{F}_\varepsilon$ (equation (3.9.3)), and a fixed perturbation band of $\pm 0.5\,\widehat{\sigma}_\varepsilon$ was computed following equation (3.9.5).

The centred training residuals from which the augmented draws are sampled were verified as symmetric, centred at zero and free of systematic bias across prediction ranges, satisfying the requirements for the perturbation analysis (Figure 50, Annexure G.4). The residual histogram is symmetric and bell-shaped with no systematic bias; the Q-Q plot shows slight tail compression consistent with the mild platykurtic behaviour noted in Section 3.6; and the residual-versus-prediction scatter is random and centred at zero across the full prediction range (R5 000 to R30 000), confirming homogeneous residual magnitude.

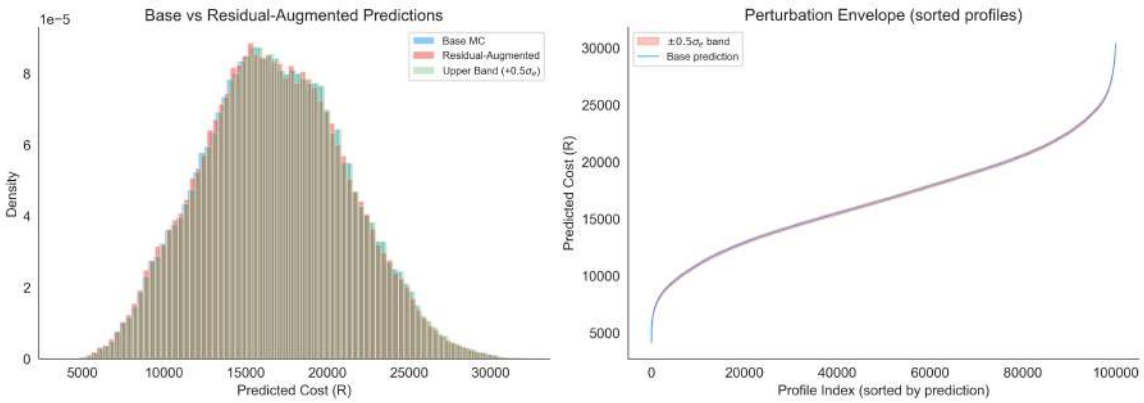


Figure 32: Residual-augmented MC sensitivity analysis.

Figure 32 (left panel) shows that the residual-augmented distribution closely overlaps the base MC distribution. The residual perturbation does not shift the central tendency or materially alter the distributional shape: the mode, spread and tail behaviour of all three distributions remain aligned. The right panel displays the sorted base predictions with the $\pm 0.5\,\widehat{\sigma}_\varepsilon$ perturbation envelope. The envelope forms a narrow, uniform band around the base prediction curve across the full range of predicted costs, from approximately R5 000 at the lower end to R30 000 at the upper end. The width of the band remains constant relative to the prediction magnitude, confirming that the fitted residual variability is small and homogeneous across the predicted-cost range.

These results indicate that the distributional conclusions drawn from the base MC analysis are not sensitive to the magnitude of the fitted model's residual error. The base predicted-cost distribution is therefore a stable foundation for the Shapley-value attribution analysis that follows.

4.7.1 Cross-seed residual-augmented sensitivity

The residual-augmented sensitivity analysis was repeated for all five DNN seeds. Figure 66 (Annexure J.3) compares the base and augmented distributions with cross-seed dispersion. The residual standard deviation $\hat{\sigma}_\varepsilon \approx$ R290 is consistent across all five seeds, and the augmentation inflates the variance of the predicted-cost distribution by approximately 0.4% (equivalently, its standard deviation by approximately 0.2%), relative to the base standard deviation of R4 417. The base-to-augmented distributional shift is negligible in every seed, confirming that the residual-augmented conclusions of Section 4.7 are not sensitive to the training initialisation.

4.8 Shapley-Value Attribution Results

Shapley values were computed for all $B = 100\,000$ simulated profiles using the gradient-based SHAP explainer described in Table 15. Dummy-level attributions were aggregated to the raw-variable level ($q = 11$) using equation (3.10.1).

4.8.1 Global feature importance

Global feature importance was assessed using the mean absolute Shapley value I_g (equation (3.10.3)) and the mean signed Shapley value $\overline{\phi}_g$ (equation (3.10.2)), computed over all B simulated profiles.

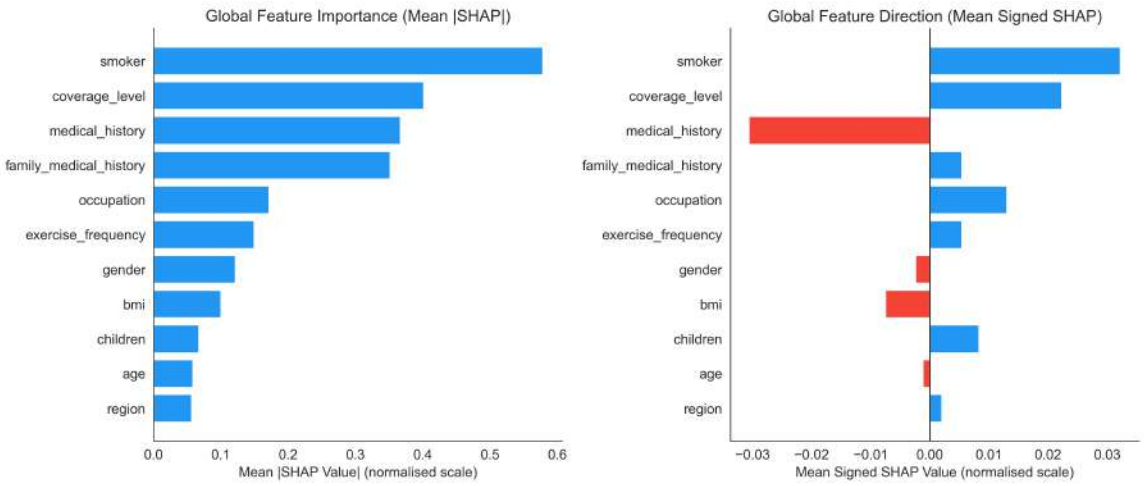


Figure 33: Global Shapley feature importance for all 11 raw variables.

Figure 33 displays the global feature-importance ranking. Table 28 reports the numerical values.

Variable	**Mean signed Shapley**	**Mean |SHAP|**
`smoker`	+0.032	0.578
`coverage_level`	+0.022	0.401
`medical_history`	−0.031	0.367
`family_medical_history`	+0.005	0.351
`occupation`	+0.013	0.171
`exercise_frequency`	+0.005	0.149
`gender`	−0.002	0.122
`bmi`	−0.007	0.100
`children`	+0.008	0.067
`age`	−0.001	0.059
`region`	+0.002	0.057

Table 28: Global Shapley feature importance for all 11 raw variables.

The four most important variables form a clear top tier: `smoker` ($I_g = 0.578$), `coverage_level` (0.401), `medical_history` (0.367) and `family_medical_history` (0.351). These four variables together account for the large majority of the fitted model's attribution magnitude. A second tier of moderate contributors includes `occupation` (0.171) and `exercise_frequency` (0.149). The remaining five variables (`gender`, `bmi`, `children`, `age` and `region`) each have mean |SHAP| values below 0.13.

The mean signed values are close to zero for all variables, indicating that the positive and

negative Shapley contributions approximately cancel out across the simulated profiles. The largest signed value in absolute terms is `medical_history` (-0.031), reflecting a mild directional asymmetry in how the model attributes cost to different medical history categories.

4.8.2 Feature-level SHAP beeswarm

The beeswarm plot of all 22 encoded features (Figure 51, Annexure G.5) reveals several patterns consistent with the aggregated importance ranking. The binary `smoker` variable produces a clear bimodal separation, with smoker = yes yielding SHAP values of approximately $+0.5$ to $+1.0$ and smoker = no yielding -0.5 to -0.7. The `coverage_level_Premium` dummy exhibits a similar bimodal pattern of smaller magnitude ($+0.3$ to $+0.7$). The Heart disease dummies for both `medical_history` and `family_medical_history` produce the next largest attributions ($+0.3$ to $+0.8$). The continuous variables (`bmi`, `children`, `age`) display more diffuse distributions centred near zero, with individual SHAP magnitudes mostly within ± 0.15.

4.8.3 Training versus MC attribution invariance

The Shapley attributions computed on the training data were compared with those computed on the MC simulated profiles (Figure 52, Annexure G.6). The Spearman rank correlation between the two feature-importance vectors is $\rho_S = 1.000$: all 11 raw variables occupy the same importance rank under both input regimes, with the mean |SHAP| values falling on the diagonal with no systematic deviation. This result extends the parametric-versus-bootstrap invariance finding of Section 3.8 (Spearman $\rho = 0.991$) and confirms that the fitted DNN's internal feature-importance structure is preserved when evaluated on the MC simulated input space.

4.9 High-Cost Conditional Shapley Analysis

The high-cost region was defined using the quantile-based thresholds u_τ (equation (2.2.18)) applied to the MC predicted-cost distribution. Table 29 reports the three threshold values and the corresponding high-cost set sizes.

Threshold	τ	u_τ (R)	$\lvert\mathcal{H}_\tau\rvert$	Proportion
τ_{90}	0.90	22 585	10 000	0.10
τ_{95}	0.95	24 249	5 000	0.05
τ_{99}	0.99	27 049	1 000	0.01

Table 29: High-cost thresholds from the MC predicted-cost distribution.

4.9.1 Conditional versus global importance

Conditional mean absolute Shapley values were computed for profiles in $\mathcal{H}_\tau$ at each threshold and compared with the global values.

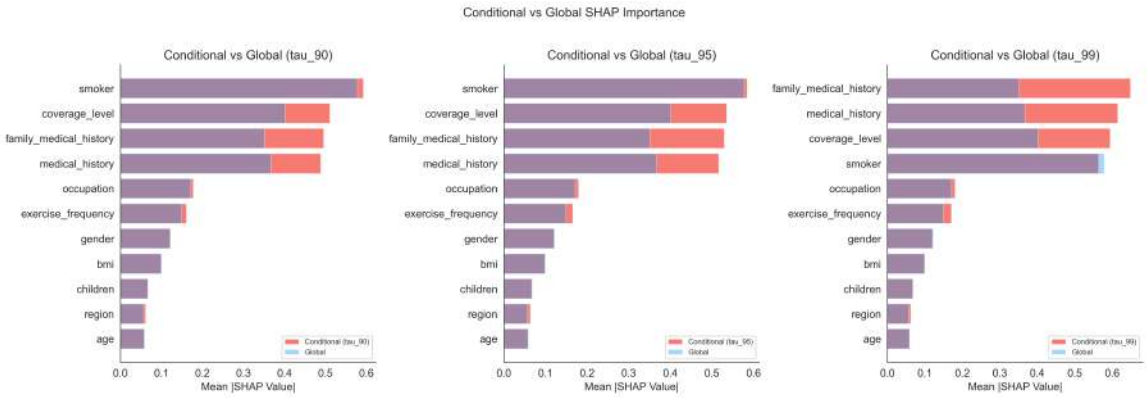


Figure 34: Conditional versus global mean |SHAP| importance at three upper-tail thresholds.

Figure 34 reveals a progressive shift in the feature-importance ranking as the threshold moves into the extreme upper tail:

- At τ_{90}, the ranking mirrors the global ordering: `smoker` remains the most important variable, followed by `coverage_level`, `medical_history` and `family_medical_history`. The conditional importance values are slightly higher than the global values for the top four variables, reflecting the more extreme nature of the high-cost profiles.
- At τ_{95}, the top four variables maintain similar positions, but `family_medical_history` and `medical_history` show increased conditional importance relative to their global values.
- At τ_{99}, a ranking reversal occurs: `family_medical_history` ($I_{g,0.99} = 0.649$) and `medical_history` ($I_{g,0.99} = 0.615$) overtake `smoker` ($I_{g,0.99} = 0.563$) as the most

important variables. `coverage_level` ($I_{g,0.99} = 0.595$) also exceeds `smoker` in the extreme tail.

Table 30 reports the conditional mean |SHAP| values at all three thresholds alongside the global values.

Variable	Global	τ_{90}	τ_{95}	τ_{99}	95% CI at τ_{99}
`smoker`	0.578	0.593	0.585	0.563	[0.559, 0.568]
`coverage_level`	0.401	0.511	0.536	0.595	[0.585, 0.604]
`medical_history`	0.367	0.488	0.517	0.615	[0.604, 0.625]
`family_medical_history`	0.351	0.496	0.530	0.649	[0.638, 0.659]
`occupation`	0.171	0.178	0.180	0.181	[0.176, 0.185]
`exercise_frequency`	0.149	0.161	0.166	0.171	[0.165, 0.178]
`gender`	0.122	0.121	0.121	0.120	[0.119, 0.121]
`bmi`	0.100	0.099	0.099	0.100	[0.096, 0.103]
`children`	0.067	0.068	0.068	0.069	[0.067, 0.072]
`age`	0.059	0.058	0.058	0.059	[0.057, 0.061]
`region`	0.057	0.062	0.063	0.063	[0.060, 0.065]

Table 30: Mean absolute Shapley values: global and conditional at each high-cost threshold (bootstrap percentile intervals).

The ranking reversal at τ_{99} is based on $|\mathcal{H}_{0.99}| = 1\,000$ profiles, providing a substantial sample for the conditional analysis. Four lines of evidence support the robustness of this finding. First, the reversal is not an isolated artefact at a single threshold: the conditional importance values for `coverage_level`, `medical_history` and `family_medical_history` increase monotonically across all four conditioning levels (global, τ_{90}, τ_{95}, τ_{99}), indicating a consistent trend rather than sampling noise. Second, the magnitude of the reversal is substantial: at τ_{99}, `family_medical_history` exceeds `smoker` by 0.086 on the normalised scale (0.649 versus 0.563), a relative difference of 15.3%. Third, the 95% bootstrap confidence intervals (Table 30) confirm that the intervals for the top three non-smoker variables do not overlap with that of `smoker`, providing formal statistical support for the reversal. Fourth, the profile-level composition analysis (Figure 36) provides a mechanistic explanation: 99% of profiles in $\mathcal{H}_{0.99}$ are smokers, so the near-constant value of the `smoker` variable within this set reduces its discriminating power, as a binary variable that takes the same value for virtually all members of a conditioning set necessarily contributes a near-constant Shapley attribution. The convergence of these four independent lines of evidence supports the conclusion that the ranking reversal reflects a genuine structural property of the fitted model's attribution surface, not a finite-sample artefact.

4.9.2 Threshold stability of the importance ranking

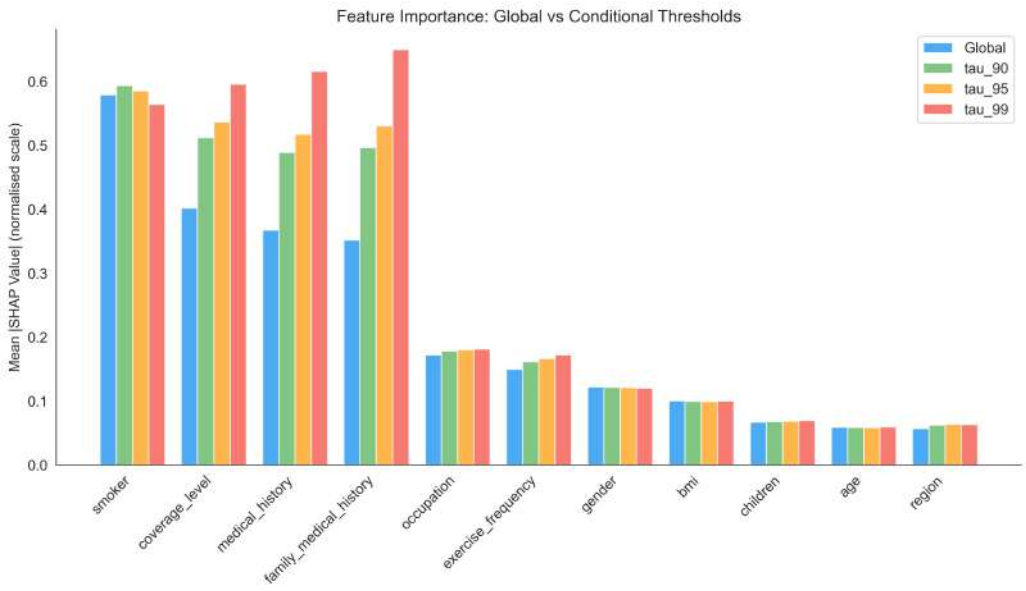


Figure 35: Feature importance across thresholds for all 11 raw variables.

Figure 35 displays the threshold-comparison bar chart. The dominant pattern is a monotonic increase in the conditional importance of `coverage_level`, `medical_history` and `family_medical_history` as the threshold moves from the global level through τ_{90} and τ_{95} to τ_{99}. In contrast, `smoker` shows a modest decline in conditional importance at the most extreme threshold (0.563 at τ_{99} versus 0.578 globally), consistent with the compositional evidence discussed above.

The second tier of variables (`occupation`, `exercise_frequency`, `gender`, `bmi`, `children`, `age` and `region`) remains stable across all thresholds, with conditional importance values close to their global values. This stability indicates that these variables contribute a consistent baseline attribution regardless of the cost region.

The ranking reversal at τ_{99} directly addresses the research gap identified in Section 2.11: global importance rankings can be misleading when the interest lies in the upper tail, because a variable that dominates on average may be nearly constant within an extreme subpopulation.

4.9.3 High-cost profile characteristics

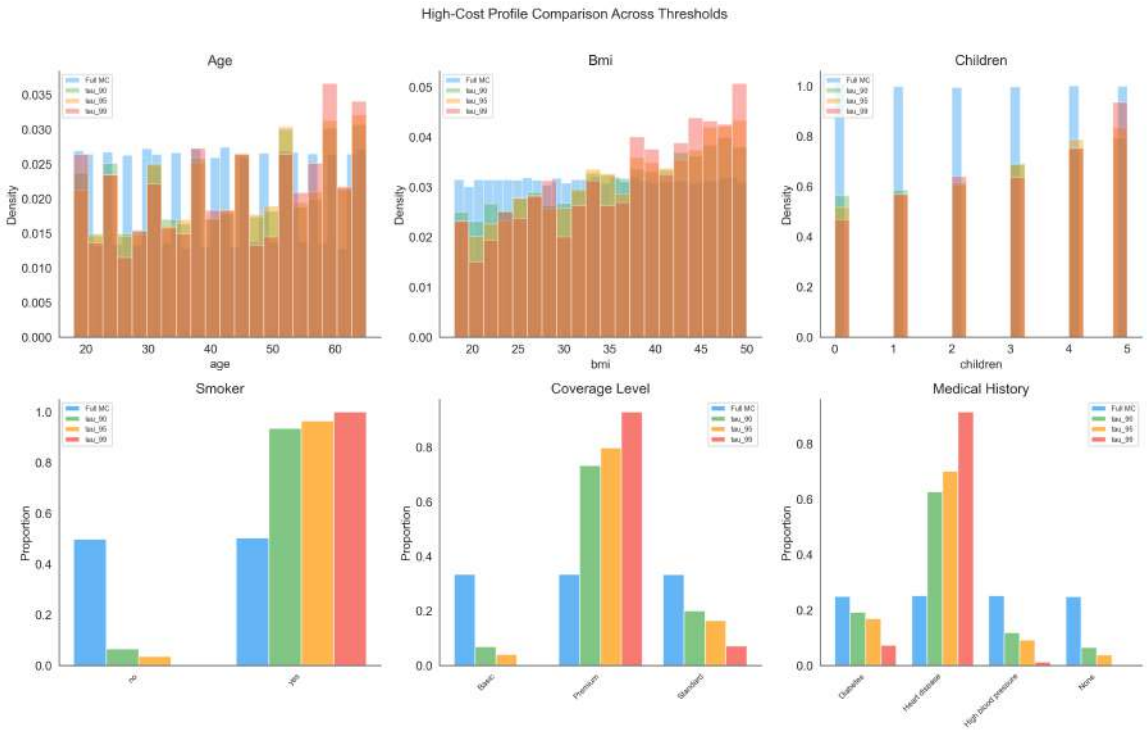


Figure 36: Distributional profiles of the high-cost sets compared with the full MC sample.

Figure 36 profiles the composition of the high-cost sets at each threshold. The categorical panels reveal a progressive concentration:

- **Smoker:** The proportion of smokers increases from approximately 50% in the full MC sample to approximately 93% at τ_{90}, 96% at τ_{95} and 99% at τ_{99}. At the most extreme threshold, virtually all high-cost profiles are smokers.
- **Coverage level:** The proportion of Premium-tier policyholders rises from approximately 33% (full MC) to 75% at τ_{90}, 80% at τ_{95} and approximately 90% at τ_{99}. Basic-tier policyholders are nearly absent from the extreme tail.
- **Medical history:** Heart disease dominates the high-cost sets: its proportion increases from approximately 25% (full MC) to 68% at τ_{90}, 77% at τ_{95} and approximately 85% at τ_{99}. Profiles with no medical history condition are progressively excluded from the upper tail.

The continuous variables (`age`, `bmi`, `children`) show little change in their distributional shapes across thresholds, consistent with their small and stable conditional Shapley values. This confirms that the high-cost region is defined primarily by the categorical risk factors rather than the continuous covariates.

4.9.4 Representative high-cost waterfall decomposition

A representative high-cost profile at the τ_{95} threshold (predicted cost R24 249) was decomposed using a SHAP waterfall plot (Figure 53, Annexure G.7). The profile is characterised by age = 34, `bmi` = 31.4, `children` = 0, `smoker` = yes, `family_medical_history` = Heart disease, `medical_history` = Heart disease and `coverage_level` = Basic.

The prediction is driven upward from the baseline ($E[f(X)] = -0.044$) to the final prediction ($f(x) = 1.699$) primarily by `medical_history` (+0.67), `family_medical_history` (+0.67) and `smoker` (+0.59), partially offset by `coverage_level` (−0.51). The remaining variables contribute smaller amounts, with the continuous covariates exerting modest negative effects (`bmi`: −0.04; `age`: −0.03; `region`: −0.01). This waterfall illustrates the additive decomposition guaranteed by the Shapley efficiency axiom: the contributions sum exactly to the difference between the prediction and the baseline, as required by equation (2.2.12).

4.10 Cross-Seed Shapley Attribution Stability

The Shapley-value attributions were recomputed for all $B = 100\,000$ MC profiles using each of the five trained networks from Section 4.3. This analysis directly addresses the single-seed delimitation stated in Section 1.8: whether the Shapley attributions are sensitive to the stochastic training process.

4.10.1 Global Shapley stability

The global mean |SHAP| values were computed for each seed and compared across all ten pairwise seed combinations. Detailed visualisations of the seed-level importance values and cross-seed dispersion are presented in Annexure J.4 (Figures 67 and 68).

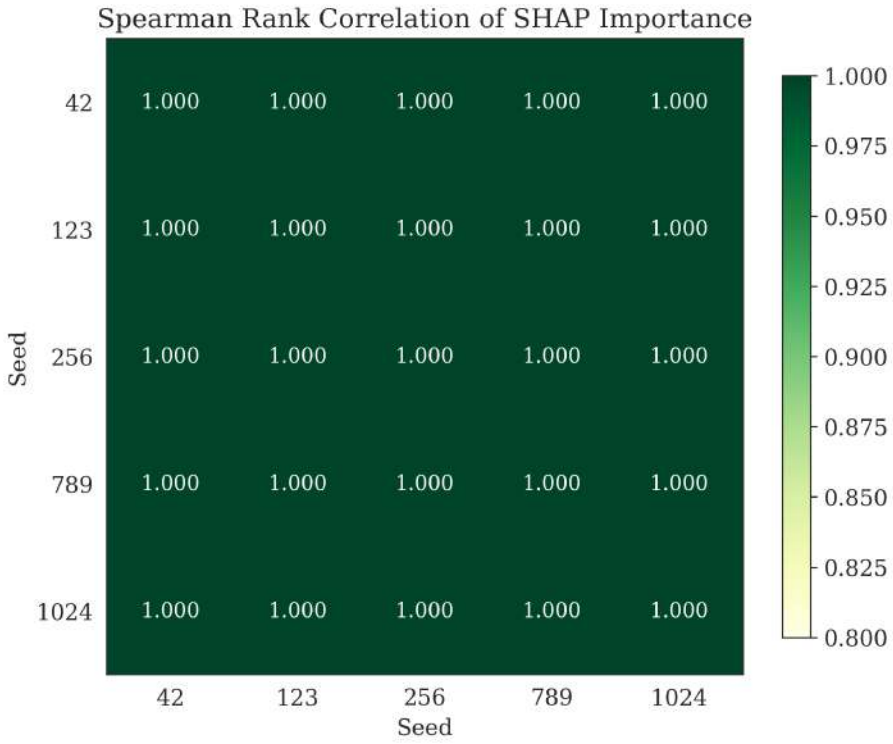


Figure 37: Spearman rank-correlation heatmap of global mean |SHAP| importance across seed pairs.

Figure 37 displays the Spearman rank-correlation matrix across all ten pairwise seed combinations. Every off-diagonal cell equals $\rho_S = 1.0000$: the global feature-importance ranking is perfectly preserved across all five seeds. The coefficient of variation of the mean |SHAP| values is below 0.1% for every variable (Figure 68), confirming that the magnitude of the attributions, not merely the ranking, is stable across seeds. This result is invariant across seeds, indicating that the model converges to a consistent attribution structure in each case despite differences in random initialisation.

4.10.2 Conditional Shapley stability at τ_{99}

The conditional Shapley analysis at the τ_{99} threshold was repeated for each seed to assess whether the ranking reversal documented in Section 4.9 is reproducible.

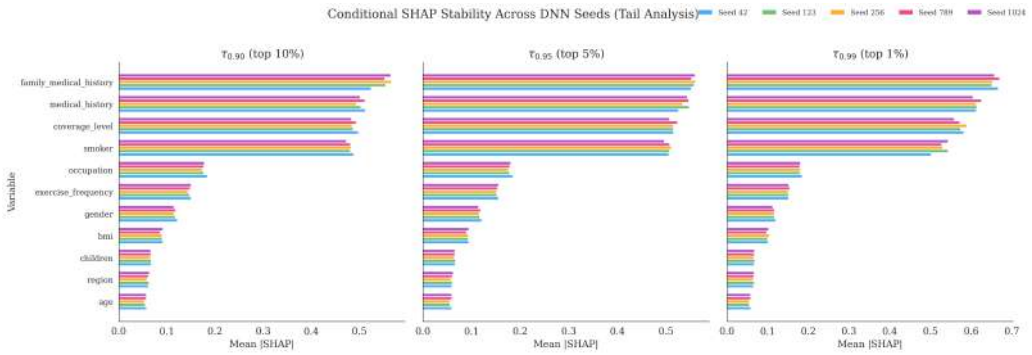


Figure 38: Conditional mean |SHAP| values at τ_{99} across five DNN seeds.

Figure 38 confirms that the τ_{99} ranking reversal is replicated in all five seeds: `family_medical_history` and `medical_history` consistently overtake `smoker` at the extreme upper tail. The pairwise Spearman correlations of the conditional importance vectors range from $\rho_S = 0.9909$ to 1.0000 across seed pairs, with minor rank swaps occurring only between adjacent variables with nearly identical importance magnitudes. This result establishes that the ranking reversal is a robust structural property of the architecture and data, not an artefact of a single random initialisation.

4.10.3 SHAP dependence patterns

The SHAP dependence plots for the `smoker` variable and the `bmi`/`coverage_level` relationship are presented in Annexure J.5 (Figures 69 and 70). The binary `smoker` variable produces a clear bimodal attribution pattern that is replicated across all five seeds. The BMI-SHAP relationship is near-perfectly linear, and no smoker-BMI interaction is evident, consistent with the additive structure of the benchmark dataset.

4.11 Shapley Attribution Correlation Analysis

To assess whether the DNN has learned meaningful pairwise interactions among the input variables, the correlation structure of the Shapley attributions was analysed by computing the pairwise Pearson correlations between all $\binom{11}{2} = 55$ pairs of raw-variable Shapley-value vectors across the $B = 100\,000$ MC profiles. Correlations between per-feature attribution vectors are a coarse proxy for interaction structure: they cannot detect interactions that are masked by dominant main effects, and a definitive statement about pairwise interactions would require formal SHAP interaction values (Shapley-Taylor indices). The results below should therefore be read as descriptive evidence rather than a formal interaction test.

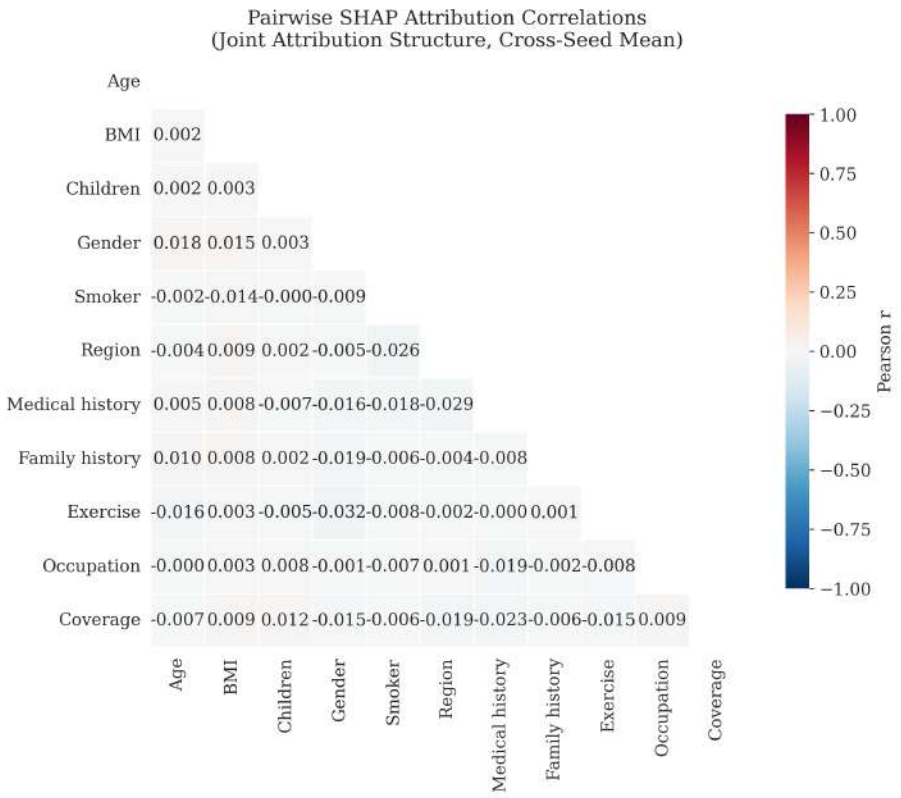


Figure 39: SHAP interaction heatmap: pairwise Pearson correlations of Shapley-value vectors.

Figure 39 displays the pairwise attribution-correlation heatmap. All 55 off-diagonal correlations fall below $|r| = 0.033$, indicating that the Shapley contributions of different variables are essentially uncorrelated across profiles. The top 15 pairwise correlations, visualised in Figure 71 (Annexure J.6), are consistent with this pattern: no pair exhibits a correlation exceeding $|r| = 0.033$.

This finding has two implications. First, it is consistent with the DNN having learned an essentially additive mapping from the input features to the predicted cost, in line with the additive data-generating process of the benchmark dataset. Second, it supports the use of main-effect Shapley summaries (global mean |SHAP| and conditional mean |SHAP|) as adequate characterisations of the model's attribution structure: with no pairwise attribution correlations of material size, the main-effect Shapley values appear to capture the attribution pattern without material loss of information, subject to the caveat that this diagnostic cannot rule out interactions masked by dominant main effects.

4.12 Integration of MC and Shapley Results

The MC and Shapley analyses provide complementary perspectives on the fitted DNN's behaviour. The MC simulation identifies the distributional regions of interest within

the predicted-cost landscape, while the Shapley-value decomposition reveals which input variables are associated with those regions.

The MC predicted-cost distribution (Figure 29) is approximately symmetric with a mean of R16 752 and an upper tail extending to R31 781. Within this landscape, three quantile-based thresholds define high-cost regions of increasing severity: $u_{0.90}$ = R22 585, $u_{0.95}$ = R24 249 and $u_{0.99}$ = R27 049.

The global Shapley analysis (Figure 33) identifies `smoker` as the dominant driver of fitted predictions across the full simulated input space. However, the conditional analysis (Figures 34 and 35) reveals that the attribution structure changes in the upper tail. As the high-cost threshold becomes more stringent, the conditional importance of `coverage_level`, `medical_history` and `family_medical_history` increases monotonically, while the importance of `smoker` declines slightly.

The profile-level analysis (Figure 36) explains this re-ordering through the ceiling-effect mechanism established in Section 4.9.

The continuous covariates (`age`, `bmi`, `children`) play a limited role at all threshold levels. Their conditional importance values remain close to their global values, and their distributional shapes within the high-cost sets are not meaningfully different from those in the full MC sample. The fitted DNN therefore assigns the dominant cost-driving role to the categorical risk factors, with the continuous variables contributing a modest, stable baseline effect.

The invariance checks reinforce the robustness of these findings. The Shapley attributions are stable whether computed on training data or MC profiles (Spearman $\rho = 1.000$, Figure 52), and the residual-augmented sensitivity analysis (Figure 32) confirms that the predicted-cost distribution is not sensitive to fitted residual variability. The attribution patterns therefore reflect the fitted model's learned internal structure rather than artefacts of the input regime or residual noise.

The multi-seed robustness analyses of Sections 4.3 to 4.11 provide a further layer of validation. The DNN's test R^2 varies by only 1.28×10^{-5} across five random seeds (Table 21); the MC mean predicted cost varies by R15 (Section 4.6.4); and the global Shapley importance ranking is perfectly preserved across all ten pairwise seed combinations ($\rho_S = 1.0000$, Figure 37). This robustness extends to the extreme tail through the cross-seed replication of the τ_{99} ranking reversal (this replication result is presented in Section 4.10), and the Shapley attribution correlation analysis is consistent with the DNN having learned an essentially additive mapping (all pairwise correlations below $|r| = 0.033$, Figure 39). These convergent findings establish that the entire chain of results, from predictive performance through the MC distribution to the global and conditional Shapley attributions, is invariant to the stochastic training process.

These results are summarised, evaluated and situated within the broader context of the study in Chapter 5.

5 Conclusion

The six research questions posed in Section 1.5 span the predictive, simulation and explanation stages of the framework. Each question is now addressed in light of the empirical results.

5.1 Research Question 1

How does a DNN regression model compare with a classical GLM, an RF regressor and an XGBoost gradient-boosted-tree regressor in predicting health insurance costs on a benchmark dataset?

The DNN outperforms the Gamma GLM (log-link) on every metric, achieving a test-set R^2 of 0.9957 against 0.9571 and reducing the test RMSE by 68.2% and the test MAE by 59.8% (Table 20); the GLM's performance gap is attributable to the log-link misspecification discussed in Section 4.2.8. The tree-based benchmarks narrow the gap substantially: after tuning, the RF reaches $R^2 = 0.9924$ and XGBoost reaches $R^2 = 0.9956$, which is not distinguishable from the DNN's 0.9957 at the reported precision (Section 4.4, Table 24). All models show near-identical training and test performance, which shows no evidence of overfitting on this dataset rather than establishing its absence, and multi-seed replication confirmed that the DNN's test R^2 varies by only 1.28×10^{-5} across five random initialisations (Section 4.3, Table 21). The hyperparameter selection underlying the final DNN configuration is documented in full in Annexure H.

5.2 Research Question 2

How adequately does parametric MC simulation of policyholder profiles reproduce the model-implied predicted-cost distribution, and how effectively does it define the high-cost conditioning regions required for tail-focused attribution?

The parametric MC simulation reproduces the model-implied predicted-cost distribution closely: the simulated distribution is approximately symmetric, unimodal and centred near the mean of the observed training charges (Figure 31). As noted in Section 1.2.2, this close agreement is expected by construction given the near-perfect DNN fit and the use of training-data marginals; the simulation's principal role is therefore to define the high-cost conditioning regions for the tail-focused Shapley analysis, a role it fulfils effectively, with convergence diagnostics confirming adequate simulation size (Section 4.6). The distributional characterisation is invariant to the network's random initialisation (Section 4.6.4).

5.3 Research Question 3

Is the product-marginal MC input distribution a defensible approximation of the training-data joint distribution for the purpose of generating simulated profiles?

Five complementary diagnostics support an affirmative answer, most directly a multivariate energy distance permutation test ($p = 0.730$) and a Shapley attribution invariance assessment comparing the parametric and bootstrap input regimes (Spearman $\rho_S = 0.991$); the individual diagnostic results are reported in Section 4.6.1. The independence assumption nonetheless broadens the simulated input space and can generate implausible covariate combinations, a limitation discussed in Section 5.9.

5.4 Research Question 4

How sensitive is the predicted-cost distribution to residual perturbation from the fitted model?

The predicted-cost distribution is robust to residual perturbation: both empirical-residual augmentation and fixed perturbation bands retained the same location, spread and shape (Section 4.7). The DNN's residuals on this benchmark dataset are sufficiently small ($\hat{\sigma}_\varepsilon \approx$ R290) that their addition does not alter the distributional conclusions, a finding consistent across all five training seeds (Section 4.7.1).

5.5 Research Question 5

Which input variables receive the largest Shapley-value attributions globally, and how does the feature-importance ranking change conditionally within high-cost regions of the predicted-cost distribution?

At the global level, four variables form a clear top tier: smoking status ($I_g = 0.578$), coverage level (0.401), medical history (0.367) and family medical history (0.351) (Table 28). The conditional analysis reveals a progressive shift across thresholds: at τ_{90} the ranking mirrors the global ordering, while at τ_{99} it reverses, with family medical history (0.649), medical history (0.615) and coverage level (0.595) overtaking smoking status (0.563) (Table 30). The reversal is driven entirely by changes in the relative importance of the top-tier categorical risk factors (Section 4.9). The multi-seed analysis (Section 4.10) confirmed that the global importance ranking is perfectly preserved across all ten pairwise seed combinations (Spearman $\rho_S = 1.0000$), and the τ_{99} ranking reversal is replicated in all five seeds with pairwise conditional Spearman correlations between 0.9909 and 1.0000. The Shapley attribution correlation analysis (Section 4.11) further established that all 55

pairwise Shapley-value correlations fall below $|r| = 0.033$, a pattern consistent with an essentially additive learned mapping.

5.6 Research Question 6

What explains any differences between the global and conditional feature-importance rankings in the upper tail?

The ranking reversal at τ_{99} is explained by the compositional structure of the extreme upper-tail profiles (Section 4.9). When smoking status is near-universal (99% of profiles), it contributes a near-constant attribution and ceases to differentiate among high-cost profiles. Two components underlie this effect: a mechanical variance-collapse component (any near-degenerate binary variable produces near-constant Shapley values within a conditioning set) and a substantive component (coverage level, medical history and family medical history retain meaningful variation and their rising conditional importance reflects genuine structure in the fitted model's attribution surface).

5.7 Contributions

The dissertation makes several contributions to the intersection of statistical learning, simulation-based inference and explainable modelling. Each of the three methodological components, namely deep neural network regression, parametric MC simulation and Shapley-value attribution, is individually well established in the statistical literature (Vapnik, 1998; Robert and Casella, 2004; Kroese et al., 2011; Lundberg and Lee, 2017; Molnar, 2022). The central methodological contribution is that the conditional Shapley analysis computed within Monte-Carlo-simulated high-cost regions produces a form of threshold-conditional feature attribution that none of these components can deliver in isolation, extending conventional global Shapley summaries into actuarially targeted subsets of the predicted-cost distribution.

The primary contribution is the integrated workflow itself. In the existing literature, these three components are typically applied in isolation: prediction studies report out-of-sample performance metrics without distributional analysis (Samiuddin et al., 2023; Drewe-Boss et al., 2022); MC studies simulate quantities of interest without connecting the results to feature-level attribution (Glasserman, 2004); and Shapley-value studies decompose predictions on observed data without examining how the attribution structure changes across specific regions of the predicted-cost distribution (Kaushik et al., 2022). Building on Blomerus (2025), the present study extends the deep-learning approach to health insurance cost prediction with an added simulation and attribution layer. The sequential dependence among the three stages means that the MC output defines the

conditioning regions for the Shapley stage, producing a connected analytical chain from fitted model through simulated population to feature attribution (see Section 1.2.4).

The conditional feature-importance summaries at high-cost thresholds represent the most distinctive contribution. The existing health insurance prediction literature reports global feature importance computed on observed data only (Kaushik et al., 2022; Rozemberczki et al., 2022), without conditioning attributions on specific regions of the predicted-cost distribution. The conditional analysis demonstrates that the variables driving extreme predicted costs differ from those that are most important on average, a finding that has not been documented in the health insurance cost-prediction literature reviewed in Chapter 2. This result contributes to the growing literature on conditional and local interpretability methods (Lundberg and Lee, 2017) by showing that global Shapley summaries can be supplemented with threshold-specific conditional summaries to provide a more complete picture of a model's attribution structure.

The MC component constructs a model-implied predicted-cost distribution from a formally stated input distribution, rather than reporting predictions only at observed data points. This connects the neural network to the simulation-based inference tradition within mathematical statistics (Glasserman, 2004; Rubinstein and Kroese, 2016; Robert and Casella, 2004): evaluating the model on a simulated population, rather than only at observed covariate combinations, yields a distributional characterisation of its output landscape (its range, dispersion and tail behaviour) under a specified input law.

The multi-seed robustness analysis constitutes a methodological strength of the study. By retraining the DNN under five independent random seeds and replicating the entire MC and Shapley pipeline for each seed, the dissertation demonstrates that the predictive performance, the model-implied cost distribution, the global Shapley ranking, and the τ_{99} ranking reversal are all invariant to the stochastic training process (Sections 4.3 and 4.10). Comprehensive cross-seed replication of this kind appears not to be routine in the applied machine-learning interpretability literature, where results are typically reported for a single trained network.

The multi-layered validation protocol, comprising joint-distribution diagnostics, convergence monitoring, residual-augmented sensitivity analysis (Saltelli, 2002), Shapley attribution invariance checks and cross-seed replication, provides a template for validating MC and Shapley analyses in other applied settings. These diagnostics address specific threats to validity that are not routinely checked in the machine-learning interpretability literature, where Shapley values are often reported without assessing their sensitivity to the input distribution or the residual structure of the fitted model. Finally, the three benchmark models (Gamma GLM, RF and XGBoost) contextualise the DNN's performance across the model-class spectrum: the DNN's predictive superiority is established over the misspecified no-interaction GLM, while tuned gradient boosting attains near-identical ac-

curacy, so the benchmarks demonstrate the adequacy, rather than the dominance, of the DNN as the prediction component of the framework.

5.8 Strengths of the Study

The study has several methodological strengths that enhance the reliability and reproducibility of its findings. First, the analysis is based on one million policyholder records, reducing sampling variability in model fitting, hyperparameter selection and Shapley-value estimation. Second, the five-phase sequential hyperparameter comparison provides a transparent, reproducible selection method documented with validation-loss curves and selection criteria (Annexure H), avoiding the opacity of automated search methods. Third, strict data-leakage prevention protocols (Section 3.3.4) ensure that all preprocessing parameters are computed exclusively on the training partition. Fourth, the multi-seed replication across five random initialisations demonstrates that neither the predictive performance nor the attribution structure depends on the specific weight initialisation, transforming the τ_{99} ranking reversal from a single-seed observation into a reproducible structural property of the fitted model class (Section 4.10). Finally, all conclusions are explicitly qualified as internal to the fitted model and benchmark data, with modelling assumptions stated formally in Table 17.

5.9 Limitations

It is important to be forthright about the limitations that qualify the scope of these findings. The delimitations set out in Section 1.8 continue to apply; three of them are qualified further by the results. First, the near-ceiling predictive performance ($R^2 = 0.9957$) is a property of the clean, approximately additive structure of the benchmark data (Kaggle, 2023), which also favours the DNN over the log-link GLM (Section 4.2.8); the benchmark data serve to demonstrate the methodology, and the specific numerical results should not be extrapolated to real-world insurance portfolios. Second, the focused model comparison is sharpened by the tuning results: consistent with evidence that gradient-boosted trees perform competitively with neural networks on tabular data (Richman, 2021), the tuned XGBoost attains near-DNN accuracy on this dataset, while stacking ensembles, Bayesian neural networks and other advanced alternatives remain unevaluated. Third, the initial conditioning of the MC and Shapley analyses on a single trained network was subsequently addressed by the multi-seed replication, in which both the training metrics (Section 4.3) and the attribution structure, including the τ_{99} reversal (Section 4.10), proved essentially unchanged across five independent seeds; five seeds do not exhaust the initialisation space, but the convergent replications provide strong evidence that the findings are not artefacts

of a single random initialisation.

The tuned XGBoost benchmark achieves $R^2 = 0.9956$, which is not distinguishable from the DNN's $R^2 = 0.9957$. This parity implies that the DNN's selection over XGBoost for the subsequent analyses is motivated by the availability of gradient-based SHAP rather than by a meaningful predictive advantage. On datasets where differentiability is not required, XGBoost may be a more parsimonious alternative that achieves equivalent accuracy with less computational cost.

The MC input distribution is constructed as a product of independent marginals. While the joint-distribution validation confirmed that the training-data dependence structure is weak ($p = 0.730$), the independence assumption broadens the simulated input space beyond what would be observed in practice. In particular, the product-marginal sampling can generate covariate combinations that are implausible in a real insurance portfolio, e.g. profiles with extreme BMI, many dependants and low exercise frequency that are rare or absent in the training data. The impact of these implausible profiles on the results is mitigated by their low probability, but it cannot be excluded entirely. The consequence bears directly on the central finding, because the high-cost sets are defined on the simulated predicted-cost distribution: profiles that reach the upper tail are, by construction, those to which the network assigns the largest predictions, and these are the profiles most likely to combine several cost-increasing attributes at once, which is precisely the region where the independence assumption places the greatest mass on combinations that a real portfolio would rarely contain. The variables identified as upper-tail drivers may therefore partly describe how the fitted network extrapolates into artificial covariate combinations rather than which characteristics drive high costs in a plausible insurance portfolio. The attribution invariance check against a bootstrap input regime that preserves the empirical dependence structure (Section 3.8) is evidence against this reading, since the attribution ordering was preserved, but it is not a substitute for repeating the analysis under a dependence-respecting input law such as the copula construction recommended in Section 5.13.

The comparison against the classical actuarial benchmark is a comparison against one particular specification rather than against classical actuarial modelling in general. The Gamma GLM fitted here uses a logarithmic link and main effects only: no interaction terms, spline or other basis expansions, or alternative link functions were considered, and no use was made of knowledge of the data-generating mechanism of the benchmark to construct a structured actuarial specification. Since the benchmark response is generated by a formula containing multiplicative structure, a GLM enriched in any of these directions would be expected to close much of the reported gap. The finding that the DNN attains $R^2 = 0.9957$ against the GLM's 0.9571 therefore supports the narrower statement that the network outperforms this specific Gamma log-link specification on this dataset, and

it does not support a general claim that flexible machine learning outperforms classical actuarial models. A fair assessment of that broader question requires the flexible model to be compared against a GLM specified with comparable care.

The upper-tail attribution analysis was carried out for the DNN alone. Because the tuned XGBoost benchmark attains predictive performance that cannot be distinguished from the network's, the ranking reversal reported at τ_{99} is established as a property of the fitted network rather than as a property shared by high-performing models of this dataset. Whether the same reversal appears when the analysis is repeated on the tuned XGBoost, whose exact tree-based attributions are available at low computational cost, has not been tested here, and the substantive finding would be considerably strengthened if it proved to be model-invariant.

The network architecture omits the additive bias vector from every layer. This is not the conventional choice for tabular regression: biases shift the activation thresholds of each unit and thereby affect the representational flexibility of the network, and the approximation theorems reviewed in Section 2.4 assume affine layer maps that include them. The choice is supported here by validation and test performance on this dataset rather than by a controlled comparison, because a bias-inclusive network of otherwise identical architecture was not fitted. The adequacy of the bias-free specification is therefore an empirical observation on one dataset rather than a justified design principle.

The hyperparameter comparison is sequential: activation, learning rate, batch size, optimiser and momentum parameters are each varied in turn with the previously selected values held fixed. This procedure is transparent and inexpensive, but it explores only a path through the hyperparameter space and cannot detect interactions between the factors, for instance that the learning rate best suited to CELU need not be the one best suited to GELU, or that the preferred batch size may depend on the learning rate and optimiser. The Gaussian process surrogate reported in Annexure H.3 was used to visualise the validation-error landscape implied by these evaluations, not to conduct a joint search. A joint procedure such as random search or Bayesian optimisation over the full space would provide a more robust selection, and the configuration retained here should be read as a good configuration found under a transparent protocol rather than as an optimum.

The reported attributions are gradient-based SHAP approximations, not exact game-theoretic Shapley values, exact computation being infeasible for the feature dimensionality considered here (Section 3.10). The approximation quality depends on the smoothness of the fitted prediction function and the adequacy of the background sample of $m_0 = 200$ observations. The gradient-based approximation is the standard approach in the SHAP literature (Lundberg and Lee, 2017), but its approximation error has not been quantified for this specific setting. The efficiency verification reported in Annexure F.3

provides only a partial check: the attributions reconstruct each prediction to within 0.5 normalised units for all B profiles, which bounds gross approximation failure but, at half a standard deviation of the standardised response, does not certify attribution-level accuracy. Quantifying the error properly would require comparison against exact Shapley values on a reduced feature set, or against a model-agnostic estimator such as KernelSHAP on a subsample, on a scale where the exact computation remains feasible.

The conditional Shapley summaries at τ_{99} rely on approximately 1 000 simulated profiles (the top 1% of $B = 100\,000$). The nonparametric bootstrap confidence intervals ($R = 10\,000$ resamples) reported in Table 30 quantify the sampling variability of the conditional attribution summaries given the fitted network and, on that conditional basis, support the ranking reversal; the monotonic trends across τ_{90}, τ_{95} and τ_{99} strengthen the finding. Nevertheless, the precision of the conditional estimates remains lower than the global estimates, which are computed over the full simulation. The upper-tail analysis in this dissertation is conducted through quantile-based thresholding of the predicted-cost distribution rather than through the formal apparatus of EVT. In the EVT framework, exceedances above a high threshold are modelled using the generalised Pareto distribution, which provides a principled basis for estimating tail probabilities and return levels. The conditional Shapley analysis developed here addresses a different question, namely which input variables contribute most to predictions in the upper tail, but it does not exploit the asymptotic distributional theory that underpins formal tail estimation. Future work could integrate EVT-based tail modelling with the conditional Shapley framework by using peaks-over-threshold methods to define the high-cost region and by assessing the sensitivity of the attribution results to the threshold selection.

A further methodological limitation concerns the scope of the uncertainty quantification around the Shapley-value estimates. The bootstrap intervals discussed above quantify the sampling variability of the attribution summaries conditional on the fitted network; uncertainty in the network parameters themselves (the weight and training uncertainty arising from the estimation of the network) is not propagated through the Shapley computation. In a Bayesian framework, one could place posterior distributions over the network weights and propagate the resulting weight uncertainty through the Shapley computation, yielding posterior distributions of the feature-importance values. Such an approach would complement the Monte Carlo simulation by adding a second layer of distributional analysis: the MC component characterises the distribution of predicted costs under input variability, while the Bayesian component would characterise the distribution of Shapley attributions under parameter uncertainty. This extension is noted as a direction for future research in Section 5.13.

Finally, the causal delimitation stated in Section 1.8 applies throughout: Shapley attributions identify model-attributed contributions, not causal determinants of healthcare

expenditure.

5.10 Implications for Actuarial Practice

The findings carry implications for actuarial practice and, as discussed in the section that follows, for the broader field of statistical methodology; both sets of implications are subject to the limitations identified above. The substantial performance advantage of the DNN over the Gamma GLM on this benchmark dataset illustrates the potential of flexible machine-learning models to improve upon the classical pricing framework, at least on datasets where the data-generating process does not conform to the log-link's multiplicative structure. The tree-based benchmarks reinforce this conclusion: both the RF and XGBoost substantially outperform the GLM, with XGBoost approaching the DNN's accuracy, confirming that the GLM's limitation on this dataset is structural rather than a general feature of simpler models. For actuarial practitioners, these results motivate the evaluation of tree-based ensembles and neural networks as supplementary or alternative pricing models alongside existing GLM-based tariffs, particularly in settings where the relationship between covariates and costs may be additive, nonlinear or interactive in ways that the log-link cannot capture. The conditional Shapley analysis has direct relevance to actuarial risk management. Insurance professionals are typically interested not only in the average cost drivers across a portfolio, but also in the variables that drive extreme costs. The finding that the global feature-importance ranking can be misleading in the upper tail suggests that conditional Shapley analysis should be considered as a standard component of model interpretation in insurance settings, particularly when the objective is to understand the composition and risk factors of high-cost claims. To illustrate: a risk manager who relies solely on the global importance ranking would focus on smoking status as the primary intervention lever; the conditional analysis reveals that within the extreme upper tail, the differentiating variables are coverage tier, medical history and family medical history, which may require different management strategies.

The MC simulation component demonstrates a method for constructing model-implied predicted-cost distributions from specified input laws. In actuarial applications, such distributions are relevant to reserve adequacy assessment, risk capital calculations and scenario analysis. The ability to generate a distributional characterisation of the model's output landscape, rather than relying on point predictions at observed data points, provides a richer basis for actuarial decision making. The benchmark-data qualifications noted in Section 5.9 apply: the methodology is transferable, but the specific performance figures are not.

5.11 Implications for Statistical Methodology

From a statistical-methodological perspective, the dissertation illustrates how the tools of simulation-based inference can be applied to the outputs of fitted machine-learning models. On datasets with greater model error or distribution shift than the benchmark used here (Research Question 2), the simulation stage would carry greater independent inferential value. The conditional Shapley analysis extends the standard global summary by partitioning the predicted-cost distribution into regions of interest and examining how the attribution structure changes within those regions. This approach is applicable to any regression model where the analyst is interested in the drivers of predictions within specific output ranges, not only in insurance pricing but also in credit scoring, healthcare resource allocation, environmental risk assessment and other domains where the upper tail of the predicted distribution is of particular importance.

There is also a connection to the broader field of extreme value analysis. The conditional Shapley analysis at upper-tail thresholds is conceptually related to the peaks-over-threshold approach in EVT, where interest centres on the behaviour of a distribution above a high quantile. In the EVT context, the focus is on estimating tail probabilities and return levels; in the conditional Shapley context, the focus is on decomposing the predictions that exceed the threshold into feature-level contributions. The two perspectives are complementary: EVT characterises the statistical properties of the tail, while the conditional Shapley analysis explains the model's behaviour within the tail; a concrete proposal for integrating them is set out in the future-work directions (Section 5.13).

5.12 Comparison with Constrained Neural-Network Approaches

A recent line of actuarial research imposes explicit shape constraints on neural networks during training, exemplified by the ICEnet of Richman and Wüthrich (2024), which was reviewed in Chapter 2. Because that study and the present dissertation differ in their response variables, evaluation metrics and datasets, a direct numerical comparison of predictive performance is not appropriate. Table 31 summarises the principal differences.

Table 31: Methodological comparison with ICEnet (Richman and Wüthrich, 2024).

Aspect	**Present dissertation**	**ICEnet**
Target variable	Continuous monetary charges	Claims frequency (count)
Response distribution	Continuous	Poisson
Models compared	GLM, RF, XGBoost, DNN	FCN, ICEnet
Best test-set result	DNN $R^2 = 0.9957$; GLM 0.9571; RF 0.9899; XGBoost 0.9955	ICEnet deviance = 0.2388; FCN 0.2387
Evaluation metric	R^2, RMSE, MAE	Poisson deviance
Dataset	Kaggle (Kaggle, 2023) ($n = 1\,000\,000$)	French MTPL ($n = 678\,007$)
Constraint mechanism	None (unconstrained models)	Smoothness and monotonicity penalties
Interpretability	Post-hoc Shapley values	In-training ICE/PDP constraints

5.12.1 Comparison of predictive performance

Richman and Wüthrich (2024) report that the ICEnet achieves a test-set Poisson deviance of 0.2388, compared with 0.2387 for the unconstrained network, a difference of 0.0001. When monotonicity constraints are applied alone, the out-of-sample deviance improves slightly to 0.2382, suggesting that these constraints act as a regulariser; the smoothness constraints account for the small overall performance decline. In the present dissertation, the unconstrained DNN achieves a test-set R^2 of 0.9957 on a continuous cost outcome. Because the response variables differ (count data versus continuous monetary charges) and the evaluation metrics differ (Poisson deviance versus R^2), these figures are not directly comparable and no cross-study ranking of accuracy is claimed. Both studies, however, demonstrate that DNNs improve upon GLM baselines within their respective insurance domains.

5.12.2 Comparison of interpretability approaches

The ICEnet enforces actuarial reasonableness during training by constraining the covariate-response relationships to be smooth and monotonic, which is important for commercial pricing and regulatory compliance. The present dissertation pursues interpretability post hoc through Shapley-value decomposition, revealing that the dominant cost drivers shift between the global level and the upper tail of the predicted-cost distribution. This dis-

tributional insight is not accessible through the ICE or PDP framework alone. The two approaches address different aspects of model interpretability and are complementary rather than competing: constrained training ensures that the learned relationships are commercially reasonable, while Shapley attribution identifies which features contribute most to individual predictions and how these contributions change in high-cost regions.

5.13 Recommendations for Future Research

Several directions for future research have been uncovered in the course of this dissertation. These can be divided into two broad categories: extensions of the modelling and simulation components, and extensions of the attribution and interpretation components. The first category, concerning the modelling and simulation components, comprises the following directions.

- Future work could extend the baseline comparison to include Bayesian neural networks, stacking ensembles and alternative deep-learning architectures (e.g. TabNet (Arik and Pfister, 2021) and the FT-Transformer (Gorishniy et al., 2021)). Such comparisons would clarify whether the Shapley attribution patterns documented here are model-dependent or model-invariant, and whether more complex architectures offer further predictive gains beyond the DNN and XGBoost benchmarks already evaluated.
- Replacing the product-marginal input distribution with a copula-based joint distribution would allow the simulation to reproduce the dependence structure of the observed data more faithfully. A Gaussian copula or vine copula fitted to the training data would generate covariate combinations that respect the empirical correlation structure, eliminating the implausible profiles that can arise under the independence assumption.
- The multi-seed robustness analysis presented in Section 4.3 addressed the cross-seed sensitivity question for this benchmark dataset by retraining under five independent seeds. Future work could extend this analysis to a larger number of seeds (e.g. 50 or 100) or to different architectures, establishing whether the observed cross-seed stability generalises beyond the FunnelDNN and the present dataset.
- Applying the framework to proprietary or regulatory insurance datasets with genuine underwriting heterogeneity would test whether the conditional Shapley findings generalise beyond the Kaggle benchmark setting (Kaggle, 2023). Real-world data would introduce challenges not present in the benchmark data, including missing values, temporal trends, policy-level effects and non-trivial dependence structures among covariates.

- The classical benchmark should be strengthened before any general comparison between flexible and classical models is attempted. Fitting a structured GLM that uses knowledge of the benchmark's data-generating mechanism, together with specifications carrying interaction terms, spline or other basis expansions and alternative link functions, would establish how much of the reported gap reflects the flexibility of the network and how much reflects the deliberate simplicity of the baseline.

- Repeating the global and upper-tail attribution analysis on the tuned XGBoost benchmark, for which exact tree-based Shapley values are computationally inexpensive, would determine whether the τ_{99} ranking reversal is specific to the fitted network or persists across high-performing model classes on this dataset. A reversal common to both would be a considerably stronger substantive finding than one observed for a single model.

- Fitting a bias-inclusive network of otherwise identical architecture would turn the bias-free specification from an assumption supported by validation performance into a controlled comparison, and would establish whether the omission of the additive bias vector affects the fitted prediction surface, the attribution ranking or only the parameter count.

- Replacing the five-phase sequential comparison with a joint search over the hyperparameter space, by random search or by Bayesian optimisation with the Gaussian process surrogate already implemented in Annexure H.3, would allow the interactions between activation, learning rate, batch size and optimiser to be detected, and would establish whether the configuration retained here is competitive with one selected jointly.

- Because the present analysis rests on a single benchmark dataset, the robustness of the framework to data quality has not been established. Future work could test this directly by injecting controlled levels of noise into the predictors and the response, or by simulating datasets from known generating processes of varying signal-to-noise ratio, and then tracking how the predictive ranking of the four model classes and the upper-tail Shapley ordering degrade as the noise level rises. Such a study would establish the conditions under which the conditional attribution findings remain stable, which a single clean, algorithmically constructed dataset cannot show.

- Coupling the MC simulation with Bayesian posterior predictive distributions or conformal prediction intervals would add formal uncertainty quantification to the predicted-cost distribution, transforming it from a point-prediction landscape into a set-valued prediction landscape with formal coverage properties.

- Quantile regression offers a complementary perspective on the conditional cost distribution that warrants investigation. Rather than modelling the conditional mean and subsequently examining quantiles of the MC output, a quantile regression neural network could directly estimate conditional quantiles of the cost distribution at specified probability levels. This approach would provide a natural alternative to the threshold-based conditioning used in the conditional Shapley analysis and would allow the analyst to study how the feature-importance structure varies across quantile levels within the fitted model itself, without the need for a separate MC simulation stage.

- As discussed in Section 5.12, integrating constrained neural-network training with the Monte Carlo-Shapley workflow would combine actuarial reasonableness with feature-level explainability. The smoothness and monotonicity constraints of the ICEnet (Richman and Wüthrich, 2024) could be incorporated into the DNN's loss function, and Shapley-value attribution could then be applied to the constrained model to verify that the imposed constraints do not distort the feature-importance ranking in the upper tail. This extension would also permit an empirical assessment of whether the negligible accuracy cost observed for frequency models extends to continuous-cost prediction.

The second category concerns the attribution and interpretation components of the framework.

- The τ_{99} ranking reversal could be traced more finely by enlarging the high-cost subsample (increasing the simulation size well beyond $B = 100\,000$ so that the top 1% contains substantially more than the $\approx$1 000 profiles available here) and by introducing intermediate thresholds (e.g. $\tau_{97.5}$, τ_{98}) to resolve the transition in the importance ranking.

- Comparing the gradient-based SHAP attributions with alternative methods, such as permutation-based SHAP, KernelSHAP or LIME (Ribeiro et al., 2016), would assess the sensitivity of the results to the choice of explanation method. Consistency across methods would strengthen the conclusion that the ranking reversal is a property of the fitted model rather than an artefact of the gradient-based approximation.

- The Shapley attribution correlation analysis in Section 4.11 established that pairwise Shapley-value correlations are negligible on this benchmark dataset (all below $|r| = 0.033$), consistent with an essentially additive learned mapping. On real-world insurance portfolios with genuine covariate interactions, the interaction structure may be more complex. Extending the analysis to formal SHAP interaction values on datasets with known non-additive relationships would clarify when main-effect

Shapley summaries remain adequate and when higher-order interaction decompositions are necessary.

- The conditioning could be extended from univariate threshold exceedances to multivariate regions defined by combinations of input variables, allowing the analyst to examine how the attribution structure changes within specific subpopulations (e.g. elderly smokers with Premium coverage) rather than within quantile-based cost regions.

- If panel or longitudinal insurance data are available, the framework could be extended to study how the predicted-cost distribution and its Shapley decomposition evolve over time. Temporal changes in the feature-importance ranking would indicate shifts in the drivers of high-cost predictions as the portfolio composition changes, providing dynamic risk-management information that static cross-sectional analyses cannot offer.

- The conditional Shapley analysis could be integrated with formal EVT by using peaks-over-threshold methods and the generalised Pareto distribution to define the high-cost region. EVT-informed threshold selection would provide a principled statistical basis for the conditioning, and the resulting Shapley decomposition would be grounded in the asymptotic distributional theory that underpins formal tail estimation. This integration would bridge the gap between the feature-attribution perspective developed in this dissertation and the tail-modelling perspective that is central to actuarial risk management and solvency assessment.

5.14 Concluding Remark

This dissertation set out to bridge two intellectual traditions that have often developed independently: the classical statistical tradition of studying estimators through their distributional properties, and the modern machine-learning tradition of training flexible prediction models and explaining their outputs. The bridge is built from three components, each with a well-established theoretical basis in mathematical statistics: supervised regression provides the prediction function, MC simulation provides the distributional characterisation, and Shapley-value decomposition provides the feature-level attribution.

The most noteworthy finding is the feature-importance ranking reversal at the 99th percentile of the predicted-cost distribution, replicated across five independent random initialisations. This result illustrates a general principle that extends well beyond insurance pricing: a variable's importance in a statistical model is not a fixed property, but depends on the region of the output space under consideration. Global importance summaries,

while useful, can obscure the attribution structure within subpopulations of particular interest.

The framework developed here is not specific to health insurance or to the particular benchmark dataset analysed. The three-stage framework of prediction, simulation and conditional explanation is applicable to any regression problem where the analyst seeks to understand the distributional behaviour of a model's predictions and to decompose those predictions into feature-level contributions within specific regions of the output space. The validation diagnostics developed for the MC and Shapley components provide a reusable template for assessing the reliability of such analyses in other applied settings, from credit scoring and healthcare resource allocation to environmental risk assessment and financial modelling.

The development of this framework reflects a conviction that the tools of mathematical statistics remain essential for understanding and interpreting the outputs of modern machine-learning models. Neural networks are powerful prediction devices, but their value to the applied statistician extends beyond the point predictions they produce. By treating the trained network as a deterministic transformation to be studied through MC experiments and decomposed through Shapley values, this dissertation grounds the machine-learning prediction paradigm in long-standing statistical practice. While this connection may seem like a modest step within the vast landscape of statistical learning and explainable modelling, it is worth remembering that each contribution to the field of interpretable machine learning is, in its own way, a contribution to the broader enterprise of understanding how statistical models learn from data, what they learn, and where their learned representations succeed or fail in capturing the structure of the phenomena they are designed to predict.

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Annexure A Extended Mathematical Details

Annexure A.1 Proof of the Universal Approximation Theorem

Theorem Annexure A.1 (Cybenko, 1989; Hornik, Stinchcombe and White, 1989)**.** *Let $\sigma : \mathbb{R} \to \mathbb{R}$ be a non-constant, bounded, continuous function (the activation function). Let $I_p = [0,1]^p$ denote the p-dimensional unit hypercube. For any continuous function $f : I_p \to \mathbb{R}$ and any $\varepsilon > 0$, there exist an integer N, real constants $\alpha_i, b_i \in \mathbb{R}$, and vectors $\mathbf{w}_i \in \mathbb{R}^p$ for $i = 1, \ldots, N$, such that the function*

$$F(\mathbf{x}) = \sum_{i=1}^{N} \alpha_i \, \sigma\big(\mathbf{w}_i^{\top} \mathbf{x} + b_i\big)$$

satisfies

$$\sup_{\mathbf{x} \in I_p} \big| F(\mathbf{x}) - f(\mathbf{x}) \big| < \varepsilon.$$

Proof. The proof proceeds by contradiction and uses the Hahn-Banach theorem from functional analysis. Define the family

$$\mathcal{S} = \big\{ \sigma(\mathbf{w}^{\top} \mathbf{x} + b) : \mathbf{w} \in \mathbb{R}^p,\ b \in \mathbb{R} \big\}$$

and let $\mathcal{M}$ denote the closure of $\operatorname{span}(\mathcal{S})$ in $C(I_p)$ with respect to the supremum norm. We wish to show $\mathcal{M} = C(I_p)$.

Suppose, for contradiction, that $\mathcal{M} \neq C(I_p)$. By the Hahn-Banach theorem, there exists a non-zero bounded linear functional $L \in C(I_p)^*$ such that $L(g) = 0$ for all $g \in \mathcal{M}$. By the Riesz representation theorem, there exists a non-zero finite signed regular Borel measure μ on I_p such that

$$L(g) = \int_{I_p} g(\mathbf{x}) \, \mathrm{d}\mu(\mathbf{x})$$

for all $g \in C(I_p)$. In particular, for all $\mathbf{w} \in \mathbb{R}^p$ and $b \in \mathbb{R}$,

$$\int_{I_p} \sigma(\mathbf{w}^{\top} \mathbf{x} + b) \, \mathrm{d}\mu(\mathbf{x}) = 0.$$

The key step, due to Cybenko (1989), shows that for discriminatory activation functions (which include all bounded, non-constant, continuous σ), the condition above forces $\mu = 0$, contradicting the assumption that L is non-zero. The discriminatory property is

established by noting that for any half-space $H = \{\mathbf{x} : \mathbf{w}^\top \mathbf{x} + b > 0\}$, the pointwise limit

$$\sigma(\lambda \mathbf{w}^\top \mathbf{x} + \lambda b) \to \begin{cases} 1 & \text{if } \mathbf{w}^\top \mathbf{x} + b > 0, \\ 0 & \text{if } \mathbf{w}^\top \mathbf{x} + b < 0, \end{cases} \quad \text{as } \lambda \to +\infty,$$

combined with dominated convergence, yields $\mu(H) = 0$ for all open half-spaces H. Since open half-spaces generate the Borel σ-algebra on I_p, it follows that $\mu = 0$, a contradiction. □

Annexure A.2 Derivation of the Adam and NAdam Update Rules

Adam update rule. Let $g_t = \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_{t-1})$ denote the gradient of the loss at iteration t. The first and second moment estimates are

$$m_t = \beta_1 \, m_{t-1} + (1 - \beta_1) \, g_t, \tag{Annexure A.2.1}$$

$$v_t = \beta_2 \, v_{t-1} + (1 - \beta_2) \, g_t^2, \tag{Annexure A.2.2}$$

where $\beta_1, \beta_2 \in [0, 1)$ are the decay rates and the squaring in (Annexure A.2.2) is element-wise. Since $m_0 = v_0 = \mathbf{0}$, these estimates are biased towards zero, especially in early iterations. The bias-corrected estimates are

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \tag{Annexure A.2.3}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}. \tag{Annexure A.2.4}$$

The parameter update is

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \eta \, \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{Annexure A.2.5}$$

where $\eta > 0$ is the learning rate and $\epsilon > 0$ (typically 10^{-8}) prevents division by zero.

NAdam update rule. NAdam incorporates Nesterov momentum by replacing the bias-corrected first moment $\hat{m}_t$ with a look-ahead estimate. The modification replaces (Annexure A.2.5) with

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \eta \, \frac{\beta_1 \hat{m}_t + \frac{(1-\beta_1)}{1-\beta_1^t} g_t}{\sqrt{\hat{v}_t} + \epsilon}. \tag{Annexure A.2.6}$$

Annexure A.3 Shapley Value Axiom Verification

Axiom 1: Efficiency.

$$\sum_{j=1}^{p} \phi_j(v) = v(N) - v(\emptyset).$$

Axiom 2: Symmetry. If two players j and k contribute equally to every coalition, i.e. $v(S \cup \{j\}) = v(S \cup \{k\})$ for all $S \subseteq N \setminus \{j, k\}$, then $\phi_j(v) = \phi_k(v)$.

Axiom 3: Dummy player. If player j adds nothing to any coalition, i.e. $v(S \cup \{j\}) = v(S)$ for all $S \subseteq N \setminus \{j\}$, then $\phi_j(v) = 0$.

Axiom 4: Additivity (linearity). For two games v and w, the Shapley value of the combined game $v + w$ is $\phi_j(v + w) = \phi_j(v) + \phi_j(w)$ for all j.

Theorem Annexure A.2 (Shapley, 1953)**.** *There exists a unique function $\phi : \mathcal{G}^N \to \mathbb{R}^p$ satisfying Axioms 1 to 4, and it is given by*

$$\phi_j(v) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(p - |S| - 1)!}{p!} \big[v(S \cup \{j\}) - v(S)\big].$$

Proof. Existence is verified by direct calculation: substituting the formula into each axiom confirms that all four hold.

Uniqueness proceeds by induction on the number of players. For $p = 1$, efficiency forces $\phi_1(v) = v(\{1\}) - v(\emptyset)$, which is unique. For the inductive step, any game v can be decomposed as a linear combination of unanimity games u_T (where $u_T(S) = 1$ if $T \subseteq S$ and $u_T(S) = 0$ otherwise). By additivity, it suffices to determine $\phi_j(u_T)$. By the dummy axiom, $\phi_j(u_T) = 0$ for $j \notin T$. By symmetry, $\phi_j(u_T) = \phi_k(u_T)$ for all $j, k \in T$. By efficiency, $\sum_{j \in T} \phi_j(u_T) = 1$. Hence $\phi_j(u_T) = 1/|T|$ for $j \in T$, which uniquely determines ϕ for all unanimity games, and by linearity, for all games. □

Annexure A.4 Gamma GLM Log-Link Score Equations

The expected response is $\mu_i = \exp(\mathbf{x}_i^\top \boldsymbol{\beta})$. The Gamma density with shape parameter α and mean μ is

$$f(y; \alpha, \mu) = \frac{1}{\Gamma(\alpha)} \left(\frac{\alpha}{\mu}\right)^{\alpha} y^{\alpha - 1} \exp\!\left(-\frac{\alpha y}{\mu}\right), \quad y > 0.$$

The log-likelihood for observation i (up to a constant not depending on $\boldsymbol{\beta}$) is

$$\ell_i(\boldsymbol{\beta}) = -\alpha \log \mu_i - \frac{\alpha y_i}{\mu_i} = -\alpha\, \mathbf{x}_i^\top \boldsymbol{\beta} - \alpha\, y_i \exp(-\mathbf{x}_i^\top \boldsymbol{\beta}).$$

The score equation with respect to the k-th coefficient is obtained by differentiating $\ell = \sum_{i=1}^{n} \ell_i$:

$$\frac{\partial \ell}{\partial \beta_k} = \alpha \sum_{i=1}^{n} \left(\frac{y_i}{\mu_i} - 1 \right) x_{ik} = 0, \quad k = 0, 1, \ldots, p. \tag{Annexure A.4.1}$$

Since $\alpha > 0$ cancels, the maximum-likelihood estimates $\hat{\boldsymbol{\beta}}$ do not depend on the shape parameter. The Fisher information matrix entry is

$$\mathcal{I}_{kl} = \alpha \sum_{i=1}^{n} \mu_i^{-1} \, x_{ik} \, x_{il},$$

and the asymptotic covariance of $\hat{\boldsymbol{\beta}}$ is $\mathcal{I}^{-1}$.

Annexure B Data Preprocessing and Encoding

The raw dataset contains 1 000 000 rows and 12 columns. The preprocessing procedure consists of four stages: data auditing, reference-category encoding, partitioning and input standardisation.

Annexure B.1 Data Audit

The data audit confirmed no missing values in numeric columns, no exact duplicate rows, and that all categorical levels matched the expected domain. Missing values in `medical_history` and `family_medical_history` were coded as `None` and treated as a valid reference level.

Annexure B.2 Reference-Category Encoding

Each categorical variable was encoded using reference-category (dummy) coding, producing 22 numeric features from the original 11 input variables. Table 32 lists the reference category for each variable.

Variable	Reference Category
region	northeast
medical_history	None
family_medical_history	None
exercise_frequency	Rarely
occupation	Unemployed
coverage_level	Basic

Table 32: Reference categories used in dummy encoding.

Binary variables (gender, smoker) were coded as single indicator columns. Numeric variables (age, bmi, children) were passed through without transformation.

Annexure B.3 Data Partitioning

The encoded dataset was split into training (60%), validation (20%) and test (20%) partitions using random sampling with `random_state=42`.

Annexure B.4 Input Standardisation

All 22 input features were standardised to zero mean and unit variance using parameters computed from the training partition only. The response variable (charges) was similarly standardised. Binary dummy columns with zero variance in the training set were protected by setting their standard deviation to 1.0 before division.

Algorithm 1 Data preprocessing and standardisation.

Require: Raw dataset $\mathcal{D}$ with n rows and 12 columns
Ensure: Standardised feature matrices $\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{val}}, \mathbf{X}_{\text{test}}$ and response vectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}, \mathbf{y}_{\text{test}}$

Fill missing values in categorical columns with reference level
Apply reference-category encoding to obtain $n \times 22$ feature matrix
Split: $\mathcal{D}_{\text{train}}$ (60%), $\mathcal{D}_{\text{val}}$ (20%), $\mathcal{D}_{\text{test}}$ (20%) with seed 42
Compute $\boldsymbol{\mu}_X, \boldsymbol{\sigma}_X$ from $\mathcal{D}_{\text{train}}$ features only
Compute μ_y, σ_y from $\mathcal{D}_{\text{train}}$ response only
for each partition $\mathcal{P} \in \{\text{train}, \text{val}, \text{test}\}$ **do**
    $\mathbf{X}_{\mathcal{P}} \leftarrow (\mathbf{X}_{\mathcal{P}}^{\text{raw}} - \boldsymbol{\mu}_X) \,/\, \boldsymbol{\sigma}_X$
    $\mathbf{y}_{\mathcal{P}} \leftarrow (\mathbf{y}_{\mathcal{P}}^{\text{raw}} - \mu_y) \,/\, \sigma_y$
end for

Annexure B.5 Response De-normalisation

Predictions on the normalised scale are converted back to the original Rand scale via $\hat{y} = \hat{y}_{\text{norm}} \cdot \sigma_y + \mu_y$, where μ_y and σ_y are the training-set response mean and standard deviation respectively.

Annexure C DNN Architecture and Training Procedure

The FunnelDNN model uses a DNN architecture with monotonically decreasing hidden layer widths: $22 \to 256 \to 128 \to 64 \to 32 \to 16 \to 1$. All layers are bias-free, and the CELU activation function is applied after each hidden layer. The output layer is linear with no activation.

Annexure C.1 Training Procedure

Training minimises the MSE loss on normalised data using the NAdam optimiser. Validation-based early stopping monitors the dot-product R^2 on the validation set every 5 epochs, with a patience of 10 checks.

Algorithm 2 FunnelDNN training with early stopping.

Require: Standardised data $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}, \mathbf{X}_{\text{val}}, \mathbf{y}_{\text{val}})$, learning rate η, batch size B, patience P, validation interval V
Ensure: Trained model state $\boldsymbol{\theta}^*$

Initialise FunnelDNN with seed 42
$\boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}_0, \quad R^{2*} \leftarrow -\infty, \quad c \leftarrow 0$
for $t = 1, 2, \ldots, T_{\max}$ **do**
    for each mini-batch $(\mathbf{X}_b, \mathbf{y}_b)$ of size B **do**
        Compute $\mathcal{L} = \text{MSE}\big(f_{\boldsymbol{\theta}}(\mathbf{X}_b),\, \mathbf{y}_b\big)$
        Update $\boldsymbol{\theta}$ via NAdam with learning rate η
    end for
    if $t \bmod V = 0$ **then**
        Compute $R^2_{\text{val}} = r^2_{\text{dot}}\big(\mathbf{y}_{\text{val}},\, f_{\boldsymbol{\theta}}(\mathbf{X}_{\text{val}})\big)$
        if $R^2_{\text{val}} > R^{2*}$ **then**
            $R^{2*} \leftarrow R^2_{\text{val}}, \quad \boldsymbol{\theta}^* \leftarrow \boldsymbol{\theta}, \quad c \leftarrow 0$
        else
            $c \leftarrow c + 1$
        end if
        if $c \geq P$ **then**
            break ▷ Early stopping triggered
        end if
    end if
end for
return $\boldsymbol{\theta}^*$

Annexure C.2 Evaluation Metrics

Three metrics are computed on the original Rand scale after de-normalisation:

- $R^2 = 1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$, where $\mathrm{SS}_{\mathrm{res}} = \sum_i (y_i - \hat{y}_i)^2$ and $\mathrm{SS}_{\mathrm{tot}} = \sum_i (y_i - \bar{y})^2$.
- $\mathrm{RMSE} = \sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2}$.
- $\mathrm{MAE} = n^{-1} \sum_i |y_i - \hat{y}_i|$.

During training, the dot-product R^2 (squared correlation between predicted and observed values) is used on normalised data for computational speed.

Annexure C.3 Checkpoint Contents

The final trained model is saved as `chapter3_final_model.pth`, a PyTorch checkpoint containing:

- Model weight tensors (state dictionary).
- Architecture specification (layer widths, activation, bias setting).
- Selected hyperparameters (learning rate, batch size, optimiser, momentum).
- Standardisation parameters ($\boldsymbol{\mu}_X$, $\boldsymbol{\sigma}_X$, μ_y, σ_y).
- Feature names and evaluation metrics.

Annexure C.4 DNN Training Algorithm (Detailed)

The following table presents the complete DNN training and model selection procedure as implemented in Section 3.5.

Algorithm C.1: DNN training and model selection

Input: model-ready training data $\{(\mathbf{z}_i, \widetilde{y}_i)\}_{i \in \mathcal{D}_{\text{train}}}$, model-ready validation data $\{(\mathbf{z}_i, \widetilde{y}_i)\}_{i \in \mathcal{D}_{\text{val}}}$, architecture specification $(d_0, d_1, \ldots, d_L)$, activation function $g(\cdot)$, loss function $\mathcal{L}_{\text{MSE}}$, optimiser, learning rate η, momentum parameters (β_1, β_2), batch size $|\mathcal{B}|$, maximum epochs $E_{\max} = 200$, validation frequency $v = 5$, early stopping patience $P = 10$, random seed $s = 42$.

Output: fitted network weights $\widehat{\theta}$, training history, selected epoch e^*.

1. Set the random seed to s.
2. Initialise network weights $\theta^{(0)}$ using Kaiming uniform initialisation (He et al., 2015).
3. For epoch $e = 1, \ldots, E_{\max}$:
   (a) Partition $\mathcal{D}_{\text{train}}$ into random mini-batches of size $|\mathcal{B}|$.
   (b) For each mini-batch $\mathcal{B}$, compute $\mathcal{L}_{\text{MSE}}(\theta; \mathcal{B})$ and update θ using the optimiser.
   (c) Record training loss $\mathcal{L}_{\text{train}}(e)$.
   (d) If $e \bmod v = 0$: evaluate validation loss $\mathcal{L}_{\text{val}}(e)$ and dot-product R^2 on both partitions. If validation R^2 improves, checkpoint the model weights; otherwise, increment the patience counter.
   (e) If the patience counter reaches P, terminate training.
4. Select $e^* = \arg\max_{e \in \mathcal{E}_{\text{val}}} R^2_{\text{val}}(e)$.
5. Retain the network weights $\widehat{\theta} = \theta^{(e^*)}$.
6. De-normalise predictions and evaluate on the held-out test set using R^2, RMSE and MAE on the original monetary scale.
7. Save $\widehat{\theta}$, architecture specification, standardisation parameters, training history, and all evaluation outputs.

Table 33: DNN training and model selection algorithm.

Annexure C.5 GLM Fitting Algorithm

Algorithm C.2: Gamma GLM with log link

Input: model-ready training data $\{(\mathbf{z}_i, y_i)\}_{i \in \mathcal{D}_{\text{train}}}$, distribution family (gamma), link function (log).

Output: fitted coefficient vector $\widehat{\boldsymbol{\beta}}$, dispersion estimate $\widehat{\phi}$.

1. Initialise the linear predictor: $\eta_i^{(0)} = \log(\bar{y}_{\text{train}})$ for all i.
2. Repeat until convergence:
   (a) Compute fitted values: $\mu_i^{(t)} = \exp(\eta_i^{(t)})$.
   (b) Compute working weights: $w_i^{(t)} = \mu_i^{(t)\,2} / (\phi\, \mu_i^{(t)\,2}) = 1/\phi$.
   (c) Compute the adjusted dependent variable: $\tilde{y}_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)})/\mu_i^{(t)}$.
   (d) Solve the weighted least squares problem: $\boldsymbol{\beta}^{(t+1)} = (\mathbf{Z}^\top \mathbf{W}^{(t)} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{W}^{(t)} \tilde{\mathbf{y}}^{(t)}$.
   (e) Update the linear predictor: $\eta_i^{(t+1)} = \mathbf{z}_i^\top \boldsymbol{\beta}^{(t+1)}$.
3. Estimate the dispersion parameter: $\widehat{\phi} = \frac{1}{n-p-1} \sum_{i=1}^{n} \big((y_i - \widehat{\mu}_i)/\widehat{\mu}_i\big)^2$.
4. Predict on the original monetary scale: $\widehat{y}_i = \exp(\mathbf{z}_i^\top \widehat{\boldsymbol{\beta}})$.
5. Evaluate R^2, RMSE and MAE on the training, validation and test partitions.

Table 34: Gamma GLM with log link fitting algorithm.

Annexure C.6 RF Training Algorithm

Algorithm C.3: RF regression

Input: model-ready training data $\{(\mathbf{z}_i, y_i)\}_{i \in \mathcal{D}_{\text{train}}}$, number of trees $B = 500$, number of candidate predictors per split $m = \lfloor p/3 \rfloor$, minimum samples per leaf $n_{\min} = 5$, random seed $s = 42$.

Output: ensemble of fitted trees $\{T_1, \ldots, T_B\}$.

1. Set the random seed to s.
2. For $b = 1, \ldots, B$:
   (a) Draw a bootstrap sample $\mathcal{D}_b^*$ of size n_{train} with replacement from $\mathcal{D}_{\text{train}}$.
   (b) Grow an unpruned regression tree T_b on $\mathcal{D}_b^*$:
      (i) At each node, select a random subset $\mathcal{S} \subset \{1, \ldots, p\}$ with $|\mathcal{S}| = m$.
      (ii) Choose the split variable j^* and split point s^* from $\mathcal{S}$ that minimise the weighted residual sum of squares.
      (iii) Split the node into two child nodes and repeat until each terminal node contains at least $n_{\min}$ observations.
   (c) Assign the sample mean of the response values in each terminal node as the node prediction.
3. For a new input $\mathbf{z}$, predict $\hat{f}_{\text{RF}}(\mathbf{z}) = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{z})$.
4. Evaluate R^2, RMSE and MAE on the training, validation and test partitions on the original monetary scale.

Table 35: RF regression training algorithm.

Annexure C.7 XGBoost Training Algorithm

Algorithm C.4: XGBoost gradient-boosted tree regression

Input: model-ready training data $\{(\mathbf{z}_i, y_i)\}_{i \in \mathcal{D}_{\text{train}}}$, model-ready validation data $\{(\mathbf{z}_i, y_i)\}_{i \in \mathcal{D}_{\text{val}}}$, maximum boosting rounds $M = 1\,000$, learning rate $\eta = 0.05$, maximum tree depth $d_{\max} = 8$, $\mathcal{L}_2$ regularisation λ, minimum split gain γ, early stopping patience $P = 50$ rounds, random seed $s = 42$.

Output: fitted additive ensemble $\hat{f}_{M^*}(\mathbf{z})$, selected round M^*.

1. Set the random seed to s.
2. Initialise the prediction: $\hat{f}_0(\mathbf{z}_i) = \bar{y}_{\text{train}}$ for all i.
3. For boosting round $m = 1, \ldots, M$:
   (a) Compute gradient statistics g_i and h_i for each training observation.
   (b) Grow a regression tree h_m of maximum depth $d_{\max}$:
      (i) At each node, evaluate all candidate split points across all p predictors.
      (ii) Select the split (j^*, s^*) that maximises the gain.
      (iii) Split only if Gain > 0; otherwise the node becomes a leaf.
      (iv) Assign optimal leaf weights $w_j^* = -G_j/(H_j + \lambda)$.
   (c) Update predictions: $\hat{f}_m(\mathbf{z}_i) = \hat{f}_{m-1}(\mathbf{z}_i) + \eta\, h_m(\mathbf{z}_i)$.
   (d) Evaluate validation loss $\mathcal{L}_{\text{val}}(m)$. If the validation loss has not improved for P consecutive rounds, terminate training.
4. Select the round $M^* = \arg\min_{m \in \{1,\ldots,M\}} \mathcal{L}_{\text{val}}(m)$ and retain the ensemble up to that round.
5. Evaluate R^2, RMSE and MAE on the training, validation and test partitions on the original monetary scale.

Table 36: XGBoost gradient-boosted tree regression training algorithm.

Annexure D MC Simulation Procedure

The parametric MC simulation generates $B = 100\,000$ synthetic policyholder profiles from marginal distributions estimated from the training partition only, in accordance with the leakage-prevention protocol of Section 3.7. Each profile is encoded, standardised and scored through the trained FunnelDNN to produce a model-implied predicted-cost distribution.

Annexure D.1 Marginal Distributions

Table 37 summarises the marginal distribution assumed for each input variable.

Variable	Distribution	Parameters
age	Discrete Uniform	$\{18, 19, \ldots, 65\}$
bmi	Continuous Uniform	$[18.00,\ 50.00]$
children	Discrete Uniform	$\{0, 1, \ldots, 5\}$
gender	Bernoulli	$\hat{p}_{\text{male}}$ from data
smoker	Bernoulli	$\hat{p}_{\text{smoker}}$ from data
region	Multinomial	4 levels, estimated proportions
medical_history	Multinomial	4 levels, estimated proportions
family_medical_history	Multinomial	4 levels, estimated proportions
exercise_frequency	Multinomial	4 levels, estimated proportions
occupation	Multinomial	4 levels, estimated proportions
coverage_level	Multinomial	3 levels, estimated proportions

Table 37: Marginal distributions used for MC profile generation.

Annexure D.2 Profile Generation and Scoring

The simulation procedure itself is specified in Section 3.7 and is not restated here. Algorithm 3 records only the concrete run configuration used in this study: the random seed, the sample size and the encoded input dimension.

Annexure D.3 Convergence Diagnostics

MC convergence was assessed using the pilot procedure described in Section 3.7, in which the running mean, standard deviation, selected quantiles and MCSE are recomputed at increasing pilot sample sizes; the pilot grid is listed in that section and is not repeated here. Stabilisation of the mean and tail quantiles confirmed that $B = 100\,000$ provides adequate precision.

Algorithm 3 Parametric MC simulation.

Require: Fitted marginals, trained DNN $f_{\boldsymbol{\theta}^*}$, standardisation parameters, sample size $B = 100\,000$

Ensure: Predicted-cost vector $\hat{\mathbf{y}}_{\mathrm{MC}} \in \mathbb{R}^B$

1: Initialise random number generator with seed 42
2: **for** $i = 1, \ldots, B$ **do**
3: Sample each variable independently from its marginal distribution
4: Apply reference-category encoding to obtain $\mathbf{x}_i \in \mathbb{R}^{22}$
5: **end for**
6: Standardise: $\mathbf{X}_{\mathrm{MC}} \leftarrow (\mathbf{X}_{\mathrm{MC}}^{\mathrm{raw}} - \boldsymbol{\mu}_X)/\boldsymbol{\sigma}_X$
7: Forward pass: $\hat{\mathbf{y}}_{\mathrm{norm}} \leftarrow f_{\boldsymbol{\theta}^*}(\mathbf{X}_{\mathrm{MC}})$
8: De-normalise: $\hat{\mathbf{y}}_{\mathrm{MC}} \leftarrow \hat{\mathbf{y}}_{\mathrm{norm}} \cdot \sigma_y + \mu_y$

Annexure D.4 Residual-Augmented Sensitivity

Sensitivity of the predicted-cost distribution to model uncertainty was assessed by augmenting point predictions with resampled centred training residuals:

$$\hat{y}_i^{\mathrm{aug}} = \hat{y}_i + \varepsilon_i,$$

where ε_i is drawn with replacement from the centred residual vector. Fixed perturbation bands of $\pm 0.5\,\hat{\sigma}_\varepsilon$ were also computed.

Annexure D.5 Encoding for MC Profiles

The same reference-category encoding scheme described in Annexure B.2 was applied to the MC profiles, producing a $B \times 22$ feature matrix that matches the trained model's input specification.

Annexure D.6 DNN Scoring

Each encoded profile is standardised using the training-set parameters ($\boldsymbol{\mu}_X$, $\boldsymbol{\sigma}_X$), passed through the trained DNN in a single forward pass (with gradients disabled), and de-normalised to the original Rand scale using the training-set response parameters (μ_y, σ_y).

Annexure E GLM Benchmark Methodology

The Gamma GLM with log-link serves as the actuarial benchmark against which the DNN is compared. An RF and an XGBoost gradient-boosted tree regressor are fitted as additional benchmarks to span the model-class spectrum from linear to piecewise-constant to smooth nonlinear (see Section 3.5.6). The GLM was fitted using `statsmodels` with an intercept term and the same 22 reference-category encoded features used by the DNN and tree-based models.

Annexure E.1 Fitting Procedure

Algorithm 4 GLM benchmark fitting and comparison.

Require: Raw feature matrices $\mathbf{X}_{\text{train}}, \mathbf{X}_{\text{val}}, \mathbf{X}_{\text{test}}$ (unstandardised), response vectors $\mathbf{y}_{\text{train}}, \mathbf{y}_{\text{val}}, \mathbf{y}_{\text{test}}$

Ensure: Comparison metrics (R^2, RMSE, MAE) for GLM, RF, XGBoost and DNN on all partitions

Prepend intercept column to each feature matrix
Fit Gamma GLM with log-link on $(\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}})$ via IRLS
Extract coefficient table: $\hat{\boldsymbol{\beta}}$, standard errors, z-values, p-values
for each partition $\mathcal{P} \in \{\text{train}, \text{val}, \text{test}\}$ **do**
    $\hat{\mathbf{y}}_{\text{GLM}} \leftarrow \exp(\mathbf{X}_{\mathcal{P}}\hat{\boldsymbol{\beta}})$
    Compute R^2, RMSE, MAE for GLM predictions
    Compute R^2, RMSE, MAE for DNN predictions (from checkpoint)
end for

Annexure E.2 Coefficient Extraction

The fitted GLM produces a coefficient vector $\hat{\boldsymbol{\beta}} \in \mathbb{R}^{23}$ (22 features plus intercept). For each coefficient, the exponentiated value $\exp(\hat{\beta}_k)$ gives the multiplicative effect on the expected cost. Statistical significance was assessed at the $\alpha = 0.05$ level using Wald z-tests.

Annexure E.3 Model Comparison

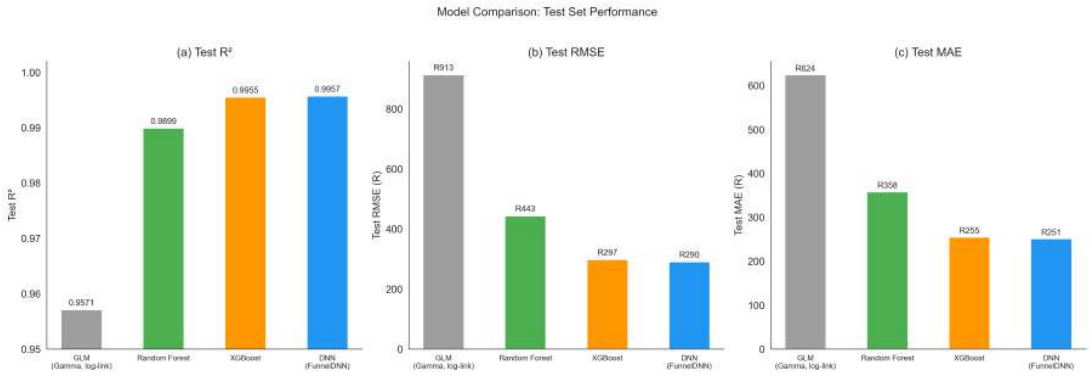


Figure 40: Test-set performance comparison across four model classes.

Annexure F Shapley-Value Computation Procedure

Approximate Shapley values were computed using the SHAP library's GradientExplainer, which uses integrated gradients as an approximation to exact Shapley values for differentiable models. All SHAP computations were performed on CPU, as GradientExplainer runs faster on CPU than on MPS for this model size.

Annexure F.1 Computation Procedure

Algorithm 5 Batch Shapley-value computation with dummy-to-raw aggregation.

Require: Trained DNN $f_{\boldsymbol{\theta}^*}$ (CPU copy), standardised training data $\mathbf{X}_{\text{train}}$, standardised MC profiles $\mathbf{X}_{\text{MC}}$, batch size $B_s = 2\,000$
Ensure: Raw-variable Shapley matrix $\boldsymbol{\Phi} \in \mathbb{R}^{B \times 11}$
Sample 200 background rows from $\mathbf{X}_{\text{train}}$ (seed 42)
Initialise GradientExplainer with $f_{\boldsymbol{\theta}^*}$ and background data
for each batch of B_s foreground profiles **do**
    Compute Shapley values $\boldsymbol{\phi} \in \mathbb{R}^{B_s \times 22}$
end for
Concatenate batches to obtain $\boldsymbol{\Phi}_{22} \in \mathbb{R}^{B \times 22}$
for each raw variable $j = 1, \ldots, 11$ **do**
    $\boldsymbol{\Phi}_{:,j} \leftarrow \sum_{k \in \mathcal{C}_j} \boldsymbol{\Phi}_{22_{:,k}}$ ▷ Sum over dummy columns $\mathcal{C}_j$
end for

Annexure F.2 Dummy-to-Raw Variable Mapping

Table 38 shows the mapping from the 22 encoded dummy columns to the 11 original input variables used for Shapley aggregation.

Raw Variable	Encoded Columns	Count
age	1	1
gender	2	1
bmi	3	1
children	4	1
smoker	5	1
region	6, 7, 8	3
medical_history	9, 10, 11	3
family_medical_history	12, 13, 14	3
exercise_frequency	15, 16, 17	3
occupation	18, 19, 20	3
coverage_level	21, 22	2

Table 38: Dummy-to-raw variable aggregation mapping.

Annexure F.3 Efficiency Verification

The Shapley efficiency axiom $\left(\sum_{j=1}^{p} \phi_j = f(\mathbf{x}) - \phi_0\right)$ was verified for all B profiles. The maximum absolute reconstruction error was below 0.5 normalised units, confirming that the gradient-based approximation satisfies the efficiency property to adequate precision. It should be noted, however, that 0.5 normalised units corresponds to half a standard deviation of the standardised response, so this check bounds gross approximation failure rather than certifying high accuracy; the relative reconstruction error on the monetary scale is the more informative measure of attribution precision.

Annexure F.4 Global Importance

Global feature importance was computed as the mean absolute Shapley value across all B MC profiles: $|\bar{\phi}|_j = B^{-1} \sum_{i=1}^{B} |\phi_{ij}|$ for each raw variable j. Variables were ranked by this measure to identify the most influential predictors of insurance cost.

Annexure F.5 High-Cost Conditional Shapley Analysis

High-cost conditional Shapley analysis was performed by subsetting the Shapley matrix to profiles with predicted costs exceeding the τ_{90}, τ_{95} and τ_{99} quantiles of the MC distribution. Global and conditional importance rankings were compared to identify features whose influence shifts in the tail of the cost distribution.

Annexure G Additional Diagnostic Figures

Annexure G.1 Data Leakage Check Results

1. Split-before-preprocessing: PASS.
2. Training-only standardisation: PASS.
3. No target-derived predictor: PASS.
4. Duplicate-row audit: PASS.
5. Cross-partition leakage: PASS.
6. Benchmark-dataset characterisation: PASS.

Annexure G.2 Partition Distribution Comparisons

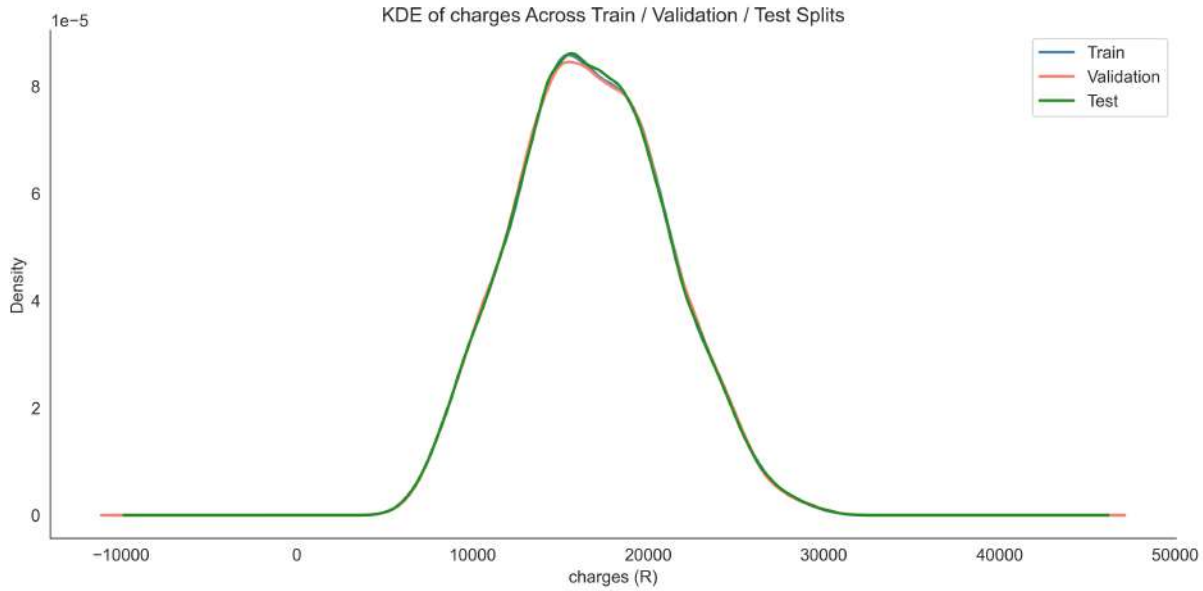


Figure 41: Charges KDE across partitions.

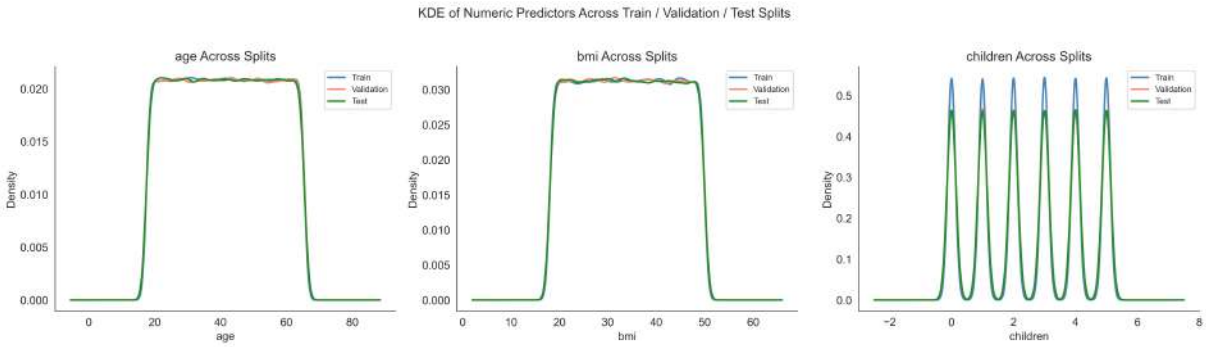


Figure 42: Numeric feature KDE across partitions.

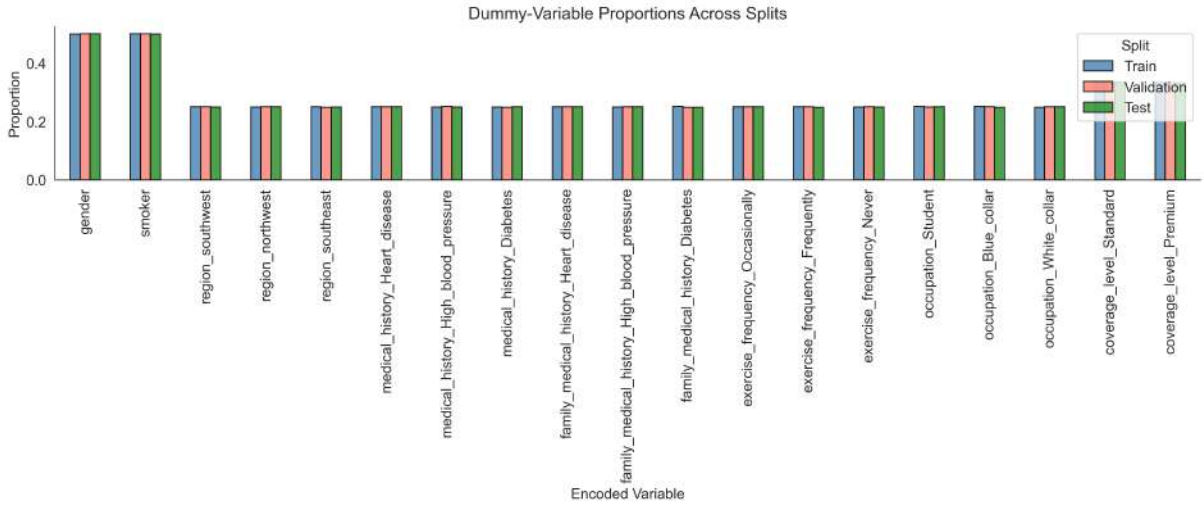


Figure 43: Dummy variable proportions across partitions.

Annexure G.3 MC Joint-Distribution Validation

The figures in this subsection provide detailed bivariate and multivariate diagnostics comparing the joint distributional structure of the training data with that of the MC sample. These diagnostics support the validation conclusions summarised in Section 4.6.1.

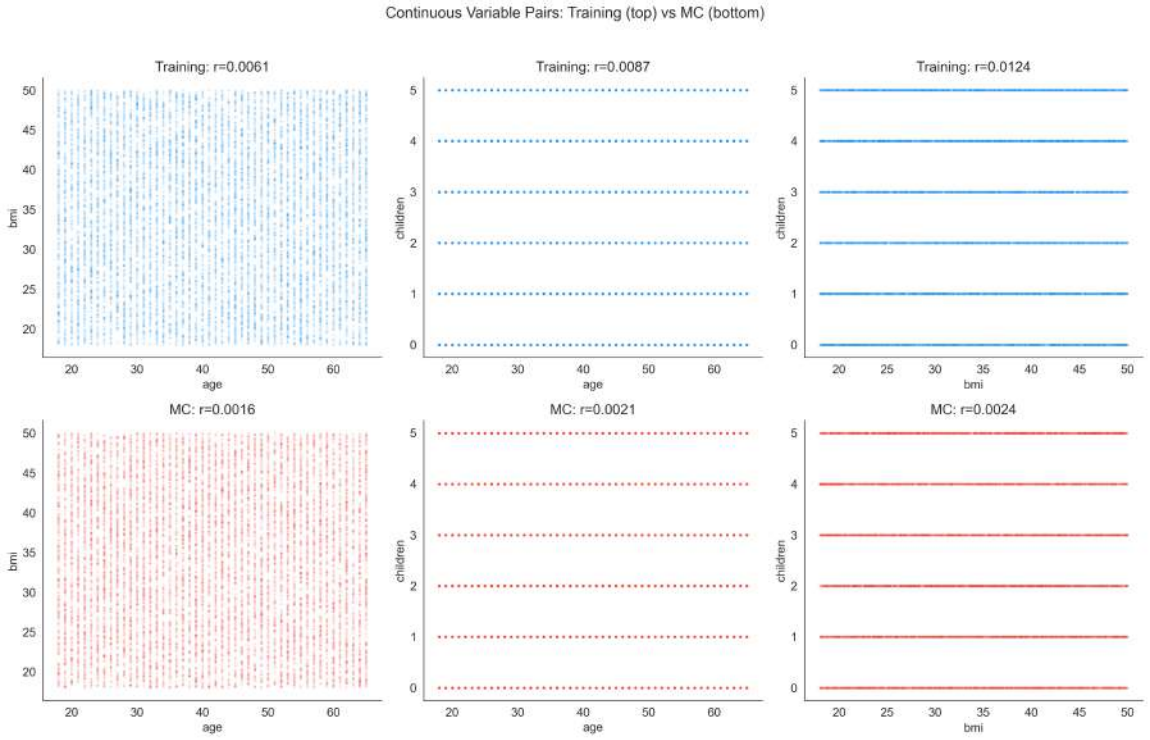


Figure 44: Continuous variable pairs: training versus MC.

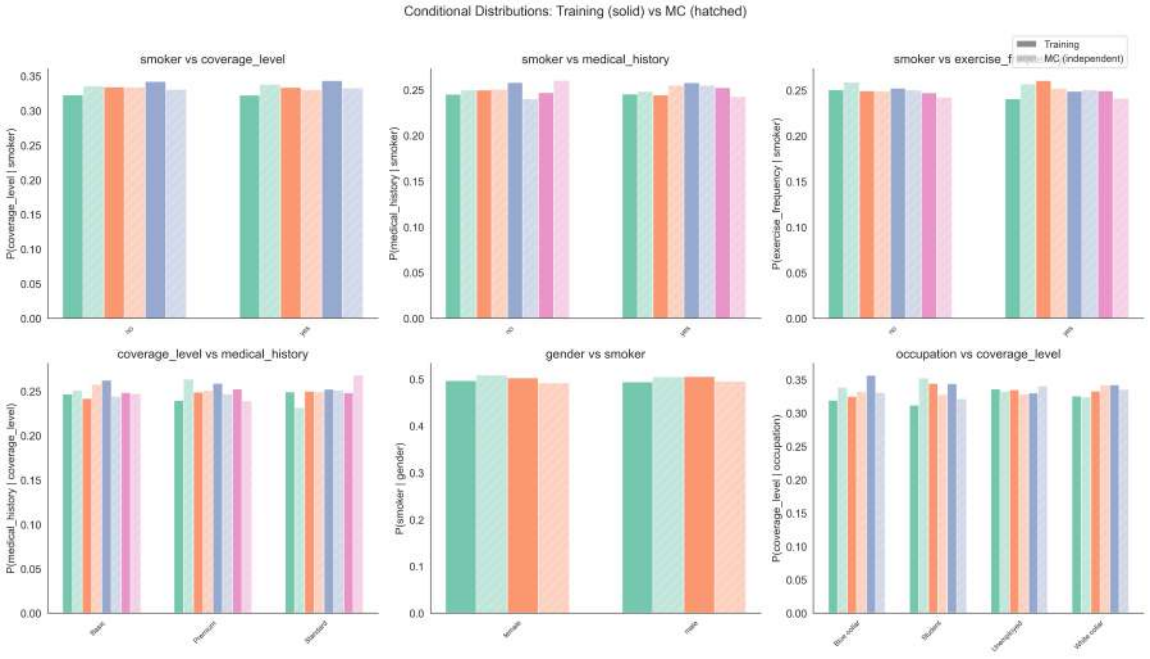


Figure 45: Conditional distributions of categorical variable pairs: training versus MC.

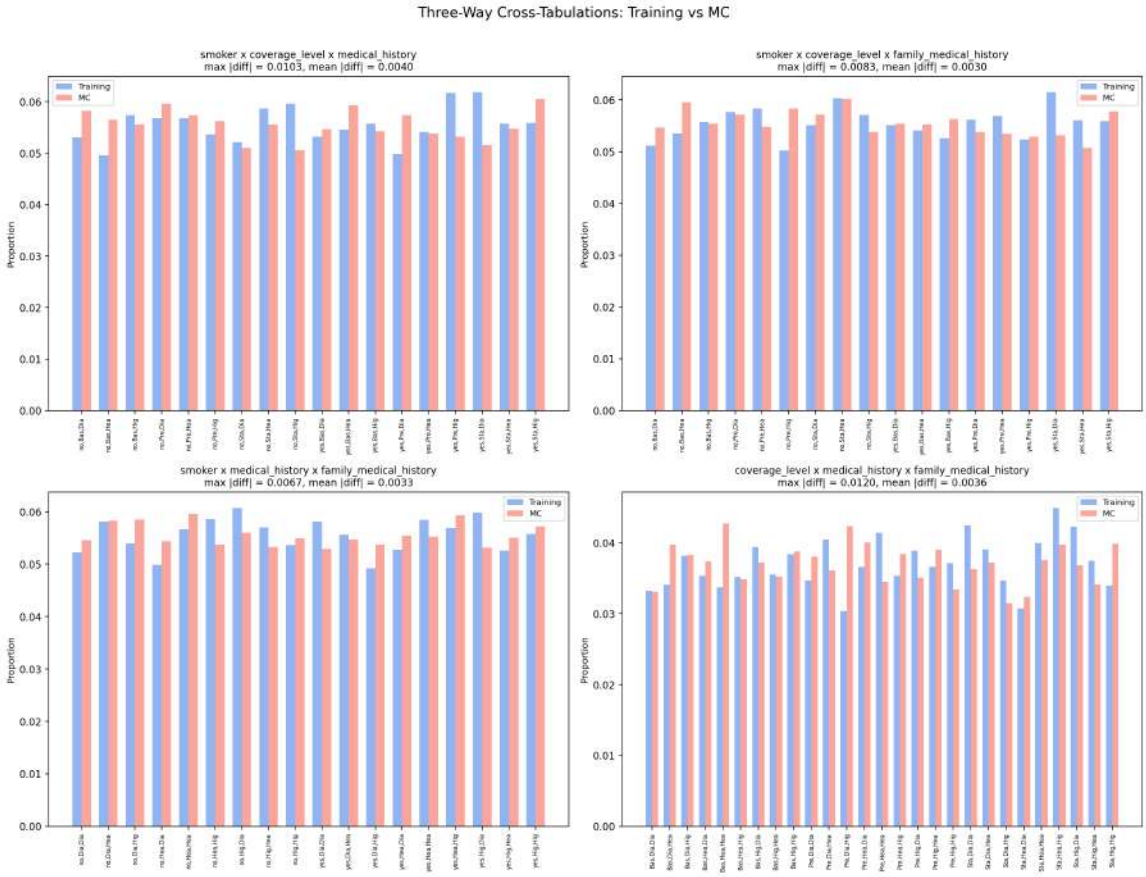


Figure 46: Three-way cross-tabulations of key categorical variables: training versus MC.

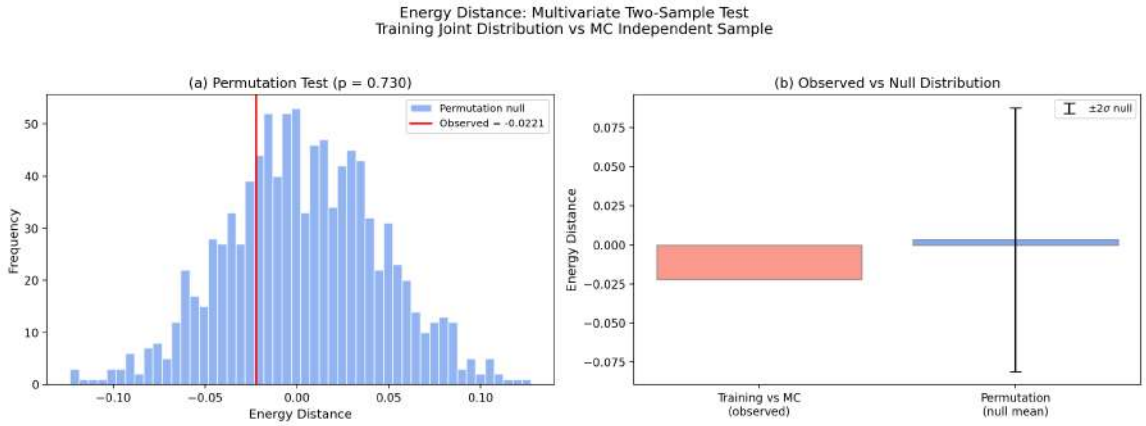


Figure 47: Energy distance permutation test: null distribution and observed statistic.

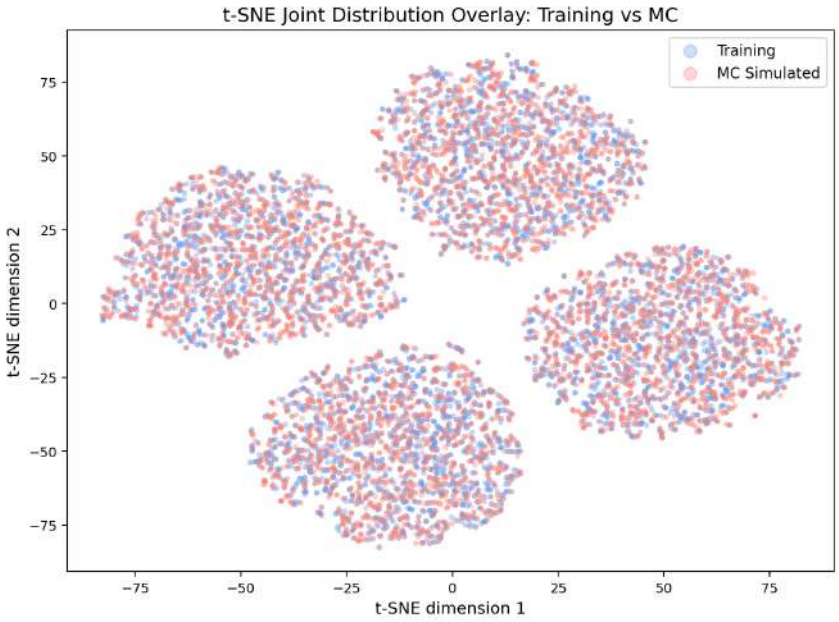


Figure 48: t-SNE projection of training versus MC joint distributions.

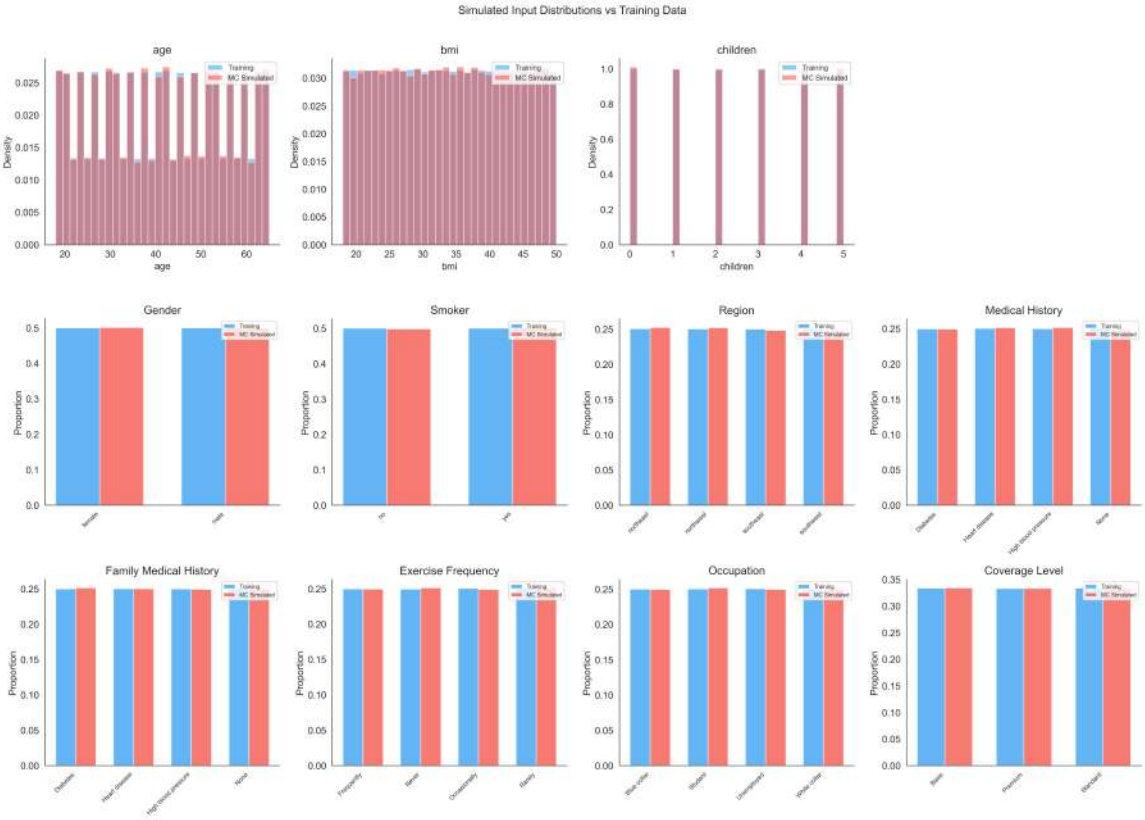


Figure 49: Marginal distributions of all 11 raw predictor variables: training versus MC.

Annexure G.4 Residual Diagnostics for Augmented Simulation

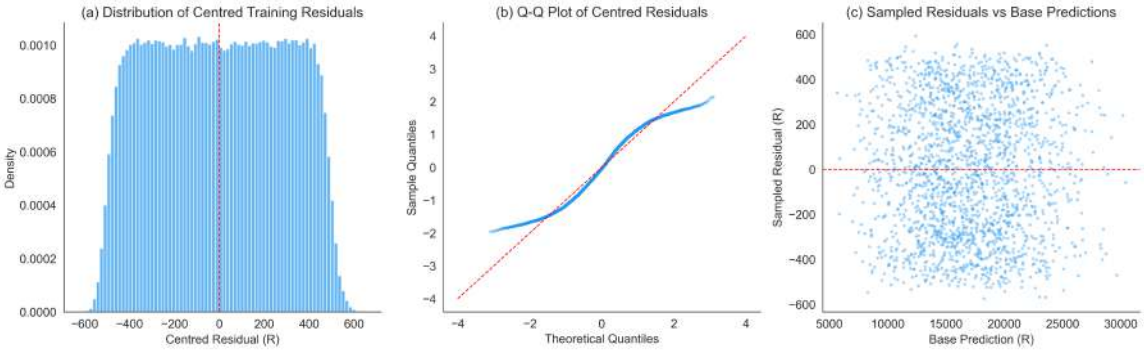


Figure 50: Centred training residual diagnostics for the augmented simulation.

Annexure G.5 SHAP Feature-Level Beeswarm

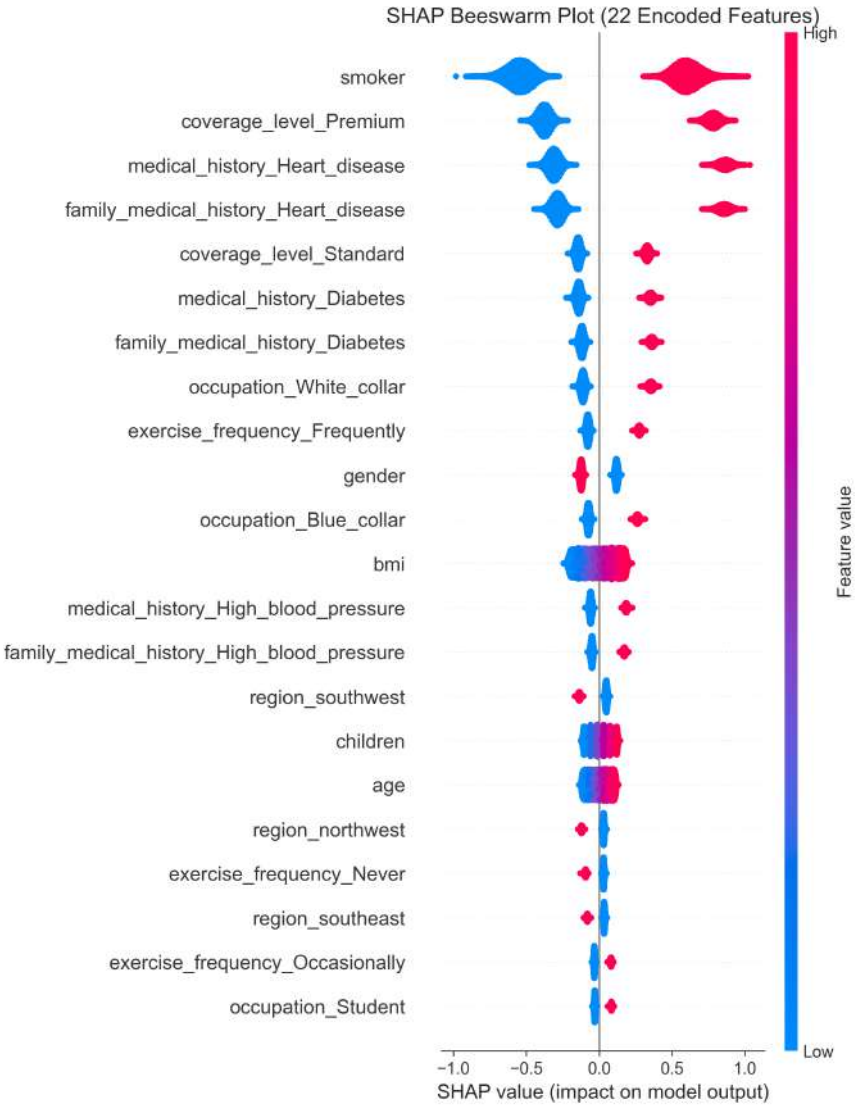


Figure 51: SHAP beeswarm plot for all 22 encoded features.

Annexure G.6 Training Versus MC Attribution Invariance

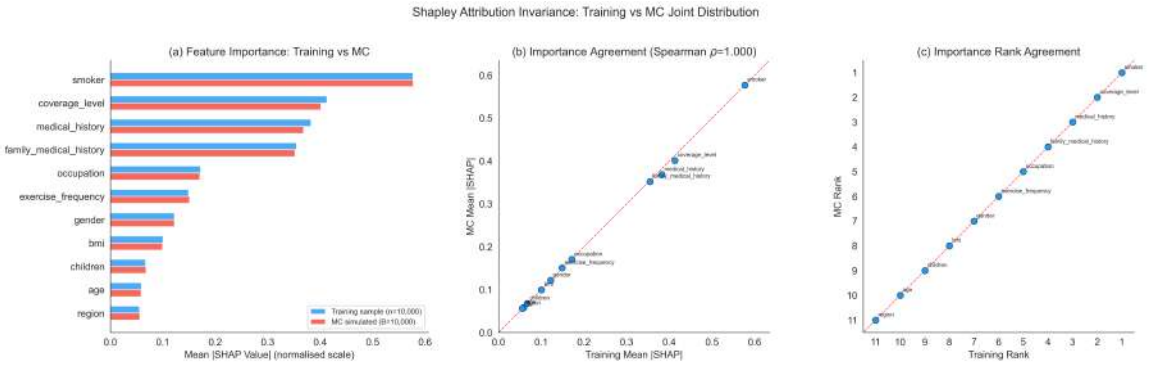


Figure 52: Shapley attribution invariance: training data versus MC profiles.

Annexure G.7 Representative High-Cost Waterfall Decomposition

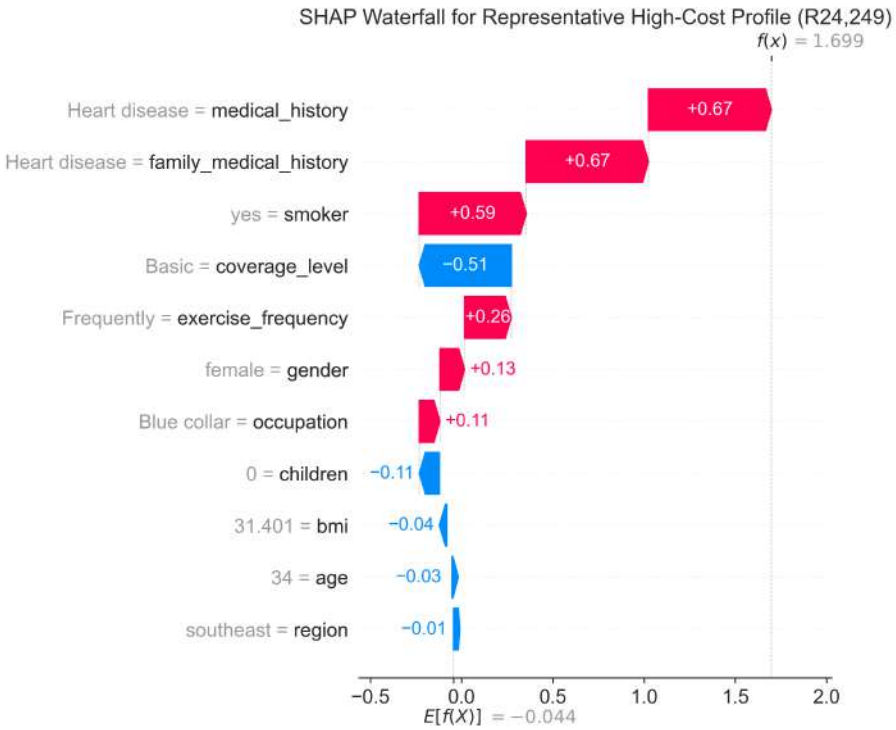


Figure 53: SHAP waterfall plot for a representative high-cost profile at τ_{95}.

Annexure H Full Hyperparameter Grid Results

Annexure H.1 Five-Phase Sequential Comparison

Hyperparameters were selected through a five-phase sequential comparison procedure. In each phase, one hyperparameter was varied while all others were held at their current best values. The phases proceeded in the following order:

1. Activation function (CELU, GELU, Tanh), with fixed lr $= 0.001$, batch size $= 256$, Adam.
2. Learning rate $(0.01, 0.001, 0.0005, 0.0001)$, with fixed best activation, batch size $= 256$, Adam.
3. Batch size $(64, 128, 256, 512)$, with fixed best activation and learning rate, Adam.
4. Optimiser (Adam vs NAdam), with all previous best values fixed.
5. Momentum β_1 $(0.90, 0.95, 0.99)$ with $\beta_2 = 0.999$ fixed, using the best optimiser.

Each configuration was trained with early stopping (patience of 10 validation checks at 5-epoch intervals) up to a maximum of 200 epochs. Table 39 reports the full results.

Phase	**Configuration**	**Best Epoch**	**Val R^2**	**Time (s)**
Phase 1: Activation	CELU	100	0.995682	556.7
	GELU	25	0.995636	313.8
	Tanh	100	0.995532	565.9
Phase 2: Learning Rate	0.01	10	0.995453	222.9
	0.001	100	0.995682	558.1
	0.0005	100	0.995692	563.8
	0.0001	100	0.995686	556.0
Phase 3: Batch Size	64	75	0.995678	1441.8
	128	30	0.995669	491.9
	256	100	0.995692	580.3
	512	160	0.995685	487.3
Phase 4: Optimiser	Adam	100	0.995692	562.7
	NAdam	85	0.995693	582.5
Phase 5: Momentum	$\beta_1 = 0.90$	85	0.995693	586.6
	$\beta_1 = 0.95$	40	0.995690	427.8
	$\beta_1 = 0.99$	95	0.995685	621.8

Table 39: Full five-phase hyperparameter comparison results.

Annexure H.2 Phase Comparison Figures

The following figures present the validation MSE loss and validation R^2 trajectories for each phase of the sequential hyperparameter comparison.

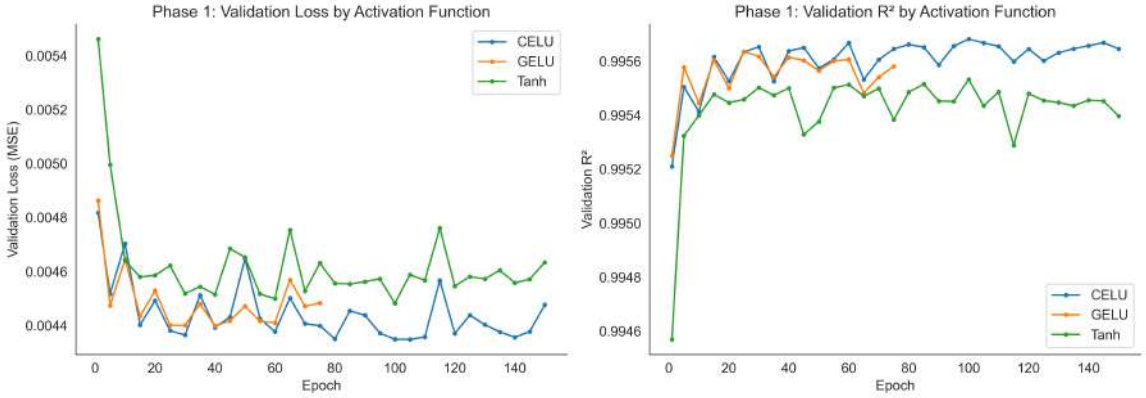


Figure 54: Phase 1: validation MSE loss (left) and validation R^2 (right) by activation function over training epochs.

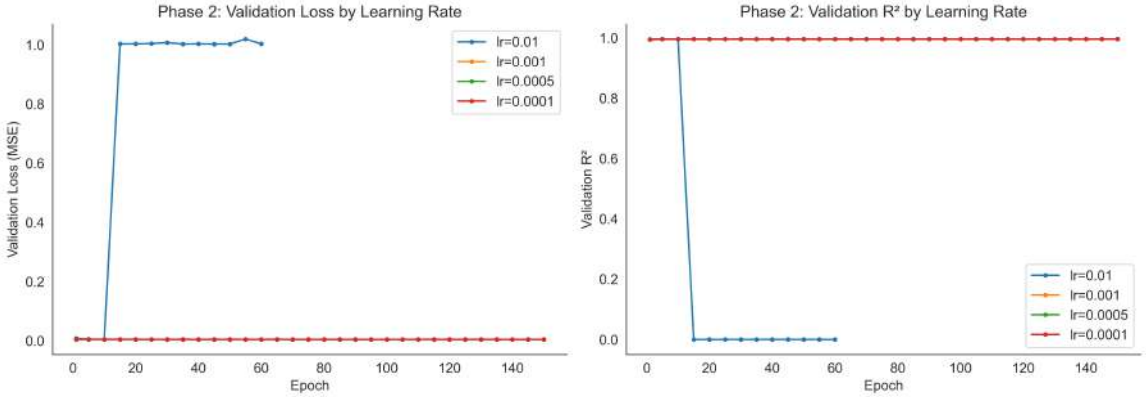


Figure 55: Phase 2: validation MSE loss (left) and validation R^2 (right) by learning rate.

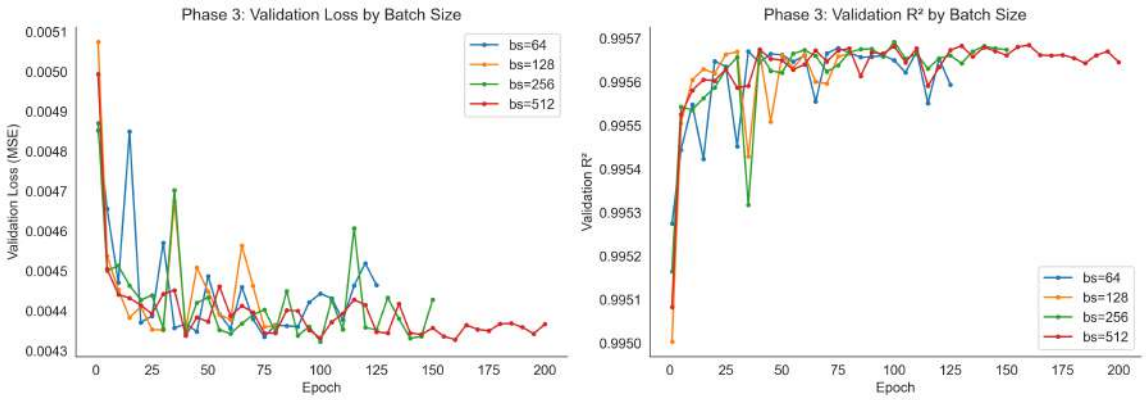


Figure 56: Phase 3: validation MSE loss (left) and validation R^2 (right) by batch size.

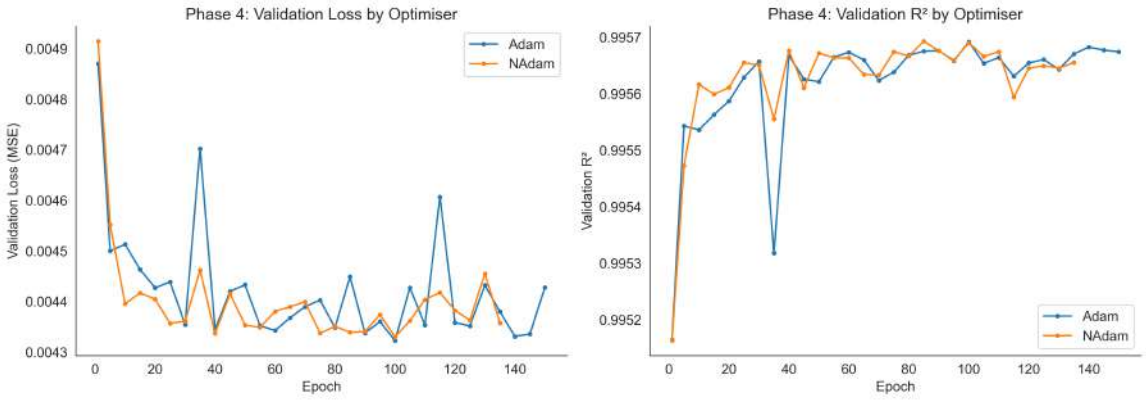


Figure 57: Phase 4: validation MSE loss (left) and validation R^2 (right) by optimiser.

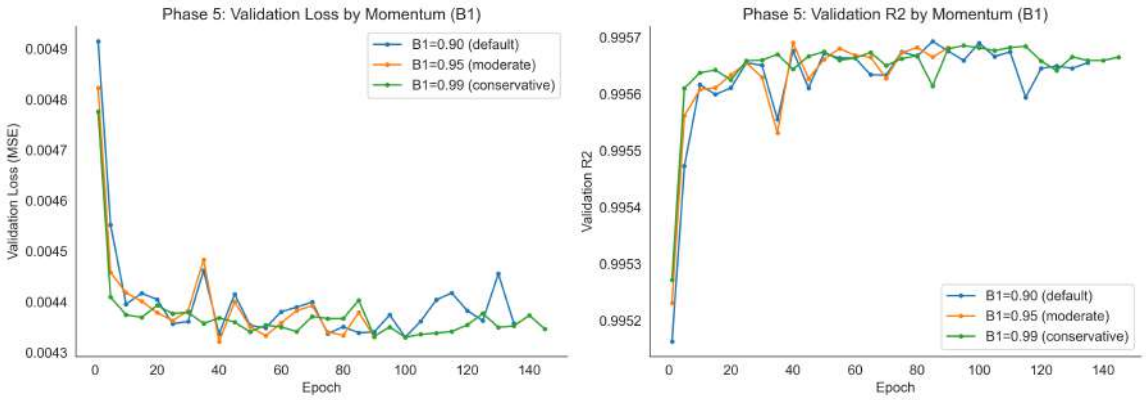


Figure 58: Phase 5: validation MSE loss (left) and validation R^2 (right) by first momentum coefficient β_1.

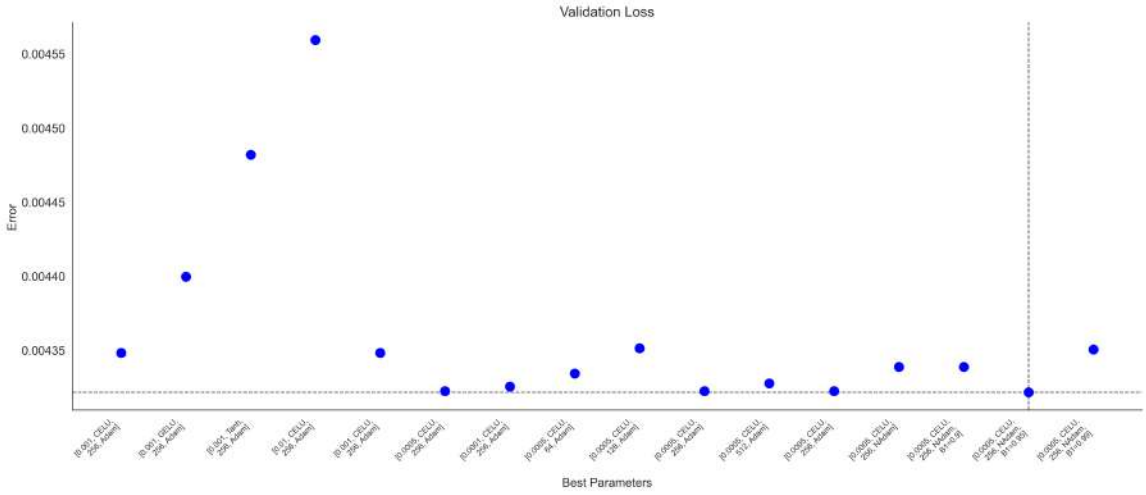


Figure 59: Best validation MSE loss across Phases 1 to 5.

Annexure H.3 Bayesian Optimisation: Gaussian Process Surrogate

A GP surrogate with a Matérn $\nu = 5/2$ kernel was fitted to the observed hyperparameter evaluations to visualise the validation-error landscape. The GP was fitted using scikit-learn's `GaussianProcessRegressor` with 15 random restarts, seed 42, and a composite kernel consisting of a constant kernel multiplied by the Matérn kernel plus a white noise kernel. Log-scaled axes were used for the learning rate dimension. The fitted surface and its uncertainty were evaluated on a 100×100 grid with 15% margin beyond the observed range.

Annexure H.4 Consolidated Best Hyperparameter Summary

Hyperparameter	**Selected Value**
Architecture	$22 \to 256 \to 128 \to 64 \to 32 \to 16 \to 1$
Activation	CELU
Learning rate	0.0005
Batch size	256
Optimiser	NAdam
β_1	0.90
β_2	0.999
Bias	False (all layers)
Max epochs	200
Early stopping	10 checks (every 5 epochs)
Loss function	MSE (on normalised scale)

Table 40: Best hyperparameter configuration selected for the final DNN model.

Annexure I Software and Environment Details

Annexure I.1 Python Distribution and Version

Python 3.11 via the Anaconda distribution. Jupyter Notebook was used for interactive development.

Annexure I.2 Core Package Versions

Package	Version	Purpose
PyTorch	2.x	DNN architecture, training and inference
SHAP	0.43+	Shapley value computation (GradientExplainer)
statsmodels	0.14+	GLM (Gamma, log-link) fitting
NumPy	1.24+	Numerical computation
pandas	2.x	Data manipulation and I/O
SciPy	1.11+	Statistical tests, distributions
scikit-learn	1.3+	RF, GP surrogate, train/test splitting, metrics
matplotlib	3.8+	Static figure generation
seaborn	0.13+	Statistical visualisation
scikit-optimize	0.10+	Bayesian Optimisation (`gp_minimize`)
XGBoost	3.x	Gradient-boosted tree benchmark
Plotly	5.x	Interactive 3D surface plots

Table 41: Python package versions used in the dissertation.

Annexure I.3 Hardware Platform

- Processor: Apple Silicon (M-series)
- Primary compute device: MPS via PyTorch
- SHAP compute device: CPU (faster than MPS for GradientExplainer)
- Fallback: CPU (used when MPS is unavailable or for SHAP computations)

The PyTorch device was selected automatically: MPS if available on Apple Silicon, CUDA if available on NVIDIA hardware, and CPU as fallback.

Annexure I.4 Reproducibility

Fixed random seed of 42:

- PyTorch weight initialisation (`torch.manual_seed(42)`)
- NumPy random operations (`np.random.seed(42)`)
- NumPy Generator objects (`np.random.default_rng(42)`)
- scikit-learn train/test split (`random_state=42`)
- RF fitting (`random_state=42`)
- XGBoost fitting (`seed=42`)
- GP surrogate fitting (`random_state=42`)

Annexure I.5 Data Files

- `insurance_dataset.csv`: 1 000 000 rows, 12 columns (age, gender, bmi, children, smoker, region, medical_history, family_medical_history, exercise_frequency, occupation, coverage_level, charges). Benchmark dataset sourced from Kaggle (Kaggle, 2023).
- `chapter3_final_model.pth`: PyTorch checkpoint containing the trained FunnelDNN model state, architecture specification, hyperparameters, standardisation parameters, feature names and evaluation metrics.
- `model_ready_charges_train_20260517_115748.csv`: reference-category encoded training partition (60% of the dataset, 22 features + charges).

Annexure I.6 Code Availability

The Python code used for data preprocessing, model training, MC simulation, GLM and tree-based benchmarking, Shapley-value computation and diagnostic figure generation is available as supplementary material. The code is organised into separate Jupyter notebooks corresponding to each stage of the analysis:

- Data preprocessing and reference-category encoding (Annexure B).
- DNN architecture definition and training (Annexure C).
- Parametric MC simulation (Annexure D).
- GLM benchmark fitting and comparison (Annexure E).
- Tree-based benchmark fitting: RF and XGBoost (Section 3.5.6).

- Shapley-value computation and conditional analysis (Annexure F).

Package versions and computing environment details are reported in Annexure I.2 to Annexure I.4 above. Full source code is available from the author on request.

Annexure J Multi-Seed Robustness Supplementary Figures

This annexure presents supplementary diagnostic figures for the multi-seed robustness analyses discussed in Chapter 4. The figures below provide detailed visual evidence supporting the cross-seed stability claims summarised in the main text.

Annexure J.1 DNN Training Convergence Across Seeds

Figures 60 to 62 display the training convergence behaviour of the FunnelDNN across five random seeds (seeds 42, 123, 456, 789 and 1024).

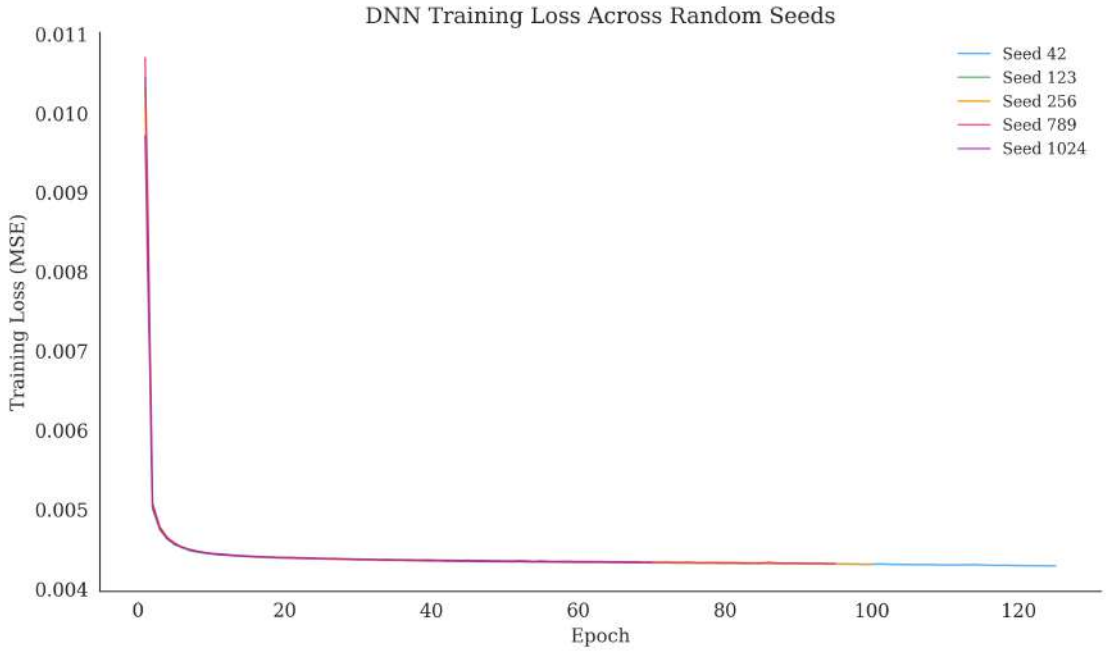


Figure 60: Training loss trajectories across five random seeds.

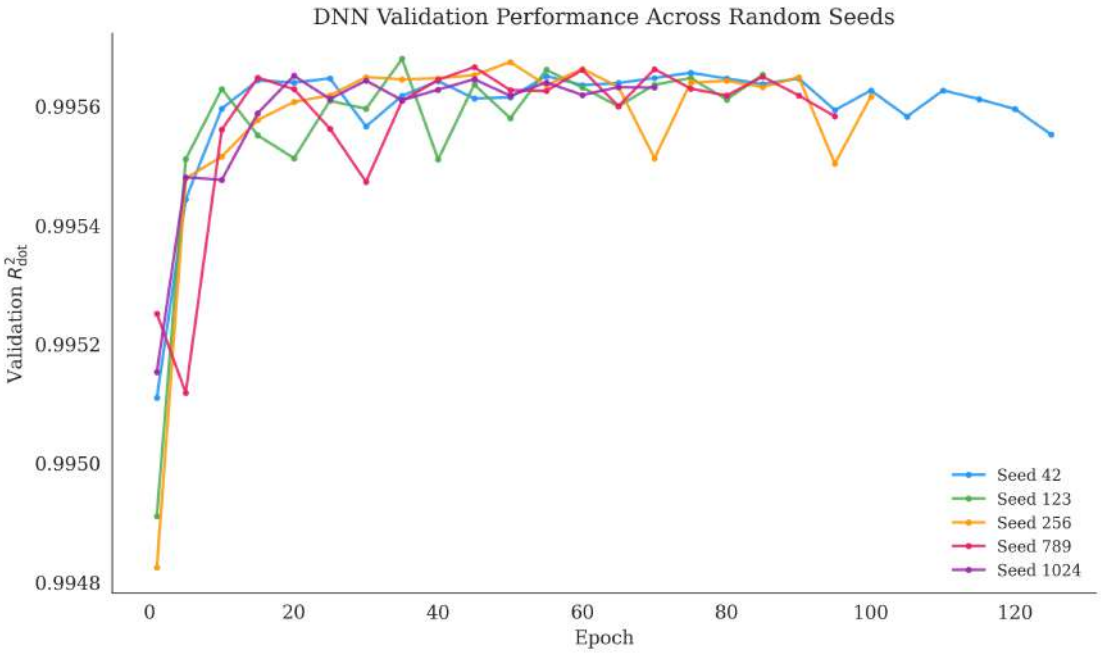


Figure 61: Validation loss and validation R^2 curves across five random seeds.

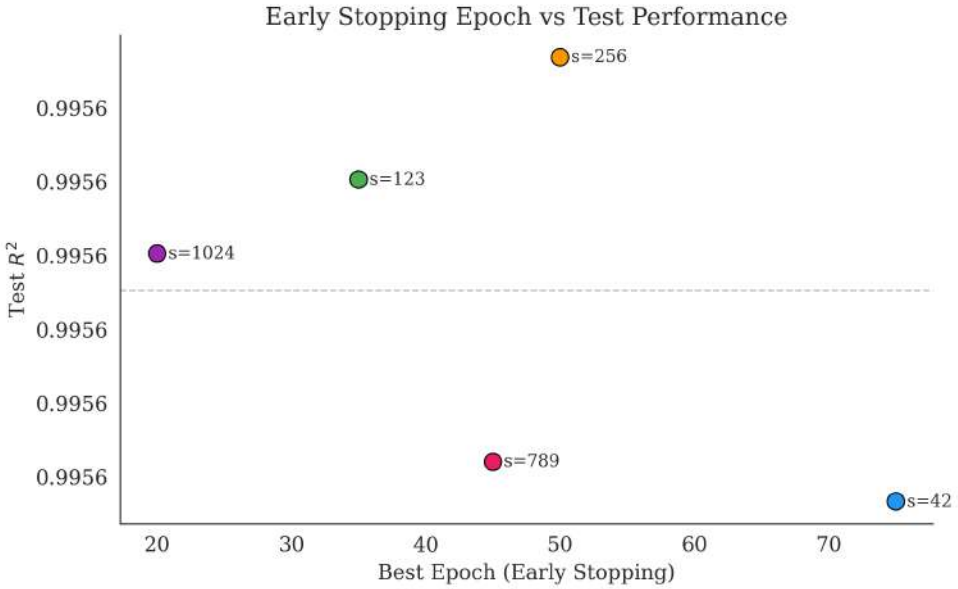


Figure 62: Epoch versus validation R^2 across five random seeds.

Annexure J.2 Cross-Seed MC Diagnostics

Figures 63 to 65 present the cross-seed MC simulation diagnostics.

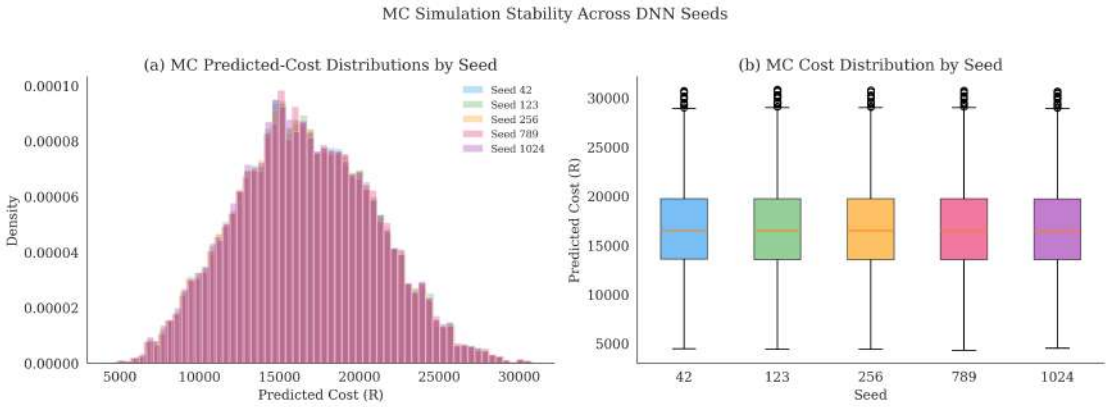


Figure 63: MC predicted-cost distributions across five DNN seeds.

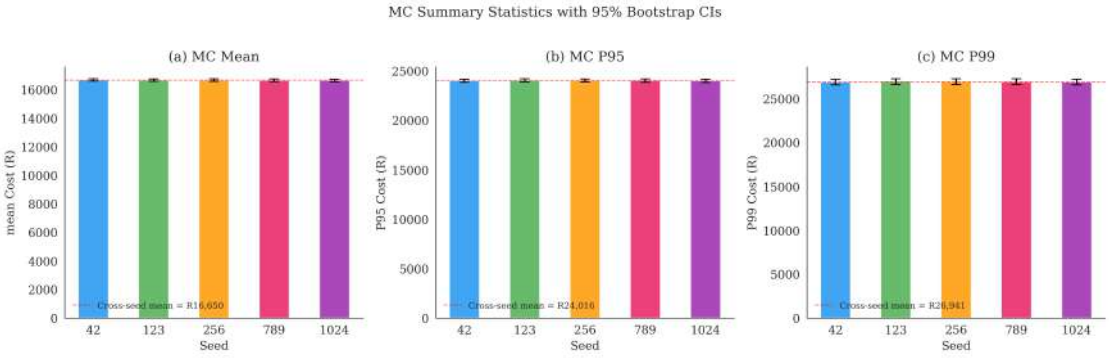


Figure 64: Bootstrap 95% confidence intervals for MC mean predicted cost across five DNN seeds.

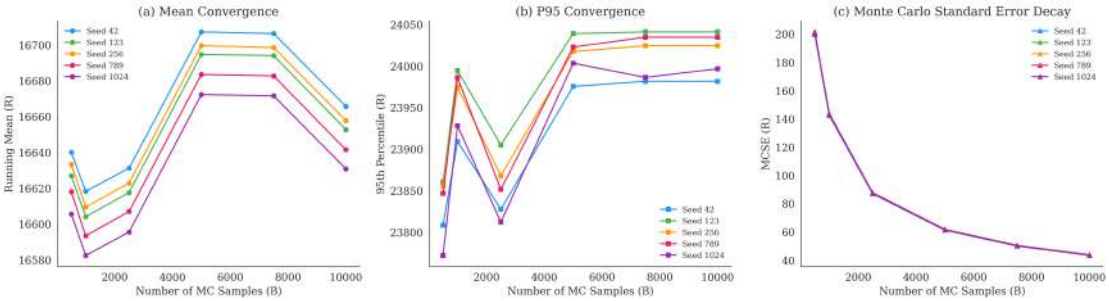


Figure 65: MC convergence rate and MCSE decay across five DNN seeds.

Annexure J.3 Cross-Seed Residual-Augmented Sensitivity

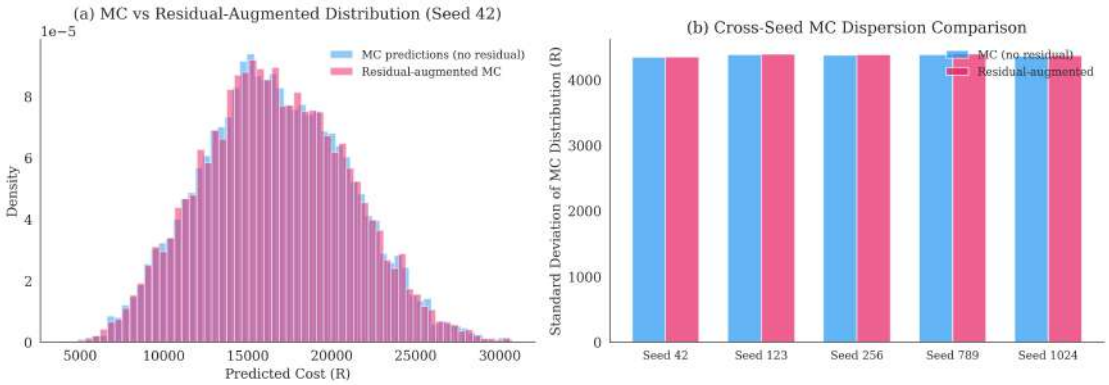


Figure 66: Base versus residual-augmented MC distributions with cross-seed dispersion.

Annexure J.4 Cross-Seed Shapley Attribution Diagnostics

Figures 67 to 68 present the cross-seed Shapley attribution diagnostics.

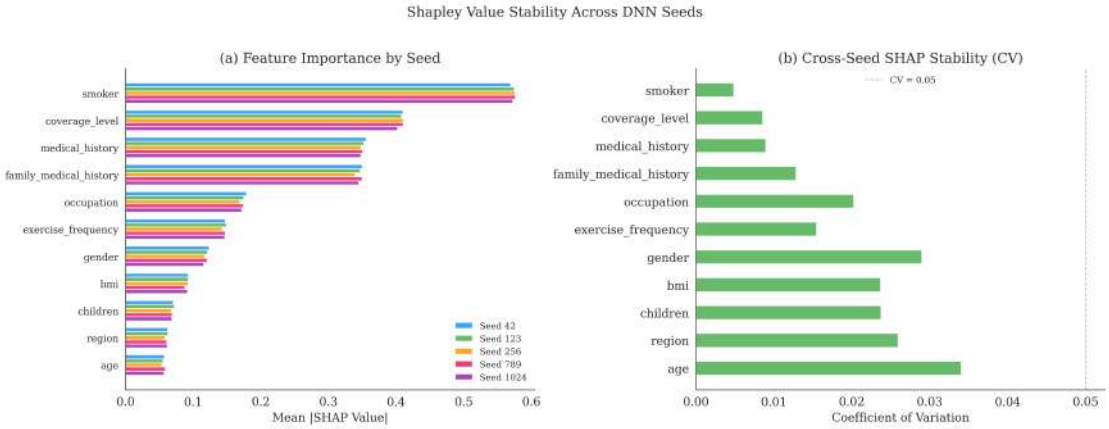


Figure 67: Global mean |SHAP| values across five DNN seeds.

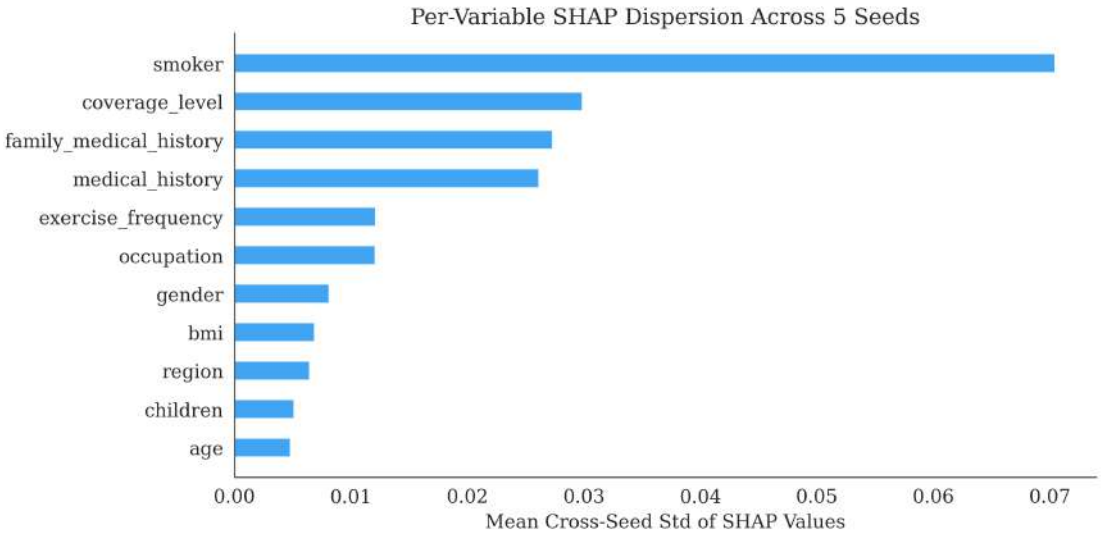


Figure 68: Cross-seed dispersion of mean |SHAP| values.

Annexure J.5 SHAP Dependence Plots

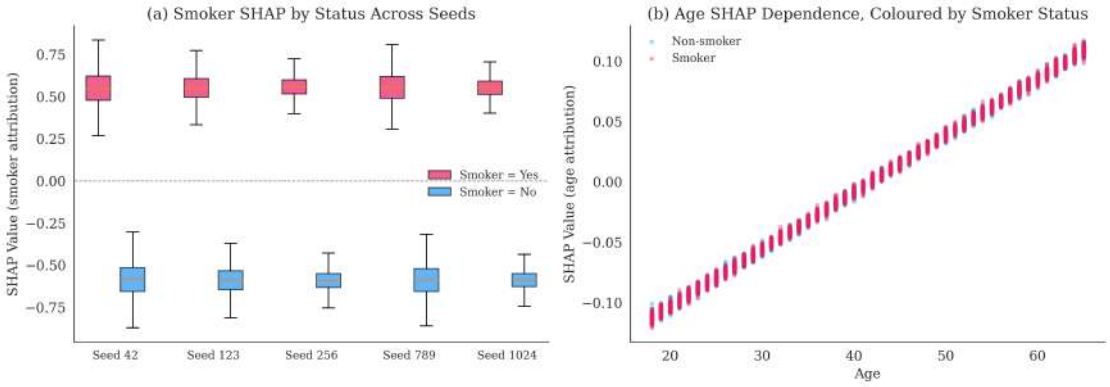


Figure 69: SHAP dependence plot for the `smoker` variable across five DNN seeds.

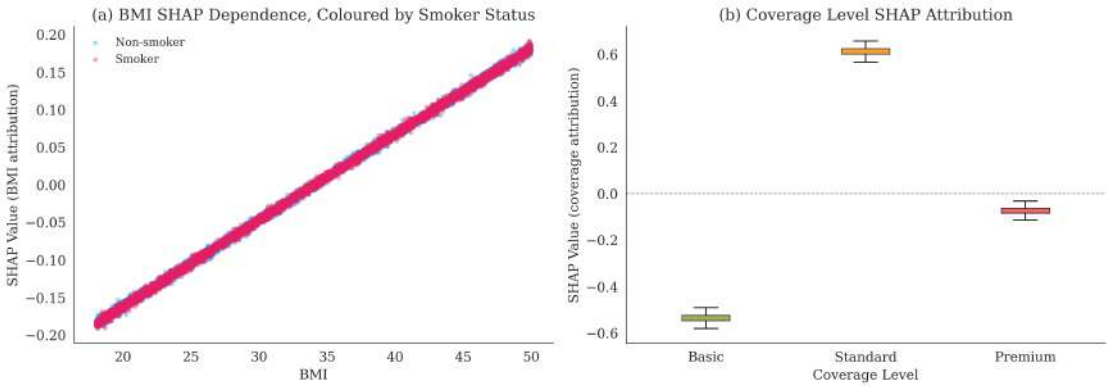


Figure 70: SHAP dependence plots for `bmi` coloured by `coverage_level`.

Annexure J.6 SHAP Interaction Diagnostics

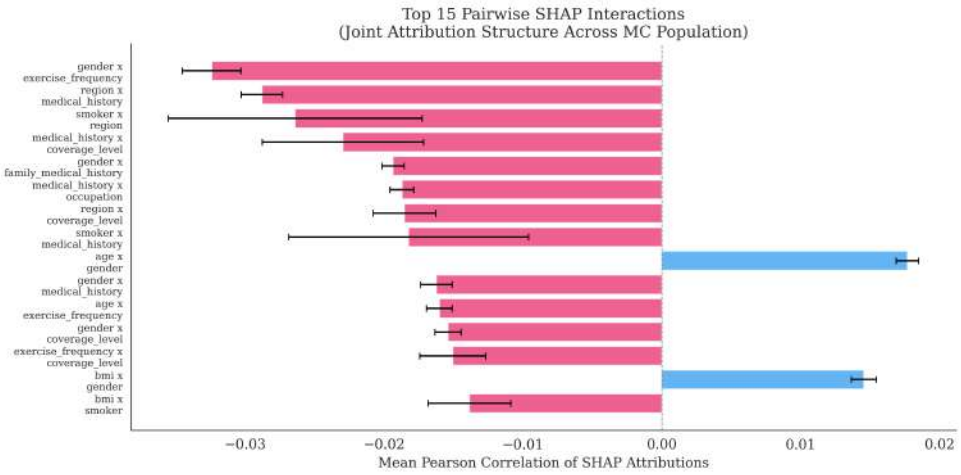


Figure 71: Top 15 pairwise SHAP interaction correlations.

Annexure K Glossary of Key Terms

The following definitions establish the meaning of key terms as they are used throughout this dissertation. Formal mathematical definitions are provided in the relevant chapters.

Deep neural network (DNN). A feed-forward artificial neural network with multiple hidden layers, each consisting of an affine transformation followed by a nonlinear activation function. In this dissertation, the DNN is used as a regression model to predict continuous insurance costs from policyholder attributes.

Funnel architecture. A network architecture in which the number of neurons decreases monotonically from the input layer to the output layer. The specific architecture used in this study has layer widths $256 \to 128 \to 64 \to 32 \to 16$.

Generalised linear model (GLM). A class of statistical models that generalise ordinary linear regression by allowing the response variable to follow any distribution in the exponential family and by relating the mean of the response to the linear predictor through a link function. In this dissertation, a Gamma GLM with a logarithmic link function serves as the actuarial benchmark.

Random forest. An ensemble learning method that constructs a collection of decision trees, each trained on a bootstrap sample of the data with a random subset of features considered at each split. The final prediction is the average of the individual tree predictions. In this dissertation, a random forest regressor serves as one of three benchmark models against which the DNN is compared.

Gradient-boosted trees (XGBoost). An ensemble method that builds an additive sequence of shallow decision trees, with each new tree fitted to the residuals of the preceding ensemble. XGBoost (Chen and Guestrin, 2016) is a widely used implementation that incorporates regularisation, approximate histogram-based splitting and early stopping. In this dissertation, an XGBoost regressor serves as one of three benchmark models.

Monte Carlo simulation. A computational method that uses repeated random sampling from a specified probability distribution to estimate the properties of a quantity of interest. In this study, Monte Carlo simulation generates synthetic policyholder profiles from fitted parametric marginals and propagates them through the trained network to obtain a predicted-cost distribution.

Model-implied predicted-cost distribution. The distribution of predicted costs obtained by evaluating the trained prediction function at a large sample of input

profiles drawn from a specified input distribution. This distribution characterises the range, central tendency, dispersion and shape of the model's output under population-level input variation.

Shapley value. A concept from cooperative game theory that assigns to each player in a coalition game a unique payoff satisfying the axioms of efficiency, symmetry, linearity and the null-player property. In the context of model interpretation, the "players" are input features and the "payoff" is the prediction for a given observation. The Shapley value quantifies the average marginal contribution of each feature across all possible feature coalitions.

SHAP (SHapley Additive exPlanations). A framework that operationalises the computation of approximate Shapley values for machine-learning models. The gradient-based variant used in this dissertation exploits the differentiability of the neural network to compute Shapley-value approximations efficiently.

Conditional Shapley analysis. The computation and summarisation of Shapley values within a subset of predictions that exceed a specified threshold. In this dissertation, conditional Shapley summaries are computed at the 90th, 95th and 99th percentiles of the predicted-cost distribution.

Feature-importance ranking reversal. The phenomenon whereby the ordering of features by their mean absolute Shapley values differs between the global summary and the conditional summary at a specific threshold. A ranking reversal indicates that the variables driving predictions in the specified region differ from those that dominate on average.

Data-leakage prevention. The set of protocols that ensure no information from the validation or test partitions contaminates the training process. In this dissertation, all preprocessing parameters are computed exclusively on the training partition and applied to the validation and test partitions without refitting.

Residual-augmented sensitivity analysis. A diagnostic procedure that adds scaled fitted residuals to the Monte Carlo predicted costs to examine whether the distributional conclusions are robust to the residual structure of the fitted model.